ATLAS: Master Build Plan

One plan, end to end: what we build, in what order, who builds which part, and how we know each part works.

Date	3 October 2026
Project	ATLAS, our entry for Huawei Custom Challenge #1, "Beyond Euclid: Interactive Experiences in Impossible Geometries"
Team	Four engineers, all working in Unity and C#
Hardware	iPhone 15 Pro Max (phone A) and iPhone 14 Pro Max (phone B) in a lens headset with the rear camera and LiDAR exposed · one MacBook Pro M3 Max (the only iOS build seat) · one Windows PC with an RTX 5070 (the Station)
This document	`documents/ATLAS_Master_Plan.md`, with figures in `documents/figures/`
Older documents	`ATLAS_Build_Pipeline_assets.md` (technology, still binding), `ATLAS_Engineering_Spec.md` and `ATLAS_Implementation_Brief.md` (background and detail; superseded where this plan differs, see Appendix D)

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0. Read me first

0.1 What this plan is

This is the single source of truth for building ATLAS. It merges the Engineering Spec, the Implementation Brief and the Build Pipeline, and it applies the decisions the team lead took on 3 October:

- The fixed, taped 3 × 3 m square is gone. The walker **scans their own room** and **walks freely** in it.
- The scanned room is rebuilt and becomes **tile zero** of a curved world.
- The build runs in **phases**. Phase 1 sets up all the technology, generates every non-Euclidean and 4D space, and connects the phone to the PC. The use cases follow.
- The work is split into **four parallel lanes**, one per engineer.
- There is **no calendar** in this plan. Work is ordered by dependency and released by gates. Tasks are written so they can be handed to an AI coding agent: each has inputs, outputs and a pass/fail test.

0.2 Order of authority

When two sources disagree, the higher one wins:

1. The team lead's decisions (recorded in §2 and used throughout this plan).
2. The Build Pipeline document (technology and implementation choices).
3. The Engineering Spec.
4. The Implementation Brief.

A few Build Pipeline lines are themselves overridden by newer decisions or by verified research (for example the taped square, and fog distance). Appendix D lists every such case.

0.3 How to read it

If you want to…	Read
Understand the project in five minutes	§1
See every decision and what it replaced	§2, Appendix D
Know what spaces exist and how the room becomes one	§3, §4
Understand the 4D and why the medical use case needs it	§5
Start building	§6 (architecture), §9 (build plan), Appendix C (set-up)
Work on the phone-to-PC connection	§7, Appendix B
Hit the frame rate and make it look good	§8
Know when something is done	§11
Build or check a use case, the demo, the risks, the open questions	§10, §13, §12, §14
Look up a term	Appendix E

0.4 Conventions

- **Must / should / may** have their usual engineering meaning.
- Source tags are short: (BP §4) is the Build Pipeline; (Spec §6.3) and (Brief §3.9) are the older documents; (R1) to (R6) are the research memos written for this plan (free-walk geometry, medical 4D, tile graphs, phone-to-PC link, performance, world generation); (derived) is our own mathematics. Every derived number in this plan was checked numerically; the scripts named in Appendix A recompute the geometry and worldtube numbers.
- **L** is the hyperbolic curvature radius in metres (the Spec calls it R). **R** in this plan is the sphere radius of the Small Planet. **ρ** is hyperbolic distance in units of L. **q** is the number of tiles that meet at a corner.
- Task ids look like `P1-B04`: phase, lane, number. Gates are `G0` to `G15`.

0.5 Ten rules

Breaking any of these is a defect.

1. **Safety runs on the phone and never touches the network.** The safety assembly cannot reference the link assembly.
2. **60 frames per second in stereo**, on both phones. Quality drops in steps before frame rate does. Never below 30.
3. **One metre walked is one metre moved**, in every space. No translation or rotation gains. The single exception is the airway's valve chamber (§10.2.3), where the world is re-aimed while the walker stands inside a closed, opaque chamber.
4. **Curvature changes only while the walker stands still.**
5. **Every action has a twin.** Each action works with the gamepad alone and with gaze dwell alone. The marker cube and any later hand input are optional extras: hand tracking is not in the Core build and no task depends on it; if the optional HoloKit spike (Appendix C.3) is ever merged, hand tracking runs only in a space that uses it (BP §0.5: it adds heat).
6. **Every curved view has a flat twin**: a map, a list or a chart of the same thing.
7. **Positions are a tile id plus local coordinates.** Never a global floating-point coordinate.
8. **One shader, one palette, fixed budgets**, checked at every gate.
9. **No clinical claims.** Every medical object on screen is labelled real, illustrative, or model-generated.
10. **No camera frames are stored.** Object photos are crops, held in memory, and deleted when the mesh is made. The one exception is the hosted fallback, which is disclosed (§7.4).

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1. ATLAS on one page

1.1 The challenge

Build an interactive application in which non-Euclidean geometry is central and meaningful. The best entries "make the user feel it, interact with it, and discover something new about space itself".

Criterion	Weight	What the judges look for
Geometric creativity	30 %	Original concept; geometry that is not cosmetic; correct mathematics
User experience and interaction	30 %	Intuitive navigation and feedback; purposeful, not disorienting for its own sake
Graphics and technical execution	25 %	Clarity and polish; smooth (30+ fps); code quality; efficient geometric computation
Documentation and presentation	15 %	Clear maths and technical explanation; commented code; a compelling demo
Bonus		Beyond gaming; educational value; exceptional polish; a novel technical approach

The challenge names four families of geometry: hyperbolic, spherical, 4D projections, and impossible or Escher-like spaces. ATLAS covers all four.

1.2 The concept

Your own room becomes the first tile of other universes, and you walk through them on your own feet.

- You scan the room with the phone. ATLAS measures the free floor and rebuilds the room around you, with your furniture regenerated as 3D models while you watch.
- The size of your walkable floor sets the curvature. A dial then asks one question: *how many copies of your room meet at each corner?* Three gives a small planet. Four is ordinary flat space. Five or six gives hyperbolic space with endless rooms.
- You feel the difference with your legs. Walk a square with four left turns: in flat space you are home. On the planet you were home after three corners. In hyperbolic space you need a fifth.
- The same engine runs glued rooms (a torus, a Klein bottle), rooms that are bigger inside than outside, and a slicer for four-dimensional objects.
- Then the geometry is put to work. In the **Lung Mission**, the airway tree is walkable because hyperbolic space has room for every branch, and a tumour that moves with breathing becomes a four-dimensional object you can slice, project and tilt.

1.3 What a first-time walker experiences

1. **Scan** (phone in hand). Sweep the room. The free floor lights up; confirm the outline.
2. **Home**. Headset on. The room is rebuilt 1:1 in a clean diagram style. Boxes stand where the furniture is; largest first, each box turns into a generated model of that object as the Station finishes it (at most 20 per room).
3. **The dial**. Stand on the marked spot and turn the dial from 4 to 5. The walls sink to kerbs, corner posts stay, and beyond them an endless tiling of rooms appears.
4. **The 3-4-5 walk**. Follow the glowing square: walk, turn left, repeat. A counter shows the corners. It takes five to come back. Turn the dial to 3 and the same square closes after three.
5. **Two ways to be hyperbolic**. In *Curved Walk* everything curves smoothly and the world slowly turns as you lap the room. Switch to *Aligned Cells*: the floor is cut into flat cells that stay locked to the real walls, and each lap round a corner post leaves you in a different room on the same piece of floor.
6. **Exhibits**. Glued rooms (see your own back; meet your mirror image), impossible rooms (a booth that contains a hall), the small planet, and the 4D toybox (push a tesseract through your room).
7. **Lung Mission** (Phase 2). Walk the airway tree, discover that the map you navigated by was one frozen slice of a breathing, four-dimensional lung, choose where and when to place a needle, and leave with a plan card.

A teammate watches all of this on the Station: a live mirror of the walker's view, a flat map of where they are in the tiling, sliders that tune the world while it runs, and a performance dashboard.

1.4 How this answers the rubric

Criterion	Our answer	Evidence we can show
Geometric creativity	The scanned room as tile zero; curvature derived from the room; smooth and corner-concentrated curvature on one tile graph; four geometry families on one kernel	The 3-4-5 walk closing to within 25 cm; unit tests of the mathematics (holonomy equals angle defect; tile counts match the known growth sequence); maths notes
User experience	Real walking with no gains; one rule and one "aha" per space; twins for every action; a flat map for every curved view; warning and haptic before any bump	Comfort scores per minute; pointing-home error; ease ratings; onboarding time
Graphics and technical execution	A consistent "lit architectural diagram" look; 60 fps stereo on a phone; exact integer tile ids; live AI-generated furniture	The metrics table (§11.2) and a 15-minute soak chart from the telemetry log
Documentation and presentation	This plan, maths notes with figures, commented code, a Spectator view that makes the demo watchable	§13
Bonus	Medical education use; teaches geometry by walking it; to our knowledge a new combination (curvature fitted to the scanned room; smooth and cell-based hyperbolic walking on one tile graph)	Lung Mission; the 3-4-5 walk

1.5 What exists at the end of each phase

Phase	Ends at gate	What exists
P1 Foundation	G7	The whole toolchain. Every space S0–S6 generated and rendered on the phone at budget, with the scanned room's walls and furniture as content: in Home, as tile zero, among the Glued copies and in Aligned Cells. The curvature dial and the 3-4-5 walk. Scan → RoomModel → Home with live AI furniture, and a built-in fallback set for when the Forge is unreachable. Bump Guard with its warning cue and haptics. The Station link with all four functions: Spectator mirror, live tuning and Console, telemetry dashboard, scan and asset round-trip. An on-device parameter menu for when the Station is unreachable
P2 Use cases I	G11	Geometry Playground (the guided experience of §1.3, steps 1–6) and Lung Mission (step 7), built in parallel
P3 Use cases II	G12, G13	Shared House / Post Circling and FieldFit
P4 Finish	G15	Polish, evaluation with outside walkers, demo operations, documentation, video, golden build

2. Decisions

Every choice that shapes the build, in one table. "Source" says who decided: **Lead** is a decision taken by the team lead on 3 October; (BP) is the Build Pipeline; (R1–R6) are the research memos; (derived) is our own mathematics. The last column names what the choice replaces in the older documents; Appendix D has the full list.

#	Decision	Choice	Source	Replaces
D1	What ATLAS is	A platform that generates walkable non-Euclidean and 4D spaces from a scanned room, plus use cases built on it	Lead	"Blades" on a taped stage
D2	Play area	Free walking in the room the walker scans. No tape, no fixed square	Lead	Taped 3 × 3 m square
D3	Phasing	P1 technology, all spaces and the Station link · P2 Geometry Playground and Lung Mission in parallel · P3 Shared House and FieldFit · P4 finish	Lead	The Spec's Phase 0–5
D4	Scheduling	No calendar. Dependency order and gates only. Tasks are written to be handed to AI coding agents	Lead	Week-based templates, hour budgets, time-based cut lists
D5	Team	Four lanes by subsystem: A Platform, Look and Flat Spaces · B Kernel and Curved Worlds · C Scan, Room and Safety · D Link, Station and Forge	Lead	Seven ownership areas left blank
D6	Engine and versions	Unity 6000.3 LTS at or above 6000.3.21f1, URP forward with Render Graph, AR Foundation 6.5.x, ARKit as the only XR loader, Xcode 26, IL2CPP, Metal, iOS 17 target	(BP §0, §0.5), (R5)	Unspecified patch; Cardboard XR plugin
D7	Stereo	Our own two-camera rig (±32 mm), VR mode, no camera background. Barrel pre-distortion for the lens headset as the single final blit	(BP §0, §1), Lead (headset)	One camera drawing both eyes; Cardboard C API
D8	Pose and latency	`TrackedPoseDriver` on Update only. Pose jumps over 0.2 m per frame are ignored by the geometry. Gyro rotation prediction is optional, behind a switch. No late warp	(BP §4), (R5)	Core Motion prediction with a late rotational warp
D9	Hyperbolic walking	**Both modes on one kernel and one tile graph.** Mode 1 *Curved Walk*: smooth $H^2 \times E$, the main world. Mode 2 *Aligned Cells*: flat cells locked to the real floor, tile identity depends on the path	Lead	"Portals plus curved wings" with tile mapping as a later option
D10	Projection	Log map (true distance and direction from the head) for everything walked. Klein and Poincaré only for maps	(BP §0, §0.5)	Klein chart in the vertex shader
D11	Curvature	Taken from the room by the **loop-fit rule**: lap square s = clamp(min(W, D) − 1.0 m, 1.0 m, 3.2 m), with W and D wall to wall in an empty rectangular room (the scan gives the free rectangle Wf × Df, already 0.30 m inside walls and furniture, so W = Wf + 0.6 m and the code uses s = clamp(min(Wf, Df) − 0.4 m, 1.0 m, 3.2 m)); L = s / 1.0613 so that five right-angle corners close the loop. Dial positions are "rooms per corner" q = 3, 4, 5, 6	Lead (room as tile zero), (derived), (R1)	A fixed curvature radius (the Spec's R) of about 0.8–0.94 m forced by 1 m cells, and about 2.2 m for the airway
D12	Addressing and precision	Exact integer tile ids for {4,5} (six integers in $\mathbb{Z}[\varphi]$). Walker pose = tile id plus a tile-local 3 × 3 matrix in double precision. Anchors are computed in double and cast to float last	(BP §0.5), (R3)	Offline lookup tables; global coordinates
D13	Culling and fog	Fog starts at ρ − 1 and is fully opaque at ρ (default ρ = 3). Every tile that reaches inside that radius and can be in view is drawn; the test runs on the CPU once per frame (§8.5, Appendix A.3). 45 tile centres lie within ρ = 3; at most 59 tiles are drawn with a level gaze and 71 at any pitch	(R3), (R5), (R6), (derived)	Fog at 5–8 units (373 tile centres within ρ = 5)
D14	Draw path	One shared shader, SRP Batcher, one pooled material instance per tile or room, no MaterialPropertyBlocks. `RenderMeshInstanced` for tile shells only if the render thread is the measured bottleneck	(BP §4), (R3), (R5)	Per-module GPU instancing
D15	Glued spaces	The universal cover is drawn fixed to the floor and the walker simply walks through the copies, including mirrored Klein rows	Lead (free walk), (R6)	"Do not cross a glued edge" (BP §4); 1.5 m cell in a taped square
D16	Portals	Stencil via Render Objects features, references 1–15, no recursion, plane clip in the shader, rooms authored in play-space coordinates so portal transforms are the identity	(BP §4), (R3)	Recursion depth 2; custom render lists
D17	4D	`Mesh4D` : tetrahedra in $\mathbb{R}^4$ cut by a hyperplane on the CPU with Burst. One code path for polytopes and for the medical worldtube. The vertex-shader slicer only above about 50,000 tetrahedra	(BP §0, §4), (R2)	Compute-shader slicer with indirect draw

#	Decision	Choice	Source	Replaces
D18	The room's role	"My room is tile zero." Home is the room rebuilt 1:1, flat and registered to reality. Turning the dial places it in curved space as the home tile of an endless generated world	Lead	Hub as a neutral Euclidean lobby
D19	Safety	**Bump Guard**: the scan's static map plus live LiDAR depth predicts contact and gives a warning cue and haptic pulses. No passthrough, no ghost of the room, no freezing of the world. LiDAR depth stays on while walking; meshing and plane detection are off after the scan	Lead	Passthrough Safe Harbour within 300 ms; taped outline; depth off after set-up (BP §5)
D20	Furniture	Scanned objects are photographed, sent to the Station, generated as meshes (TripoSG on the RTX 5070, hosted API as fallback), cleaned and returned live. At most 20 objects per room receive a generated mesh, largest boxes first, and together they never exceed 60,000 triangles; further objects stay placeholder boxes. A placeholder box, tinted by category, stands in until its mesh arrives, and a built-in fallback set replaces the boxes when the Forge is unreachable (§4.5, §7.4)	Lead (live, per object); the cap of 20 (derived)	Kit props; pre-generated assets
D21	Scanning	Both paths: the walker self-scans (headline), or the team pre-scans and the app relocalises from a saved world map (fast path). Scan routes in order of robustness: manual outline, LiDAR mesh plus planes, RoomPlan boxes, Swift RoomPlan bridge	Lead, (R1)	Venue scanned the day before
D22	Phone ↔ Station link	LiteNetLib on both ends. A .NET Hub on the Station terminates it. Phase 1 ends with all four functions: Spectator mirror, live tuning and Console, telemetry, scan and asset round-trip. Assets and uploads travel over HTTP to the Forge	(BP §0.5), Lead, (R4)	Python FastAPI link, WebSocket, raw UDP, MessagePack
D23	Look	"Lit architectural diagram": one palette texture, one material, vertex-colour ambient occlusion, edge and tile lines in the shader, fog in the room's hue, emissive accents. Dark palette. No post-processing, no shadows, no bloom	(BP §2), (R5)	Lit shader with probes, LUTs and trim sheets
D24	Budgets	At most 300 draw calls and 500,000 triangles per frame (both eyes). Render scale 0.8 with 4× MSAA to start. A quality ladder steps down before the frame rate does	(BP §5), (R5)	150k triangles and 100 draws per eye
D25	Medical 4D	Four pieces: 4D toybox, breathing worldtube, breath lane walk, scanner-artefact demo. One real public case (TCIA 4D-Lung patient 101) beside a labelled illustrative twin. Airway: procedural and labelled, with an optional real proximal tree	Lead, (BP §0.5), (R2)	Time-only 4D; case-card deck; segmentation on a cluster
D26	Claims	Education and research visualisation. "Not for clinical use" on screen. Measured, not asserted	(Brief §2.4)	unchanged

Three consequences worth stating plainly.

- *Curved Walk cannot stay registered to the real room.* A flat floor cannot be laid on a curved one without slipping, so in Mode 1 the virtual room and furniture drift away from the real ones as the walker laps. This is a theorem, not a bug (§4.6). Bump Guard, not the picture, is what protects the walker there. In every flat space (Home, Glued, Impossible Rooms) and in Aligned Cells, what you see is where things really are.
- *Bump Guard gives a warning and a haptic, nothing else.* That is the lead's decision. It puts weight on scan quality, conservative thresholds and keeping LiDAR depth on, and we recommend a human spotter whenever a stranger wears the headset (§4.10, §12).
- *Live furniture depends on the Station.* When the Forge is unreachable each placeholder box is replaced by a built-in fallback mesh of its category; nothing else changes (§7.4).

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3. The spaces we generate

Phase 1 delivers seven spaces, S0 to S6. They share one geometry kernel, one shader and one host scene (SpaceLab), and every one of them is generated from the scanned room rather than authored for a fixed stage.

Two layers exist at all times and must not be confused:

- **The world layer** is everything the walker sees. It is generated, and it may be flat, curved, glued or four-dimensional.
- **The physical layer** is the scan: where the real floor, walls and furniture are, in ARKit's tracking coordinates. Only Bump Guard reads it. It is never drawn.

In a flat space the two layers coincide, so what you see is where things are. In a curved space they cannot coincide (§4.6).

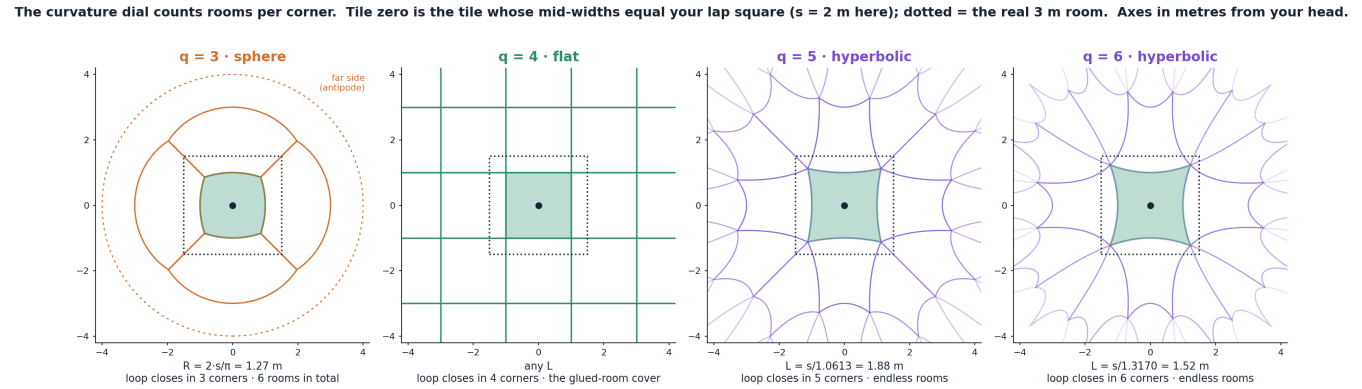
3.1 Catalogue

Id	Space	Geometry	What is generated, and from what	Matches the real room?	The "aha"	Flat twin
S0	Home	Flat	Floor, walls and door openings rebuilt from the scan on a 0.25 m grid; one AI-generated mesh per scanned object, for at most 20 objects, largest first (§4.5)	Yes, 1:1	"This is my room." It is the calibration and the place every journey starts and ends	Floor plan
S1	Glued	Flat, wrapped: torus T^2, Klein bottle, 3-torus T^3	One open-frame cell cut from the free floor (corner posts, floor grid, one chiral landmark); copies of it laid out as a lattice fixed to the floor (5 × 5 around the walker for T^2 and the Klein bottle, 3 × 3 × 3 for T^3); mirrored rows for the Klein bottle; a self-avatar in every copy but the walker's own. The room itself is not repeated: its walls sink to kerbs and its furniture is drawn once, where it really is	Yes	Look ahead and see your own back. Walk east and re-enter from the west. In the Klein bottle your copy raises the other hand	Lattice plan with the fundamental cell
S2	Curved Walk (Mode 1)	$H^2 \times E$: hyperbolic floor, ordinary height	{4,5} tiling generated lazily by exact tile id; tile zero carries the room; other tiles get an archetype and props from a hash of their id	Only at the start	Five rooms meet at every corner. A closed square on the floor is not closed in the world. There is far more room than there should be	Poincaré disc map
S3	Aligned Cells (Mode 2)	The same tiling, built from flat pieces	The free floor cut into square cells; each cell shows one flat tile; which tile depends on the path walked	Yes	Walk once round a corner post, through its four cells, and you are standing in a fifth room. All the curvature sits at the corners	The same disc map
S4	Small Planet	$S^2 \times E$: spherical floor, ordinary height	{4,3} tiling: six tiles, the faces of a cube blown onto a sphere; all always loaded	Only at the start	Three right-angle turns make a closed triangle. The world is finite: six rooms, and the map counts them	Unfolded cube net
S5	Impossible Rooms	Flat rooms joined by portals	Rooms sized from the free rectangle and stitched by stencil portals: a booth that contains a hall; a corridor with four right turns that arrives somewhere new	Yes, each room	The building is bigger than the room	Room graph
S6	Slicer	Flat $\mathbb{R}^4$, shown as 3D slices and 3D shadows; and $\mathbb{R}^3 \times S^1$	Tetrahedral meshes of 4D polytopes (5-cell, tesseract, 16-cell, 24-cell) and periodic tubes, cut by a hyperplane every frame	Yes (the object floats in the room)	A solid changes shape as it moves in a direction you cannot point to	Slice film strip and the shadow

The challenge's four families map onto these directly: hyperbolic (S2, S3), spherical (S4), 4D projections (S6), impossible and Escher-like spaces (S1, S5).

3.2 The dial: how many rooms meet at a corner?

One question drives the curved spaces. Take the lap square, the largest square the walker can comfortably lap inside the scanned room (side s, §4.4). Tile the world with four-sided rooms that measure s across the middle, so that their centres are s apart. How many of them meet at each corner?



Dial q	Space	Curvature radius from the lap square	A square lap closes after	World
3	Small Planet (S4)	R = 2s / π	3 corners	Finite: six rooms
4	Flat (S0, and the cover of S1)	any	4 corners	The ordinary plane
5	Hyperbolic (S2, S3), the default	L = s / 1.0613	5 corners	Endless
6	Hyperbolic, stronger	L = s / 1.3170	6 corners	Endless, grows faster

Where the numbers come from (derived, checked numerically; formulas in Appendix A):

- In a regular tiling with four-sided tiles and q tiles per corner, the distance between neighbouring tile centres is 2·asinh(√cos(2π/q)) in units of L. For q = 5 that is 1.0613, and its hyperbolic cosine is exactly the golden ratio φ.
- A walker who goes from one tile centre to the next and turns left 90° each time circles one corner post and passes through the q tiles around it. So a square lap of side s closes after exactly q corners when s equals the centre spacing. Setting them equal gives the table.
- On the sphere the same argument gives the triangle with three right angles, whose sides are a quarter of the circumference.

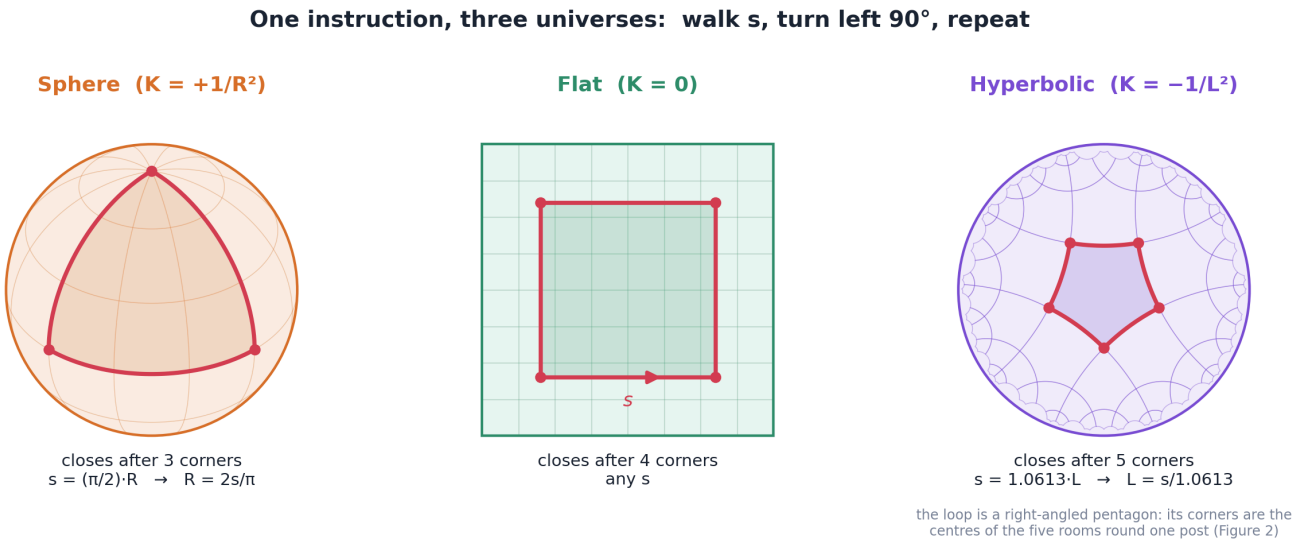
Typical values (derived):

Real room (wall to wall, empty)	Lap square s	L at q = 5	R at q = 3	L at q = 6	Tile edge at q = 5	Fog radius (ρ = 3)
2.5 × 2.5 m	1.5 m	1.41 m	0.95 m	1.14 m	1.77 m	4.2 m
3 × 3 m	2.0 m	1.88 m	1.27 m	1.52 m	2.36 m	5.7 m
3 × 4 m	2.0 m	1.88 m	1.27 m	1.52 m	2.36 m	5.7 m
4 × 5 m	3.0 m	2.83 m	1.91 m	2.28 m	3.54 m	8.5 m
6 × 8 m	3.2 m (cap)	3.02 m	2.04 m	2.43 m	3.78 m	9.0 m

The table is for empty rectangular rooms. Furniture, or a room that is not a rectangle, makes the free rectangle smaller, and the lap square follows the free rectangle (§4.4).

A bigger room gives gentler curvature for the same effect. That is a direct benefit of dropping the fixed square: the old plan's 1 m cells forced a radius near 0.9 m, which its own authors called strong and disorienting (Brief §3.7).

3.3 The 3-4-5 walk



The instruction is the same in every space: *walk s, turn left 90°, repeat.* The guide is a beacon s metres straight ahead and, when you reach it, the next one to your left. A flag marks where you started. The beacons live in the world, not on the real floor.

- At q = 4 the flag is back under your feet after four corners.
- At q = 3 it is there after three.
- At q = 5 you return to where you physically started after four corners, but the flag is one more side away: you need a fifth.

What one physical lap does in hyperbolic space at the loop-fit curvature (derived, verified to 1e-12):

- You end displaced by exactly s, one tile spacing, from where you began, and the world has turned a quarter-turn (90°) round you.
- Equivalently, the whole world has turned 72° about the centre of the five-sided loop. In the room that centre lies 0.092·L from the room centre (0.17 m at L = 1.88 m); it is a corner post only when the loop runs through tile centres, as Figure 2 below draws it. Five laps bring everything back.

For smaller loops the turn per lap is smaller and approaches area ÷ L^2 (a 1 m square at L = 1.88 m turns the world by about 16°). This is curvature measured with your feet, and the read-out shows it live: corners walked, distance from the flag, and the turn accumulated.

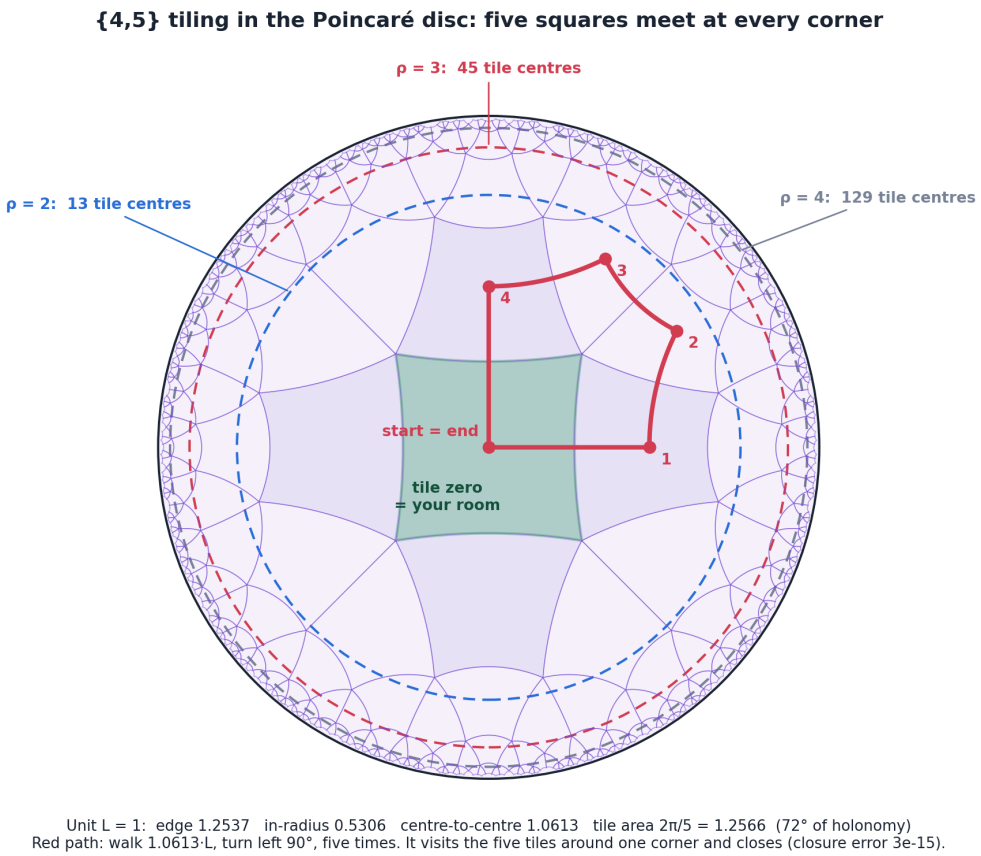


Figure 2 draws the loop through tile centres, so that it circles one corner post. In the room the same loop is walked on the lap square, whose corners lie 0.17 m inside tile zero's corner posts (at L = 1.88 m).

3.4 How the curved worlds are built

Tiles are generated in code, not modelled. The curvature radius depends on the scanned room, so tile meshes cannot be authored for one fixed L. The generator (task `P1-B05`) builds each tile like this:

1. Lay a regular grid on the tile's square in the Klein model, where the tile really is a straight-edged square (half-width 0.4859 for {4,5}).
2. Move every vertex to its position in the log chart: same direction from the tile centre, distance atanh of the Klein radius, times L.
3. Add kerbs along the edges, a wedge of a corner post at each corner, palette colours and vertex ambient occlusion. Each wedge is bounded by the tile's own two edges: 72° in the curved metric, which is 65.8° in mesh coordinates for {4,5} (Appendix A.6). It is built in Klein coordinates and mapped by step 2, and five wedges from five tiles make one post.

Because the grid lines are geodesics in the Klein model, neighbouring tiles meet exactly along their shared edge. The warp shader then needs nothing special: it receives tile-local positions and applies the log map centred on the walker's head (§4.6).

Three levels of detail keep the triangle count flat as the tile count grows. They are set by grid count, not by edge length: n = 16 grid cells per side on the walker's current tile and the 12 tiles that touch it; n = 8 on every other drawn tile; n = 4 only on the high tier, beyond ρ = 3 (far tiles are a few pixels wide). Every level keeps the same 16 boundary segments per edge, so neighbours of different level share their edge vertices exactly. The three levels hold at most 1,200, 500 and 250 triangles, for any L. They are accepted by error, not by length: the worst screen-space chord error is at most 0.15° from an eye height of 1.6 m at L = 1.4, 1.9 and 3.0 m. The rule "longest edge at most 0.25 m" stays for the Home shell, furniture and authored props.

Content is a pure function of the tile id. `archetype = table[hash(tileId, seed)]`, constrained by region. The same seed gives the same world on the phone and on the Station, which is what lets the Spectator redraw the walker's view from a few numbers.

Archetype	What it is	Purpose
Home	Tile zero and any tile under the room's footprint: no generated props	The room itself is the content
Plaza	Open floor with the four corner wedges	Default; keeps sight lines open
Corridor	Low kerb walls (0.9 m) on two opposite edges	Shows "parallel" walls diverging
Doorway	One edge with a framed opening	Region gates
Tower	A tall emissive beacon at the tile centre	Navigation; at least one within ρ = 2.5 of every tile
Span A / Span B	Two neighbouring tiles that each carry half of one long object	A straight beam visibly bends across tiles
Garden	Low props only (under 0.5 m)	Variety without blocking the view

Regions are the four sectors of the tiling: a tile belongs to the edge of tile zero through which its shortest path leaves; when two edges tie, the lower edge index wins. Each has its own hue; the fog and the background take the hue of the region the walker is standing in (§8.1). The world reads as four "lands" that touch only along their borders.

WorldLint is an editor check that every generated world must pass before it reaches the phone: every tile reachable; a tower visible from every tile; every tile has its four wedges; at least one spanning object near the start; draw calls and triangles inside budget from every tile centre and eight headings; no coincident faces between neighbours; the region of every tile to depth 6 is the same on the phone and on the Station; the same seed gives the same world twice.

3.5 What makes a curved world legible

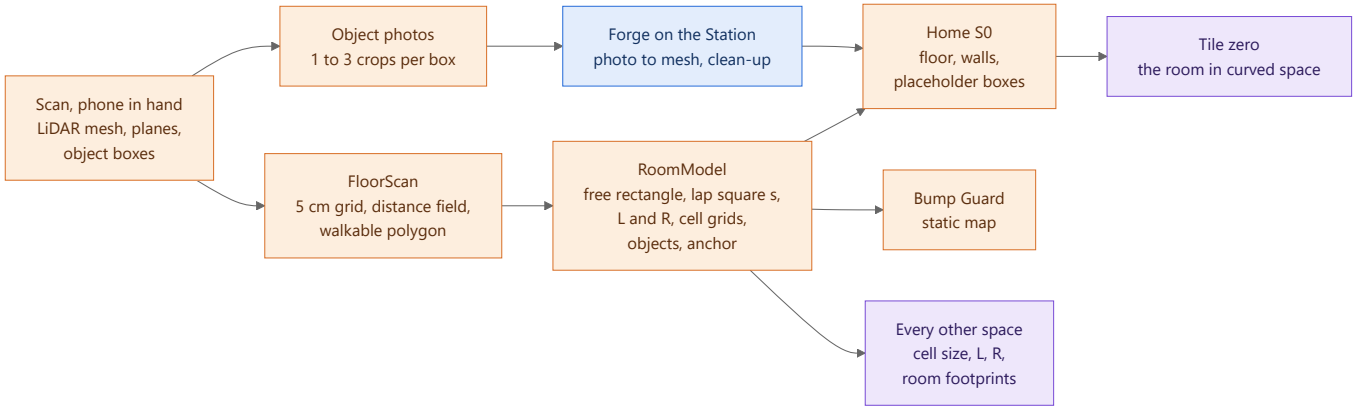
These rules come from the one user-tested hyperbolic walking system we found (Holonomy, TU Delft) and from our own comfort constraints (R1, R6).

Rule	Value
A post at every tile corner	Always. It is how "five rooms meet here" is read at a glance
Tile edges drawn on the floor	Always, in the shader
One object that spans two tiles, visible from the start	At least one
A tall landmark visible from every tile	Tower within ρ = 2.5; a pole beacon on the Small Planet
Open sight lines	Kerbs of 0.9 m, not walls; no tile closed on four sides
Fog in the hue of the walker's region	Starts at ρ = 2, fully opaque at ρ = 3; every tile that reaches inside that radius and can be in view is drawn (§8.5)
Flat twin shown after the first lap, not before	A map shown too early distracts from noticing the geometry (Spec §10.2)
A home beacon	Arrow and distance to tile zero; people steer home as if space were flat (R1)
Solid-looking things near the head	Fade within 0.3 m: virtual walls in Curved Walk are not real walls

4. Scan your room into curved space

This section is the design that replaces the taped square. It covers the scan, the data it produces, how the room becomes Home and then tile zero, how the two hyperbolic modes work, and how the walker is kept from bumping into things.

4.1 The pipeline



Everything downstream reads the **RoomModel**. No space, and no use case, contains a room size of its own.

4.2 Scanning

Two paths, both supported (D21):

- **Self-scan (the headline).** The walker holds the phone, sweeps the room, confirms the outline shown on the floor, and puts the phone in the headset.
- **Pre-scan (the fast path).** A team member scans beforehand. The app saves the `ARWorldMap` with the RoomModel and relocalises when the next walker starts.

Four scan routes, from most robust to most detailed. Each later route adds to the earlier ones; the app always has a result from the highest route that succeeded.

Route	What it uses	What it gives	Status
1 Manual outline	Taps on the floor corners, or the path walked along the walls	Floor polygon	Core; always available
2 FloorScan	LiDAR mesh with per-triangle classes, plus plane detection	Free floor with holes for furniture, wall directions, obstacle blobs	Core
3 Object boxes	RoomPlan through AR Foundation's bounding-box manager (iOS 17+)	One oriented box per piece of furniture, with a category	Core; needed for live furniture
4 Wall bridge	A Swift plugin around `RoomCaptureSession(arSession:)`	Clean walls, doors, windows and openings	Extended

Facts that constrain the design (R1, verified against Apple and Unity documentation): AR Foundation exposes RoomPlan only as furniture boxes, so walls come from planes and the mesh unless route 4 is built; the LiDAR reaches about 5 m; glass and mirrors produce missing or phantom floor, which is why unknown cells are never treated as free and the walker confirms the outline; meshing and plane detection are switched off as soon as the scan is confirmed because they are a major source of heat.

What each of the phone's sensors does in ATLAS:

Sensor	During the scan	While walking
Rear camera	ARKit tracking; key frames from which object photos are cropped	ARKit tracking

Sensor	During the scan	While walking
LiDAR	Classified room mesh, planes, furniture boxes; meshing and plane detection stop when the scan is confirmed	Live depth image for Bump Guard: 256 × 192, one with every camera frame (60 Hz), about 0.3 to 5 m, forwards only (§4.10)
IMU (gyroscope, accelerometer)	Part of ARKit's visual-inertial tracking	The same; plus the gyro rate for the optional rotation prediction (§8.7)
Taptic engine		Bump Guard pulses
Screen, through the headset's lever	Confirm the outline (in hand)	One-button input twin
Wi-Fi	Uploads to the Forge	Link to the Station

FloorScan, in outline (R1 §C.3):

```
y0    = height of the largest plane classified Floor
grid  = 5 cm cells over the mesh's footprint
mark FLOOR  under every Floor triangle within 8 cm of y0, and under the walker's own path
mark OBST   under every triangle with a vertex between 0.10 m and 2.0 m above y0
FREE  = FLOOR minus OBST grown by one cell, intersected with the outline the walker confirmed; small holes closed; unseen cells
stay not-free
SDF   = signed distance to the edge of FREE, in metres                    # reads 0.05 m less than the true distance to a wall
or obstacle
WALK  = the connected part of {SDF > 0.25 m} that contains the walker     # 0.30 m from every wall and obstacle = body half-
width + tracking error
axis  = dominant wall direction (length-weighted wall-plane normals)
rect  = largest rectangle inside WALK aligned to axis                      # the free rectangle, Wf × Df
boxes = oriented boxes of the OBST blobs, merged with the RoomPlan boxes
anchor = ARAnchor at the centre of rect; everything is stored relative to it
```

It runs off the main thread, once when the sweep ends and again if the walker corrects the outline.

4.3 The RoomModel

Field	Content	Used by
`anchor`	Pose of the ARAnchor at the centre of the free rectangle; axis direction	Everything
`sdf`	Distance field on the 5 cm grid (positive inside free floor)	Bump Guard
`walkable`	Polygon with holes	Home floor, room fitting, Station view
`freeRect`	Wf × Df, centred on the anchor; already 0.30 m inside walls and furniture	Loop-fit, Glued cell, Aligned Cells grid, Impossible Rooms, breath lane
`lapSquare`	Side s; centred on the anchor (§4.4)	3-4-5 walk, loop-fit
`curvature`	L for q = 5 and 6, R for q = 3	S2, S3, S4
`cells`	Glued cell size (rule below); Aligned Cells grid (n × m, side g; §4.7)	S1, S3
`standing`	True when the free rectangle is smaller than 1.5 × 1.5 m (§4.4)	Dial, S1, S6
`walls`	Wall segments with height; door and window openings where known	Home
`objects`	Per object: id, category, oriented box, state of its mesh (placeholder, requested, ready, built-in), content hash	Home, Forge, asset client
`quality`	Share of unknown cells inside the outline; scan route reached	Onboarding, risk checks

Glued cell (S1). Along each side of the free rectangle: n = max(1, round(Wf / 1.5 m)), cell = Wf / n; likewise for Df (R6 §3.2). Cells come out between 1.12 and 2.25 m in any room large enough to walk (derived). For empty rooms measured wall to wall: a 3 × 3 m room (free rectangle 2.4 × 2.4 m) gives 2 × 2 cells of 1.20 m; a 4 × 5 m room gives 2 × 3 cells of 1.70 × 1.47 m; a 6 × 8 m room gives 4 × 5 cells of 1.35 × 1.48 m.

The schema and three sample rooms (3 × 3 m; a 3 × 4 m L-shape; 4 × 5 m with a table in the middle) are part of the contracts frozen at G0, so every lane can build against them before the first real scan exists. The same contracts hold a performance fixture, `budget_room`: 4 × 5 m with twelve stand-in objects of exactly 5,000 triangles each and no edge over

0.25 m. Budgets are measured on it; the three sample rooms stay for functional tests.

4.4 The loop-fit rule

```
s = clamp( min(W, D) − 1.0 m , 1.0 m , 3.2 m )      # the rule (D11): W, D wall to wall in an empty rectangular room
  = clamp( min(Wf, Df) − 0.4 m , 1.0 m , 3.2 m )    # as computed from the scan: W = Wf + 0.6 m, D = Df + 0.6 m
L = s / 1.0613      (q = 5)        L = s / 1.3170   (q = 6)        R = 2 s / π   (q = 3)
```

The scan never measures W and D. It produces the free rectangle Wf × Df (§4.3), which already lies 0.30 m inside walls and furniture, so the code always uses the second line. In an empty rectangular room the two lines give the same s. The lap square is centred on the anchor and lies at least 0.20 m inside the free rectangle, so a walker on the lap keeps at least 0.5 m from the nearest wall or scanned obstacle. The table in §3.2 gives the resulting values.

- **Comfort floor.** The Curved-walk spike (gate G2) measures comfort at the loop-fit value. If the measured floor is higher than the loop-fit value for a small room, the dial uses the floor value and offers the exact loop-fit as an opt-in "strong" setting. Until G2 has run, the provisional floors are L ≥ 1.4 m and R ≥ 1.2 m.
- **Small rooms.** With less than 1.5 × 1.5 m of free floor (min(Wf, Df) < 1.5 m) the app selects standing mode and sets the RoomModel's `standing` flag. The walker looks and leans. The dial, the Slicer and the Glued view still work in place, with no lap markers and no prompt to walk (tasks `P1-B09`, `P1-A15`).
- **Large rooms.** The lap square is capped at 3.2 m so that a lap stays a short walk. The rest of the floor is still free to walk.

4.5 Home, and furniture that arrives while you watch

Home (S0) is the room rebuilt from the RoomModel: floor, walls and door openings as clean meshes on a 0.25 m grid in the shared palette, placed at the anchor so they coincide with the real ones. It is flat and registered: touching a virtual wall is touching the real wall.

Each scanned object starts as a **placeholder box** of the scanned size, tinted by category. In parallel:

1. During the scan the phone keeps, for each object box, the one to three camera frames in which the box is best seen (largest on screen, centred, sharp, from different sides), and crops them.
2. The crops go to the Forge on the Station with the box's size and category and the object's triangle target.
3. The Forge makes a mesh from the photo, then cleans it: oriented and scaled to the box, decimated to the triangle target, no edge longer than 0.25 m so it can bend, one palette colour taken from the photo, baked vertex shading.
4. The phone downloads the result, checks it against the asset contract (§6.4), and swaps the box for the mesh.

The furniture budget. At most 20 objects receive generated meshes, largest boxes first, which is also the order of the Forge's queue. Any further objects stay placeholder boxes. The triangle target for each object is clamp(60,000 ÷ n, 3,000, 5,000), where n is the number of objects that receive meshes, so the furniture of one room never exceeds 60,000 triangles. Whatever the target, the asset validator accepts only 1,500 to 5,000 triangles and no edge longer than 0.25 m.

The walker sees their room fill in. If the Forge cannot be reached, the placeholder boxes are replaced by the built-in fallback set: one CC0 mesh of at most 1,500 triangles per RoomPlan category, scaled to the box (task `P1-D12`). An object with no usable photo, or whose job fails, takes its mesh from the same set. §7.4 specifies the round-trip.

4.6 Mode 1 · Curved Walk

The idea. The walker's real displacement each frame is replayed in the hyperbolic plane. Walk one metre north in the room, move one metre along a geodesic in the world. Nothing is scaled and nothing is rotated behind the walker's back, so there is no conflict between eyes and balance.

The update (BP §4, R1 §F.4), once per frame, after the pose has been read:

```
d = head.xz − previousHead.xz
if |d| > 0.2 m or tracking is not Normal:  d = 0          # ignore relocalisation jumps
U = U · Boost(d / L)                                       # 3×3, double precision; U is the walker's frame in the current tile
U = orthonormalise(U) with respect to diag(1, 1, −1)
```

```
if the walker has left the current tile through edge i:
    U = inverse(G[i]) · U ;  tile = Neighbour(tile, i) ;  rebuild the visible set
for each visible tile t:
    anchor[t] = inverse(U) · F[t]                          # product in double, cast to float last
    draw t only if it passes the draw rule of §8.5         # it reaches inside the fog radius and can be in view
```

`G[i]` is the fixed step to the neighbour across edge i, `F[t]` is the chain of steps from the current tile to tile t, and `inverse(M)` is $\eta \cdot M^{\intercal} \cdot \eta$. Because U is always expressed in the tile the walker stands in, its entries stay near 1 however far they travel, and precision never degrades.

The picture is made by the vertex shader, which lays every vertex out at its true distance and bearing from the head:

```
float3 p = mul(_Anchor, ExpH2(posOS.xz / _HypL));          // tile-local position → walker's frame
float2 t = LogH2(p) * _HypL;                               // true distance and direction, in metres
posWS    = float3(_HeadPosWS.x + t.x, posOS.y, _HeadPosWS.z + t.y);
```

Height is untouched: the floor stays level and at the real floor's height. Ordinary stereo projection then gives correct depth for each eye.

What the walker sees. Nearby things behave normally. Farther away, straight edges bow, rooms shrink quickly (at L = 1.88 m an object 5 m away looks about 0.4 of the size it would in flat space, and 0.3 at the fog edge), and there are many more of them than there should be: 45 tile centres lie within the fog radius ρ = 3, where a flat grid of the same spacing has 21.

What cannot be true at the same time. A flat floor cannot be mapped onto a curved surface without distortion (Gauss). We keep every step exactly 1:1, so we must give up registration: *the virtual world slips against the real room as the walker moves.* A lap of the room leaves the world turned (§3.3). Consequences, all designed for:

- The rebuilt room is content of tile zero, not a boundary. It starts exactly on the real room and drifts off as the walker laps.
- Virtual walls cannot be solid in this mode. When curvature turns on, the walls of tile zero sink to kerbs and corner posts, and anything solid-looking fades within 0.3 m of the head.
- Virtual furniture is no longer where the real furniture is. Bump Guard, which works in real coordinates, is what warns the walker (§4.10).
- "Return to Home" (one button) fades out, resets U and curvature, and fades in with the room registered again.

How the room enters curved space. The Home meshes are already stored relative to the room centre, in metres. Those coordinates are used directly as tile-local coordinates of tile zero, so the same shader places them with the exp map about the room centre: distances from the centre are exact, and at zero curvature the result is identical to Home. Each furniture mesh gets its own anchor at its scanned position so that it stays rigid instead of stretching (the exp map widens things by 11 % at 1.5 m from the centre when L = 1.88 m). Tiles that lie under the room's footprint are flagged Home and receive no generated props.

Turning curvature on. The walker stands on the mark at the room centre. Curvature ramps from zero to the dial value while they stand still, and never changes while the head moves faster than 0.1 m/s.

Getting around. Laps in one direction only carry the walker round a circle of tiles. To travel, mix loops: a lap round the left half of the room, then one round the right half, or a figure of eight. The world offers help: a post at every corner, towers, region colours, a home beacon, and the flat map.

4.7 Mode 2 · Aligned Cells

The idea (after Holonomy, TU Delft 2024). Cut the free floor into square cells. Each cell shows one tile of the same {4,5} tiling, drawn flat. Stepping across a cell edge moves to the neighbouring tile. On the real floor four cells meet at a corner; in the world five tiles do. So walking once round a corner post brings you back to the same cell but a different tile, and a fifth quarter-turn is needed to close the loop.

The grid.

```
n = max(3, ceil(Wf_short / 1.5))     g = Wf_short / n     m = floor(Wf_long / g + 1e-6)
if g < 0.75 m:  n = 2, g = Wf_short / 2      # corner demo only: a 2 × 2 grid, five tiles round one post
```

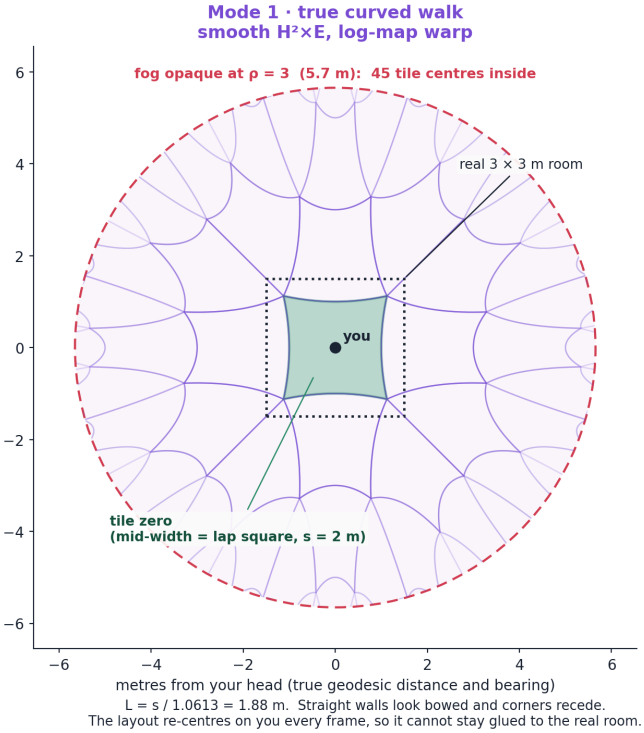
Wf_short and Wf_long are the shorter and the longer side of the free rectangle (§4.3). For empty rooms measured wall to wall: a 3 × 3 m room (free rectangle 2.4 × 2.4 m) gives 3 × 3 cells of 0.8 m; a 4 × 5 m room gives 3 × 3 cells of 1.13 m; a 6 × 8 m room gives 4 × 5 cells of 1.35 m. Two interior corners are enough to reach every tile of the infinite tiling.

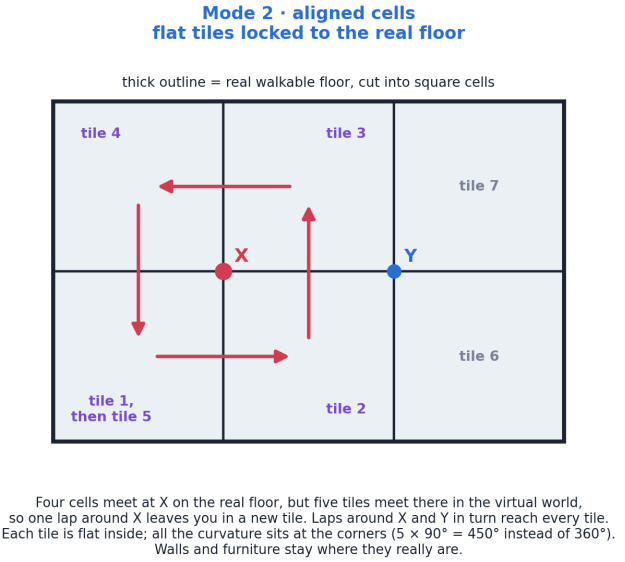
Drawing it. The tile under the walker and its row and column are placed directly. A cell seen "round a corner" can show different tiles depending on which side of the corner post the line of sight passes, so each such view is a separate instance of the cell, clipped per pixel against the lines from the eye through the posts ("gates"). No stencil buffer is needed, each eye uses its own position, and the corner post hides the seam (R1 §F.7). On a 3 × 3 grid this is 13, 15 or 17 instances depending on the walker's cell, and up to 57 on the largest grids (§8.5). An instance is one draw: the flat tile shell and its props are merged. Furniture is real and identical in every instance of its cell, so each object is drawn once per eye, unclipped, whenever any instance of its cell is drawn; it is never duplicated per instance (task `P1-B12`). Holonomy's own solution, seven stencil views, is the documented alternative.

What stays true. Everything is flat inside a cell, so the world is registered: the real walls are the edge of the grid, and furniture stays where it really is, drawn in whichever tile currently occupies its cell. This is the mode with published comfort evidence (65 users, no session ended for sickness).

What it costs. Each cell looks Euclidean. The geometry shows itself only at the posts and on the map. And the equivalent curvature is strong and not adjustable: it behaves like L ≈ 0.89 g, about 0.7 to 1.2 m.

4.8 The two modes side by side





	Mode 1 · Curved Walk	Mode 2 · Aligned Cells
Where the curvature is	Everywhere, evenly	Only at the corner posts
Curvature per tile	−2π/5	−2π/5: the same total, concentrated in four corners
Each step 1:1	Yes	Yes
World matches the real room	No, it slips as you walk	Yes, always
Furniture touchable where seen	No	Yes

	Mode 1 · Curved Walk	Mode 2 · Aligned Cells
Curvature radius	From the room: 1.4 to 3 m	Fixed by the cell: about 0.7 to 1.2 m
Renderer	Log-map vertex warp	Flat tiles, clip gates
Meshes	Generated in the log chart; 16 or 8 grid cells per tile side (§3.4)	Plain flat squares
Comfort evidence	None published; our G2 spike	Holonomy: no sickness aborts
Role	The main world	The registered comparison, the comfort fallback, and the natural way to walk a tree (§10.2)

One kernel serves both: the same tile ids, the same neighbour function, the same content hash, the same map. Switching mode keeps the walker in the same tile. The switch goes through a short fade because Mode 2 must re-register to the room.

For the documentation this pair is a result in itself. By the Gauss–Bonnet theorem a {4,5} tile carries total curvature $-2\pi/5$ however it is drawn. Mode 1 spreads it smoothly; Mode 2 is the polyhedral model that puts all of it into cone points of 450° at the corners. The walker can feel both.

4.9 Which spaces match the real room

Space	Registered?	Walls and furniture you see are where they really are?
S0 Home	Yes	Yes
S1 Glued	Yes (the lattice is fixed to the floor)	Yes. Walls (as kerbs) and furniture are drawn once, where they really are; the copies of the cell fill the space around them
S2 Curved Walk	Only until the first step	No
S3 Aligned Cells	Yes	Yes
S4 Small Planet	Only until the first step	No
S5 Impossible Rooms	Yes, room by room	The real room's, yes; a virtual room's walls are virtual
S6 Slicer	Yes	Yes

4.10 Bump Guard

The lead's decision (D19): the world is VR, nothing of the real room is drawn as a ghost, and the response to an imminent bump is **a warning cue and a haptic pulse**. Nothing freezes and nothing is replaced by a camera image. Bump Guard therefore has one job: predict contact early and say so clearly.

Inputs (all on the phone, all in real coordinates):

Input	Source	Covers
Static clearance	The RoomModel distance field	Walls and furniture present at scan time, in every direction
Live depth	LiDAR depth image (256 × 192), processed in a Burst job. The subsampled image stamps the dynamic layer; seven sectors across ±35° of the view direction give the direction of a hazard and the fallback in the head frame	Anything new in front of the walker: a person, a moved chair, an opened door
Head velocity	Raw ARKit pose, low-passed over 0.2 s	Where the walker will be
Tracking state	ARKit	Tracking that is not Normal (Limited or NotTracking)

Logic, every frame, before anything else:

```
every depth frame:
  each depth sample that lands between 0.10 m and 2.0 m above the floor, and more than 0.3 m nearer
  than the static map predicts, marks its 5 cm cell in the dynamic layer and keeps the highest hit
  a dynamic cell is cleared only when a later depth ray passes over it below its stored height
  and lands farther away (it has been seen empty), or after 30 s without a new hit;
```

```
  leaving the depth image does not clear it
clear(p)  = min( static SDF at p , distance from p to the nearest dynamic cell )
now       = clear(head)
path(T)   = the smallest clear() along the straight line the head would cover in the next T seconds
level = STRONG  if now < 0.20 m or path(0.4 s) < 0.05 m, or ARKit tracking is not Normal for 0.5 s
        WARN    if now < 0.30 m or path(0.8 s) < 0.05 m
        CLEAR   otherwise          (relax only after 0.05 m and 0.3 s of margin)
direction = from the head towards the nearest hazard
```

Worked numbers (derived). The static SDF reads 5 cm less than the true distance to a wall, because the free floor stops one cell short of every obstacle (§4.2).

- Walking straight at a wall at 0.7 m/s: WARN starts 0.66 m from the wall, about 0.9 s before the head would reach it, and STRONG starts at 0.38 m. At 1.0 m/s WARN starts at 0.90 m and STRONG at 0.50 m. At 0.5 m/s WARN starts at 0.50 m.
- The lead time to predicted contact is at least 0.85 s at every speed from 0.3 to 1.0 m/s. Contact is predicted where the straight path reaches clear() = 0, which is 5 cm before the wall.
- Walking along a wall at half a metre raises nothing.
- Turning a corner of the lap square, where a wall is 0.5 m ahead, raises nothing if the walker is slower than 0.5 m/s at the corner. Faster than that, a WARN at the corner is a true warning (a WARN that starts within 0.5 m of a lap corner counts as a corner cue, §11.4.5). STRONG does not fire on the lap below 1.0 m/s.

These constants are starting values. They are confirmed, or changed in code, by the `BumpGuard_Replay` test (§11.1) on a recorded lap before G5. If no constants give both a lead of at least 0.6 s and a quiet lap, the lead chooses between a wider lap inset and accepting corner cues (open item 19, §14).

Outputs.

Level	Warning cue	Haptic
WARN	A chevron at the edge of the view on the side of the hazard; a soft tick from that side	Two pulses per second
STRONG	The chevron grows and pulses; a clear tone	Continuous pulses, six per second
Tracking lost (ARKit tracking not Normal for 0.5 s)	A distinct "stand still" icon	Continuous pulses
Rescan advised (the static map disagrees with live depth; see the rules below)	A distinct "rescan" icon	As WARN: two pulses per second

Haptics use the phone's own engine (felt through the headset) and the gamepad's rumble when a gamepad is in hand.

Rules.

- **Constants, not parameters.** The thresholds and the depth rate are constants in code. Bump Guard has no parameter in the registry and no off switch, so the Console cannot change it, and Bump Guard never reads the link.
- **Raw pose.** Bump Guard uses the raw pose. The curved-space frame ignores pose jumps; safety must not. On a pose jump of more than 0.2 m in one frame the new position is taken at once and the velocity filter is reset to zero, so a relocalisation never produces a look-ahead spike.
- **Tracking.** While ARKit tracking is not Normal (Limited or NotTracking) the static map is not trusted. From the first such frame Bump Guard uses live depth alone, in the head frame: the nearest return in the seven sectors is compared with the same thresholds, using the last valid speed. The "stand still" cue is raised at 0.5 s. The static map is used again only when tracking is Normal and the anchor is tracked.
- **Static map.** It follows the ARAnchor. Its error after walking is measured at G5 (§11.2). If the median disagreement between live depth and the static map on the walls in view exceeds 0.15 m for 2 s, Bump Guard shows the "rescan" icon with the WARN haptic and the session is rescanned.
- **Live depth.** LiDAR depth stays on while walking. It is read every frame the phone delivers one, which is 60 Hz with the chosen video format. The processing rate is a constant in code. It may be lowered in a rebuild, never below 15 Hz (at 1.0 m/s the walker moves 6.7 cm between samples), and protocol 11.4.5 must pass at the rate that ships. Processing fewer depth frames saves CPU only; it does not reduce the LiDAR's heat. That heat is measured at G4, by a soak with depth on against a soak with depth off and nobody in the headset, and it is budgeted, not avoided.

- **Depth sectors.** The seven sectors (±35°) give the direction of the cue and the head-frame fallback. The dynamic layer is stamped from the subsampled depth image.
- **Operator.** The Console's STOP button fires the same cue through the same local entry point, `BumpCue.Fire` (§7.3.2, task `P1-C14`). It is an extra source, never the only one.

What it does not do, stated honestly. It shows no picture of the obstacle. The LiDAR looks only where the head looks, so something approaching from the side or behind is covered only if it was in the scan. A LiDAR in a headset that looks level cannot see low obstacles close by: a chair back 0.9 m high drops out of a level view when the walker is about 1.6 m from it (for a head at 1.6 m, taking the depth image's vertical field of view as about 48°; protocol 11.4.5 records the real value). That is why the dynamic layer remembers what it has seen. Anything lower than 10 cm (cables, rug edges, door thresholds) is not covered. Glass and mirrors can be missed by the scan, and floor seen through glass or in a mirror may be counted as free until the walker corrects the outline. Sessions are therefore run only on one level floor with no stairs or step-down in or beside the walked area, with glass doors and mirrors left outside the confirmed outline, and with cables and loose objects removed. Wherever someone outside the team wears the headset, this plan also recommends a scan the walker has confirmed and a human spotter within reach. None of that changes what the software does; it is how we run sessions (§12, §13).

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5. The fourth dimension

The challenge describes 4D like this: "Just as a 3D object casts a 2D shadow, a four-dimensional object can be projected into our 3D world… rotations happen in planes we cannot directly perceive." ATLAS uses 4D twice: once as a toybox that teaches three verbs, and once where those verbs are real clinical objects. This section explains what the medical 4D is, why it is useful, and how one piece of code serves both.

5.1 Three verbs

We never draw a 4D world in 4D perspective. Every view is one of three operations on a 4D object, and the walker holds the control for each.

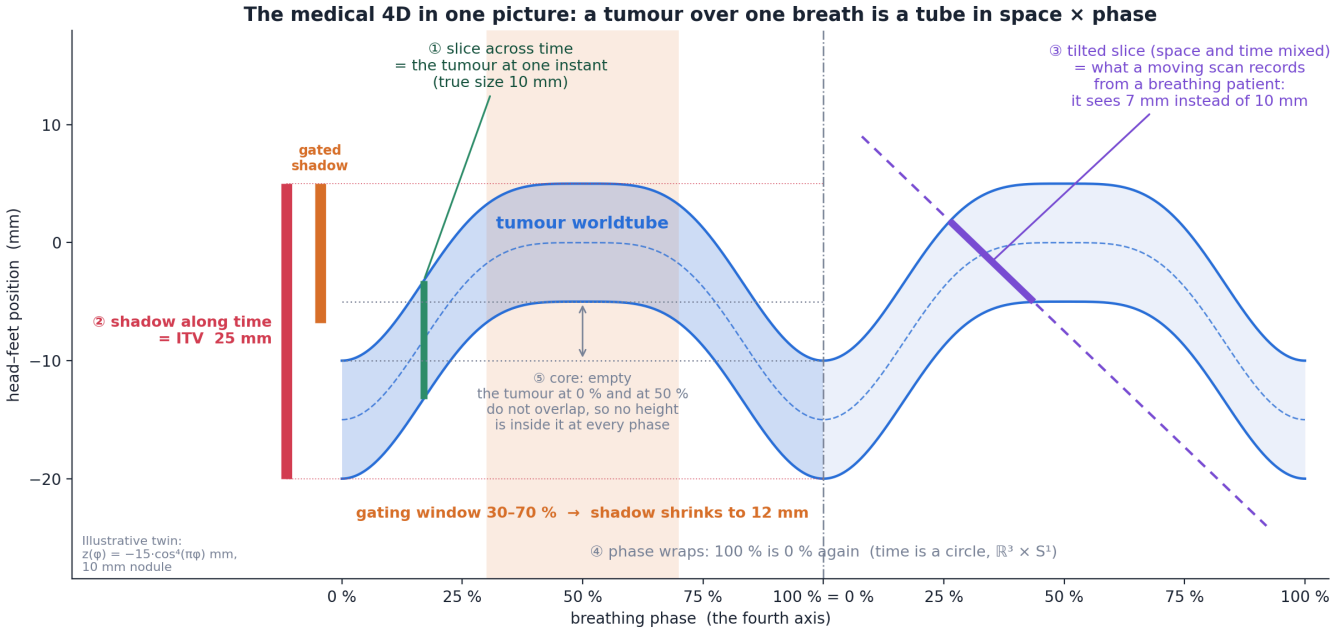
Verb	What it does	On a tesseract
Slice	Cut the object with a flat 3D space and show the cross-section. Move the cut along the fourth direction and the cross-section changes	Corner-first: a tetrahedron appears, grows, becomes a truncated tetrahedron, then an octahedron, and shrinks back
Shadow	Flatten the object along the fourth direction and show everything it ever occupies	The familiar cube-within-a-cube
Rotate	Turn the object, or the cut, in a plane that contains the fourth direction	The cross-section morphs in a way no ordinary rotation can produce

The **4D toybox** (space S6) lets the walker do all three on a tesseract, a 5-cell and a 16-cell floating in their room. Each cell has its own colour, so the faces of a cross-section can be traced back to the cell they came from.

5.2 The medical 4D: a breathing lung

Radiotherapy and lung biopsy already use the word. A **4D-CT** is a set of ten 3D scans, one for each phase of the breathing cycle. Stack them along a fourth axis, the phase, and every structure in the chest becomes a 4D object. Because breathing repeats, the phase axis is a circle: phase 100 % is phase 0 % again. The space is $\mathbb{R}^3 \times S^1$.

In that space a tumour is a **worldtube**: the set of all points it occupies, at all phases.



The three verbs now have clinical names:

Operation on the worldtube	What it is	Clinical name
Slice across time	The tumour at one instant of the breath	One phase of the 4D-CT
Shadow along time	Everywhere the tumour ever is	The internal target volume (ITV; strictly, the union of the tumour over all phases is the IGTV) and the maximum-intensity projection
Shadow with density	How much of the breath the tumour spends at each point	Occupancy; its centre is the "mid-ventilation" position
Core	The points that are inside the tumour at every phase	No standard name. For a small, mobile nodule it can be empty
Shadow of part of the tube	Treat only during a chosen arc of the breath	Gating, and its duty cycle
Tilted slice	A cut that mixes position and time	What a scanner with a moving table records from a breathing patient

The worldtube and the $\mathbb{R}^3 \times S^1$ language are ours. The objects in the right-hand column are standard practice (R2, with sources in Appendix F).

Is this non-Euclidean? Not by curvature. $\mathbb{R}^3 \times S^1$ is flat. What is unusual is its topology: one direction wraps, exactly as in the Glued space. It belongs in ATLAS because the challenge lists 4D projections as a geometry family in its own right, and because the three verbs are the same ones the toybox teaches. The curvature in the Lung Mission comes from the airway (§10.2).

5.3 Why it is useful

A static 3D model of a lung teaches several things that are wrong. The 4D view corrects each of them, and each correction is something the walker sees happen.

What a static 3D model suggests	What is true (source)	What the walker sees in 4D
"The CT is a snapshot of the tumour."	A scan taken during free breathing can shorten or lengthen a moving object by centimetres: a sphere moving with an amplitude of 1 cm and a 4 s period was imaged up to 2 cm short and 0.8 cm off-centre (Chen 2004)	Tilt the cut and the cross-section stretches or shrinks; it is striped by the phase at which each part was recorded
"A generous margin all round is safe."	A uniform 10 mm margin doubled the treated volume and still under-dosed the two most mobile tumours in one study (Underberg 2004)	The shadow is long in one direction and tight in the others. A sphere both overshoots and misses
"Tumours move a lot" (or "they don't move")	It depends where they are: 12 ± 2 mm head-to-feet for free lower-lobe tumours, 2 ± 1 mm sideways (Seppenwoolde 2002); only about one tumour in ten moves more than 10 mm (Liu 2007)	Two tubes side by side: a real one that is nearly straight, and an illustrative one that swings
"The average position is halfway."	The tumour dwells near the breathed-out position	The shadow is darker at one end
"It goes out and comes back the same way."	About half of tumours trace a loop 1 to 5 mm wide (Seppenwoolde 2002)	Look along the time axis: the centre line of the tube is a loop
"If the needle tip is inside the nodule on the scan, it is inside the nodule."	Between the planning scan and normal breathing during the procedure, nodules shift by 17.6 mm on average (Chen 2015). For a 10 mm nodule moving 15 mm, no point is inside at every phase (derived)	The core is small or empty. The best point to aim at is where the shadow is darkest
"Gating is free."	Treating only between 30 % and 70 % of the cycle removes 88 % of the motion in our illustrative breath, but the beam is on for only 40 % of the time (derived)	Drop two flags on the tube: the shadow shrinks and the duty cycle is printed

These are teaching claims. ATLAS is for education and research visualisation, and says so on screen.

5.4 The four pieces

The lead chose all four. They appear in this order, and each reuses what the previous one taught.

Piece	What the walker does	What they learn	Number on screen
1 · 4D toybox	Pushes a tesseract through the room; looks down to see its shadow; rotates the cut with the gamepad (or with the optional marker cube)	Slice, shadow, rotate	Name of the cross-section; offset along w
2 · Breathing worldtube	Stands beside a tumour tube in the lesion chamber of the airway. Scrubs the phase; switches to the shadow and its dark end; places a needle point	The "map" they navigated by was one frozen slice of a 4D object	Shadow volume ÷ tumour volume; share of the breath the needle point spends inside
3 · Breath lane walk	Walks along a lane on the floor where position is time: one breath is 1.5 m, and the lane repeats. Plants two flags to bracket a gating window	Time is a circle. Choosing when to treat changes what must be treated	Gated shadow volume; duty cycle
4 · Scanner-artefact demo	Tilts the cut from "snapshot" towards "slow table"	Why an ordinary scan of a breathing patient shows a stretched, shortened or split tumour, and why 4D-CT exists	Apparent length against true length; the equivalent table speed

Details that make them work:

- **The breath lane** is the free-walk form of the old Breath Room. It is laid along the longest clear run of the scanned floor. Because phase wraps, the lane has no end: at the real wall the walker turns round and time runs backwards. The tube is drawn once per period with the Glued engine's copy code along one axis.
- **The tube is built in front of the walker.** Shown cold, people read it as a sausage-shaped tumour (R2). So it starts as an ordinary breathing tumour; on a button press each phase slides out to its place along the lane and leaves a trail, like a multiple-exposure photograph; a "now" plane then travels along it in step with a breathing chest.
- **The tilt has a physical meaning.** A scanner whose table moves at speed v sees position z at a time that depends on z. In space-time that is a tilted cut whose angle θ satisfies tan θ = c / v, where c is the scale that turns seconds into metres on the lane. θ = 0 is an infinitely fast scanner (a true snapshot); a larger θ is a slower table. A sphere of diameter D moving at constant speed u along the table appears with length D·v / (v − u). For constant u that formula is a unit test; for a breathing tumour the slicer gives the range directly, never more than D + b, where b is the distance the tumour travels (the published phantom effect, Chen 2004).
- **Honest scale.** Real tumours in the public dataset move little (median 4.3 mm head-to-feet across its 20 patients), so the real tube is nearly straight. It is shown beside a clearly labelled illustrative nodule (10 mm, moving 15 mm) on which every effect is plain.

5.5 One slicer for both

`Mesh4D` holds tetrahedra with four-dimensional vertices. A Burst job cuts every tetrahedron with the hyperplane $n \cdot x = w_0$ and writes triangles into a preallocated mesh (a tetrahedron cut by a hyperplane gives a triangle or a quadrilateral).

Object	How it becomes tetrahedra	Size	Cost on the phone
Tesseract	8 cubic cells, each split into 6	48	about 0.02 ms
5-cell, 16-cell, 24-cell	Directly	5, 16, 96	negligible
Tumour worldtube	A 600-triangle surface carried through 10 phases: each triangle between two phases is a prism, split into 3 tetrahedra	18,000 per breath	under 1 ms; nothing when the cut is not moving

The same tests therefore cover both: the tesseract's corner-first slice has area 0.8660 at offset 0.5 and 6.9282 at offset 2 (the octahedron); a 4D ball of radius 1 cut at offset c is a sphere of radius $\sqrt{(1 - c^2)}$; a sphere moving at constant speed u, cut at a tilt, has length D·v / (v − u).

The shadow, its density and the core are not computed on the phone. The Station's pipeline produces them once per case as small meshes and a 64^3 occupancy texture (§10.2).

5.6 Data

Case	Source	Label on screen
Real	TCIA 4D-Lung, patient 101 (public, CC BY 3.0): tumour contoured on all ten phases, with lungs, heart, oesophagus, cord and trachea	"Public research data: TCIA 4D-Lung, CC BY 3.0. Meshes derived by ATLAS."
Illustrative twin	A 10 mm sphere in the lower lobe moving 15 mm with a published breathing-curve shape (Lujan) and a 3 mm loop	"Illustrative model"
Airway	Procedural tree (branching rules from the literature); optionally the first generations from a public segmented scan (AeroPath, CC BY 4.0)	"Model-generated airway" on every generated branch; "AeroPath, CC BY 4.0, not this patient" on the optional real tree

These are the label strings of §10.2.1. Every screen of the Lung Mission also carries the footer "Education and research visualisation. Not for clinical use."

5.7 How the 4D joins the hyperbolic airway

One sentence carries the Lung Mission: **the lung defeats an ordinary 3D picture twice. It has too many branches to fit in flat space, and it will not hold still.**

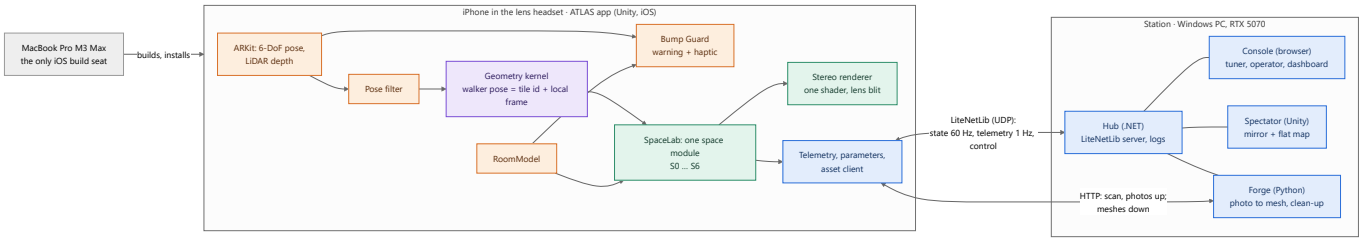
- Negative curvature fixes the first. An airway tree doubles at every generation (about 16 generations of conducting airway, 23 counting the gas-exchange zone). A hyperbolic floor grows exponentially, so every branch gets a corridor of the same width.
- A fourth, circular dimension fixes the second. The walker reaches the lesion along that tree and then learns that the map was one slice of a 4D object.

The same shape of argument both times: a flat, static 3D model is the wrong container, and the right one is a geometry chosen to match the structure of the data.

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6. System architecture

6.1 Devices and processes



The colours are the four lanes (§9.2): green A, purple B, orange C, blue D. The principle from the older design stands and is now stricter: **the phone renders everything and decides everything that matters within a frame. The Station watches, tunes, measures and manufactures assets. If the Station is unplugged the walker notices nothing except that furniture is the built-in fallback set (§7.4), not generated meshes.**

6.2 Assemblies

One assembly definition per area. An assembly references only rows above it in this table.

Assembly	Contains	May reference	Lane
`Atlas.Kernel`	`Iso2` (3 × 3 double isometries for negative, zero and positive curvature), exp and log maps, `TileId`, tile graph, `Mesh4D` and the slicing job	Mathematics, Burst, Collections. No scene types, so it runs in EditMode tests	B (4D part: A)
`Atlas.Link`	Message types, binary packing, clock sync, watchdog, client and host over LiteNetLib, `ParamRegistry` (the engine-free half)	Nothing from Unity. The same source compiles into the Hub	D
`Atlas.World`	`ISpaceModule`, SpaceLab, `RoomModel` types, world generator, archetypes, WorldLint	Kernel, Link (message and parameter types only)	A (`ISpaceModule`, SpaceLab) · B (generator, WorldLint) · C (RoomModel types)
`Atlas.Rendering`	Stereo rig, lens blit, shared shader and its includes, tile renderer, portal features, quality ladder	Kernel, World, URP	A (warp and gates includes, tile renderer: B)
`Atlas.Scan`	AR session set-up, scan routes, FloorScan and loop-fit (a core with no AR types, so it runs in EditMode tests), Home rebuild, photo capture, world map	Kernel, World	C
`Atlas.Safety`	Bump Guard and its warning cue and haptics (`BumpCue`)	World (RoomModel types only). **Never Link or Telemetry**; it uses no link type	C
`Atlas.Input`	Action map and twins	none	C
`Atlas.Telemetry`	Unity-side parameter binding (`AtlasParamAttribute` scanner), perf probe, HUD, event log, asset client	Link, World	D
`Atlas.App`	The boot scene's composition root: `AtlasFrameLoop` (§6.3), SpaceLab start-up, and the handler that turns operator commands into local calls (for example `cue` into `BumpCue.Fire`), so that Safety never sees a link type	Kernel, World, Rendering, Scan, Safety, Input, Telemetry	A
`Atlas.Spaces.*`	One assembly per space S0–S6	Kernel, Link (message and parameter types only), World, Rendering, Input, Telemetry	S0: C · S1, S5, S6: A · S2, S3, S4: B
`Atlas.Blades.*`	The use cases, Phase 2 onwards ("blade" is the older documents' word for a use case)	All of the above	per use case

Assembly	Contains	May reference	Lane
`Atlas.Spectator`	Windows only: consumes `State`, drives the mirror and the map	Rendering, World, Link, Spaces	D

6.3 The phone's frame, in order

1. **Read**: ARKit pose (once, in Update), gamepad, the newest LiDAR depth image.
2. **Bump Guard**: uses the raw pose and the depth image. Sets the warning level. Runs before anything can be skipped.
3. **Pose filter**: rejects jumps over 0.2 m for the geometry.
4. **Kernel**: updates the walker's frame, handles a tile crossing, rebuilds the list of candidate tiles when the walker changes tile, and applies the draw rule of §8.5 every frame.
5. **Space module**: interaction, simulation, the slicer job.
6. **Upload**: per-tile anchors and parameters to materials. Where the hyperbolic warp is on, the draw rule of §8.5 is applied here, once for both eyes.
7. **Render**: each eye camera into its own render texture, then one lens blit to the screen that samples both.
8. **After the frame**: sample performance counters, send `State`, drain incoming parameter changes. Nothing in steps 1–7 waits for the network.

How the order is enforced: steps 2–6 are plain C# calls made in this order by one driver, `AtlasFrameLoop`, in the boot scene (assembly `Atlas.App`, lane A). It is marked `[DefaultExecutionOrder(100)]` so that it runs after the `TrackedPoseDriver` (Update only) and after every other script's Update. No other script reads the head pose or steps the kernel in its own Update (BP §4).

6.4 Contracts frozen at G0

Lanes work in parallel because these are agreed first and changed only with notice to all four lanes. Each lane writes the pages for the contracts it owns straight from this plan (`P1-X05` to `P1-X08`; no code is needed), and gate G0 (`P1-X02`) freezes them once the stubs for every interface compile in every assembly.

Contract	What it fixes	Owner	Main consumers
`ISpaceModule`	How a space is configured from the RoomModel, entered, ticked, left, and how it describes itself to the Spectator	A	all
Kernel types	`Iso2`, `TileId`, `WalkerPose`, `ITileGraph`; unit hyperboloid with metric diag(1, 1, −1); lengths divided by L at the boundary	B	A, C, D
RoomModel schema	The fields of §4.3, coordinate frame, units; three sample rooms for functional tests, and the performance fixture `budget_room` (4 × 5 m, twelve stand-in objects of exactly 5,000 triangles each, no edge over 0.25 m) for budget tests	C	all
Shader include contract	Entry points and signatures of `AtlasCore.hlsl`, `AtlasWarp.hlsl`, `AtlasPortalClip.hlsl`, `AtlasGates.hlsl`; the single `UnityPerMaterial` block; every property id in `AtlasShaderIDs.cs` (edited by lanes A and B only; an id is added or changed only after all four lanes are told)	A and B	all
Parameter registry	How a tunable is declared (id, group, range, default) and reaches the Console. Bump Guard declares none: its thresholds and depth rate are constants in code (§4.10)	D	all
Link schema	Message ids, the `State` layout, the event format (Appendix B)	D	all
Asset contract	What a generated GLB must satisfy: one mesh, no textures or materials (the phone assigns the shared one), metres, Y up, origin at the centre of the box's base, longest edge ≤ 0.25 m, 1,500–5,000 triangles (the validator's hard range) and never more than the triangle target sent with its job. The target is clamp(60,000 ÷ n, 3,000, 5,000) when n objects receive meshes (§4.5)	D	C
Layer and stencil table	One row per room layer and stencil reference (1–15)	A	B
Frame loop	`AtlasFrameLoop` and its call order (§6.3, steps 2–6)	A	B, C
Scene ownership	Who may edit which scene and prefab (§9.3)	A	all

6.5 Core interfaces

Names are binding. Bodies are free.

```
// Atlas.Kernel -------------------------------------------------------------
public readonly struct Iso2 {                       // 3×3 double; κ < 0 hyperbolic, 0 flat, > 0 spherical
    public static Iso2 Boost(double2 step, double kappa);   // move by a tangent vector (already divided by L)
    public static Iso2 Rot(double angle);
    public Iso2 Inverse();                                  // η·Mᵀ·η in H², transpose on S²
    public Iso2 Orthonormalised();
    public static Iso2 operator *(Iso2 a, Iso2 b);
}
public readonly struct TileId : IEquatable<TileId> {        // six Int64: X, Y, T in ℤ[ϕ] for {4,5}
    public TileId Step(int edge);                           // neighbour across edge 0..3; additions only
    public ulong ContentSeed(uint worldSeed);
}
public struct WalkerPose { public TileId Tile; public Iso2 Local; }   // never a global coordinate
public interface ITileGraph {
    int Q { get; }                                          // tiles per corner
    void VisibleSet(TileId centre, double rho, List<VisibleTile> into);   // id, step chain F[t], graph distance
}

// Atlas.World (ParamRegistry and StateMessage are declared in Atlas.Link) ---------
public interface ISpaceModule {
    SpaceId Id { get; }
    bool Registered { get; }                                // does this space match the real room?
    void Configure(RoomModel room, ParamRegistry registry);  // every size comes from the RoomModel
    void Enter(in WalkerPose start);
    void Tick(in FrameContext frame);                       // after Bump Guard and the pose filter
    void Exit();
    void WriteState(ref StateMessage state);                // what the Spectator needs to redraw this frame
}

// Atlas.Safety (RoomModel types from Atlas.World only; no type from Atlas.Link) ---
public enum BumpLevel { Clear, Warn, Strong, TrackingLost }
public interface IBumpGuard {
    BumpLevel Level { get; }
    float2 Direction { get; }                               // towards the hazard, in head space
    bool RescanAdvised { get; }                             // live depth and the static map disagree (§4.10)
    void Tick(in SensorFrame sensors);                      // first call of the frame; never awaits
}

// Atlas.Telemetry ----------------------------------------------------------
[AttributeUsage(AttributeTargets.Field)]
public sealed class AtlasParamAttribute : Attribute { /* id, group, min, max, default */ }
```

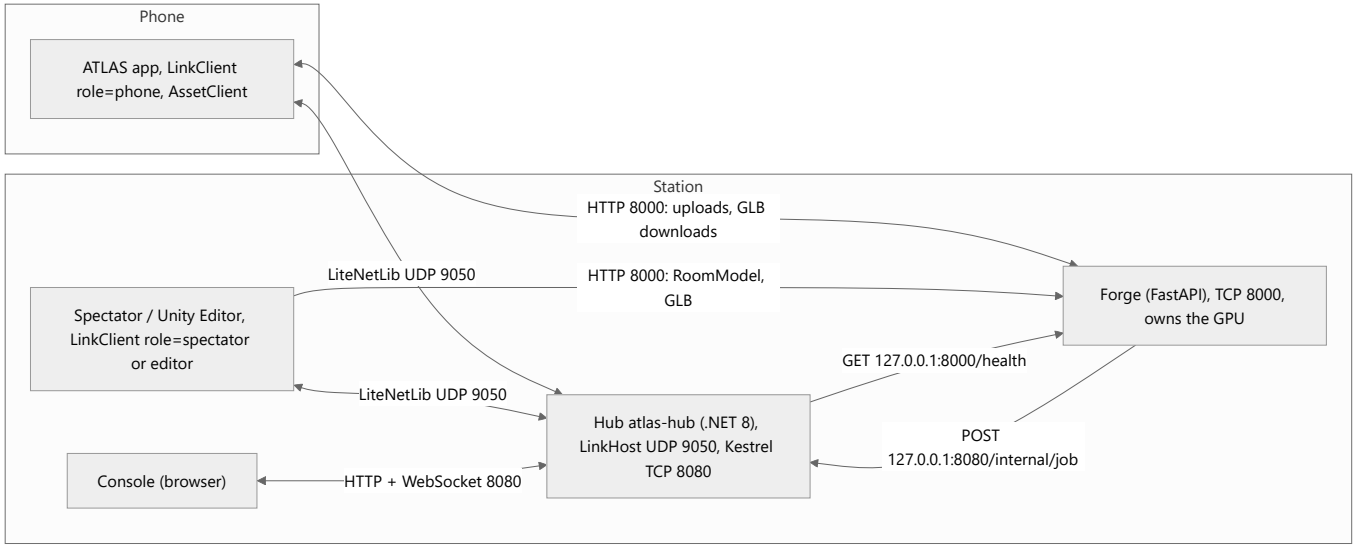
`ParamRegistry` and `StateMessage` are declared in `Atlas.Link` (§7.1). That is why `Atlas.World`, and every assembly that implements `ISpaceModule`, references `Atlas.Link`, for its message and parameter types and nothing else. `Atlas.Safety` uses no link type: from `Atlas.World` it takes only the RoomModel types, so it compiles with no reference to `Atlas.Link` or `Atlas.Telemetry`. The assembly half of the test `Safety_Independence` (§11.1) checks both from G0.

---

7. Phone ↔ Station link

Phase 1 ends with four link functions working together (D22): Spectator mirror, live tuning and operator Console, performance telemetry, and the scan and asset round-trip with furniture generated live per object (D20). The link is an accessory: the ATLAS app is complete without it (§7.5).

7.1 Topology and processes



Process	Listens on	Role
Hub	UDP 9050 (the only LiteNetLib listener); TCP 8080 on `0.0.0.0`	Relays `State` to spectators and commands to the phone; writes logs; serves the Console
Forge	TCP 8000 on `0.0.0.0`	Stores RoomModels, runs object jobs, serves GLBs by hash; never sees live pose
ATLAS app, Spectator, Editors	nothing (clients)	Same `LinkClient`; the phone rejects every connection request

- `Atlas.Link` is one plain C# library (asmdef with no engine references; no `UnityEngine` types) compiled into the app, the Spectator and the Hub (linked sources). It holds `LinkClient`, `LinkHost`, `ClockSync`, the message types of Appendix B (`StateMessage` among them) with their readers and writers, and `ParamRegistry`: the engine-free half of the parameter registry (ids, ranges, defaults, current values and the clamp). The Unity-side half, the `AtlasParamAttribute` scanner that binds fields to the registry at boot, is in `Atlas.Telemetry` (§6.2). (R4 §5.9)
- LiteNetLib 2.1.4, connection key `atlas-v1`; unicast to a known address only. Pairing: `http://<station>:8080/pair` shows a QR of `atlas://<ip>:9050`, read by the iOS Camera app; a typed-IP field and a DHCP-reserved default are the fallbacks. (R4 §3.7)
- The first `Connect` is made in hand during onboarding to raise the Local Network prompt, retried every 1 s. (BP §0.5, R4 §3.3)

7.2 Channels, encoding, clock and watchdog

Ch	Delivery	Direction	Messages	Rate	Payload
0	ReliableOrdered	both	`Hello`, `Welcome`, `ParamManifest`, `ParamSet`, `SceneCmd`, `OperatorCmd`, `RoomOutline`, `Event`, `Log`, `JobStatus`, `AssetReady`, `AssetAck`, `Bye`, `Telemetry`	on demand; `Telemetry` 1 Hz	< 1 kB each

Ch	Delivery	Direction	Messages	Rate	Payload
1	Sequenced	phone → Hub → spectators	`State`	60 Hz, one per frame (30 Hz when the thermal ladder locks to 30 fps, §8.6; `link.stateHz` can lower it to 30 or 10)	202 B (derived)
–	Unreliable	both	`TPing`, `TPong`	8 at connect, then 1 Hz	13 B, 29 B
HTTP	TCP	phone ⇄ Forge	RoomModel, crops, GLB	on demand	§7.4

- **Encoding.** `State`, `Telemetry`, `ParamSet`, `RoomOutline`, `TPing`, `TPong`: hand-packed little-endian through one reused `NetDataWriter` (zero allocation). Everything else: one type byte plus a UTF-8 JSON string (`JsonUtility` on the phone, `System.Text.Json` on the Hub), forwarded to the Console unchanged. `NetSerializer` is not used. (R4 §4)
- **Bandwidth.** `State` on the wire ≈ 234 B × 60 Hz ≈ 14 kB/s; no datagram reaches the 1,024-byte initial MTU (derived from R4 §5.2).
- **Send point.** `SendState` runs once per frame in `LateUpdate` after the kernel step, then `TriggerUpdate()`. LiteNetLib settings: `ChannelsCount 2`, `PingInterval 500`, `DisconnectTimeout 3000`, `UpdateTime 5`, `IPv6Enabled false`; Hub adds `AllowPeerAddressChange true`. (R4 §5.5)
- **Clock sync.** Four-timestamp exchange on `Stopwatch` microseconds: offset = ((t2 − t1) + (t3 − t4)) / 2. Eight exchanges 50 ms apart at connect, then 1 Hz; keep the minimum-RTT sample of the last 16; low-pass α = 0.1. The Hub clock is the reference for phone and spectators. (R4 §5.4)

Watchdog state	Condition	Behaviour on the phone
CONNECTED	a packet within 500 ms	normal
DEGRADED	silent 500 ms–3 s (enter after 2 missed pings, leave after 2 s healthy)	`ParamSet` ignored
OFFLINE	peer disconnected	app unchanged; reconnect at 0.5, 1, 2, 4, 4 … s; asset client falls back to polling (§7.4)
RESYNC	just reconnected	`Hello` → `Welcome` → 8 pings → `ParamManifest`; Hub replays the last parameter snapshot and any unacknowledged `AssetReady`

On `OnApplicationPause(false)` with no peer, the app calls `Connect` again. (R4 §3.5)

Link indicator. In hand: state dot (green, amber, grey), Hub address, RTT, loss, Forge reachability, "Reconnect", "Change IP". In the headset: one dot at the lower edge of both eyes, refreshed at 2 Hz, hidden while CONNECTED under demo lock. (R4 §5.7)

7.3 The four functions

7.3.1 Spectator mirror

- **Flow.** `State` (channel 1) carries scene id, space id, the tile id as six `i64` for {4,5}, the tile-local double 3×3, eight parameters, and the head pose in the RoomModel anchor frame. The Spectator is the app's own render scene built with `ATLAS_SPECTATOR` (no ARKit); it sets the same kernel state and draws one mono camera. No video is sent.
- **Determinism.** The Spectator never integrates motion; it applies the received transform and re-anchors exactly as the phone does. `Hello` and `Welcome` carry `worldHash` = SHA-256 of (space id, tiling, generator seed and version, RoomModel hash, L in millimetres), truncated to 16 hex characters. `State.worldVer` bumps on any regeneration. On mismatch the Spectator shows "world mismatch" and re-fetches `GET /rooms/{roomId}`; it never draws a guessed world. (R4 §5.3)
- **Beside the mirror.** Flat-twin mini-map (the space's flat model with the walker's path); the RoomModel outline from `RoomOutline` with the walker's physical position and object boxes; the Bump Guard level (`State.bumpLevel`: 0 clear, 1 warn, 2 strong, 3 tracking lost) as a coloured bar.
- **Furniture.** The Spectator gets the relayed `AssetReady` and downloads the same GLBs.
- **Acceptance.** Tile ids equal on 100% of samples; photon-to-photon ≤ 150 ms (tests L5, L7, L8).

7.3.2 Live tuning and operator Console

- **Registry.** Each parameter is registered once at boot: `{id u16, name, group, min, max, def, val, onRelease}`. The phone sends `ParamManifest` after `Welcome`; the Console builds its sliders from it with no further code. First set: `_HypL`, ρ (`curved.cullRho`: the fog is opaque there and the draw rule of §8.5 follows it), 4D slice w and rotation angles, render scale, MSAA, line width, IPD. (R4 §6.3)
- **Path.** Slider → WebSocket `{id, value}` → Hub → `ParamSet` → phone clamps to `[min, max]` and applies → the applied value returns in the next `State.p[]` or `Event param`. Parameters that regenerate geometry have `onRelease = true` and bump `worldVer`.
- **Presets and Bake** (`P1-D03`). A preset is saved and loaded as JSON on the Station (`/api/presets`, Appendix B.3); loading one replays all its values in one `ParamSet`. "Bake" writes a preset into the project defaults, so a build started with the Hub off has those values and the phone is correct with no Station.
- **On-device menu** (`P1-D13`). The same registry drives an in-headset list of the §12.4 parameters, opened by a gamepad chord. It needs no link, so each of them can be changed with the Station unreachable; every change is written to the phone's log.
- **Operator commands.** Exactly six: switch space (`SceneCmd`), and `OperatorCmd` `home`, `recentre`, `quality`, `cue`, `lock`. `cue` is the Console's **STOP** button (D22). It fires the Bump Guard cue at STRONG level (indicator, tone, continuous haptic) through `BumpCue.Fire`, the same local entry point Bump Guard uses (`P1-C14`). The world keeps rendering; nothing freezes. There is no "panic → passthrough" and no freeze (D19).
- **Lock.** `demoLock` (build flag or `OperatorCmd lock`) makes the phone ignore `ParamSet`; operator commands still work.
- **Acceptance.** Slider visible on the phone ≤ 100 ms; clamped; ignored under lock (L9, L10). With the link off, the on-device menu still changes every parameter of §12.2 and §12.4 (L22).

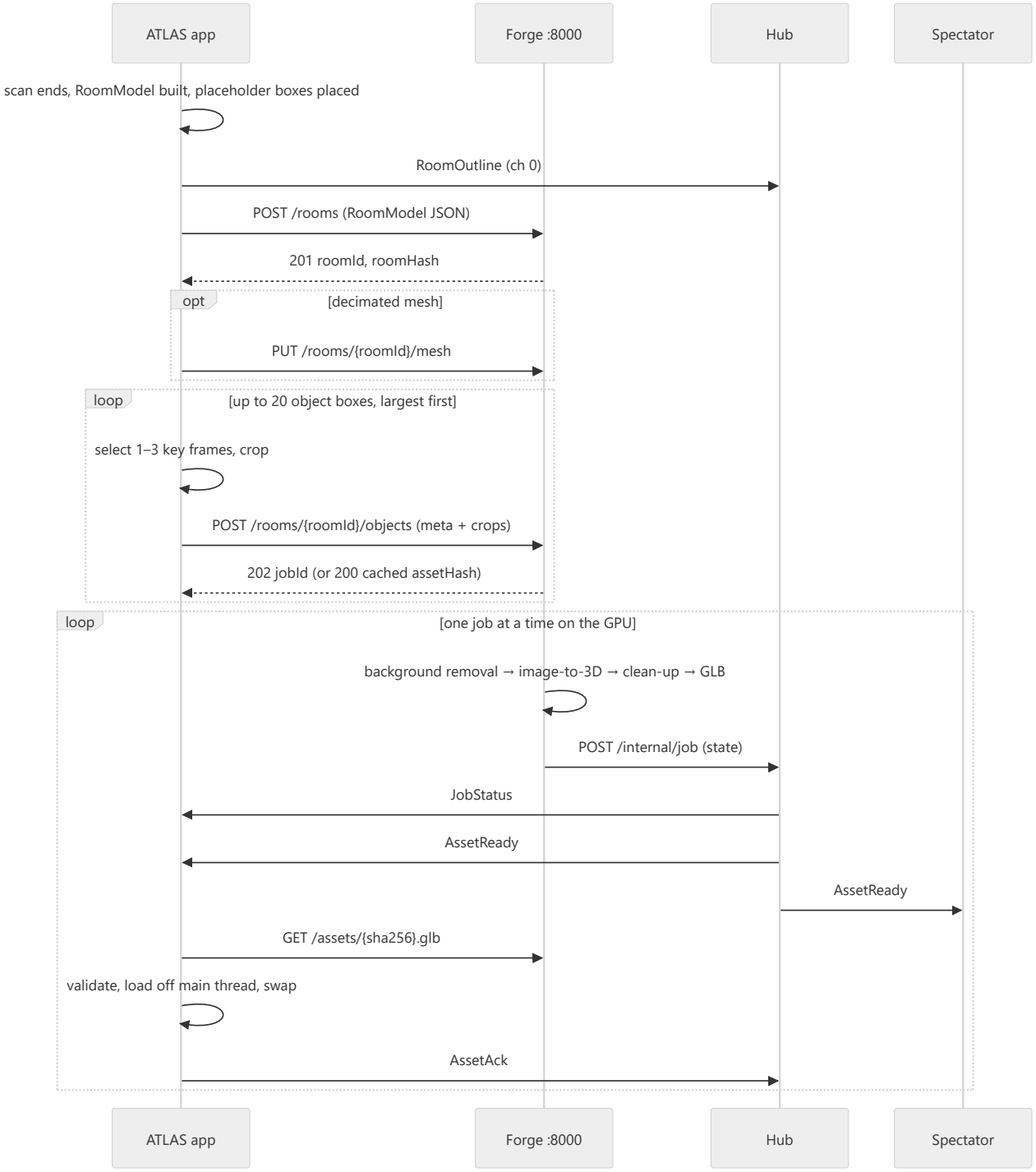
7.3.3 Telemetry dashboard and CSV

- **Flow.** `AtlasPerfProbe` keeps a 3,600-frame ring buffer and emits one `Telemetry` row per second (91 B, derived), reliable, so the CSV has no holes. (R5 D3)
- **Fields.** Listed in Appendix B.2: frame-time percentiles, CPU and GPU times, draw calls, triangles, memory, thermal state, battery, tracking, quality settings, link health, Bump Guard and asset counters.
- **Storage.** The Hub appends `logs/<session>/perf.csv` (header row, one line per second), plus `state.bin`, `events.jsonl`, `phone.log`. The phone writes the same rows to `Application.persistentDataPath/perf/perf_<session>.csv` from a writer thread (`UIFileSharingEnabled` on), so a lost link loses nothing. What is kept from a session, and how it is deleted, is set out in §13.5.
- **Console.** A tile per field; line charts of frame-time p95 and fps; a thermal colour bar; Forge queue length.
- **Acceptance.** 15-minute soak: ≥ 895 rows, no gap > 2 s, PC and phone copies identical (L11).

7.3.4 Scan and asset round-trip

See §7.4. Acceptance: every box that takes part (at most 20 per room) ends as a generated mesh or as a built-in fallback mesh; ≤ 60 s from job start to `AssetAck` per object (L15–L21).

7.4 Scan and asset round-trip



Phone: capture and selection.

- During the scan the app keeps a key-frame ring in memory: a new frame when the camera has moved ≥ 0.25 m or turned ≥ 15°, and gyro speed < 30°/s; at most 96 frames, each with camera pose and intrinsics (thresholds derived; tune at G6).
- When the RoomModel is final, the phone picks the objects that receive meshes: all of them, or the 20 largest boxes by volume when there are more. The rest keep their placeholder box (§4.5). Each job carries a triangle target in its `meta` (`triTarget`, Appendix B.4): clamp(60,000 ÷ n, 3,000, 5,000), where n is the number of objects that receive meshes. That is 5,000 up to twelve objects and 3,000 at twenty. The furniture of one room therefore never exceeds 60,000 triangles.
- Each picked box is projected into each key frame. A frame is a candidate when all eight corners are inside the image with a 5% margin, the projected bounding box covers 10–70% of the image, the camera is 0.6–3.0 m from the box,

and LiDAR depth at the box centre is within 0.15 m of the expected distance (not occluded).

- Score = 0.4·min(area / 0.35, 1) + 0.3·centredness + 0.3·sharpness, where centredness = 1 − (offset of the box centre from the image centre ÷ half-diagonal) and sharpness = variance of the Laplacian on a 160-pixel thumbnail ÷ the scan's median (derived).
- Take the best frame, then up to two more whose azimuth about the box differs by ≥ 30° from every frame already taken. Zero candidates → no upload; the box is reported in the log and takes its built-in fallback mesh at once.
- Crop to the projected box padded by 12%, squared, ≤ 768 × 768, JPEG quality 85, on a worker thread. View 0 feeds generation; views 1–2 are retry inputs and colour samples (TripoSG takes one image: verify).

Forge: job per object.

State	Work	Timeout
`queued`	Cache lookup by `sha256(view 0)`, the triangle target and `pipelineVer`; a hit goes straight to `ready`	–
`matting`	Background removal (`rembg`)	10 s
`generating`	TripoSG on the RTX 5070 (BP §0.5)	90 s, then one retry with view 1, then the hosted API if a key is configured (120 s)
`cleaning`	Up axis and the yaw in {0°, 90°, 180°, 270°} whose bounds best match the scanned box; scale to the box (≤ ±15% per axis); decimate to the triangle target sent with the job and regrid so no edge exceeds 0.25 m, ending at or below the target; nearest palette entry to the photo's dominant colour; vertex ambient occlusion; export GLB	20 s
`ready` / `failed`	`AssetReady`, or `JobStatus failed` → the box takes its built-in fallback mesh	–

- GLB contract: one mesh, `POSITION`, `NORMAL`, `TEXCOORD_0` (palette cell), `COLOR_0` (ambient occlusion), no textures, no material; box-local frame, origin at the base centre, +Y up, metres. The phone assigns the shared look material.
- One GPU job at a time; the queue is ordered by box volume, largest first.
- The `forge.backend` setting (§12.4) is sent in each job's `meta`. `local`, the default, runs the table above. `hosted` goes to the hosted API without trying the local generator. `cache` answers from the cache only: a miss fails the job at once and nothing is generated, on the Station or by a third party. With `off` the phone sends nothing. A picked box whose job fails, or is not sent because of this setting, takes its built-in fallback mesh.

Phone: placement.

- Placeholder: a box of the scanned dimensions, tinted by category, gridded at 0.25 m, present from the moment Home is built.
- On `AssetReady`: download to `persistentDataPath/assets/<sha256>.glb`, verify the SHA-256, load with glTFast asynchronously, instantiate at most one asset per frame, validate triangles ≤ 5,000 and bounds within ±10% of the box, then cross-fade placeholder to mesh over 0.3 s. Send `AssetAck {ok, loadMs, worstFrameMs}`. A rejected asset is not shown: the phone reports `ok: false` and the box takes its built-in fallback mesh.
- Without the Hub, the phone polls `GET /jobs?room=` every 2 s.
- Without the Forge, the placeholder box of every picked object is replaced by a mesh from the built-in fallback set (`P1-D12`): one CC0 mesh of at most 1.5k triangles per RoomPlan category, palette-mapped, shipped in the build and scaled to the box. The set is placed after 3 failed uploads (back-off 1, 2, 4 s). An attempt also counts as failed when no connection is made within 2 s (proposed), so the set is in place at most 9 s after the first attempt (derived: 3 × 2 s + 1 s + 2 s). A box with no candidate frame, a failed job or a rejected asset takes its built-in mesh at once. Each built-in mesh is tagged "built-in" in the log and on the Console. Uploads resume when the Forge returns, and each built-in mesh is then replaced by the generated one.
- The saved RoomModel stores `boxId → sha256`, so the pre-scan path (D21) loads furniture from the phone's cache with no Station.

Privacy. Only crops leave the phone; full camera frames are never uploaded or written to disk and the ring is cleared when cropping ends. The Forge holds crops in memory and deletes them when the job finishes. `FORGE_DEBUG_KEEP=1` keeps them for debugging and is for team sessions only: the Forge reports the flag in its status (`GET /health`) and the

Console shows a red "crops kept" banner. The hosted fallback (D20) sends the crop to a third party whose retention we do not control. The booth sign's data line says so, the operator tells a walker from outside the team before the scan, and if they decline the operator sets `forge.backend` to `cache`, so their objects use the cache or the built-in set. Scan with no other person in view; a crop in which ARKit's human stencil covers > 5% of pixels is discarded (Extended: the human stencil is off by default, §8.2, and this check switches it on for the scan only). The room record, its mesh and the GLBs made for it are removed from the Forge with `DELETE /rooms/{roomId}` (Appendix B.4); §13.5 says what is kept from a session and when it is deleted.

Budget	Value
RoomModel JSON	≤ 300 kB (160 × 160 grid, base64; derived from R1 C.3)
Decimated room mesh	≤ 2 MB; optional
Crop	≤ 200 kB each; ≤ 3 per object
GLB	≤ 1 MB; 1.5–5k triangles (§6.4), and no more than the target sent with its job
Furniture of one room	≤ 20 meshes; ≤ 60,000 triangles in total (§4.5)
Worst frame during a swap	< 20 ms

7.5 Link rules

1. Safety never touches the network. Bump Guard, tracking, the kernel and rendering never read link state. `OperatorCmd cue` is an extra trigger, never the only one.
2. Nothing in `Update`, `LateUpdate` or a render callback awaits the link. Sockets run on LiteNetLib's threads; HTTP, hashing, JPEG encoding, GLB parsing and file writes run off the main thread; received commands are queued and applied on the next frame.
3. Every remote parameter is clamped on the phone. Bump Guard has no registry parameter and no off switch: its thresholds and its depth rate are constants in code (§4.10).
4. With the Hub unplugged the app behaves identically apart from the link indicator; with the Forge unplugged, apart from the furniture, which is then the built-in fallback set (§7.4).
5. A receiver ignores unknown message ids and trailing bytes; a `proto` major mismatch is refused with a readable reason.
6. `Atlas.Link` allocates 0 B per frame in steady state.

7.6 Acceptance tests for gate G6

Own router, Station wired, phone in the headset. Loss and latency simulation is enabled on the Hub side (R4 §5.10). In L15 to L21 "every box" means every box that takes part in the round-trip: all of them, or the 20 largest when the room has more (§7.4).

#	Test	Pass
L1	Cold connect, permission already granted	CONNECTED ≤ 3 s p95 over 10 launches
L2	First connect with the iOS prompt	CONNECTED ≤ 5 s after "Allow"; no restart
L3	`State` rate at the Hub, 5-minute walk	≥ 50 Hz mean (should: ≥ 58 Hz); loss < 2%
L4	Phone → Hub latency, clock-corrected	p95 ≤ 25 ms, p99 ≤ 50 ms (should: p50 ≤ 10 ms)
L5	Photon to Spectator (slow-motion video)	≤ 150 ms
L6	Clock offset over 10 min	jitter ≤ 2 ms p95
L7	Closed five-corner loop in S2	tile ids equal on 100% of samples; `worldHash` equal
L8	`worldHash` forced to differ	Spectator shows "world mismatch"; draws no world

#	Test	Pass
L9	Sliders for L, fog, ρ, w	visible ≤ 100 ms at p95; ≤ 250 ms at 10% loss; clamped; ignored under lock
L10	Six operator commands	each takes effect ≤ 250 ms; `cue` (STOP) gives indicator, sound and haptic; the world does not change
L11	Telemetry, 15-minute soak	≥ 895 rows; no gap > 2 s; PC and phone CSV identical
L12	Router off 10 s; Hub restarted; app backgrounded 10 s	OFFLINE ≤ 3.5 s; CONNECTED ≤ 5 s after return; parameters restored; no phone restart
L13	Link on against Hub unplugged, 5 min each	main-thread p95 differs ≤ 0.2 ms; 0 B per frame from `Atlas.Link`
L14	Station cable pulled while walking at an obstacle	Bump Guard cue identical; no frame over 33.8 ms (two vsync intervals + 0.5 ms, as in §11.2)
L15	Room with ≥ 4 object boxes, Station up	100% placeholders at Home entry; every box ends as mesh or fallback; ≤ 60 s job start → `AssetAck` each
L16	Forge output check	every GLB has 1,500–5,000 triangles and no more than the target sent with its job; edges ≤ 0.25 m; longest side within 10% of the box (should: 5%); the room's furniture ≤ 60,000 triangles in total
L17	Mesh swap	worst frame < 20 ms over 10 swaps
L18	Same crops uploaded twice with the same triangle target	second run: 100% cache hits, no GPU job
L19	Forge stopped before the scan	app enters Home; within 10 s of Home entry (proposed) every box holds its built-in fallback mesh (`P1-D12`: one CC0 mesh per RoomPlan category, ≤ 1.5k triangles, no edge over 0.25 m after scaling); after a Forge restart every box is replaced by its generated mesh
L20	Privacy	after all jobs the Forge holds 0 crop files with the debug flag off; with `FORGE_DEBUG_KEEP=1` the Console shows the "crops kept" banner
L21	No hosted key configured; local generator stopped	0 requests leave the Station (packet capture on the Station); every box takes its built-in mesh
L22	On-device menu with `link.enabled` off	every parameter of §12.2 and §12.4 can be changed from the gamepad; every change is in the phone's log

8. Graphics and performance

The aim is a picture that reads clearly in both eyes at 60 frames per second for a whole session, on a phone with no fan. The Build Pipeline sets the method; the performance research (R5) checked its budgets against the hardware and corrected three numbers. Both are folded in here.

8.1 The look: a lit architectural diagram

Colour and line define form, not textures (BP §2). This is also the cheapest thing to draw in stereo.

Rule	Detail
One palette	A single 128^2 texture of flat colour blocks and a few gradient strips. Compression off, mipmaps off
Other textures	Any image texture or UI sprite: ASTC 6×6 (4×4 for UI), mipmaps on, at most 1024 (BP §5). Data textures (sky clearance maps, lens lookups): uncompressed, no mipmaps, clamped, sRGB off. TextMeshPro atlases keep their defaults
One material	One shader for every opaque surface, so the SRP Batcher works and draw submission stays cheap
Shading baked into vertices	Ambient occlusion in vertex colours. No lightmaps (they do not follow warped or copied geometry), no real-time shadows
Lines in the shader	Tile edges and a 0.5 m floor grid drawn with `fwidth`, anti-aliased and stable in both eyes. They show the geometry directly
Fog in the room's hue	Lightness steps down with distance or copy depth, so repetition reads and the edge of the drawn world disappears
Background	Both eye cameras clear to a two-stop vertical gradient made in the shader, with no texture (BP §3c). At and below the horizon its colour equals the global fog colour, and every surface fades to that one colour, so geometry at the fog radius and beyond cannot be told from the background. The region hue tints surfaces and lines; the fog colour is the hue of the region the walker stands in and cross-fades over one second when the region changes
Emissive accents	Portals, cross-sections, beacons. No bloom
Dark base palette	The display is OLED: dark pixels cost less power, which delays thermal throttling
Outlines	Inverted-hull outlines only on a few hero objects (portal frames, the slicer)
Transparency	Sparingly. Dithered, or a rim term, instead of sorted blending
Text	TextMeshPro for every label and sign

8.2 Render set-up

Setting	Value	Why
Pipeline	URP, forward, Render Graph	(BP §4)
Cameras	Two eye cameras, ±32 mm, one shared renderer, no camera stacking, no AR camera background	(BP §0, §1)
Target	One render texture per eye, 1118 × 1032 (0.8 × the 1398 × 1290 half-screen), 4× MSAA. Multisampled colour and depth are memoryless; only the resolved colour is stored. Each eye camera targets its own texture with a full viewport. The thermal ladder swaps between texture sets allocated once; an MSAA change outside the ladder (`render.msaa`) re-creates the eye textures	Fill rate is not the limit on this GPU; line art needs the resolution (R5). With a target of its own, each eye's pass resolves in place and no multisampled surface is stored or reloaded
Render scale	URP Render Scale stays 1.0; the size of the eye textures carries the render scale. Texture sets for 0.8, 0.7 and 0.6, each with the MSAA level of its rung, are allocated once, and the thermal ladder swaps between them (§8.6)	A change of rung allocates nothing
Final pass	One full-screen blit to the screen that samples both eye textures: per-eye barrel pre-distortion for the lenses, combined with the upscale	The only full-screen pass in the app
Off	HDR, depth texture, opaque texture, depth priming, post-processing, FXAA, STP, shadows, additional lights, GPU Resident Drawer, baked occlusion culling, static batching of warped meshes	(BP §4)
On	SRP Batcher, native render pass	

Setting	Value	Why
Frame rate	`targetFrameRate = 60`; ProMotion off; Unity's thermal frame-rate adjustment off (needs Unity ≥ 6000.3.21f1)	Unity otherwise forces 30 fps at thermal state Serious (R5)
ARKit	Lowest-resolution 60 Hz video format. Environment depth stays on, through `AROcclusionManager`: mode Fastest, temporal smoothing off, human stencil and human depth off, read only with `TryAcquireEnvironmentDepthCpuImage` (256 × 192). Meshing, planes, light estimation and probes off after the scan	Heat. Bump Guard needs only the small depth image, read on the CPU (§4.10)

A check that must be made once with an Xcode GPU capture (task `P1-A05`): for each eye, the opaque pass, the portal stencil passes and the tile passes are merged into one native pass; that pass resolves its colour (store action MultisampleResolve) and discards depth and stencil; no pass loads a multisampled attachment; and there is exactly one blit to the screen. If a pass break forces the MSAA surface to be stored and reloaded, a 10 % cost becomes several milliseconds.

8.3 One shader, four includes

File	Owner	Does
`AtlasCore.hlsl`	A	Palette lookup, vertex ambient occlusion, grid and edge lines, fog, the single per-material constant block
`AtlasWarp.hlsl`	B	One entry point: object position in, world position out. Modes: none, hyperbolic, spherical (log map centred on the head)
`AtlasPortalClip.hlsl`	A	Clip against a portal plane
`AtlasGates.hlsl`	B	Per-pixel clip against up to four sight-line gates (Aligned Cells)

Each include is edited only by its owning lane. The entry points, the per-material block and the property ids in `AtlasShaderIDs.cs` are a contract frozen at G0 (§6.4).

Rules that keep it fast:

- Per-tile or per-room data (the anchor matrix) lives in the per-material block, one pooled material instance per tile or room. Camera-wide data (head position, L, fog) are globals. **No MaterialPropertyBlocks.**
- No new shader keyword without telling lane A. Prefer a per-material number or a global mode.
- The warp evaluates the exponential once per vertex and derives sinh and cosh from it. Positions stay in full `float`.
- Portal rooms are flat and curved spaces have no portals, so the clip and the warp never meet in one material.
- Warped renderers get huge `localBounds`. Unity then culls nothing. Wherever the hyperbolic warp is on, the CPU draw rule of §8.5 (a distance test and a view test, once per frame) is therefore the only culling, and it is mandatory. The Small Planet always draws its six tiles.

A note on the reference code: HyperEngine's own shader passes its per-object matrix through a MaterialPropertyBlock (R3). The port must move that matrix into the per-material block; it is not a copy-and-paste.

8.4 Geometry and asset budgets

Asset	Made by	How	Triangles
Tile shell, three levels of detail	Lane B	Generated in C# at run time (§3.4). Levels by grid count: n = 16 per side on the walker's current tile and the 12 tiles that touch it; n = 8 on every other drawn tile; n = 4 only on the high tier, beyond ρ = 3. Every level keeps the same 16 boundary segments per edge, so neighbours of different levels share their edge vertices exactly	≤ 1,200 · ≤ 500 · ≤ 250 for the three levels, at any L
Home shell (floor, walls)	Lane C	Generated from the RoomModel on a 0.25 m grid	≤ 3,000

Asset	Made by	How	Triangles
Furniture	Lane D	Forge: photo to mesh, then clean-up to the triangle target sent with the job (§4.5). At most 20 objects receive generated meshes, largest boxes first; any further objects stay placeholder boxes	Target per object: clamp(60,000 ÷ n, 3,000, 5,000), n being the number of objects that receive meshes, so the room's generated furniture never exceeds 60,000. The asset validator accepts 1,500–5,000; longest edge ≤ 0.25 m
Built-in fallback furniture	Lane D	One CC0 mesh per RoomPlan category, shipped in the build; replaces the placeholder boxes when the Forge is unreachable (task `P1-D12`)	≤ 1,500 each; no edge over 0.25 m after scaling to the box
Glued cell	Lane A	An open frame cut from the free floor: corner posts, a floor grid drawn in the shader, one chiral landmark. Repeated 5 × 5 (T^2, Klein bottle) or 3 × 3 × 3 (T^3); every copy except the walker's own carries a self-avatar. The room itself is not repeated: its walls sink to kerbs and its objects are drawn once, where they really are (§4.9)	≤ 1,500 per cell
Impossible Rooms	Lane A	Parametric from the free rectangle (ProBuilder or C#)	≤ 2,000 per room
Landmarks, spans, garden props, avatar, door frame	Lanes A and B	Blender, CC0 kits (Kenney, Quaternius, KayKit) recoloured to the palette	Hero 5,000–15,000 · small 300–1,500 · self-avatar ≤ 600
Polytopes and tubes	Lane A	Generated	Tesseract 48 tetrahedra; tube 18,000

Tile levels are set by grid count, not by edge length, and are accepted by error: the worst screen-space chord error is at most 0.15° from an eye height of 1.6 m at L = 1.4, 1.9 and 3.0 m (task `P1-B05`). The 0.25 m longest-edge rule applies to the Home shell, furniture and authored props.

Authored meshes (BP §2, §3a): export FBX, metric, scale 1.0, transforms applied, Apply Scalings = FBX Units Scale, Forward −Z, Up Y. Pivot at the centre of the base for props and at a floor corner for flat rooms. Ambient occlusion is baked to the colour attribute in Blender (Cycles, or the VertexOven add-on). Anything placed in a curved space is subdivided so that no edge is longer than 0.25 m.

Licence rules: Asset Store packs never enter the repository; CC-BY assets and any hosted AI output are credited; Hunyuan3D stays off the default path (territory limits); TripoSG is MIT.

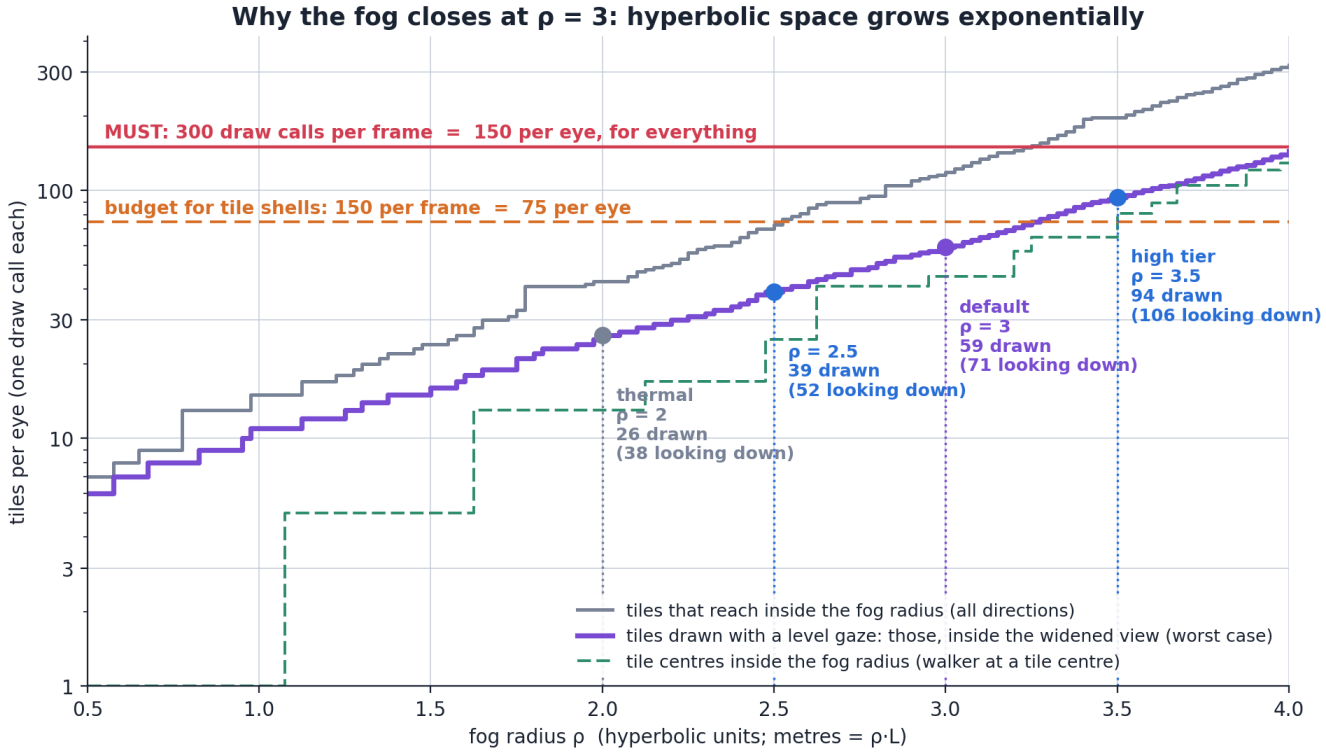
8.5 Budgets

Per frame, both eyes (must): at most 300 draw calls, 500,000 triangles, 60 set-pass calls, 1.5 GB of memory and 1 kB of managed allocation per frame (§11.2). The aim in steady state is no managed allocation at all.

Per space (per frame, both eyes; planning figures, to be replaced by values measured on `budget_room` at G4):

Space	Draw calls per frame	Triangles per frame	What drives it
S0 Home	≤ 60	≤ 130k	Up to 20 furniture meshes, 60,000 triangles in total
S1 Glued, 5 × 5 (T^2, Klein)	≤ 150	≤ 250k	25 cell frames, 24 avatars, the room drawn once
S1 Glued, T^3 3 × 3 × 3	≤ 160	≤ 250k	27 cell frames, 26 avatars, the room drawn once
S2 Curved Walk, ρ = 3	≤ 210	≤ 250k	Up to 71 tiles drawn, plus the room
S3 Aligned Cells	≤ 170	≤ 200k	Up to 57 cell instances; furniture drawn once
S4 Small Planet	≤ 70	≤ 190k	Six tiles plus the room
S5 Impossible Rooms	≤ 60	≤ 150k	Two rooms through one portal, plus the room's content (Home shell and furniture, 63,000 per eye)
S6 Slicer	≤ 70	≤ 160k	Home plus the dynamic cross-section

`budget_room` is the performance fixture frozen with the contracts at G0 (§4.3, task `P1-X02`): a 4 × 5 m room with twelve stand-in objects of exactly 5,000 triangles each and no edge over 0.25 m, which is the full furniture load of §8.4 (60,000 triangles). Budgets are measured on it; the three sample rooms stay for functional tests. In Glued the room is drawn once, among the copies of the cell. In Aligned Cells an instance is one draw call (the flat tile shell and its props merged), and each piece of furniture is drawn once per eye, however many instances of its cell are drawn (§4.7).



The figure shows why the fog radius ρ is the main performance control in hyperbolic space: the number of tiles grows exponentially with distance. It plots three counts per eye against ρ: the tile centres within ρ of the walker, the tiles that reach inside ρ, and the tiles drawn after the view test below. Counted from a tile centre, 45 tile centres lie within ρ = 3, 129 within ρ = 4 and 373 within ρ = 5. The Build Pipeline's "fog at 5–8 units" is the numerical precision horizon, not a drawable distance; the working rule is **fog fully opaque at ρ = 3; every tile that reaches inside that radius and can be in view is drawn**.

Fog starts at ρ − 1 and is fully opaque at ρ; ρ = 3 by default, so the fog runs from 2 to 3. ρ is the registry parameter `curved.cullRho`; the fog and the draw rule both follow it. Culling tiles by their centres at ρ would leave holes, because a tile whose centre lies just outside ρ still reaches up to 0.8425 inside it, where the fog is not yet opaque. The draw rule is therefore:

```
d = hyperbolic distance from the walker to the tile's centre
c = position of the tile's centre on the floor, in metres from the head: L·d along its bearing
draw the tile when both hold:
  (a) d ≤ ρ + 0.8425                              # it can reach inside the fog radius
  (b) the disc of radius 0.8425·L round c is not wholly outside any of the four side planes
      of the view frustum, widened by 12° to each side and by 4° up and down
      (a disc of radius r round c is wholly outside a plane with inward normal n
       when  n·c + r·|n_floor| < 0, n_floor being the part of n that lies in the floor)
```

0.8425 is the circumradius of a {4,5} tile. The log map never stretches a length, so the whole tile lies inside the disc of radius 0.8425·L round the image of its centre; the test is therefore safe at any head pitch, including straight down. The view is about 50° to each side and 48° up and down; the extra 12° and 4° cover a fast head turn between the test and the display, and the optional rotation prediction of §8.7. The test runs once per frame on the CPU, for both eyes. A simulation over walker positions, headings, pitches from 20° up to straight down and L from 1.4 to 2.8 m found no floor point that is in view, nearer than ρ and in an undrawn tile (`research/scripts/tile_draw_rule.py`). Task `P1-B06` repeats the check against the camera frustums on a recorded lap.

Counts per eye (derived by enumeration, same script):

Fog radius ρ	Tile centres within ρ (walker at a tile centre)	Tiles that reach inside ρ	Tiles drawn, gaze level	Tiles drawn, any pitch	Shell draw calls per frame, worst case
2 (Serious rung)	13	34 to 43	at most 26	at most 38	76
2.5 (fallback step, §12.2)	25	62 to 73	at most 39	at most 52	104
3 (default)	45 (40 to 50 elsewhere in the tile)	104 to 119	at most 59	at most 71	142
3.5 (high tier)	81	179 to 197	at most 94	at most 106	212

Looking down draws more tiles than looking ahead, because the floor near the feet fills the view. The figure marks the level-gaze worst cases. A tile shell is one draw call per eye, so the default costs at most 142 draw calls per frame, inside the 150 budgeted for tile shells (the dashed line in the figure, drawn at 75 per eye because the figure counts per eye; task `P1-B06`). The high tier can reach 212, over that budget: it is only for a cool phone and for video capture (§8.6). It is still below the solid line, the must of 300 draw calls per frame (150 per eye, as the figure draws it) for everything that is drawn.

In Glued the fog is fully opaque at the distance up to which every cell is certain to be drawn: $(n - 1)/2$ cells for an $n \times n$ lattice round the walker's current cell, so 2 cells at 5×5 and 1 cell at 3×3. It starts one cell nearer (half a cell at 3×3). The end of the lattice is therefore never seen. The vertical period of T^3 is not fogged, so the copy above the walker stays visible.

If the view test is not built, the fallback is to cull tile centres at $\rho = 3$ and make the fog opaque at $\rho = 2.15$, so that every tile that reaches inside the fog is still drawn.

Frame time at 60 fps (16.7 ms), cold targets (R5):

Stage	Thread	Target	Cap
ARKit update with live depth, pose, input	main	1.5 ms	2.5
Bump Guard, kernel, visible set, slicer, space logic	main	0.8 ms	1.5
Link, telemetry, HUD	main	0.2 ms	0.5
URP set-up and culling for two cameras, engine overhead	main	1.8 ms	3.0
Main thread		**4.3 ms**	**7.5**
Draw submission (≤ 300 draws), encode, present	render	3.0 ms	5.0
Opaque, warp, fog, portals, resolve, lens blit, both eyes	GPU	4.6 ms	7.6

The ARKit row is R5's figure plus the live depth image: 0.5 ms of its target and 1.0 ms of its cap, the "should" and "must" values of the depth-cost row in §11.2. The GPU can lose 30–50 % of its speed when hot, which is why the cold GPU target is under 5 ms.

8.6 Heat

The plan is never to reach thermal state Serious during a session. Levers, in order of benefit for effort (R5):

1. Dark palette and dark fog.
2. Screen brightness fixed near 40 %.
3. Lowest-resolution 60 Hz ARKit format; environment depth in its cheapest mode (§8.2); meshing and planes off after the scan.
4. No charging while running; case off; phone out of the headset and in front of a fan between sessions.
5. The quality ladder below.
6. Two phones: swap between sessions at Fair, during a session at Serious.
7. Hand tracking, if it is ever added, is on only in a space that uses it (BP §0.5).

Trigger	Eye-texture set (render scale)	MSAA	Fog radius ρ	Glued copies	Note
Nominal	0.8	4×	3	5 × 5	
Fair, or GPU p95 over 9 ms	0.7	4×	3	5 × 5	Console shows amber; swap at the next change of walker
Serious, or GPU p95 over 12 ms	0.6	2×	2	3 × 3	Console says "swap phone"; still 60 fps
Still missing 60 fps after 10 s, or Critical	0.6	off	2	3 × 3	Lock to 30 fps and end the session

The ladder moves one step at a time, at most every 20 seconds. A step in render scale is a swap to another eye-texture set (§8.2); the sets are allocated once, so a step allocates nothing. ρ is the fog radius of §8.5, and the draw rule follows it. A "high" tier (ρ = 3.5, 7 × 7 copies) exists for a cool phone and for video capture.

One cost is accepted rather than avoided: LiDAR depth stays on for Bump Guard whenever someone walks. The cost has two parts, and each is measured.

- **Frame time.** Reading and processing the depth image is budgeted inside the ARKit row of §8.5. It is measured at G5 with the probe, depth on against depth off; the must is a rise of at most 1.0 ms in frame-time p95 (§11.2).
- **Heat.** The LiDAR and ARKit's depth processing warm the phone. Task `P1-A14` measures this at G4: the 15-minute soak is run with depth on, then once more with depth off, driven by a recorded pose track with nobody in the headset, and the difference in time-to-Fair is recorded.

Processing fewer depth frames saves CPU time only; it does not reduce the LiDAR's heat. (The processing rate is a constant in code and never goes below 15 Hz, §4.10.) So if the heat cost proves too high, the lever is to start the ladder one rung lower. Depth is never switched off while someone walks.

8.7 Latency

ARKit reports the pose of the moment the camera frame was captured, not a prediction. The expected motion-to-photon time on this stack is 55–95 ms (R5, an estimate until measured). Three decisions follow:

- Keep `TrackedPoseDriver` on Update only. One pose sample per frame feeds the kernel and both cameras, so the virtual and real motion stay consistent. Sampling again before render would gain about 1 ms and add judder.
- Gyro rotation prediction is the one cheap improvement: rotate the rig by the gyro rate times 30–60 ms after the kernel has used the unpredicted pose. It ships behind a switch and stays only if the slow-motion measurement improves and three people see no jitter.
- Measure it once (240 fps video of the phone and its screen, §11.4) and publish the number.

8.8 Measuring

A single component, `AtlasPerfProbe`, samples frame timings and render counters every frame, keeps one minute of history and, once per second, computes percentiles and sends one telemetry row to the Station. The row is also appended to a CSV on the phone, so a lost link loses nothing. A HUD in the headset shows the same numbers. §11.2 lists every metric with its must, should and stretch value; §11.4 gives the measurement protocols. Published numbers come from a Release build, in the headset, after the soak.

8.9 Where to spend optimisation effort

In order of expected gain:

1. Frame-rate settings: `targetFrameRate`, thermal adjustment off, Release configuration, debugger detached.
2. The URP switches of §8.2.
3. The ARKit configuration and switching managers off after the scan.
4. One mesh, and so one draw call per eye, for each tile shell, room shell or Glued cell; one shader; pooled materials.
5. The draw rule of §8.5: fog opaque at ρ = 3, and only the tiles that reach inside that radius and can be in view are drawn.

6. Tile levels of detail by graph distance: the finest grid (n = 16) only on the walker's current tile and the 12 tiles that touch it, n = 8 on every other drawn tile (§8.4).
7. The MSAA pass-merge check.
8. Zero allocations per frame: pooled materials, preallocated slicer buffers, no string building in Update.
9. Dark palette and brightness.
10. The thermal ladder.
11. Decimating every generated mesh to its triangle target in the Forge, so that the room's generated furniture never exceeds 60,000 triangles (§8.4).

Not worth effort: baked occlusion culling (broken by the warp), static batching, the GPU Resident Drawer, a compute-shader slicer, single-camera stereo. Stretch only: a hidden-area mask for the lens circle, instanced tile shells, 30 fps rendering with reprojection.

Profiling tools, in the order to reach for them: the HUD and the CSV; the Metal Performance HUD for an independent reading; the Unity Profiler over USB for a CPU problem; the Frame Debugger for batch breaks; the Render Graph Viewer for pass structure; an Xcode GPU capture for a GPU problem.

8.10 Sound

Cheap polish (BP §3c), done in Phase 4: one ambient loop per space, a chime at portals pitched by depth, a tick at tile edges and a rising tone as the 4D cut moves. The Bump Guard tick and tone are not polish: they are built in Phase 1 with the warning cue (task `P1-C14`), and Phase 4 only does their final mix. Plain Unity audio; sources are CC0 packs.

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9. Build plan

9.1 How the plan is organised

- **By dependency, not by date.** Every task lists what it needs and what proves it is done. A task starts when its inputs exist. A phase ends when its exit gate passes.
- **Contracts first.** The start is wide: three set-up tasks and four sets of contract pages have no dependency, so every engineer and their agents have work from the first minute. They meet in gate G0, where the interfaces between lanes are frozen and stubbed. After that the four lanes do not wait for each other's code until the integration gates.
- **Editor first, phone last.** Only one Mac can build to the phones, so everything that can be proved in the Unity Editor or in a unit test is proved there. The phone is for acceptance.
- **Two levels.** People plan in **work areas**: sixteen in Phase 1, a handful per lane, each a piece one person owns (§9.5.2); later phases have one to three areas per lane and use case (§9.6–9.8). Agents work on **tasks**: each task fits one coding agent, touches one lane's files, and has a binary acceptance test with numbers.
- **The plan's tables and diagrams come from one data set.** The Phase 1 graph has 70 tasks and 166 dependency edges, checked to be free of cycles; the diagrams below are generated from the same data as the tables.

9.2 The four lanes

Lane	Phase 1 · Foundation	Phase 2 · Use cases I	Phase 3 · Use cases II	Phase 4 · Finish
A · Platform, Look and Flat Spaces	Project, build seat, stereo rig, lens blit, shared shader, SpaceLab, Glued (S1), Impossible Rooms (S5), Slicer and `Mesh4D` (S6), room content in Glued, thermal ladder, soak	Slicer toybox, equation sources, look of the Playground; the whole medical 4D runtime (worldtube, breath lane, scanner tilt, tasks) and the Lung director with its labels and credits	Shared House: sheets, door, second home, visitor	Look and audio, performance hardening, golden build
B · Kernel and Curved Worlds	Kernel, holonomy rig, warp shader, exact tile ids, tile meshes and renderer, world generator, Small Planet (S4), dial and 3-4-5 walk, Aligned Cells (S3), room as tile zero, room content in Aligned Cells	Lung Mission: airway wing in hyperbolic space, valve chambers, Two Compasses, needle sky	FieldFit: lens mathematics, lookup, composed final blit, blind-spot finder	Maths notes, correctness report, equation-card audit, figures
C · Scan, Room and Safety	AR session, FloorScan core (Editor), scan routes, RoomModel and loop-fit, Home (S0), object photos, Bump Guard with its warning cue and haptics, input, pose pipeline, onboarding, pre-scan path	Geometry Playground: the shell (director, guided walk, captions, twins, menu), the pointing probe, and the Glued and Impossible Rooms exhibits	Shared House: second scan, door fit, Bump Guard on the second sheet, peer link	Onboarding with outside walkers, comfort protocol, failure drills, rehearsals
D · Link, Station and Forge	`Atlas.Link`, Hub, tuner and Console, perf probe and telemetry, Spectator, flat-twin map, Forge and mesh clean-up, asset client, built-in fallback set, on-device menu, pairing	Lung Mission: the Station pipeline (`atlas case`), case packages, plan card, Lung panels on Console and Spectator	FieldFit: field presets, Console lens panel, Spectator heatmap, search task and logging; Shared House: peer relay (Extended) and two-plan flat twin	Judges' dashboard, session logs, metrics report, video capture, deck
Work packages	A 15 · B 12 · C 15 · D 13 (+ 8 shared, 7 gates)	A 17 · B 12 · C 17 · D 13	A 7 · B 8 · C 5 · D 8	A 5 · B 5 · C 6 · D 6

The lane owner decides inside the lane. Anything that touches a contract (§6.4) is announced to all four before it is committed.

9.3 Shared resources and the rules that protect them

The build seat (one Mac).

1. Lane A's first jobs are the build chain (`P1-A02`, `P1-A13`) and the URP settings and the stereo rig (`P1-A03`, `P1-A04`). They take priority over every other lane-A task until G1 passes. Editor-only lane-A task `P1-A06` and the Editor part of `P1-A12` run alongside them as the graph in §9.5.5 allows.

2. Everyone runs the Unity Editor on their own machine. Native code is wrapped so the Editor and the Station build compile without it.
3. Device builds go out as a build train: one integration build at every gate, plus short on-request slots from a visible queue (who, branch, what is tested).
4. Prefer iteration that needs no rebuild: live tuning from the Console, replay of recorded RoomModels and pose logs in the Editor, kernel and graph logic as EditMode tests.
5. One bundle id and one signing account. A build signed by a different account cannot install over the existing one.

The two phones.

1. Phone A is the reference device for every published number. Phone B is lane C's working device (every one of lane C's Phase-1 tasks except the Editor-only FloorScan core, `P1-C15`, needs a phone), the hot-swap spare, and the second walker in Phase 3.
2. Hand-held before head-mounted: scan, link and input are debugged with the phone in the hand; the headset is for acceptance.
3. Record once, replay many times: the first good scan and a few walks are saved and replayed in the Editor by every lane.
4. A soak leaves a phone hot. Soaks are scheduled, not squeezed between someone else's measurements, and the thermal state is noted at the start of every measurement.

The Station's GPU. The RTX 5070 has 12 GB. The Forge's generator needs about 8 GB and owns the card while jobs are queued. The Spectator runs at modest settings beside it. The medical pipeline (`atlas case`, Phase 2) runs when the Forge is idle.

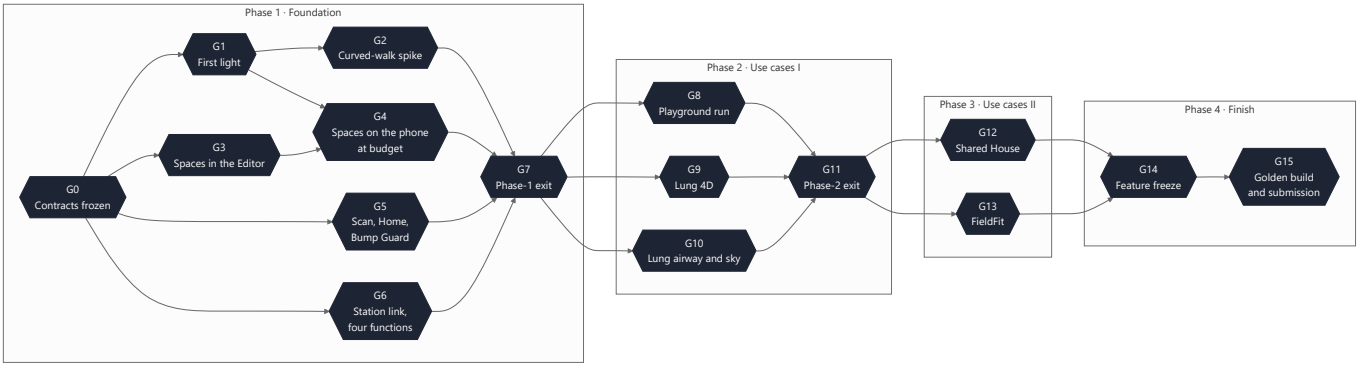
Scenes and prefabs. Unity scenes are single files, so each has exactly one owner.

Asset	Owner	Rule
`Atlas_Boot` **(XR origin, stereo rig, SpaceLab, bootstraps)**	A	Nobody else edits it. Systems register themselves from their own scenes
`Space_Home`	C	One additive scene per space, one owner each
`Space_Glued`, `Space_Portal`, `Space_Slicer`	A	
`Space_CurvedWalk`, `Space_AlignedCells`, `Space_SmallPlanet`	B	
`Sys_Scan`, `Sys_BumpGuard`, `Sys_Onboarding`	C	Loaded additively by the boot scene
`Sys_Link`, `Sys_Hud`, `PC_Spectator`	D	The Spectator scene is never in the iOS build list
Use-case scenes (Phase 2 onwards)	The lane that owns the use case's runtime	Reference spaces through `ISpaceModule`; never copy their objects
`Sandbox_<name>`	Each engineer	Never in the build list
URP assets, Project Settings, layers, stencil table, package manifest	A	Changed only by A, announced first
The shared material and palette	A	Per-tile and per-room material instances are created at run time from a pool, never saved as assets

The shared shader. Ownership of the four include files is in §8.3. A contract check runs at every gate: the shader is still reported SRP-Batcher compatible, and the draw-call count of the stress scene has not grown.

Source control. Text serialisation, visible meta files, UnityYAMLMerge as the merge tool, Git LFS for binaries. Small merges to one main branch. Tunables live in the parameter registry and in ScriptableObjects, one file per space, so tuning never touches a scene. Asset Store packs stay out of the repository.

9.4 Phases and gates



Gate	Passes when (summary; full checklists in §11.3)
G0 Contracts frozen	Every contract of §6.4 is written and stubbed; each lane's sandbox runs against the stubs
G1 First light	Empty stereo scene at 60 fps on the phone; telemetry visible on the Station; anyone can install a build
G2 Curved-walk spike	Five-corner loop closes within 0.25 m and three walkers from the team are comfortable at the loop-fit curvature (with a spotter: Bump Guard does not exist yet). Decides the default mode and the comfort floor
G3 Spaces in the Editor	S1–S6 run in SpaceLab on the three sample rooms; all mathematics tests green
G4 Spaces on the phone at budget	Every space S0–S6 meets the Must column cold, with the room's content, on the `budget_room` fixture; the 15-minute soak passes
G5 Scan, Home and Bump Guard	Both scan paths end in a registered Home; Bump Guard passes its scripted test in the headset
G6 Station link, four functions	Mirror, tuning and Console, telemetry, and the scan and asset round-trip all pass §7.6
G7 Phase-1 exit	A first-time walker goes from self-scan to every space, watched and tuned from the Station
G8 Playground run	A first-time walker completes the guided run in two different rooms
G9 Lung 4D	The four 4D pieces run on both case packages at budget
G10 Lung airway and sky	Four forks walked without rescue; sky and plan card correct
G11 Phase-2 exit	Playground and Lung Mission in one session by an outsider, hot, at budget
G12 Shared House	Post Circling walked by three people with no contact
G13 FieldFit	Blind spot found; search is faster with the lens than without
G14 Feature freeze	Full path twice in a row on one phone without cooling; only fixes after this
G15 Golden build and submission	Golden build runs with the Station unplugged; documents, video and metrics delivered

Phase 3's packages depend technically only on Phase 1. They are placed after G11 because every lane is occupied until then.

9.5 Phase 1 · Foundation

9.5.1 What is sequential and what is parallel

```
THE START  (wide from the first minute: seven tasks with no dependency)
   X01 Build seat, signing, Unity (A)  ‖  X03 Station ready (D)  ‖  X04 Phones and network ready (C)
   X05 ‖ X06 ‖ X07 ‖ X08   Contract pages, one set per lane, written from this plan
   A01  Unity project and repository skeleton              after X01
   X02 = G0  Contracts frozen, stubs compile               after A01 and X05–X08

THEN FOUR LANES IN PARALLEL
   ‖  same wave: can run at the same time          →  a later wave: cannot start before the group on its left
Each entry is one work area: a piece a person owns and hands to their agents task by task.
Under it: its tasks in order, then what it waits for from outside the lane.
```

```
LANE A  Platform, Look and Flat Spaces
  A1 Get the app running in stereo on both phones
       A02 ‖ A03  →  A04 ‖ A13  →  A05
       waits for A01, X04 (start)
  A2 Build the shared look shader and the SpaceLab host
       A06  →  A08
  A3 Implement the flat and 4D spaces
       A10 ‖ A12  →  A09 ‖ A11  →  A15
       waits for C07 (lane C)
  A4 Hold the frame rate: thermal ladder, performance runs and soak
       A07  →  A14
       waits for G3 (gates) · B11, B12 (lane B) · C07 (lane C) · D04, D06 (lane D)

LANE B  Kernel and Curved Worlds
  B1 Implement the geometry kernel and the curved-walk rig
       B01  →  B02
       waits for A01 (start)
  B2 Implement hyperbolic terrain generation
       B03 ‖ B04  →  B05  →  B06 ‖ B07
       waits for A06 (lane A)
  B3 Implement Small Planet and Aligned Cells
       B08 ‖ B10  →  B12
       waits for C07 (lane C)
  B4 Put the scanned room into the curved world
       B09  →  B11
       waits for C06, C07 (lane C)

LANE C  Scan, Room and Safety
  C1 Bring up AR tracking, input and the pose pipeline
       C01  →  C11  →  C10
       waits for A02, A04, A08 (lane A)
  C2 Implement room scanning and the RoomModel
       C15  →  C02 ‖ C03 ‖ C04 ‖ C05  →  C06
       waits for A01, X07 (start)
  C3 Rebuild the room as Home and add Bump Guard
       C14  →  C08  →  C07 ‖ C09
       waits for A04, A06 (lane A)
  C4 Build onboarding and the pre-scan path
       C13  →  C12
       waits for D11 (lane D)

LANE D  Link, Station and Forge
  D1 Build the phone-to-Station link
       D01  →  D02  →  D11
       waits for A02 (lane A) · X03, X08 (start)
  D2 Build live tuning, the operator Console and telemetry
       D03  →  D04  →  D06  →  D13
       waits for A04 (lane A) · C10, C14 (lane C)
  D3 Build the Spectator mirror and the flat-twin map
       D05  →  D07
       waits for A06, A08 (lane A) · B04 (lane B)
  D4 Build the Forge: object photos to furniture meshes
       D08  →  D09  →  D10  →  D12
       waits for C07 (lane C) · X03, X08 (start)

GATES  (where lanes meet)
  G1 First light ............... A04, A13, C01, D04, D11
  G2 Curved-walk spike ......... G1, B02, C11
  G3 Spaces in the Editor ...... A09, A11, A12, B06, B07, B08, B10
  G4 Spaces on the phone ....... G1, G3, A14
  G5 Scan, Home, Bump Guard .... C07, C08, C09, C10, C11, C12, C13, C14, D04
  G6 Station link .............. B07, C08, C09, D03, D04, D05, D06, D07, D10, D12, D13
  G7 Phase-1 exit .............. G2, G4, G5, G6, A15, B09, B11, B12
```

Three facts from the graph:

- **Fourteen lane tasks need neither a phone nor the build seat**: URP settings, the shared look shader, the kernel, the holonomy rig, the warp shader, tile ids, tile meshes, the world generator, the FloorScan core, the link library, the Hub,

the Spectator, the Forge and mesh clean-up. Thirteen of them never wait for a device task; the Spectator (`P1-D05`) waits for the SpaceLab host (`P1-A08`), which needs the stereo rig on the phone.

- **The longest chains have eleven tasks, and every one of them runs through the scan and Home**: X01 → A01 → A02 → C01 → C02 or C03 → C06 → C07, then either (A15, B11 or B12) → A14 → G4 → G7, or D10 → D12 → G6 → G7. So the first device build, the scan and the Home rebuild are the critical path. Start them as early as the graph allows and keep a phone free for lane C.
- **Lane B's integration tasks (B09, B11, B12) and lane A's (A15) wait for lane C's RoomModel and Home.** Until those exist, they are built against the three sample rooms and `budget_room` from G0.

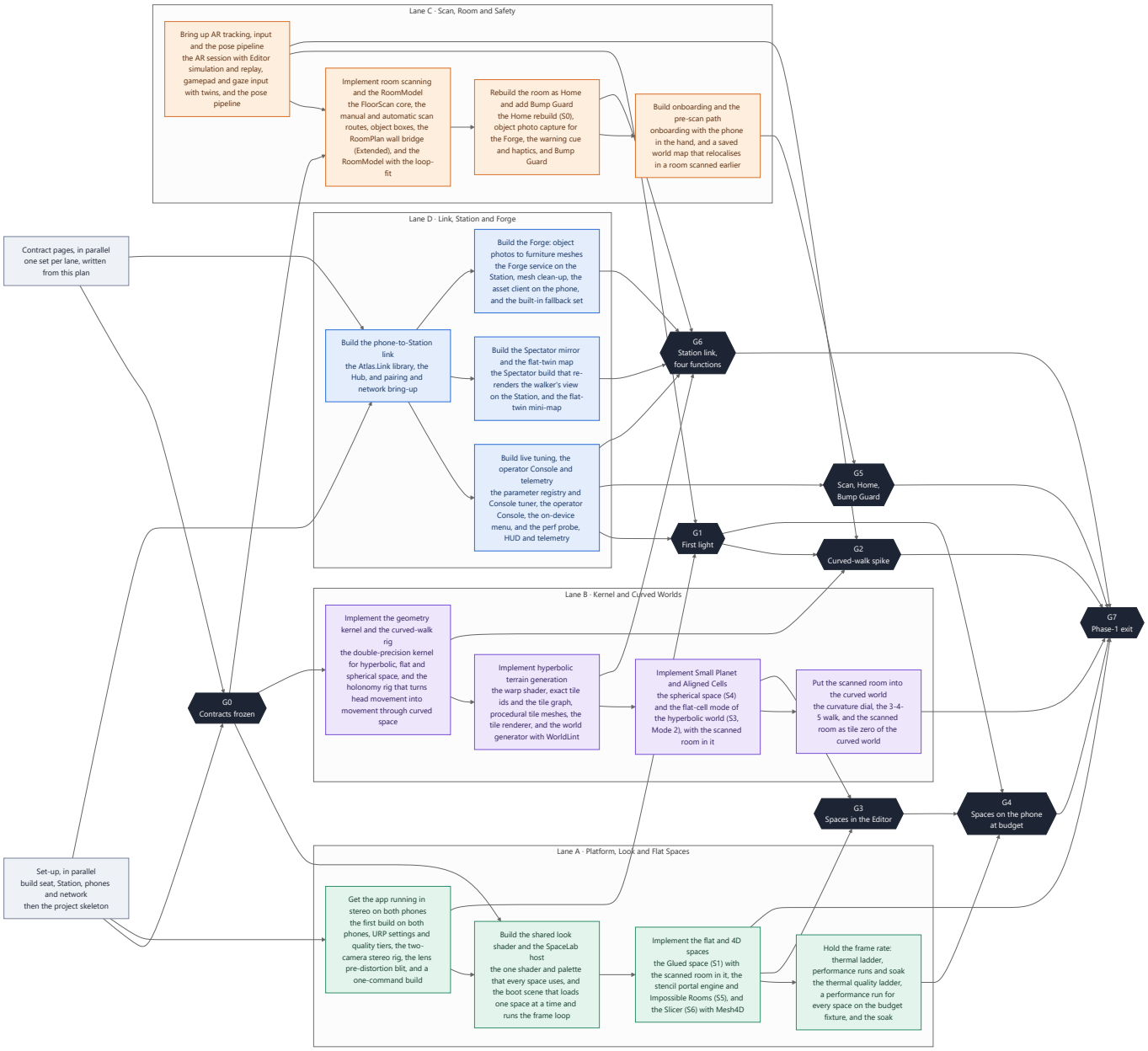
Waves. A task's wave is one more than the highest wave among the tasks it depends on. Tasks in the same wave do not depend on each other, so they can all be in progress at once; a task may start as soon as its own dependencies are done, even if the rest of its wave is not. Read a column downwards to see one lane's order; read a row across to see what the four lanes (and their agents) can be doing at the same time.

Wave	Shared	Lane A	Lane B	Lane C	Lane D	Gate reached
1	`X01` Build seat, signing, Unity `X03` Station ready `X04` Phones and network ready `X05` Contract pages, lane A `X06` Contract pages, lane B `X07` Contract pages, lane C `X08` Contract pages, lane D					
2		`A01` Project skeleton			`D01` Atlas.Link library `D08` Forge service	
3		`A02` First build on phones `A03` URP settings and tiers	`B01` Geometry kernel	`C15` FloorScan core (Editor)	`D02` Hub `D09` Mesh clean-up	G0 Contracts frozen
4		`A04` Stereo rig `A06` Shared look shader `A13` One-command build	`B02` Holonomy rig `B03` Warp shader include `B04` Exact tile ids and graph	`C01` AR session, simulation	`D03` Registry and tuner `D11` Pairing and network	
5		`A05` Lens distortion blit `A08` SpaceLab host `A10` Portal engine `A12` Slicer S6 and Mesh4D	`B05` Tile meshes	`C02` Scan route 1 manual `C03` Scan route 2 FloorScan `C04` Object boxes `C05` RoomPlan walls (Extended) `C11` Pose pipeline `C14` Warning cue and haptics	`D04` Perf probe, telemetry	
6		`A09` Glued space S1 `A11` Impossible Rooms S5	`B06` Tile renderer `B07` World generator, WorldLint `B08` Small Planet S4 `B10` Aligned Cells S3	`C06` RoomModel and loop-fit `C08` Object photo capture `C10` Input and twins	`D05` Spectator `D06` Operator Console	G1 First light
7		`A07` Thermal ladder	`B09` Dial and 3-4-5 walk	`C07` Home rebuild S0 `C09` Bump Guard `C13` Pre-scan and relocalise	`D07` Flat-twin mini-map `D13` On-device menu	G2 Curved-walk spike G3 Spaces in the Editor
8		`A15` Room in Glued, standing mode	`B11` Room becomes tile zero `B12` Room in Aligned Cells	`C12` Onboarding in hand	`D10` Asset client	

Wave	Shared	Lane A	Lane B	Lane C	Lane D	Gate reached
9		`A14` Perf runs and soak			`D12` Built-in fallback set	G5 Scan, Home, Bump Guard
10						G4 Spaces on the phone at budget G6 Station link, four functions
11						G7 Phase-1 exit

9.5.2 Overview: the work areas

Sixteen work areas, a handful per lane. An area is the unit a person plans in, about the size of "Implement hyperbolic terrain generation" or "Build the Forge: object photos to furniture meshes"; the tasks inside it (§9.5.4 onwards) are the units handed to agents. An arrow means the area at its head needs something from the area at its tail.



Area	Owner	Tasks inside	Needs from other lanes	Hands over to	Gates it feeds
A1 · Get the app running in stereo on both phones the first build on both phones, URP settings and quality tiers, the two-camera stereo rig, the lens pre-distortion blit, and a one-command build	Lane A	`A02`, `A03`, `A04`, `A05`, `A13`	nothing	C1 Bring up AR tracking, input and the pose pipeline; C3 Rebuild the room as Home and add Bump Guard; D1 Build the phone-to-Station link; D2 Build live tuning, the operator Console and telemetry	G1
A2 · Build the shared look shader and the SpaceLab host the one shader and palette that every space uses, and the boot scene that loads one space at a time and runs the frame loop	Lane A	`A06`, `A08`	nothing	B2 Implement hyperbolic terrain generation; C1 Bring up AR tracking, input and the pose pipeline; C3 Rebuild the room as Home and add Bump Guard; D3 Build the Spectator mirror and the flat-twin map	none directly
A3 · Implement the flat and 4D spaces the Glued space (S1) with the scanned room in it, the stencil portal engine and Impossible Rooms (S5), and the Slicer (S6) with Mesh4D	Lane A	`A09`, `A10`, `A11`, `A12`, `A15`	C3 Rebuild the room as Home and add Bump Guard	nothing	G3, G7
A4 · Hold the frame rate: thermal ladder, performance runs and soak the thermal quality ladder, a performance run for every space on the budget fixture, and the soak	Lane A	`A07`, `A14`	B3 Implement Small Planet and Aligned Cells; B4 Put the scanned room into the curved world; C3 Rebuild the room as Home and add Bump Guard; D2 Build live tuning, the operator Console and telemetry	nothing	G4
B1 · Implement the geometry kernel and the curved-walk rig the double-precision kernel for hyperbolic, flat and spherical space, and the holonomy rig that turns head movement into movement through curved space	Lane B	`B01`, `B02`	nothing	nothing	G2
B2 · Implement hyperbolic terrain generation the warp shader, exact tile ids and the tile graph, procedural tile meshes, the tile renderer, and the world generator with WorldLint	Lane B	`B03`, `B04`, `B05`, `B06`, `B07`	A2 Build the shared look shader and the SpaceLab host	D3 Build the Spectator mirror and the flat-twin map	G3, G6
B3 · Implement Small Planet and Aligned Cells the spherical space (S4) and the flat-cell mode of the hyperbolic world (S3, Mode 2), with the scanned room in it	Lane B	`B08`, `B10`, `B12`	C3 Rebuild the room as Home and add Bump Guard	A4 Hold the frame rate: thermal ladder, performance runs and soak	G3, G7
B4 · Put the scanned room into the curved world the curvature dial, the 3-4-5 walk, and the scanned room as tile zero of the curved world	Lane B	`B09`, `B11`	C2 Implement room scanning and the RoomModel; C3 Rebuild the room as Home and add Bump Guard	A4 Hold the frame rate: thermal ladder, performance runs and soak	G7

Area	Owner	Tasks inside	Needs from other lanes	Hands over to	Gates it feeds
C1 · Bring up AR tracking, input and the pose pipeline the AR session with Editor simulation and replay, gamepad and gaze input with twins, and the pose pipeline	Lane C	`C01`, `C10`, `C11`	A1 Get the app running in stereo on both phones; A2 Build the shared look shader and the SpaceLab host	D2 Build live tuning, the operator Console and telemetry	G1, G2, G5
C2 · Implement room scanning and the RoomModel the FloorScan core, the manual and automatic scan routes, object boxes, the RoomPlan wall bridge (Extended), and the RoomModel with the loop-fit	Lane C	`C02`, `C03`, `C04`, `C05`, `C06`, `C15`	nothing	B4 Put the scanned room into the curved world	none directly
C3 · Rebuild the room as Home and add Bump Guard the Home rebuild (S0), object photo capture for the Forge, the warning cue and haptics, and Bump Guard	Lane C	`C07`, `C08`, `C09`, `C14`	A1 Get the app running in stereo on both phones; A2 Build the shared look shader and the SpaceLab host	A3 Implement the flat and 4D spaces; A4 Hold the frame rate: thermal ladder, performance runs and soak; B3 Implement Small Planet and Aligned Cells; B4 Put the scanned room into the curved world; D2 Build live tuning, the operator Console and telemetry; D4 Build the Forge: object photos to furniture meshes	G5, G6
C4 · Build onboarding and the pre-scan path onboarding with the phone in the hand, and a saved world map that relocalises in a room scanned earlier	Lane C	`C12`, `C13`	D1 Build the phone-to-Station link	nothing	G5
D1 · Build the phone-to-Station link the Atlas.Link library, the Hub, and pairing and network bring-up	Lane D	`D01`, `D02`, `D11`	A1 Get the app running in stereo on both phones	C4 Build onboarding and the pre-scan path	G1
D2 · Build live tuning, the operator Console and telemetry the parameter registry and Console tuner, the operator Console, the on-device menu, and the perf probe, HUD and telemetry	Lane D	`D03`, `D04`, `D06`, `D13`	A1 Get the app running in stereo on both phones; C1 Bring up AR tracking, input and the pose pipeline; C3 Rebuild the room as Home and add Bump Guard	A4 Hold the frame rate: thermal ladder, performance runs and soak	G1, G5, G6
D3 · Build the Spectator mirror and the flat-twin map the Spectator build that re-renders the walker's view on the Station, and the flat-twin mini-map	Lane D	`D05`, `D07`	A2 Build the shared look shader and the SpaceLab host; B2 Implement hyperbolic terrain generation	nothing	G6
D4 · Build the Forge: object photos to furniture meshes the Forge service on the Station, mesh clean-up, the asset client on the phone, and the built-in fallback set	Lane D	`D08`, `D09`, `D10`, `D12`	C3 Rebuild the room as Home and add Bump Guard	nothing	G6

Which areas can run at the same time. Areas in the same step do not need each other. An area can be finished only after the areas it needs in earlier steps; work inside it often starts sooner (see the lane lines in §9.5.1).

Step	Lane A	Lane B	Lane C	Lane D
Start	Build seat, signing, Unity (`X01`) contract pages (`X05`) then the project skeleton (`A01`)	Contract pages (`X06`)	Phones and network (`X04`) contract pages (`X07`)	Station (`X03`) contract pages (`X08`)
1	A1 Get the app running in stereo on both phones	B1 Implement the geometry kernel and the curved-walk rig		
2	A2 Build the shared look shader and the SpaceLab host			D1 Build the phone-to-Station link
3		B2 Implement hyperbolic terrain generation	C1 Bring up AR tracking, input and the pose pipeline	
4			C2 Implement room scanning and the RoomModel	D3 Build the Spectator mirror and the flat-twin map
5			C3 Rebuild the room as Home and add Bump Guard	
6	A3 Implement the flat and 4D spaces	B3 Implement Small Planet and Aligned Cells	C4 Build onboarding and the pre-scan path	D2 Build live tuning, the operator Console and telemetry D4 Build the Forge: object photos to furniture meshes
7		B4 Put the scanned room into the curved world		
8	A4 Hold the frame rate: thermal ladder, performance runs and soak			

Colours: green lane A, purple lane B, orange lane C, blue lane D, grey shared. Dashed boxes are inputs from other lanes or from the other diagram; dark hexagons are gates.

Cross-lane handovers are not drawn in the overview; they are listed in §9.5.3 and drawn in the lane diagrams.

9.5.3 Cross-lane handovers

Every point where one lane's area unblocks another lane's area, with the tasks that carry it. These are the moments to talk.

From	To	What is handed over (task → task)
A · Get the app running in stereo on both phones	C · Bring up AR tracking, input and the pose pipeline	`P1-A02` First build on phones → `P1-C01` AR session, simulation; `P1-A04` Stereo rig → `P1-C10` Input and twins; `P1-A04` Stereo rig → `P1-C11` Pose pipeline
A · Get the app running in stereo on both phones	C · Rebuild the room as Home and add Bump Guard	`P1-A04` Stereo rig → `P1-C14` Warning cue and haptics
A · Get the app running in stereo on both phones	D · Build the phone-to-Station link	`P1-A02` First build on phones → `P1-D11` Pairing and network
A · Get the app running in stereo on both phones	D · Build live tuning, the operator Console and telemetry	`P1-A04` Stereo rig → `P1-D04` Perf probe, telemetry
A · Build the shared look shader and the SpaceLab host	B · Implement hyperbolic terrain generation	`P1-A06` Shared look shader → `P1-B05` Tile meshes
A · Build the shared look shader and the SpaceLab host	C · Bring up AR tracking, input and the pose pipeline	`P1-A08` SpaceLab host → `P1-C10` Input and twins
A · Build the shared look shader and the SpaceLab host	C · Rebuild the room as Home and add Bump Guard	`P1-A06` Shared look shader → `P1-C07` Home rebuild S0
A · Build the shared look shader and the SpaceLab host	D · Build the Spectator mirror and the flat-twin map	`P1-A06` Shared look shader → `P1-D05` Spectator; `P1-A08` SpaceLab host → `P1-D05` Spectator

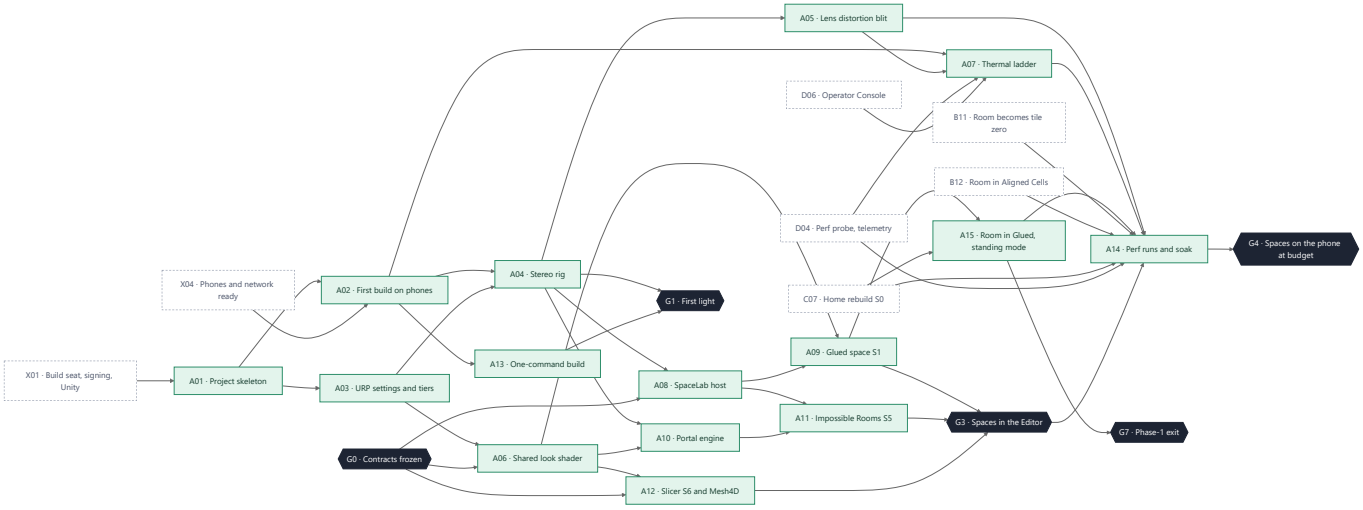
From	To	What is handed over (task → task)
B · Implement hyperbolic terrain generation	D · Build the Spectator mirror and the flat-twin map	`P1-B04` Exact tile ids and graph → `P1-D07` Flat-twin mini-map
B · Implement Small Planet and Aligned Cells	A · Hold the frame rate: thermal ladder, performance runs and soak	`P1-B12` Room in Aligned Cells → `P1-A14` Perf runs and soak
B · Put the scanned room into the curved world	A · Hold the frame rate: thermal ladder, performance runs and soak	`P1-B11` Room becomes tile zero → `P1-A14` Perf runs and soak
C · Bring up AR tracking, input and the pose pipeline	D · Build live tuning, the operator Console and telemetry	`P1-C10` Input and twins → `P1-D13` On-device menu
C · Implement room scanning and the RoomModel	B · Put the scanned room into the curved world	`P1-C06` RoomModel and loop-fit → `P1-B09` Dial and 3-4-5 walk
C · Rebuild the room as Home and add Bump Guard	A · Implement the flat and 4D spaces	`P1-C07` Home rebuild S0 → `P1-A15` Room in Glued, standing mode
C · Rebuild the room as Home and add Bump Guard	A · Hold the frame rate: thermal ladder, performance runs and soak	`P1-C07` Home rebuild S0 → `P1-A14` Perf runs and soak
C · Rebuild the room as Home and add Bump Guard	B · Implement Small Planet and Aligned Cells	`P1-C07` Home rebuild S0 → `P1-B12` Room in Aligned Cells
C · Rebuild the room as Home and add Bump Guard	B · Put the scanned room into the curved world	`P1-C07` Home rebuild S0 → `P1-B11` Room becomes tile zero
C · Rebuild the room as Home and add Bump Guard	D · Build live tuning, the operator Console and telemetry	`P1-C14` Warning cue and haptics → `P1-D06` Operator Console
C · Rebuild the room as Home and add Bump Guard	D · Build the Forge: object photos to furniture meshes	`P1-C07` Home rebuild S0 → `P1-D10` Asset client
D · Build the phone-to-Station link	C · Build onboarding and the pre-scan path	`P1-D11` Pairing and network → `P1-C12` Onboarding in hand
D · Build live tuning, the operator Console and telemetry	A · Hold the frame rate: thermal ladder, performance runs and soak	`P1-D04` Perf probe, telemetry → `P1-A07` Thermal ladder; `P1-D04` Perf probe, telemetry → `P1-A14` Perf runs and soak; `P1-D06` Operator Console → `P1-A07` Thermal ladder

9.5.4 All hands

ID	Task	Depends on	Output	Acceptance test	Runs on
`P1-X01`	**Build seat, signing and Unity (lane A leads, everyone takes part)**	none	Appendix C.1 and C.2: the Mac with Xcode 26; Unity ≥ 6000.3.21f1 with iOS and Windows build support on all four machines; one signing owner for the Personal Team	All four machines open an empty Unity project of the pinned version; the Mac produces an Xcode project from it without errors	Mac
`P1-X02`	**Contracts frozen (this is gate G0)**	`P1-A01`, `P1-X05`, `P1-X06`, `P1-X07`, `P1-X08`	The four sets of contract pages merged under `docs/contracts/`; stubs for every interface: identity warp, fake tile provider, loopback link, recorded pose log, the sample rooms as RoomModel files	Stub implementations compile in every assembly; each lane's sandbox scene runs against the stubs in the Editor with no other lane's code; a change to any contract after this point is announced to all four lanes before it is committed	Editor
`P1-X03`	**Station ready (lane D)**	none	Appendix C.7: .NET 8, Python 3.11 and a working CUDA build of PyTorch on the RTX 5070 PC; the `station/` folders reserved for the Hub and the Forge	`nvidia-smi` and a PyTorch CUDA test pass; `dotnet --version` reports 8.x	Station
`P1-X04`	**Phones and network ready (lane C)**	none	Appendix C.1, C.6 and C.7 (the phone and router lines): both phones in Developer Mode and trusted by the Mac; the team's own Wi-Fi router with a fixed address for the Station; both phones and the Station joined to it	Both phones report LiDAR (iPhone 15 Pro Max and 14 Pro Max, not Plus models); each phone opens a test page served from the Station's fixed address	phone in hand, Station

ID	Task	Depends on	Output	Acceptance test	Runs on
`P1-X05`	**Contract pages, lane A**	none	Written from §6 and §8.3 of this plan, needing no code: `ISpaceModule`; the shader include contract and the list of ids for `AtlasShaderIDs.cs` (with lane B); the layer and stencil table; the frame-loop order; scene ownership	Each page names every type, signature, unit and owner an implementer needs; the other three lanes have read it and no question is open	no device
`P1-X06`	**Contract pages, lane B**	none	Written from §6.5 and Appendix A: the kernel types (`Iso2`, `TileId`, `WalkerPose`, `ITileGraph`), the metric and the unit conventions, and the `AtlasWarp` entry point of the shader contract	Each page names every type, signature, unit and owner an implementer needs; the other three lanes have read it and no question is open	no device
`P1-X07`	**Contract pages, lane C**	none	Written from §4.3: the RoomModel JSON schema with its coordinate frame and units; three sample rooms (3×3 m, 3×4 m L-shape, 4×5 m with a table) and the performance fixture `budget_room` (4×5 m with twelve stand-in objects of exactly 5,000 triangles each, no edge over 0.25 m; used for budget tests only)	The sample rooms and `budget_room` validate against the schema; the other three lanes have read the page and no question is open	no device
`P1-X08`	**Contract pages, lane D**	none	Written from §7 and Appendix B: the parameter registry (how a tunable is declared and reaches the Console); the link message ids and the `State` layout; the GLB asset contract	A hand-packed `State` example of 202 B is given byte by byte; the other three lanes have read the pages and no question is open	no device

9.5.5 Lane A · Platform, Look and Flat Spaces



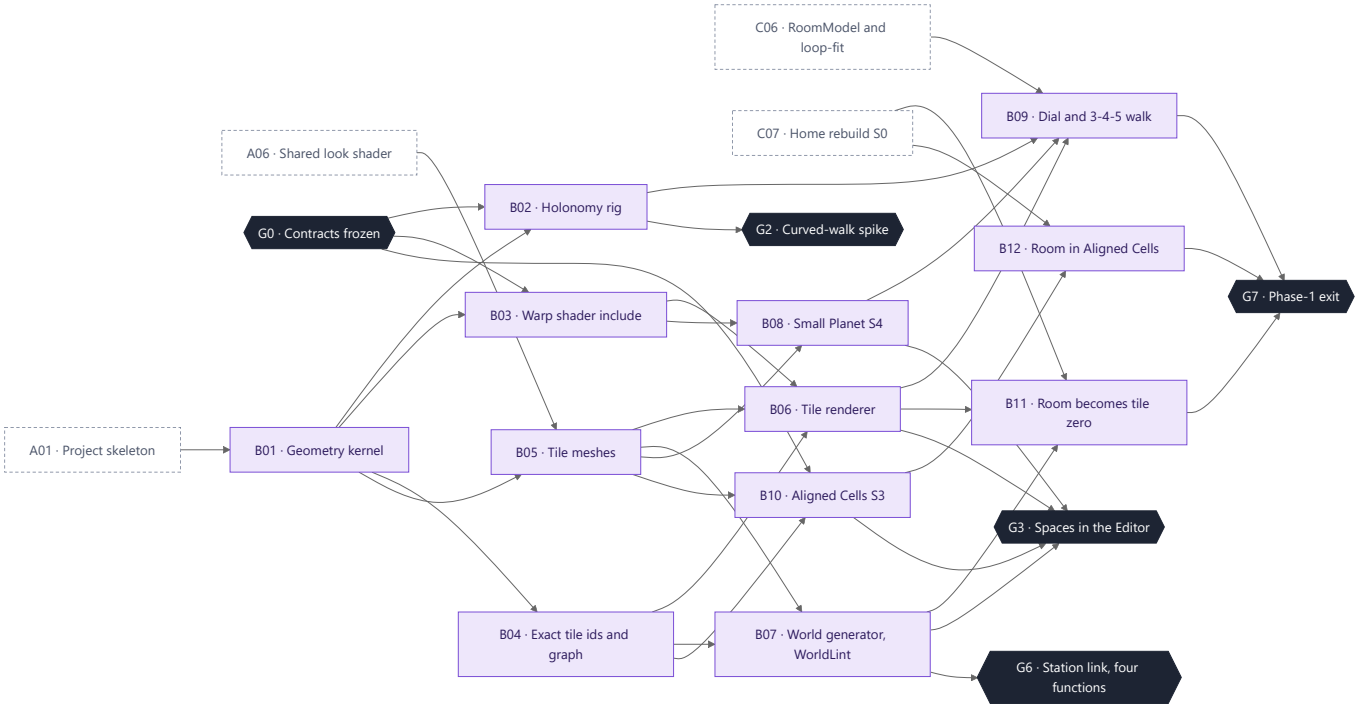
Colours: green lane A, purple lane B, orange lane C, blue lane D, grey shared. Dashed boxes are inputs from other lanes or from the other diagram; dark hexagons are gates.

ID	Task	Depends on	Output	Acceptance test	Runs on
`P1-A01`	**Unity project and repository skeleton**	`P1-X01`	Project from the AR Mobile template; packages pinned (Appendix C); one assembly definition per area; text serialisation, Git LFS, UnityYAMLMerge; `station/` folders; scene-ownership file	A clean clone opens with zero console errors on all four machines; the EditMode test runner runs (0 failing)	Editor, Mac
`P1-A02`	**First build on both phones**	`P1-A01`, `P1-X04`	ARKit as the only XR loader; signing with the Personal Team; post-build script for Info.plist keys; `targetFrameRate = 60`; thermal frame-rate adjustment off; Frame Timing Stats on; native `Assets/Plugins/iOS/AtlasThermal.mm` returning `NSProcessInfo.thermalState`, with an Editor stub that returns 0	An empty AR scene runs on phone A and phone B with tracking state Normal; the build recipe is one paragraph in the README	Mac, phone in hand
`P1-A03`	**URP asset and quality tiers**	`P1-A01`	Forward, Render Graph, HDR off, depth and opaque textures off, depth priming off, no post-processing, shadows off, SRP Batcher on; tier assets Nominal / Fair / Serious	Render Graph Viewer shows one native pass per eye camera on the test scene; settings equal the Appendix C table	Editor

ID	Task	Depends on	Output	Acceptance test	Runs on
P1-A04	**Two-camera stereo rig**	P1-A02, P1-A03	Eye cameras at ±IPD/2 under the tracked head, half-screen viewports (until P1-A05 gives each eye its own texture), one shared renderer, no camera background; IPD in the parameter registry; the AR camera only carries the pose	Phone A in the headset, empty scene: 60 fps on the Metal HUD, CPU main ≤ 3 ms, GPU ≤ 2 ms; a cube at 1 m fuses comfortably for three testers	phone in headset
P1-A05	**Lens pre-distortion as the single final blit**	P1-A04	Each eye renders into its own 1118 × 1032 target (0.8 × the half-screen) with 4× MSAA; multisampled colour and depth are memoryless and only the resolved colour is kept; one full-screen pass samples both eye textures, applies per-eye barrel distortion (headset coefficients) and upscales; switchable; texture sets for render scales 0.8, 0.7 and 0.6 are allocated once	A straight-line grid looks straight through the lenses to three testers; the pass adds ≤ 1.0 ms GPU; Xcode GPU capture: each eye pass resolves its colour (store action MultisampleResolve) and discards depth and stencil, no pass loads a multisampled attachment, and there is exactly one blit to the screen	phone in headset
P1-A06	**Shared look shader**	P1-X02, P1-A03	AtlasSurface with AtlasCore.hlsl: palette lookup, vertex-colour ambient occlusion, fwidth grid and edge lines, distance fog in the room hue, emissive accents; calls the AtlasWarp entry point; one UnityPerMaterial block; 128^2 palette texture (dark base); gradient background whose horizon colour equals the fog colour	Frame Debugger reports the shader SRP-Batcher compatible; ≤ 8 variants; a test room renders in the Editor's stereo preview	Editor
P1-A07	**Thermal quality ladder**	P1-A02, P1-A05, P1-D04, P1-D06	Poll of the thermal plugin (from P1-A02) every 2 s; ladder controller that steps the eye-texture set (render scale), MSAA, fog radius ρ and glued copies from thermal state and GPU p95	Each tier can be forced from the Console and takes effect within 1 s; at most one step per 20 s with hysteresis; thermal state is visible on the HUD and in the CSV	phone in headset
P1-A08	**SpaceLab host**	P1-X02, P1-A04	Boot scene that loads one ISpaceModule scene additively at a time, with a fade, and registers that space's parameters; the AtlasFrameLoop driver ([DefaultExecutionOrder(100)]) that calls Bump Guard, pose filter, kernel, space module and upload in the §6.3 order	Switching among all loaded spaces 20 times leaves memory within ±5 %; works in the Editor (XR Simulation) and on the phone	Editor, phone in headset
P1-A09	**Glued space S1**	P1-A06, P1-A08	Universal cover fixed to the floor: an open-frame cell (corner posts, floor grid, one chiral landmark; ≤ 1,500 triangles) cut from the free floor at the RoomModel cell size; 5×5 copies around the walker's current cell; mirrored rows for the Klein bottle (negative scale); T^3 as 3×3×3 copies with a vertical period of 2.6 m; a self-avatar (≤ 600 triangles) in every copy but the walker's own	Walking across a cell edge shows no seam in either eye; Klein rows pass the determinant-sign test; cells and avatars alone cost ≤ 110 draw calls and ≤ 110k triangles per frame at 5×5 (T^2, Klein) and ≤ 120 draw calls and ≤ 120k triangles at 3×3×3 (T^3)	Editor, phone in headset
P1-A10	**Stencil portal engine**	P1-A06, P1-A04	Render Objects features per room layer, stencil references 1–15 from one allocation table, plane clip in AtlasPortalClip.hlsl, room swap only when the old room's bounds fail GeometryUtility.TestPlanesAABB for both eye frusta and the head is outside the overlap volume, held for 3 consecutive frames (BP §4), portal box for the straddling eye; no recursion	Two-room test: no seam or pop when the head crosses the doorway in stereo; ≤ 1.0 ms GPU per visible portal; passes stay merged so MSAA is resolved on-tile	Editor, phone in headset
P1-A11	**Impossible Rooms S5**	P1-A10, P1-A08	Booth-to-hall (bigger inside) and a four-turn loop corridor, generated from the RoomModel free rectangle and authored in play-space coordinates so every portal transform is the identity; room graph with a reachability check	Layouts fit inside the eroded free floor for all three sample rooms; four right turns arrive in a different room; every room is reachable (graph test)	Editor, phone in headset
P1-A12	**Slicer S6 and Mesh4D**	P1-X02, P1-A06	Mesh4D (tetrahedra in $\mathbb{R}^4$), Burst job that cuts it with a hyperplane, generators for the 5-cell, tesseract (48 tetrahedra), 16-cell and 24-cell, shadow (projection) view, rotations in planes that contain w, a colour per cell, and a periodic tube generator in $\mathbb{R}^3 \times S^1$ (moving sphere)	EditMode: tesseract corner-first slice areas 0.8660 at 0.5 and 6.9282 at 2.0; 4D ball slice radius $\sqrt{(1 - c^2)}$; moving-sphere length $D \cdot v/(v - u)$. Phone A: tesseract ≤ 0.5 ms (should: ≤ 0.2 ms), 18k-tetrahedron tube ≤ 1.0 ms at p95, zero allocations per frame	Editor, phone in headset

ID	Task	Depends on	Output	Acceptance test	Runs on
`P1-A13`	**One-command build and the build-seat rules**	`P1-A02`	Script: clean checkout → Unity batch build → Xcode build → install on a connected phone; written rules for the build queue and phone booking	One command installs the current main branch on a phone; a second engineer has done it from the README alone	Mac
`P1-A14`	**Per-space performance runs and the soak**	`P1-D04`, `P1-A07`, `P1-A05`, `P1-G3`, `P1-C07`, `P1-B11`, `P1-B12`, `P1-A15`	Scripted tour that visits every space S0–S6 on `budget_room`; one CSV per run; the 15-minute in-headset soak, alternating S2 Curved Walk (ρ = 3, room as tile zero) and S1 Glued 5×5; one more soak with depth off, driven by a recorded pose track with nobody in the headset	Metrics table of §11.2 filled for every space at the Must level, cold; soak: no Critical thermal state, mean ≥ 45 fps, never below 30; the difference in time-to-Fair with depth on and off is recorded	phone in headset
`P1-A15`	**Room content in Glued, and standing mode**	`P1-A09`, `P1-C07`	In S1 the Home walls sink to kerbs and every object from the RoomModel (placeholder box or generated mesh) is drawn once, at its true position, as a solid among the copies; with the RoomModel's standing flag set, S1 and S6 run in place (no wrap-walk prompt; the 4D cut is moved with the stick)	On phone A in S1 a virtual table edge is within 5 cm of the real one at three touch points; S1 on `budget_room` stays within its row of §8.5; with a 1.4 × 1.4 m RoomModel, S1 and S6 load and run	Editor, phone in headset

9.5.6 Lane B · Kernel and Curved Worlds



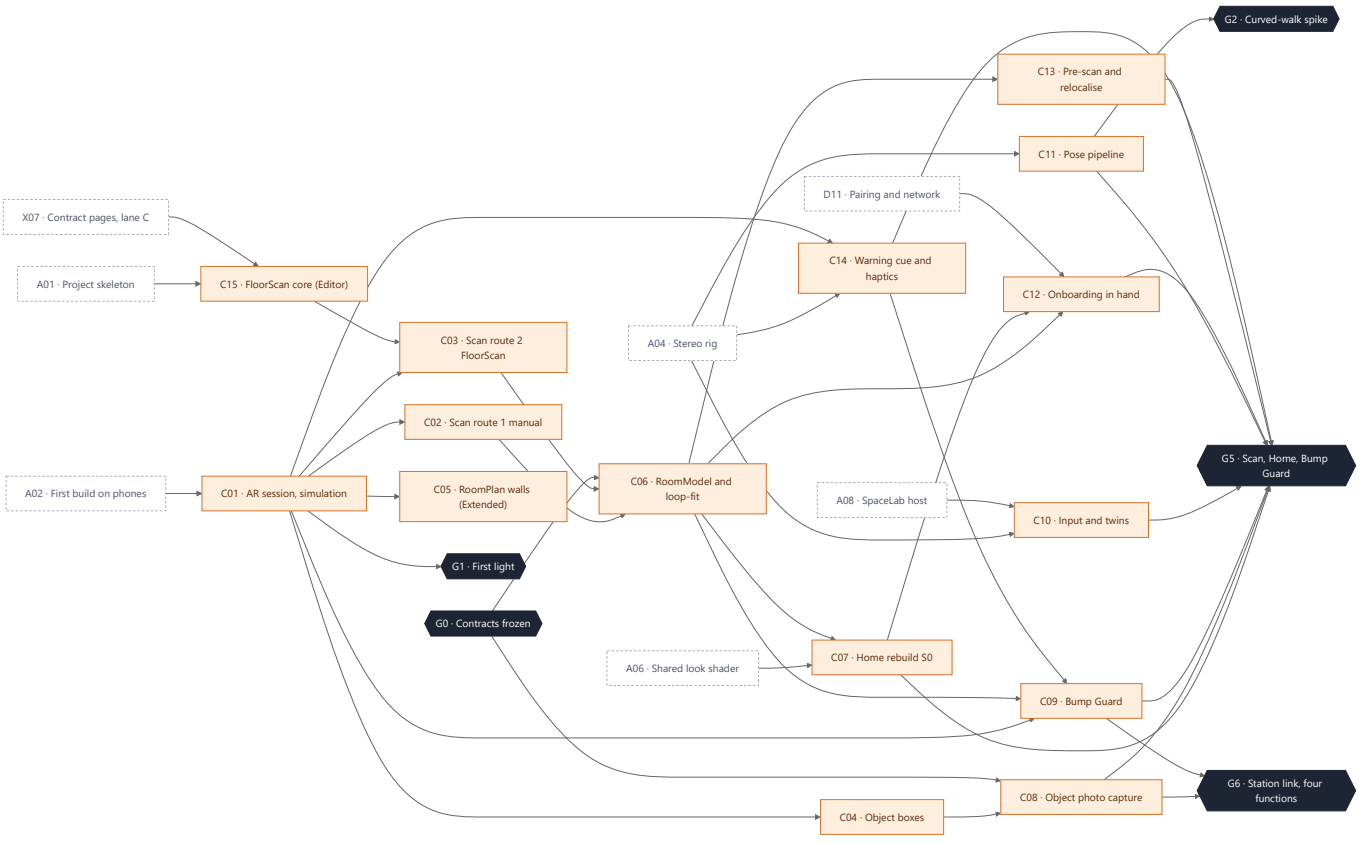
Colours: green lane A, purple lane B, orange lane C, blue lane D, grey shared. Dashed boxes are inputs from other lanes or from the other diagram; dark hexagons are gates.

ID	Task	Depends on	Output	Acceptance test	Runs on
`P1-B01`	**Geometry kernel**	`P1-A01`	`Atlas.Kernel` in double precision: boosts and rotations for κ < 0, = 0, > 0, exp and log maps, distance, η-Gram–Schmidt, inverse as $\eta M^{T} \eta$; no scene types	EditMode: exp∘log = identity to 1e-12; holonomy of a geodesic triangle equals its angle defect to 1e-9; five steps of 1.0613 with 90° turns return to the identity, three steps of π/2 on the sphere, four flat; $\|U^{T} \eta U - \eta\| < 1e-12$ after 100,000 random boosts	Editor

ID	Task	Depends on	Output	Acceptance test	Runs on
P1-B02	**Holonomy rig**	P1-B01, P1-X02	Per-frame update U ← U·Boost(Δ/L) from the filtered head displacement, tile-crossing hook, curvature ramp that only runs while the walker stands still	Replay of a recorded square lap at loop-fit L closes after five sides to 1 mm; a recorded stand-still of 60 s moves the frame by < 5 mm; the kernel step runs after the pose update and exactly once in each of 600 consecutive frames (logged frame counter)	Editor
P1-B03	**Warp shader include**	P1-X02, P1-B01	AtlasWarp.hlsl: one entry point (object position in, world position out) for none / hyperbolic / spherical, log map centred on the head, one per-material anchor matrix computed in double on the CPU and cast last; component that sets huge localBounds	GPU positions match the kernel's CPU reference to 1 mm for ρ ≤ 3; the shared shader is still SRP-Batcher compatible; exp is evaluated once per vertex	Editor
P1-B04	**Exact tile ids and the tile graph**	P1-B01	TileId as six integers in $\mathbb{Z}[\varphi]$, neighbour step S_k = R^k·T·H by additions only, breadth-first visible set, re-root when the walker crosses an edge, content seed = hash of the id; tolerance-hash variant for {4,6}	Per-depth counts equal OEIS A377322 to depth 10; a walk round a vertex closes after exactly 5 steps; $\varphi(X^2+Y^2) - T^2 = -1$ for every tile (evaluated in Int128, because the products overflow 64 bits long before the ids do); tile centres within ρ = 2 / 3 / 3.5 / 4 of a tile centre number 13 / 45 / 81 / 129, and 105 within 3.8425; a crossing costs ≤ 0.5 ms in the Editor	Editor
P1-B05	**Procedural tile meshes**	P1-B01, P1-A06	Generator in C#: Klein-square grid mapped to the log chart; levels of detail by grid count (n = 16 per side on the walker's tile and the tiles that touch it, n = 8 on every other drawn tile, n = 4 on the high tier beyond ρ = 3), every level keeping the same 16 boundary segments per edge; kerbs; corner-post wedges bounded by the tile's own two edges (72° in the curved metric, 65.8° in mesh coordinates for {4,5}); palette UVs; vertex ambient occlusion; a flat variant for Aligned Cells	Edges of neighbouring tiles coincide to 1 mm for every pair of levels; five wedges round a post leave no gap and no overlap; worst screen-space chord error ≤ 0.15° from an eye height of 1.6 m at L = 1.4, 1.9 and 3.0 m (raise n if it fails); ≤ 1,200 / 500 / 250 triangles for the three levels at any L	Editor
P1-B06	**Tile renderer**	P1-B03, P1-B04, P1-B05	Pool of tile objects and material instances; per-tile anchor each frame; level of detail by graph distance; the CPU draw rule of §8.5 (centre within ρ + 0.8425, and the tile's floor disc not outside the widened view frustum; one test per frame for both eyes); fog from ρ − 1, opaque at ρ	Phone A at ρ = 3: tile shells cost ≤ 150 draw calls per frame at any head pitch; 60 fps; no allocation on a tile crossing; over a recorded lap with head turns and looks down at the floor, no undrawn tile has a point that is nearer than ρ and inside a camera frustum (EditMode check over the pose log); in a screenshot the edge of the drawn world cannot be told from the background	Editor, phone in headset
P1-B07	**World generator and WorldLint**	P1-B04, P1-B05	Archetype from hash(tile id, seed): Plaza, Corridor, Doorway, Tower, Span A/B, Garden, Home; four regions with their own hue; corner posts; one spanning object near the start; WorldLint editor check	WorldLint passes for three seeds (reachable, a tower within ρ ≤ 2.5 of every tile, budgets, no coincident faces, the same region for every tile to depth 6 on two runs); the same seed gives the same world hash in two Editor runs and under IL2CPP on the phone (checked at G6)	Editor
P1-B08	**Small Planet S4**	P1-B03, P1-B05	{4,3} tiling: six tiles always loaded, spherical branch of the warp, fragments beyond 2.6 rad discarded, fog opaque at 0.8π, pole beacon	Three steps of (π/2)·R with 90° turns close to 1 mm in the Editor; no smeared triangle in a 360° look-around on the phone	Editor, phone in headset
P1-B09	**Curvature dial and the 3-4-5 walk**	P1-B02, P1-B06, P1-B08, P1-C06	Dial with snaps q = 3, 4, 5, 6 that ramps curvature only while the walker stands still; guided loop markers from the RoomModel lap square; corner counter, closure-error and world-turn read-outs	On phone A the loop closes after 3, 4 and 5 corners with closure error < 0.25 m each; curvature never changes while head speed exceeds 0.1 m/s; with the standing flag set the dial still snaps q = 3 to 6 and no lap markers are shown	phone in headset

ID	Task	Depends on	Output	Acceptance test	Runs on
`P1-B10`	**Aligned Cells S3 (Mode 2)**	`P1-B04`, `P1-B05`, `P1-X02`	Cell grid from the RoomModel (sample rooms until the live scan exists), tile per cell chosen by the walked path, sight-line instances clipped by per-fragment gates (no stencil), a post at every interior corner, switch from Mode 1 at the same tile id	EditMode: 13 / 15 / 17 instances for centre / edge / corner cell of a 3×3 grid; in the Editor and on the phone four quarter-turns round an interior corner end in a fifth tile and a fifth returns to the first; tile instances alone cost ≤ 120 draw calls per frame on the 4×5 m sample room	Editor, phone in headset
`P1-B11`	**The room becomes tile zero**	`P1-B06`, `P1-B07`, `P1-C07`	Home rebuild placed in curved space by the exp map about the room centre; walls sink to kerbs and corner posts when curvature is on; tiles under the room's footprint are flagged home and kept clear; return-to-Home re-registers; the lap line starts 0.17 m from a corner post of tile zero (at L = 1.88 m) and runs along tile zero's edges at a shallow angle, so the post fade and the tile-crossing rule are checked on a replayed lap	At κ = 0 the curved-space room matches Home to 1 cm; a ramp to loop-fit L and back restores registration to 2 cm; WorldLint finds no generated prop inside the room footprint; a replayed lap of the lap square shows no flicker of the tile id along the edges	phone in headset
`P1-B12`	**Room content in Aligned Cells**	`P1-B10`, `P1-C07`	Home walls on the edge of the cell grid; every object from the RoomModel (placeholder box or generated mesh) drawn once per eye at its true position whenever any instance of its cell is drawn, never duplicated per instance (§4.7)	On phone A a virtual table edge is within 5 cm of the real one at three touch points after ten cell crossings; S3 on `budget_room` stays ≤ 170 draw calls and ≤ 200k triangles per frame at 60 fps	phone in headset

9.5.7 Lane C · Scan, Room and Safety

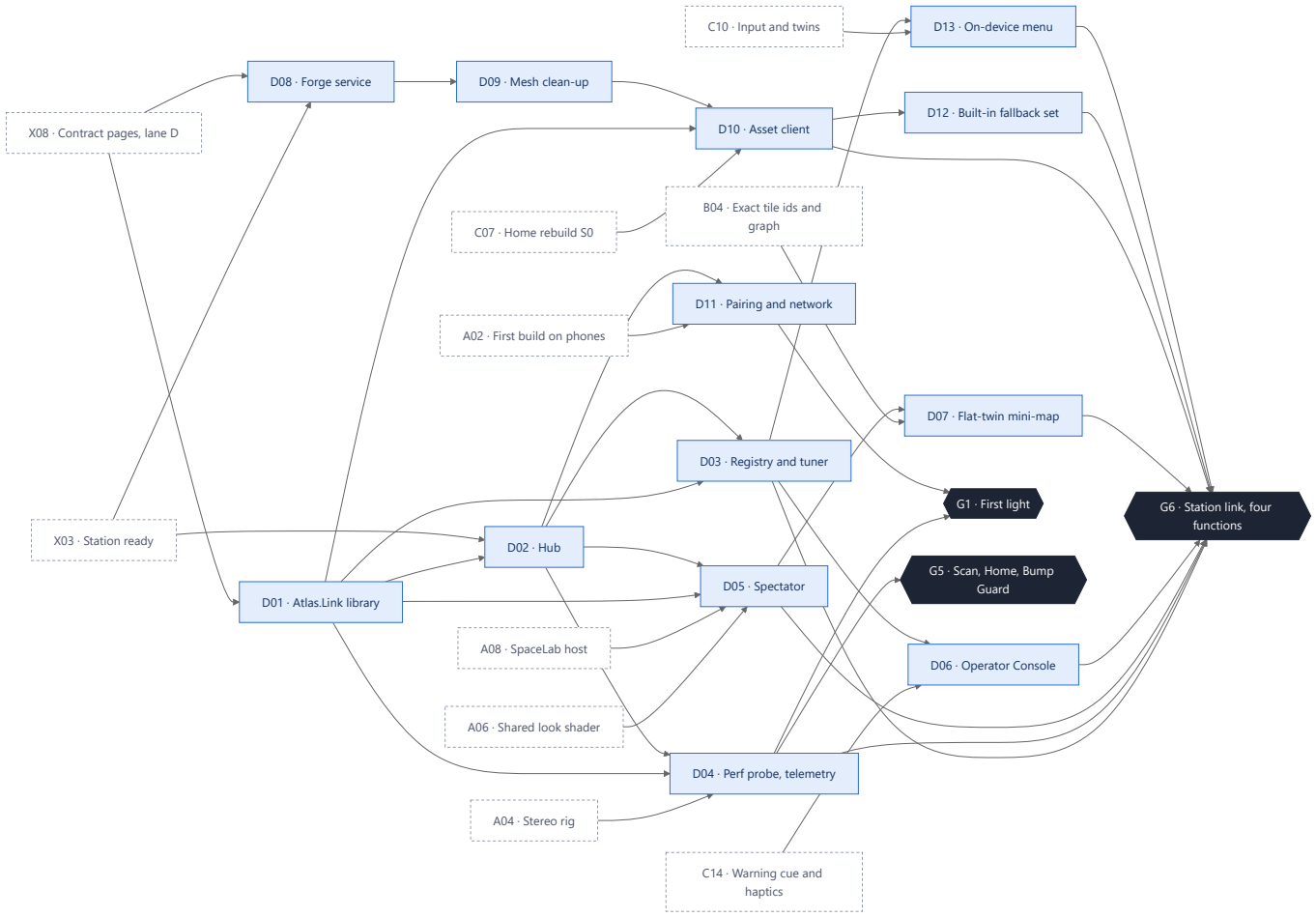


Colours: green lane A, purple lane B, orange lane C, blue lane D, grey shared. Dashed boxes are inputs from other lanes or from the other diagram; dark hexagons are gates.

ID	Task	Depends on	Output	Acceptance test	Runs on
P1-C01	**AR session and Editor simulation**	P1-A02	Lowest-resolution 60 Hz video format; environment depth through AROcclusionManager (mode Fastest, temporal smoothing off, human stencil and human depth off), read with TryAcquireEnvironmentDepthCpuImage (256 × 192); meshing and plane detection on only during the scan; XR Simulation environments; pose-log and RoomModel replay in the Editor	The session holds 60 Hz on the phone and the depth image is non-null at the chosen video format; a recorded walk drives the rig in the Editor	Editor, phone in hand
P1-C02	**Scan route 1: manual outline**	P1-C01	Tap or walk the floor corners to make the floor polygon	Polygon within 5 cm of a tape measure in three rooms	phone in hand
P1-C03	**Scan route 2: automatic FloorScan on the phone**	P1-C01, P1-C15	The classified LiDAR mesh and the planes of a live scan fed into the FloorScan core of P1-C15, with a progress view of the floor found so far	On the phone the free rectangle is within 10 cm of a tape measure and scan-to-walkable takes ≤ 90 s	phone in hand
P1-C04	**Object boxes**	P1-C01	RoomPlan furniture boxes through ARBoundingBoxManager (category, size, pose), merged with the FloorScan obstacles in one coordinate frame	Boxes overlay the mesh to ≤ 10 cm; in three test rooms every piece of furniture wider than 0.4 m has a box (proposed)	phone in hand
P1-C05	**Swift RoomPlan bridge for walls, doors and windows (Extended)**	P1-C01	Native plugin using RoomCaptureSession(arSession:); walls, doors, windows and openings as JSON in session coordinates; AR configuration re-applied afterwards	Wall positions within 3 cm of the mesh-derived ones; scene depth is available again after capture	phone in hand
P1-C06	**RoomModel and the loop-fit**	P1-C02, P1-C03, P1-X02	RoomModel JSON from either scan route: polygon, distance field, free rectangle, lap square s, L and R from the loop-fit rule, cell grids for S1 and S3, object list, anchor; outline confirmation in hand; save and load	Validates against the schema; a manual-outline scan and an automatic scan of the same room give free rectangles within 10 cm of each other; a room with less than 1.5 × 1.5 m of free floor selects standing mode; the log confirms meshing and planes are off after the scan	Editor, phone in hand
P1-C07	**Home rebuild S0**	P1-C06, P1-A06	Floor, walls and door openings as 0.25 m-grid meshes in the palette, registered at the anchor; a placeholder box per object, tinted by category, on an object component that later takes a generated mesh	In the headset a virtual wall is within 5 cm of the real wall at three touch points (should: 3 cm); Home on budget_room costs ≤ 60 draw calls and ≤ 130k triangles per frame	phone in headset
P1-C08	**Object photo capture**	P1-C04, P1-X02	For each object box, one to three key frames chosen by projected area, centredness, sharpness and spread of view angle; cropped and JPEG-encoded off the main thread; handed to the asset client	Every box that has a candidate frame under the rule of §7.4 gets at least one crop; boxes with none are reported (they take the built-in fallback mesh in P1-D12); capture adds < 2 ms per frame; nothing is written to disk	phone in hand
P1-C09	**Bump Guard**	P1-C06, P1-C01, P1-C14	The logic of §4.10: static clearance from the distance field; a persistent dynamic layer stamped from live LiDAR depth (Burst); straight-path prediction over 0.4 s and 0.8 s; live depth alone while tracking is not Normal; a check of the static map against live depth; a level and a direction passed to BumpCue. Thresholds and the depth rate are constants in code, not in the parameter registry	Scripted approaches at 0.5 and 1.0 m/s to a wall, to a chair and to a 0.45 m stool placed after the scan, with the walker looking level: warning ≥ 0.6 s before predicted contact in 20 of 20 trials; during a 5-minute lap of the lap square: at most 2 STRONG cues or straight-side WARNs (should: none), corner WARNs counted and reported; ≤ 0.3 ms per frame; identical with the Station unplugged	Editor, phone in headset
P1-C10	**Input and twins**	P1-A04, P1-A08	Gamepad through the Input System, gaze reticle with 600–800 ms dwell, headset lever tap; one action map	Every SpaceLab action can be done with the gamepad alone and with gaze dwell alone	Editor, phone in headset

ID	Task	Depends on	Output	Acceptance test	Runs on
P1-C11	**Pose pipeline**	P1-A04	TrackedPoseDriver on Update only; jumps over 0.2 m per frame rejected for geometry while safety keeps the raw pose; tracking-state events; gyro rotation prediction behind a switch	A 0.5 m jump injected in replay is absent from the geometry pose stream and present in the safety pose stream; loss of tracking raises the tracking-lost event within 0.5 s	Editor, phone in headset
P1-C12	**Onboarding in hand**	P1-C06, P1-C07, P1-D11	Permissions (camera, Local Network), pairing, scan route, outline confirmation, headset prompt	Three first-time users reach Home in the headset with at most three prompts each	phone in hand, phone in headset
P1-C13	**Pre-scan path: world map save and relocalise**	P1-C06	Save ARWorldMap with the RoomModel; relocalise at launch	Relocalised RoomModel aligns to ≤ 5 cm within 10 s in the room it was scanned in (should: 3 cm within 5 s)	phone in hand
P1-C14	**Warning cue and haptics**	P1-C01, P1-A04	BumpCue in Atlas.Safety: head-locked chevron on the hazard side, stand-still and rescan icons, spatialised tick and tone, CoreHaptics pulses at 2 and 6 per second, gamepad rumble if reachable (open item 9); one local entry point Fire(level, direction) used by Bump Guard and by the Console's STOP command	For WARN, STRONG and tracking-lost the indicator, sound and haptic all start within one frame of the call in 20 of 20 trials; ≤ 0.2 ms per frame and no allocation (proposed); identical with the Station unplugged	phone in headset
P1-C15	**FloorScan and loop-fit core in the Editor**	P1-A01, P1-X07	The grid pipeline as plain C# with no AR types: classified mesh in → 5 cm occupancy grid → signed distance field → walkable polygon with holes, dominant wall direction, free rectangle, obstacle boxes; then the loop-fit (s, L, R) and the cell grids for S1 and S3	EditMode FloorScan_Synthetic passes on synthetic rooms (rectangle, L-shape, table in the middle); s, L and R equal the §3.2 table for empty 3 × 3 m and 4 × 5 m rectangles, and the reference values on lane C's contract page for the two furnished sample rooms; cell grids equal the examples of §4.3 and §4.7	Editor

9.5.8 Lane D · Link, Station and Forge



Colours: green lane A, purple lane B, orange lane C, blue lane D, grey shared. Dashed boxes are inputs from other lanes or from the other diagram; dark hexagons are gates.

ID	Task	Depends on	Output	Acceptance test	Runs on
`P1-D01`	**Atlas.Link library**	`P1-X08`	Plain C# (no Unity references): message types, hand-packed `State` and `Telemetry`, JSON control, clock sync, watchdog, `LinkClient` and `LinkHost` over LiteNetLib; loopback tests. It is built and tested as a .NET library first, from lane D's own contract pages, and compiles into the Unity project once `P1-A01` exists (checked at G0)	Round trip for every message; 60 Hz `State` survives 2 % simulated loss; clock offset error ≤ 2 ms p95 on a simulated link	Station, Editor
`P1-D02`	**Hub**	`P1-D01`, `P1-X03`	`atlas-hub` (.NET 8 console): LiteNetLib server, relay between phone, Spectators and Console, web server, JSONL and binary logs	Relay adds < 2 ms; the Hub can be restarted and every client reconnects within 5 s	Station
`P1-D03`	**Parameter registry and Console tuner**	`P1-D01`, `P1-D02`	Registry on the phone with min, max and default per parameter; `ParamManifest` and `ParamSet`; clamps on the phone; demo lock; web page with sliders grouped by space; presets saved and loaded as JSON on the Station (`/api/presets`); Bake writes a preset into the project defaults	A slider change is visible in the headset within 100 ms at p95 (proposed); with the lock on, `ParamSet` is ignored; a saved preset reloads all its values in one `ParamSet`; after Bake, a build started with the Hub off has those values	Editor, Station, phone in headset
`P1-D04`	**Perf probe, HUD and telemetry**	`P1-D01`, `P1-D02`, `P1-A04`	`AtlasPerfProbe` (FrameTimingManager and ProfilerRecorders), in-headset HUD at 2 Hz, `Telemetry` at 1 Hz, dashboard charts, CSV on the Station and a local copy on the phone	Every `Telemetry` field of Appendix B.2 appears in the CSV (the probe rows of §11.2 from frame time to managed allocation, plus the thermal state from the `P1-A02` plugin); counters whose source is not built yet read 0; the HUD costs < 0.2 ms and allocates nothing; a 15-minute run yields a complete CSV	phone in headset, Station

ID	Task	Depends on	Output	Acceptance test	Runs on
P1-D05	**Spectator**	P1-D01, P1-D02, P1-A06, P1-A08	The same Unity project built for Windows with ATLAS_SPECTATOR; re-renders the walker's view from State (space, tile id, tile-local transform, parameters, head pose); world-hash check	The mirror lags the phone by ≤ 150 ms at p95; a world mismatch shows a banner instead of a wrong picture	Station
P1-D06	**Operator Console**	P1-D03, P1-C14	Buttons: switch space, return to Home, recentre, quality tier, STOP (fires the warning cue and haptic), lock; comfort-score entry and notes	Every command acts on the phone within 250 ms and is written to the log	Station, phone in headset
P1-D07	**Flat-twin mini-map**	P1-B04, P1-D05	One renderer for the headset and the Station: Poincaré disc of the tile graph centred on the walker with the walked trail and the room outline; cube net for S4; lattice for S1; room graph for S5	In replay the map shows the same tile-id sequence as the log; ≤ 0.3 ms on the phone	Editor, Station, phone in headset
P1-D08	**Forge service**	P1-X03, P1-X08	FastAPI on the Station: scan upload, object upload, job queue with one GPU worker, background removal, image-to-3D with TripoSG, job status, asset download by content hash	One object photo becomes a raw mesh on the RTX 5070 in ≤ 60 s (proposed; record the measured value); a corrupt image fails its job cleanly	Station
P1-D09	**Mesh clean-up**	P1-D08	Orient and scale to the scanned box, decimate to the triangle target sent with the job (1,500–5,000), regrid so no edge exceeds 0.25 m, palette colour from the photo, vertex ambient occlusion, GLB, content hash	Every output passes the asset validator (triangle count, longest edge, longest side within 10 % of the box (should: 5 %), one material slot); the same input gives the same hash	Station
P1-D10	**Asset client on the phone**	P1-D01, P1-D09, P1-C07	Uploads (RoomModel, crops), JobStatus and AssetReady handling, download with a cache, glTFast load off the main thread, material remap to the shared shader, placeholder-to-mesh swap; each job carries its object's triangle target (§4.5)	Worst frame during a swap < 20 ms; a repeated photo is served from the cache; with the Station unplugged the app behaves normally	phone in hand, phone in headset
P1-D11	**Pairing and network bring-up**	P1-D02, P1-A02	Router set-up, QR page from the Hub, Local Network prompt triggered in hand with retry, link indicator	A fresh install pairs by QR in ≤ 30 s (proposed); after a Wi-Fi drop the link returns within 5 s	phone in hand, Station
P1-D12	**Built-in fallback set**	P1-D10	One CC0 mesh per RoomPlan category, ≤ 1.5k triangles, palette-mapped, shipped in the build; placed scaled to the box after 3 failed uploads (back-off 1, 2, 4 s) or when a box has no candidate frame; a box with no RoomPlan category keeps its placeholder; tagged "built-in" in the log and on the Console; replaced when the Forge returns	Test L19: with the Forge stopped before the scan, every box that has a RoomPlan category holds a fallback mesh within 10 s of entering Home (proposed); after a Forge restart each of them is replaced by its generated mesh; every fallback mesh has ≤ 1.5k triangles and no edge over 0.25 m after scaling	phone in headset, Station
P1-D13	**On-device parameter menu**	P1-D03, P1-C10	In-headset list of the parameters of §12.2 and §12.4 (the fallback ladders and the kill switches), opened by a gamepad chord and driven by the same registry; every change written to the phone's log	With link.enabled off, each parameter of §12.2 and §12.4 can be changed from the gamepad and the change is in the phone's log	phone in headset

9.5.9 Phase 1 gates

ID	Task	Depends on	Output	Acceptance test	Runs on
P1-G1	**G1 First light**	P1-A04, P1-A13, P1-C01, P1-D11, P1-D04	Stereo at 60 fps on the phone, live telemetry on the Station, the build recipe and the phone-booking rule in force	Empty stereo scene at 60 fps with the HUD on; telemetry row per second on the dashboard; every engineer has installed a build; the depth image is non-null at the chosen video format	phone in headset, Station
P1-G2	**G2 Curved-walk spike**	P1-B02, P1-C11, P1-G1	Rolling update on the phone with line-drawn tiles (no shader needed); three walkers from the team. Safety for the spike: Bump Guard does not exist yet, so the floor is cleared, the lap square is marked on the real floor for the spotter (a test aid, not part of the product) and a spotter stays within arm's reach throughout	Five-corner closure error < 0.25 m at loop-fit L; comfort score ≤ 3 of 20 after three minutes at loop-fit L for all three walkers; zero wall contacts and zero spotter stops. If it fails: raise L one step, or make Aligned Cells the default mode	phone in headset

ID	Task	Depends on	Output	Acceptance test	Runs on
P1-G3	**G3 Spaces in the Editor**	P1-A09, P1-A11, P1-A12, P1-B06, P1-B07, P1-B08, P1-B10	Every space module S1–S6 runs in SpaceLab in the Editor on the three sample rooms	All EditMode tests green; each space selectable from the gamepad menu; shader contract unchanged since G0 or re-agreed	Editor
P1-G4	**G4 Spaces on the phone at budget**	P1-G1, P1-G3, P1-A14	Per-space metric table and the soak CSV	Every space S0–S6 meets the Must column of §11.2, cold, on budget_room; the 15-minute soak passes	phone in headset
P1-G5	**G5 Scan, Home and Bump Guard**	P1-C07, P1-C08, P1-C09, P1-C10, P1-C11, P1-C12, P1-C13, P1-C14, P1-D04	Self-scan and pre-scan paths end in a registered Home; Bump Guard verified by someone who did not write it	Both scan paths reach Home; the Bump Guard scripted test passes in the headset; a 1.4 × 1.4 m free floor selects standing mode	phone in headset
P1-G6	**G6 Station link, all four functions**	P1-D03, P1-D04, P1-D05, P1-D06, P1-D07, P1-D10, P1-D12, P1-D13, P1-C08, P1-C09, P1-B07	Spectator mirror, live tuning and Console, telemetry dashboard and CSV, scan and asset round-trip	§7.6 table passes: state ≥ 50 Hz with < 2 % loss; mirror latency ≤ 150 ms p95; a parameter change visible ≤ 100 ms; one CSV row per second; photo to placed mesh for every object box in a test room with no frame over 20 ms; with the Forge stopped the fallback set is placed; with the Station unreachable a parameter can still be changed from the on-device menu; the app is unaffected when the Hub is unplugged	phone in headset, Station
P1-G7	**G7 Phase-1 exit**	P1-G2, P1-G4, P1-G5, P1-G6, P1-B09, P1-B11, P1-B12, P1-A15	End-to-end run by a first-time user, watched on the Spectator	Self-scan → Home with furniture arriving live → dial → 3-4-5 walk → Curved Walk → Aligned Cells (a virtual table edge stays within 5 cm of the real one) → Glued → Small Planet → Impossible Rooms → Slicer, with tuning and telemetry live; the §11.2 table is filled; every Phase-1 gate checklist in §11.3 is ticked	phone in headset, Station

9.6 Phase 2 · Geometry Playground and Lung Mission

Two tracks run in parallel from G7. The specifications and the work-package tables are in §10.1 (Playground) and §10.2 (Lung Mission); the diagrams here show the order.

Track	Lanes	Packages	Exit gate
Geometry Playground	C (shell; Glued and Impossible Rooms exhibits) and A (Slicer toybox, equation sources, look)	P2-C01 – P2-C17, P2-A01 – P2-A04	G8
Lung Mission	D (Station pipeline, case packages, plan card), A (medical 4D runtime), B (airway wing, needle sky)	P2-D01 – P2-D13, P2-A20 – P2-A32, P2-B01 – P2-B12	G9 (4D), G10 (airway and sky)

Lane A serves both tracks, which departs from a strict "two lanes each" (open item 17 in §14): the Slicer toybox and the medical 4D runtime both build on lane A's Mesh4D. Lane A does P2-A01 and P2-A02 first, because P2-A01 unblocks lane C's exhibit menu (P2-C08), then starts P2-A20 onwards; P2-A03 and P2-A04 follow when their lane-C inputs land.

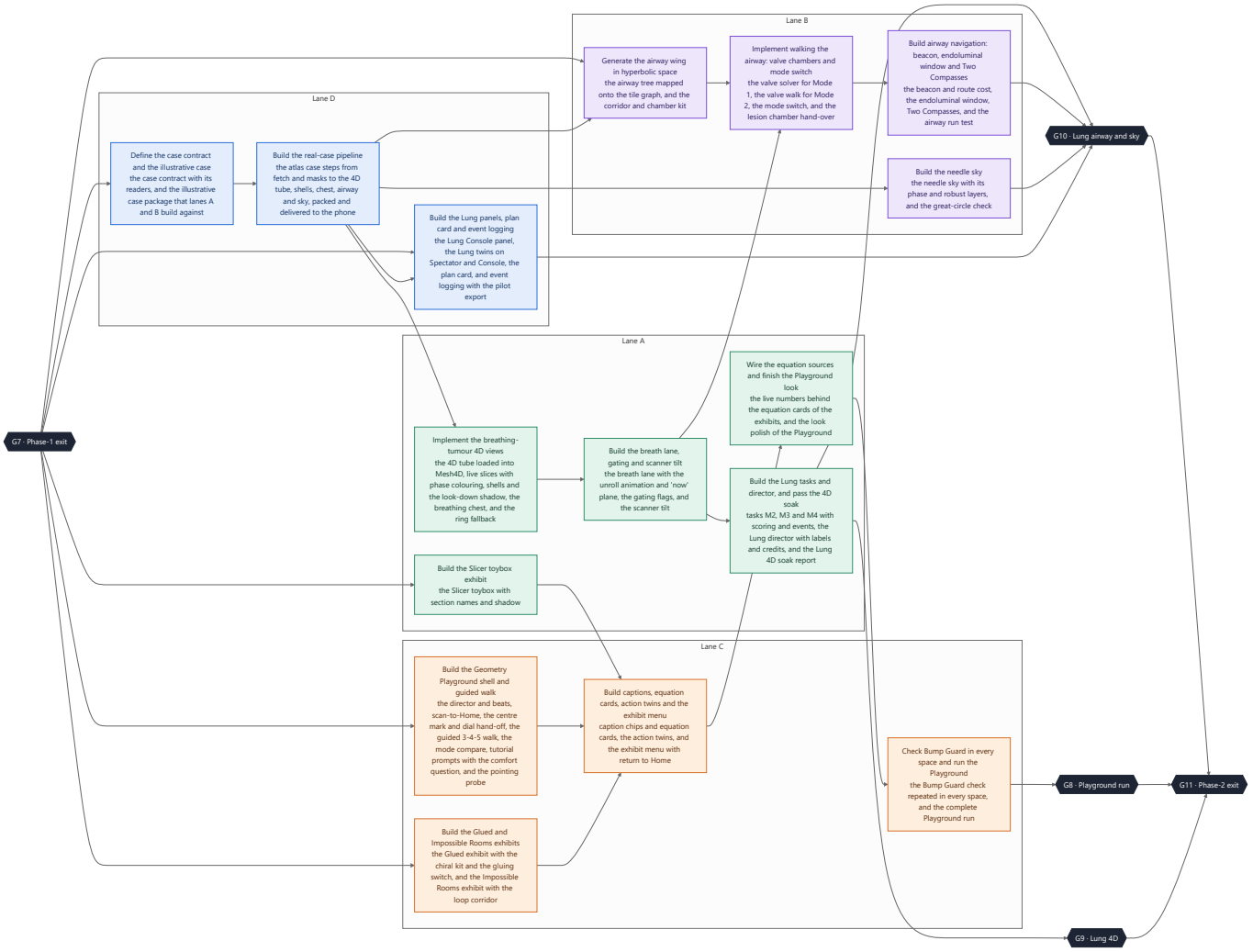
What is sequential:

- In the Playground everything hangs off the director (P2-C01); the exhibits wrap Phase 1 spaces and are independent of one another; the run (P2-C11) is last.
- In the Lung Mission the case contract (P2-D01) comes first. Lane D then produces the **illustrative** package (P2-D02) before the real one, so lanes A and B have data to build against without waiting for the full pipeline. The 4D runtime (lane A) and the airway wing (lane B) are independent of each other until the lesion chamber hand-over (P2-B09). The Lung director (P2-A32) stubs the beats that belong to lanes B and D, so G9 does not wait for the airway.

- G11 needs G8, G9 and G10, and then one combined session by someone from outside the team.

Work areas, Phase 2

The areas are for people; the packages inside them are for agents. An arrow from a gate into an area means work in that area can start at that gate; an arrow between areas means the second cannot be finished without the first.



Colours: green lane A, purple lane B, orange lane C, blue lane D; dark hexagons are gates.

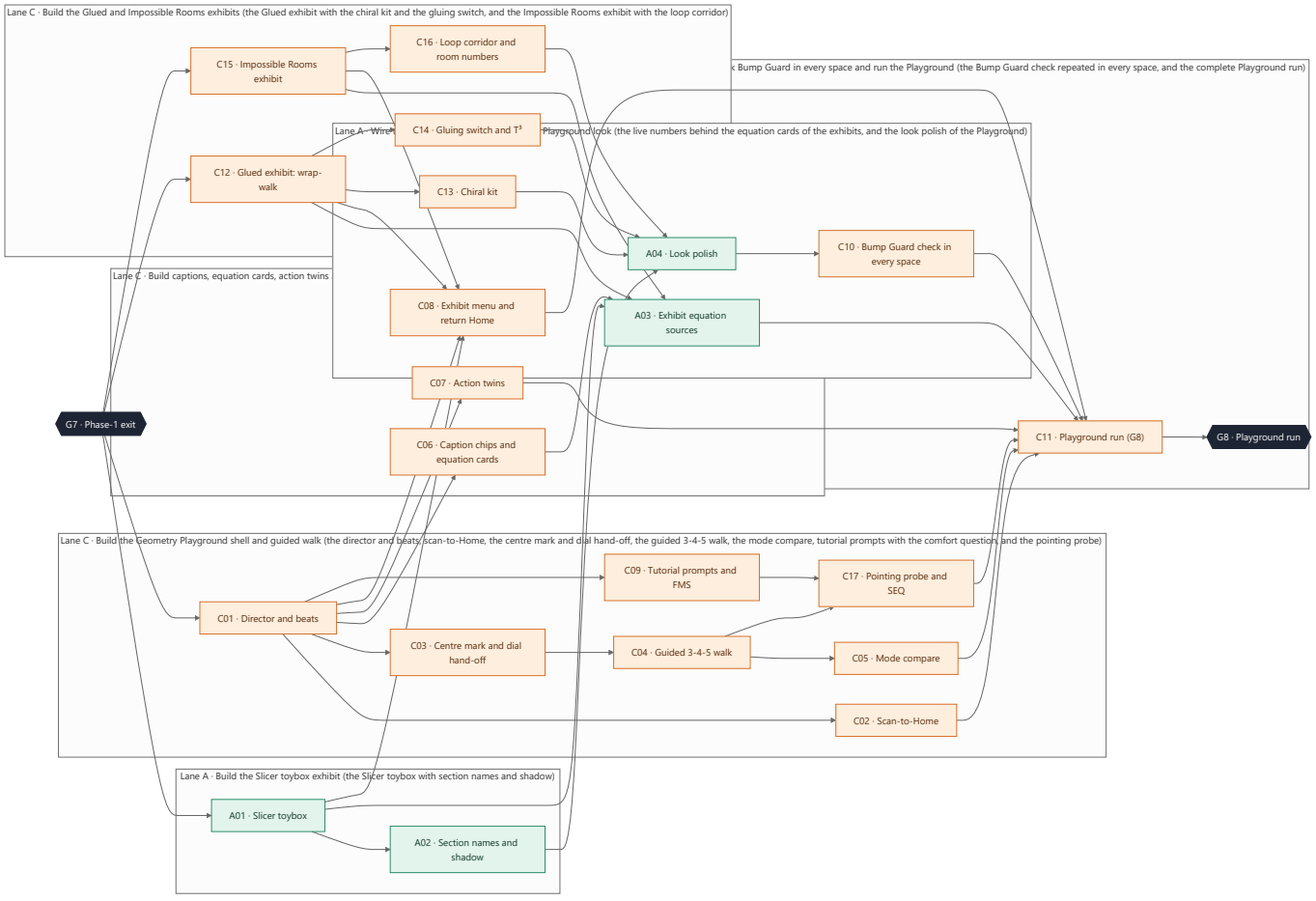
Area	Owner	Tasks inside	Needs from other lanes	Hands over to	Gates it feeds
Build the Slicer toybox exhibit the Slicer toybox with section names and shadow	Lane A	`P2-A01`, `P2-A02`	nothing	Build captions, equation cards, action twins and the exhibit menu	none directly
Wire the equation sources and finish the Playground look the live numbers behind the equation cards of the exhibits, and the look polish of the Playground	Lane A	`P2-A03`, `P2-A04`	Build captions, equation cards, action twins and the exhibit menu; Build the Glued and Impossible Rooms exhibits	Check Bump Guard in every space and run the Playground	none directly
Build the Geometry Playground shell and guided walk the director and beats, scan-to-Home, the centre mark and dial hand-off, the guided 3-4-5 walk, the mode compare, tutorial prompts with the comfort question, and the pointing probe	Lane C	`P2-C01`, `P2-C02`, `P2-C03`, `P2-C04`, `P2-C05`, `P2-C09`, `P2-C17`	nothing	nothing	none directly

Area	Owner	Tasks inside	Needs from other lanes	Hands over to	Gates it feeds
Build captions, equation cards, action twins and the exhibit menu caption chips and equation cards, the action twins, and the exhibit menu with return to Home	Lane C	`P2-C06`, `P2-C07`, `P2-C08`	Build the Slicer toybox exhibit	Wire the equation sources and finish the Playground look	none directly
Build the Glued and Impossible Rooms exhibits the Glued exhibit with the chiral kit and the gluing switch, and the Impossible Rooms exhibit with the loop corridor	Lane C	`P2-C12`, `P2-C13`, `P2-C14`, `P2-C15`, `P2-C16`	nothing	Wire the equation sources and finish the Playground look	none directly
Check Bump Guard in every space and run the Playground the Bump Guard check repeated in every space, and the complete Playground run	Lane C	`P2-C10`, `P2-C11`	Wire the equation sources and finish the Playground look	nothing	G8
Implement the breathing-tumour 4D views the 4D tube loaded into Mesh4D, live slices with phase colouring, shells and the look-down shadow, the breathing chest, and the ring fallback	Lane A	`P2-A20`, `P2-A21`, `P2-A22`, `P2-A23`, `P2-A24`, `P2-A30`	Define the case contract and the illustrative case; Build the real-case pipeline	nothing	none directly
Build the breath lane, gating and scanner tilt the breath lane with the unroll animation and "now" plane, the gating flags, and the scanner tilt	Lane A	`P2-A25`, `P2-A26`, `P2-A27`, `P2-A28`	nothing	Implement walking the airway: valve chambers and mode switch	none directly
Build the Lung tasks and director, and pass the 4D soak tasks M2, M3 and M4 with scoring and events, the Lung director with labels and credits, and the Lung 4D soak report	Lane A	`P2-A29`, `P2-A31`, `P2-A32`	Build the real-case pipeline	nothing	G9, G10
Generate the airway wing in hyperbolic space the airway tree mapped onto the tile graph, and the corridor and chamber kit	Lane B	`P2-B01`, `P2-B02`	Build the real-case pipeline	nothing	none directly
Implement walking the airway: valve chambers and mode switch the valve solver for Mode 1, the valve walk for Mode 2, the mode switch, and the lesion chamber hand-over	Lane B	`P2-B03`, `P2-B04`, `P2-B05`, `P2-B09`	Build the breath lane, gating and scanner tilt	nothing	G10
Build airway navigation: beacon, endoluminal window and Two Compasses the beacon and route cost, the endoluminal window, Two Compasses, and the airway run test	Lane B	`P2-B06`, `P2-B07`, `P2-B08`, `P2-B12`	Build the real-case pipeline	nothing	G10
Build the needle sky the needle sky with its phase and robust layers, and the great-circle check	Lane B	`P2-B10`, `P2-B11`	Build the real-case pipeline	nothing	G10
Define the case contract and the illustrative case the case contract with its readers, and the illustrative case package that lanes A and B build against	Lane D	`P2-D01`, `P2-D02`	nothing	Implement the breathing-tumour 4D views	none directly

Area	Owner	Tasks inside	Needs from other lanes	Hands over to	Gates it feeds
Build the real-case pipeline the `atlas case` steps from fetch and masks to the 4D tube, shells, chest, airway and sky, packed and delivered to the phone	Lane D	`P2-D03`, `P2-D04`, `P2-D05`, `P2-D06`, `P2-D07`, `P2-D08`, `P2-D09`	nothing	Implement the breathing-tumour 4D views; Build the Lung tasks and director, and pass the 4D soak; Generate the airway wing in hyperbolic space; Build airway navigation: beacon, endoluminal window and Two Compasses; Build the needle sky	none directly
Build the Lung panels, plan card and event logging the Lung Console panel, the Lung twins on Spectator and Console, the plan card, and event logging with the pilot export	Lane D	`P2-D10`, `P2-D11`, `P2-D12`, `P2-D13`	nothing	nothing	G10

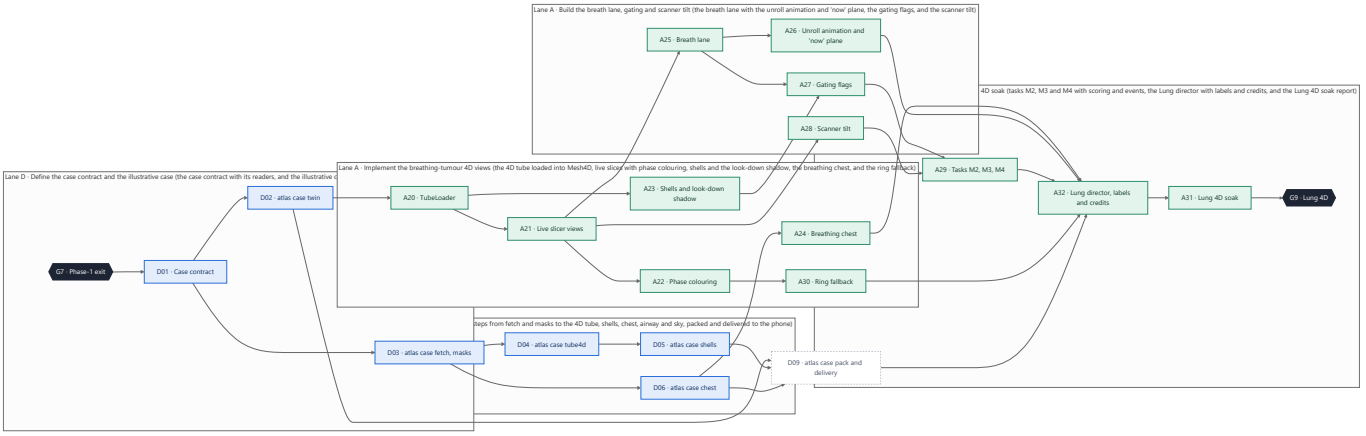
The three diagrams below show the same phase package by package.

Track: Geometry Playground



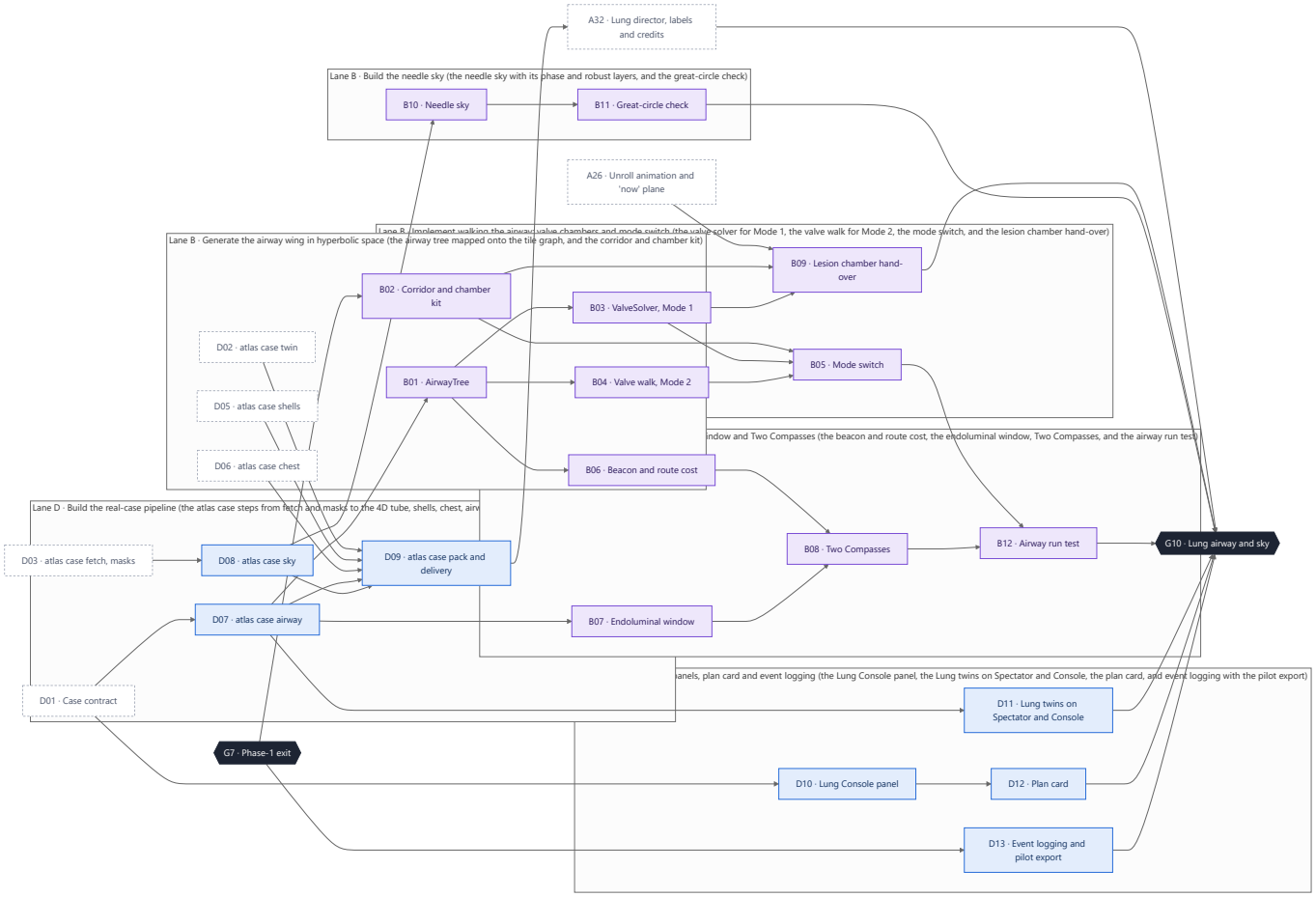
Colours: green lane A, purple lane B, orange lane C, blue lane D, grey shared. Dashed boxes are inputs from other lanes or from the other diagram; dark hexagons are gates.

Track: Lung Mission, data and 4D (lanes D and A, to G9)



Colours: green lane A, purple lane B, orange lane C, blue lane D, grey shared. Dashed boxes are inputs from other lanes or from the other diagram; dark hexagons are gates.

Track: Lung Mission, airway and sky (lanes B and D, to G10)



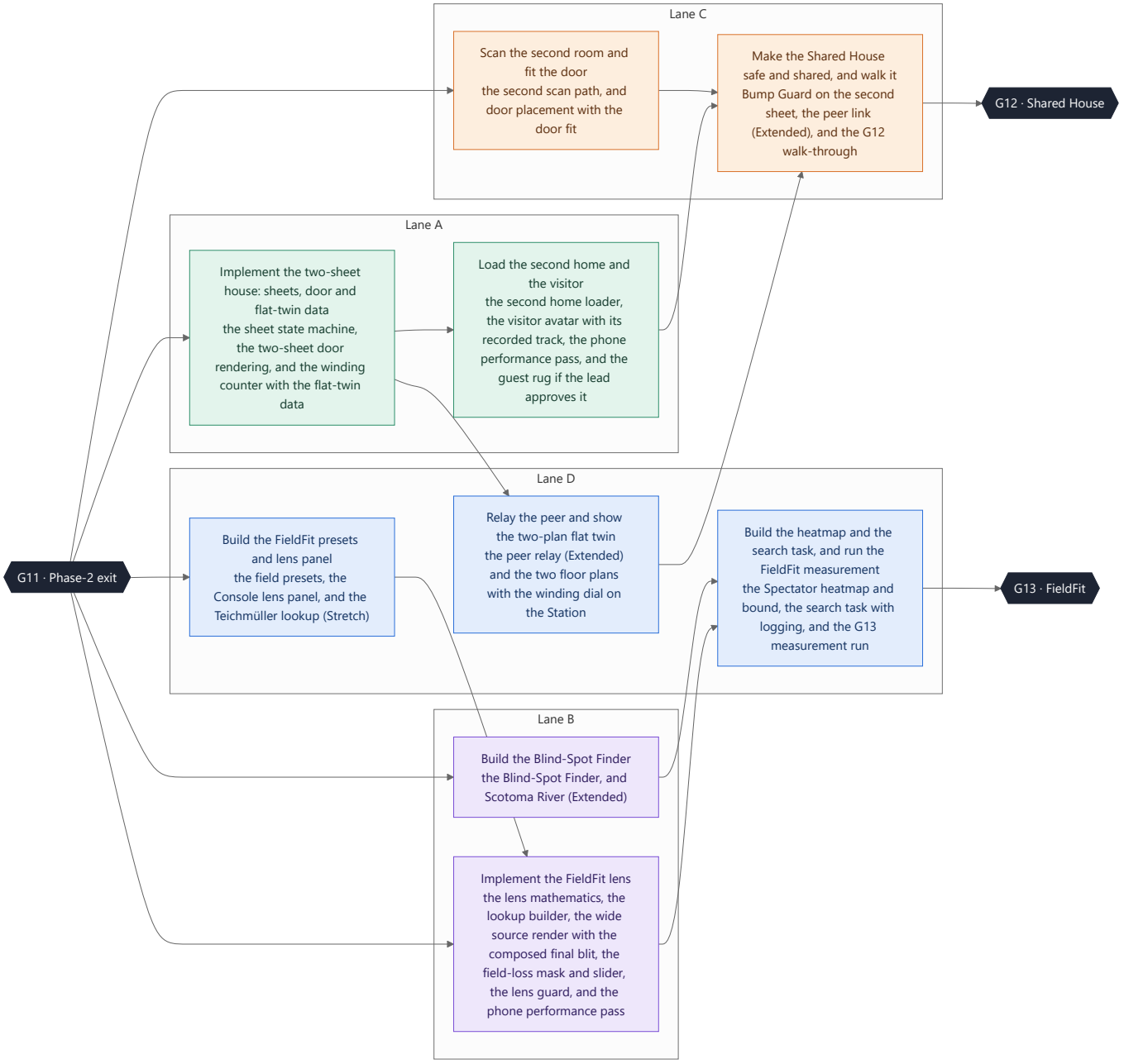
Colours: green lane A, purple lane B, orange lane C, blue lane D, grey shared. Dashed boxes are inputs from other lanes or from the other diagram; dark hexagons are gates.

9.7 Phase 3 · Shared House and FieldFit

Two independent use cases, two lanes each; lane D adds two small packages to the Shared House (the peer relay and the two-plan flat twin). Specifications and work-package tables: §10.3 and §10.4. Packages marked Extended, Stretch or Blocked are drawn but feed no gate.

Work areas, Phase 3

An arrow from a gate into an area means work in that area can start at that gate; an arrow between areas means the second cannot be finished without the first.

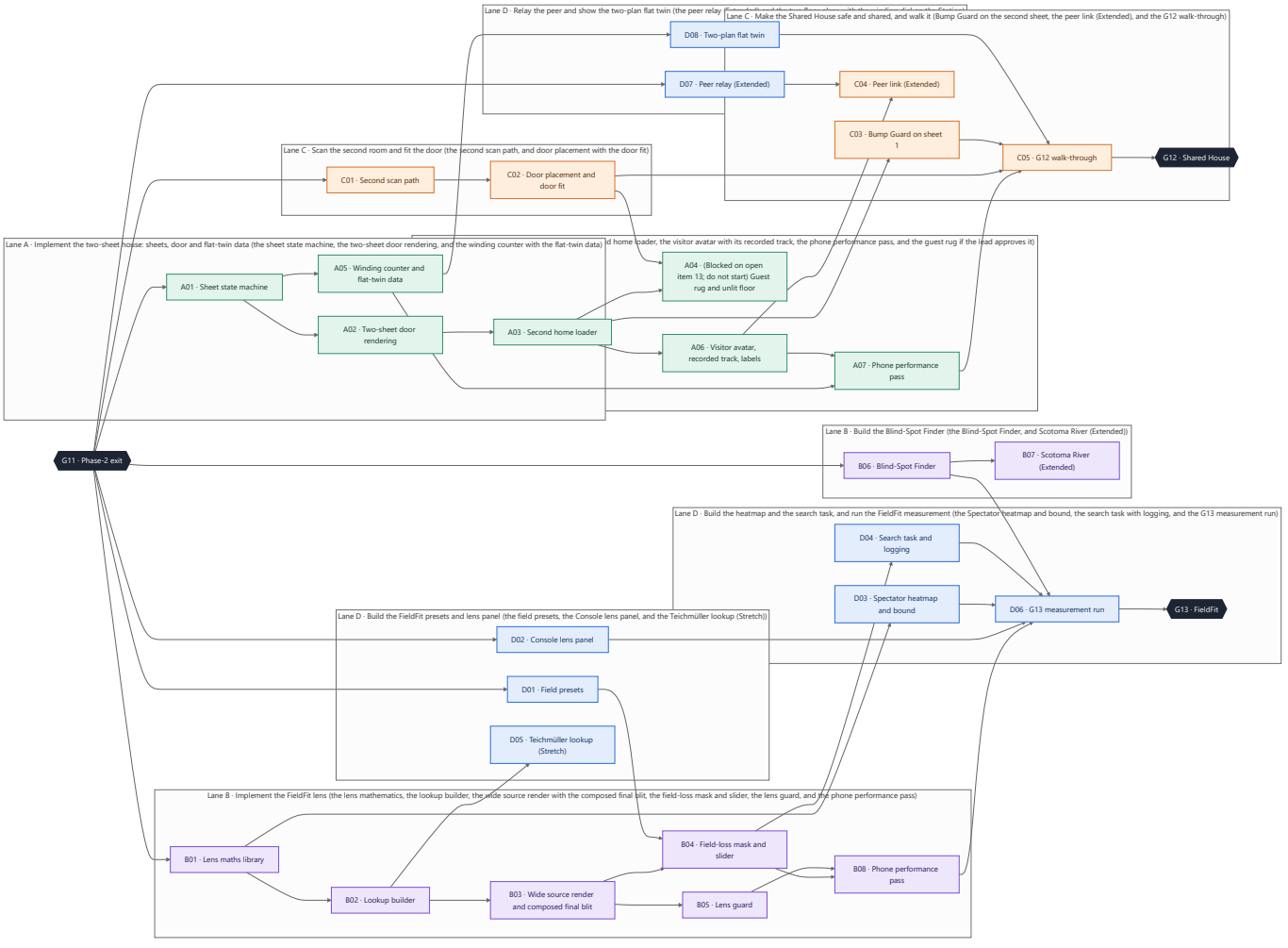


Colours: green lane A, purple lane B, orange lane C, blue lane D; dark hexagons are gates.

Area	Owner	Tasks inside	Needs from other lanes	Hands over to	Gates it feeds
Implement the two-sheet house: sheets, door and flat-twin data the sheet state machine, the two-sheet door rendering, and the winding counter with the flat-twin data	Lane A	`P3-A01`, `P3-A02`, `P3-A05`	nothing	Relay the peer and show the two-plan flat twin	none directly
Load the second home and the visitor the second home loader, the visitor avatar with its recorded track, the phone performance pass, and the guest rug if the lead approves it	Lane A	`P3-A03`, `P3-A04`, `P3-A06`, `P3-A07`	nothing	Make the Shared House safe and shared, and walk it	none directly
Scan the second room and fit the door the second scan path, and door placement with the door fit	Lane C	`P3-C01`, `P3-C02`	nothing	nothing	none directly

Area	Owner	Tasks inside	Needs from other lanes	Hands over to	Gates it feeds
Make the Shared House safe and shared, and walk it Bump Guard on the second sheet, the peer link (Extended), and the G12 walk-through	Lane C	`P3-C03`, `P3-C04`, `P3-C05`	Load the second home and the visitor; Relay the peer and show the two-plan flat twin	nothing	G12
Relay the peer and show the two-plan flat twin the peer relay (Extended) and the two floor plans with the winding dial on the Station	Lane D	`P3-D07`, `P3-D08`	Implement the two-sheet house: sheets, door and flat-twin data	Make the Shared House safe and shared, and walk it	none directly
Implement the FieldFit lens the lens mathematics, the lookup builder, the wide source render with the composed final blit, the field-loss mask and slider, the lens guard, and the phone performance pass	Lane B	`P3-B01`, `P3-B02`, `P3-B03`, `P3-B04`, `P3-B05`, `P3-B08`	Build the FieldFit presets and lens panel	Build the heatmap and the search task, and run the FieldFit measurement	none directly
Build the Blind-Spot Finder the Blind-Spot Finder, and Scotoma River (Extended)	Lane B	`P3-B06`, `P3-B07`	nothing	Build the heatmap and the search task, and run the FieldFit measurement	none directly
Build the FieldFit presets and lens panel the field presets, the Console lens panel, and the Teichmüller lookup (Stretch)	Lane D	`P3-D01`, `P3-D02`, `P3-D05`	nothing	Implement the FieldFit lens	none directly
Build the heatmap and the search task, and run the FieldFit measurement the Spectator heatmap and bound, the search task with logging, and the G13 measurement run	Lane D	`P3-D03`, `P3-D04`, `P3-D06`	Implement the FieldFit lens; Build the Blind-Spot Finder	nothing	G13

The same phase package by package:



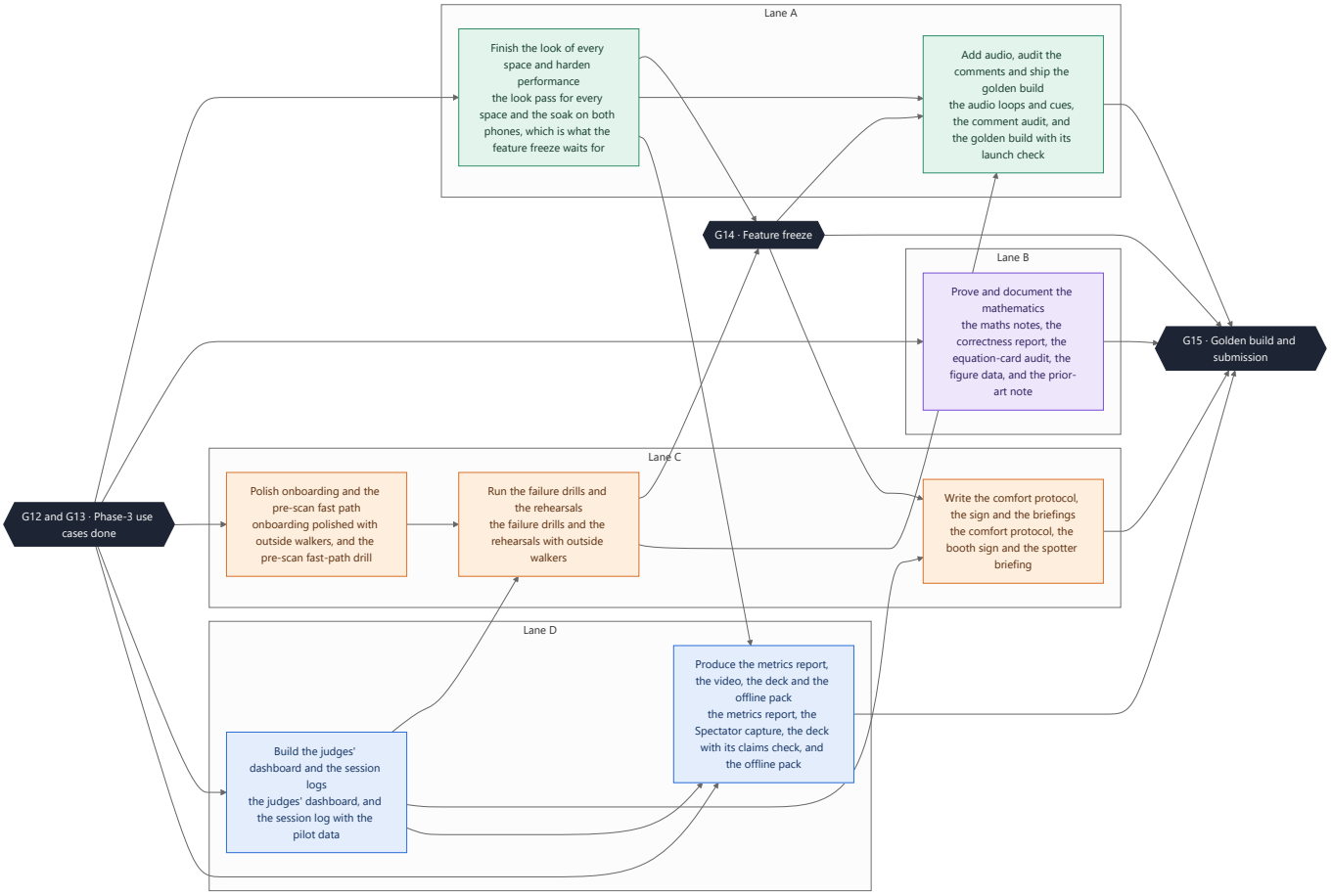
Colours: green lane A, purple lane B, orange lane C, blue lane D, grey shared. Dashed boxes are inputs from other lanes or from the other diagram; dark hexagons are gates.

9.8 Phase 4 · Finish

Polish, evaluation, demo operations and documentation. Work-package table: §13.6. Feature freeze (G14) follows the look pass, the performance hardening and the failure drills; after it, only fixes, audio, documents and rehearsals. In the diagrams of §9.7 and §9.8 the start gate shows when a lane is free; the tables give the technical dependency (§9.4).

Work areas, Phase 4

An arrow from a gate into an area means work in that area can start at that gate; an arrow between areas means the second cannot be finished without the first.

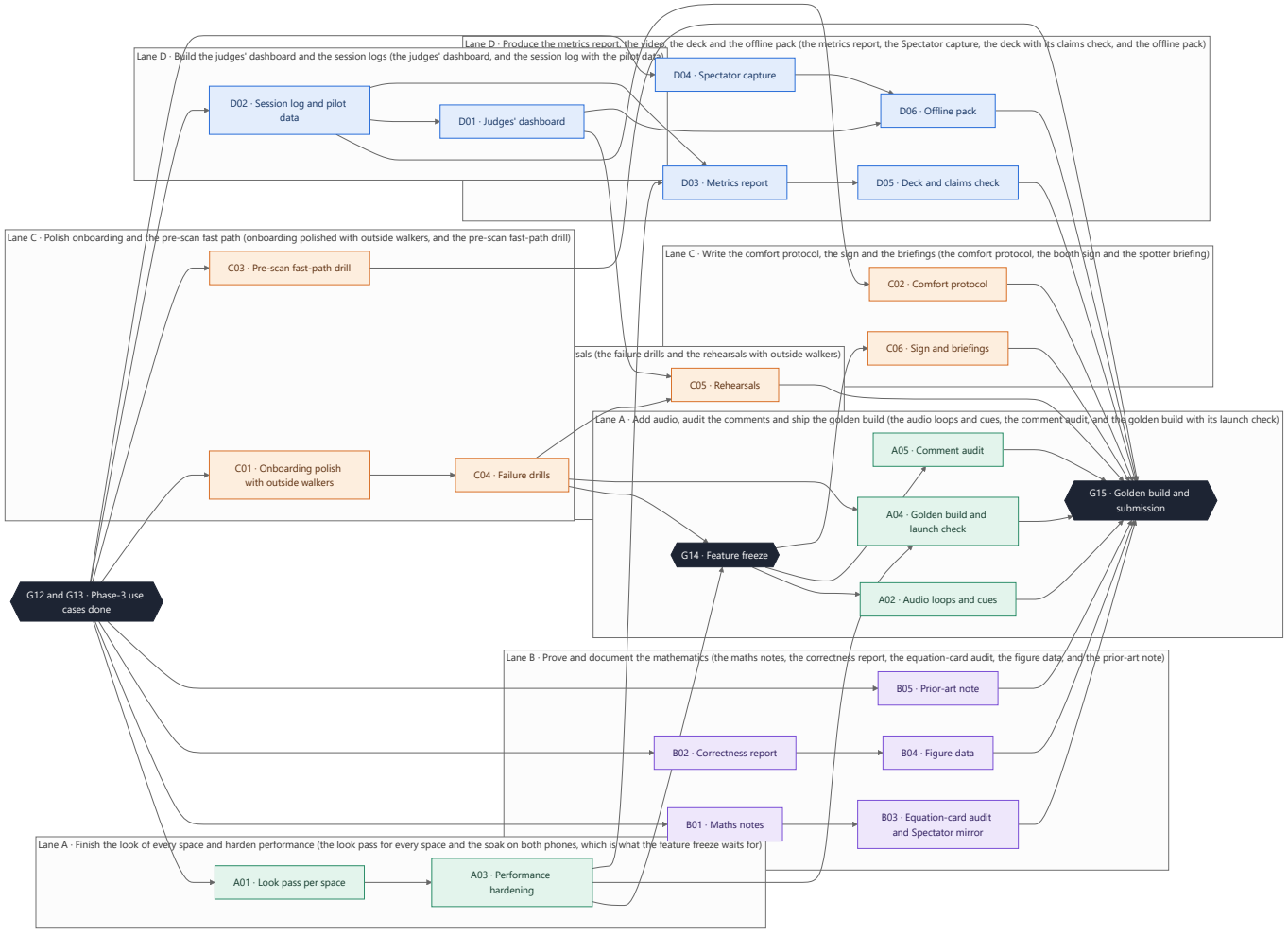


Colours: green lane A, purple lane B, orange lane C, blue lane D; dark hexagons are gates.

Area	Owner	Tasks inside	Needs from other lanes	Hands over to	Gates it feeds
Finish the look of every space and harden performance the look pass for every space and the soak on both phones, which is what the feature freeze waits for	Lane A	`P4-A01`, `P4-A03`	nothing	Produce the metrics report, the video, the deck and the offline pack	G14
Add audio, audit the comments and ship the golden build the audio loops and cues, the comment audit, and the golden build with its launch check	Lane A	`P4-A02`, `P4-A04`, `P4-A05`	Run the failure drills and the rehearsals	nothing	G15
Prove and document the mathematics the maths notes, the correctness report, the equation-card audit, the figure data, and the prior-art note	Lane B	`P4-B01`, `P4-B02`, `P4-B03`, `P4-B04`, `P4-B05`	nothing	nothing	G15

Area	Owner	Tasks inside	Needs from other lanes	Hands over to	Gates it feeds
Polish onboarding and the pre-scan fast path onboarding polished with outside walkers, and the pre-scan fast-path drill	Lane C	`P4-C01`, `P4-C03`	nothing	nothing	G15
Run the failure drills and the rehearsals the failure drills and the rehearsals with outside walkers	Lane C	`P4-C04`, `P4-C05`	Build the judges' dashboard and the session logs	Add audio, audit the comments and ship the golden build	G14, G15
Write the comfort protocol, the sign and the briefings the comfort protocol, the booth sign and the spotter briefing	Lane C	`P4-C02`, `P4-C06`	Build the judges' dashboard and the session logs	nothing	G15
Build the judges' dashboard and the session logs the judges' dashboard, and the session log with the pilot data	Lane D	`P4-D01`, `P4-D02`	nothing	Run the failure drills and the rehearsals; Write the comfort protocol, the sign and the briefings	none directly
Produce the metrics report, the video, the deck and the offline pack the metrics report, the Spectator capture, the deck with its claims check, and the offline pack	Lane D	`P4-D03`, `P4-D04`, `P4-D05`, `P4-D06`	Finish the look of every space and harden performance	nothing	G15

The same phase package by package:



Colours: green lane A, purple lane B, orange lane C, blue lane D, grey shared. Dashed boxes are inputs from other lanes or from the other diagram; dark hexagons are gates.

9.9 Working with AI coding agents

The tasks are sized and worded to be handed over. The rules that make that safe:

1. **One task, one branch, one merge request.** The branch name is the task id.
2. **An agent stays in its lane's folders.** It may read contracts and other lanes' public interfaces; it may not edit them. A contract change is made by a human and announced.
3. **The acceptance test comes first.** For anything provable in the Editor the agent writes the test from the acceptance cell, sees it fail, then implements.
4. **Parallel agents take tasks marked ||** in §9.5.1: they touch different files by construction.
5. **A human runs every phone test** and records the numbers. An agent prepares the build, the test scene and the checklist.
6. **No agent merges on a red test or a broken shader contract.**
7. **Numbers in a task are part of the task.** If a budget cannot be met, the agent reports the measured value and stops; it does not relax the number.

A task brief, ready to paste:

```
Task        P1-B04  Exact tile ids and the tile graph            Lane B
Read first  docs/contracts/kernel.md; this plan §3.4, §6.5, Appendix A.5
Depends on  P1-B01 (merged)
You own     Assets/Atlas/Kernel/Tiling/**  and its tests. Touch nothing else.
Deliver     TileId.cs, TileGraph.cs, Tests/TileIdTests.cs
Accept      - per-depth tile counts equal OEIS A377322 to depth 10
            - a walk round a vertex closes after exactly 5 steps
            - ϕ(X²+Y²) − T² = −1 for every generated tile (evaluated in Int128)
            - tile centres within ρ = 2 / 3 / 3.5 / 4 of a tile centre: 13 / 45 / 81 / 129; within 3.8425: 105
            - a tile crossing costs ≤ 0.5 ms in the Editor
Out of scope  rendering, materials, scenes, shader code
Report      test log; measured crossing cost; anything in the contract you could not satisfy
```

---

10. Use cases

Four use cases are built on the Phase-1 platform. Each is specified here with its work packages; §9.6–9.8 show the order in which the packages are built.

Use case	Phase	Lanes	Geometry it needs, and why	Exit gate
10.1 Geometry Playground	2	C, A	All of S0–S6: the room bent by the dial, walked, and compared	G8
10.2 Lung Mission	2	B, D, A	Hyperbolic space for a tree that doubles at every generation; $\mathbb{R}^3 \times S^1$ for a tumour that moves with breathing; a sphere for the directions a needle can take	G9, G10
10.3 Shared House / Post Circling	3	A, C (two packages from D)	A two-sheet branched cover: two homes joined at one door, flat everywhere except two posts	G12
10.4 FieldFit	3	B, D	The sphere of view directions: a lens that trades sharpness for field	G13

10.1 Geometry Playground

10.1.1 Purpose

Promise: **"your own room, bent"**. The walker scans a room, sees it rebuilt 1:1, turns one dial, and walks the same floor in spherical, flat-wrapped, hyperbolic, portal-stitched and 4D space. The Playground adds no new geometry: it packages the Phase-1 spaces S0–S6 into one guided run in which every claim is backed by a live number. Built in Phase 2 by lanes C and A: lane C owns the shell and the Glued and Impossible Rooms exhibits, lane A the Slicer toybox, the equation sources and the look; lane B hands over S2, S3, S4, the dial and the 3-4-5 walk; lane D supplies `Event` transport, the flat-twin mini-map and the Station views. Exit gate: G8 Playground run.

10.1.2 The flow

`PlaygroundDirector` (lane C) is a state machine over the beats below, stored in `Beats.asset`. A beat advances on its exit condition or on an operator `SceneCmd` from the Console, and each change emits `Event{ev:"pg.beat"}`.

#	Beat	What happens	Exit condition
1	Scan	Phone in hand; walker sweeps the room and confirms the floor outline (D21). Pre-scan + ARWorldMap is the fast path.	RoomModel valid
2	Home appears	S0 drawn 1:1. Each object box shows a placeholder tinted by category; Forge meshes arrive and replace them one by one (the built-in fallback set of `P1-D12` if the Forge cannot be reached).	Walker confirms; Local Network prompt already answered (BP §0.5)
3	Headset on	Phone into the headset; stereo starts.	Head pose stable
4	Centre	A floor disc marks the free-rectangle centre c0.	Head within 0.3 m of c0 (derived)
5	Dial	Rooms per corner q = 3, 4, 5, 6: Small Planet (R = 2s/π), flat, {4,5} (L = s/1.0613), {4,6} (L = s/1.3170). Walls sink to kerbs when q ≠ 4.	q = 5 selected
6	3-4-5 walk	Beacons lead the loop of side s (§3.3): corner counter, distance from the start flag, and "world turned by N°". The walker repeats it at q = 3 and q = 4. The pointing probe is taken after the third leg at q = 4 and at q = 5 (below).	Loop closed at q = 5
7	Free Curved Walk	S2, Mode 1. Mini-map revealed after the first lap; home beacon on.	Walker opens the mode chip
8	Compare	Switch to S3, Mode 2. Caption: "smooth curvature" against "curvature at the corners".	Walker opens the menu
9	Exhibit menu	Signs in Home for Glued, Impossible Rooms, Slicer.	Exhibit chosen, or done
10	Return Home	Kernel reset to tile zero, κ ramps to 0, S0 registered again.	End of run

Two measures are taken during the run without adding a beat (`P2-C17`). **Pointing probe:** after the third leg of the walk at q = 4 (flat, S0) and again at q = 5 (S2) the walker looks at where they started and presses A, or dwells; the app logs the head ray beside the geodesic and the Euclidean prediction as a `probe` event. **Single Ease Question:** at the end of the run the walker is asked for a score from 1 to 7 and the operator types it into the Console; it is logged as `seq` . Both fill the comfort and usability rows of §11.2.

10.1.3 Exhibit cards

What the walker meets

Exhibit	Caption chip (the rule)	The "aha"	Tasks	Registered to the real room?
S1 Glued	"Every wall is a door to its opposite wall. Flat, but it wraps."	You see your own back; in the Klein rows your copy is mirrored.	Wrap-walk through one cell edge; wave at your copies; switch to Klein and compare the chiral landmark; switch to T^3 and look up.	Yes. Universal cover fixed to the floor: an open-frame cell, sized from the RoomModel (§4.3), repeated 5 × 5 (T^2, Klein) or 3 × 3 × 3 (T^3, vertical period 2.6 m; Spec §11.2). The room is drawn once: its walls (as kerbs) and furniture stand where they really are, and the copies of the cell fill the space around them.
S2 Curved Walk	"Five rooms meet at every corner. Constant negative curvature."	Four right turns do not bring you home; a fifth does.	Walk the loop; follow a spanning object across a tile edge; reach a region tower; use the home beacon to return.	Only at entry. Tile zero is the room, then it drifts, so walls are 0.9 m kerbs and corner posts (five wedges per post, each 72° in the curved metric). Bump Guard is the only protection.
S3 Aligned Cells	"Each cell is flat. All the curvature sits at the corners."	Left of a corner post is a different room from right of it.	Circle one corner post: four cells pass, a fifth room appears.	Yes, permanently. Cell grid from the RoomModel; furniture stays put; a post at each interior corner. Needs ≥ 3×2 cells for unbounded exploration; below that, the five-tile corner demo (R1 §D.4).
S4 Small Planet	"Three rooms meet at every corner. Six tiles, then it repeats."	Three right turns close the loop; the whole world is six rooms.	Three-corner loop; watch the cube-net twin light each room you enter and count how many of the six you reach; find the pole beacon.	No. Six tiles always loaded; fog starts 0.5π, opaque 0.8π; fragments beyond 2.6 rad discarded; props ≤ 1.2 m (R6 §2.5).
S5 Impossible Rooms	"Flat rooms joined so that they overlap."	A 1 m booth contains a hall; four right turns arrive somewhere new.	Enter the booth; walk the loop corridor and read the room number.	Yes. Authored in play-space coordinates; hall = free rectangle; corridor width clamp(Wf/3, 0.7, 1.0 m). Loop only when min(Wf, Df) ≥ 2.4 m, else booth → hall (R6 §3.5).
S6 Slicer toybox	"A 3D slice of a flat 4D shape."	A solid changes shape when moved along a direction you cannot point to.	Slice the tesseract corner-first: tetrahedron → truncated tetrahedron → octahedron; look down for the shadow; rotate in a plane containing w; repeat on the 5-cell and 16-cell.	Pedestal at c0, registered. This exhibit is the onboarding for the Lung Mission's three verbs (R2).

What is measured

Exhibit	Equation card (live numbers)	Events (ev)	Acceptance test
S1	`x mod a` , `z mod b` ; cell a × b; wraps; copy parity (−1)^row	`glued.wrap` , `glued.mode`	No seam in either eye on 10 of 10 crossings; avatar in all 24 other copies of 5×5; Klein copy determinant = (−1)^row in 100% of copies.
S2	`cosh(s/2L) = √2·cos(π/q)` with live s, L, q; laps; world turned N°	`walk.corner` , `walk.lap` , `walk.closed` , `tile.cross`	Five-corner loop closes within 0.25 m; read-out 90° ± 5° after one lap at loop-fit L (R1 §D.1).
S3	corner curvature `2π − q·π/2 = −90°` ; `L_eff = 0.89·g`	`cells.cross` , `cells.postLoop`	After four cell crossings round one corner post the tile id differs from the start; after five it matches.
S4	`R = 2s/π` ; triangle angle sum 270°; predicted: straight ahead, home is 4s away (2πR = 4s)	`walk.*`	Three-corner loop closes within 0.25 m; no geometry smear at the antipode in 8 headings.
S5	virtual floor walked ÷ free-rectangle area	`portal.cross` , `rooms.loop`	Hall visible through the booth door before entry; no pop when either eye crosses the portal, 10 of 10.

Exhibit	Equation card (live numbers)	Events (ev)	Acceptance test
S6	w_o, rotation angle, section name, face count	`slicer.w`, `slicer.rot`, `slicer.shadow`, `slicer.poly`	Section named "octahedron" with area 6.9282 ± 0.001 at x+y+z+w = 2.0 (R2); tesseract slice ≤ 0.5 ms at the 95th percentile (§11.2).

Twins. Hand tracking is not in the Core build (rule 5 of §0.5). Every action has a gamepad input and a gaze-dwell input, and every action but the shadow also has a headset-lever input; all are mapped in `ActionMap.cs`. Gaze dwell is 700 ms (Spec §4.5).

Action	Gamepad	Gaze dwell	Headset lever
Confirm / next beat	A	prompt chip	click
Dial q	D-pad left/right	floor dial, four detents	click cycles 3→4→5→6
Mode 1 ↔ Mode 2	Y	mode chip	double click
Exhibit menu, pick	Start, stick + A	exhibit sign	long press opens, click steps, long press picks
Gluing T^2 / Klein / T^3	bumpers	gluing sign	click
Slice w	bumpers	ends of the w-ruler	hold to sweep
Rotate in a w-plane	triggers (or marker cube pitch)	floor dial, five detents	double click steps 15°
Shadow	X	look down past −35° (R2)	— (gaze twin only)
Return Home	B held	home beacon	triple click

10.1.4 Comfort and legibility rules

Rule	Value	Source
Real walking only; gains at identity; no vignette	Always	BP §0.5
Curvature changes only while standing	Dial locked when head speed > 0.1 m/s (§4.6); κ ramps over 4 s	R1 §E.3
Fog fully opaque at a fixed distance	S2: ρ = 3, and every tile that reaches inside that radius and can be in view is drawn. S1: 2 cells for the 5 × 5 lattice and 1 cell for 3 × 3, with the vertical period of T^3 not fogged (§8.5). S4: 0.8π	§8.5; R6 §5.1
Tiles spawn only beyond the fog	Always	R6 §5.2
Corner landmark at every tile vertex; ≥ 1 spanning object within ρ ≤ 2 of entry; tower within ρ ≤ 2.5; ≥ 4 region colours	WorldLint pass	R6 §5.1, §5.3
Walls ≤ 0.9 m in curved spaces	Always	R6 §5.1
Caption chip at every entrance; a flat space is never called curved	Always	Brief §4.0
Mini-map and UI anchored to the floor or body, never warped	Always	R1 §E.3
FMS	Asked aloud once per minute; operator types it into the Console; logged as `fms`; stop value `pg.fmsStop` set from the G2 spike (must be measured)	BP §0.5
Bump Guard	Warning cue + haptic only, in every space	D19

10.1.5 Work packages

Files live under `Assets/Atlas/Blades/Playground/Exhibits/` and `Assets/Atlas/Blades/Playground/Shell/`. Exhibits wrap the Phase 1 spaces through `ISpaceModule`; a change needed inside a space is requested from the lane that owns it. Lane A's later ids (`P2-A20` onwards) belong to the Lung Mission (§10.2.8).

ID	Work package	Lane	Depends on	Output (files, components, interfaces)	Acceptance test (binary, with numbers)	Needs (Editor / phone in hand / phone in headset / Station)
P2-A01	Slicer toybox	A	P1-A12	`ToyboxExhibit.cs`, w-ruler, cell colours (8 / 5 / 16)	Area test 6.9282 ± 0.001; tesseract slice ≤ 0.5 ms at the 95th percentile on phone A (should: ≤ 0.2 ms)	Editor, phone in headset
P2-A02	Section names and shadow	A	P2-A01	`SectionNamer.cs` (face-count table), look-down shadow	Correct name at five test offsets per polytope; shadow toggles at −35° ± 2°	Editor
P2-A03	Exhibit equation sources	A	P2-C12, P2-C15, P2-A01, P2-C06	`IEquationSource` for S1, S5, S6	Card values equal logged values to 3 significant figures	Editor
P2-A04	Look polish	A	P2-C12, P2-C13, P2-C14, P2-C15, P2-C16, P2-A01, P2-A02	Kerb and post meshes, fog tints, emissive signs, audio cues	Frame p95 ≤ 17.5 ms, ≤ 300 draw calls, ≤ 500k triangles in every beat	Phone in headset, Station
P2-C01	Director and beats	C	G7, P1-D01	`PlaygroundDirector.cs`, `Beats.asset`, `pg.beat`	All 10 beats advance by condition and by `SceneCmd`; 10 events in `events.jsonl`	Editor, Station
P2-C02	Scan-to-Home	C	P2-C01, P1-C06, P1-D10	Beats 1–3; placeholder → mesh swap	Every object box shows a placeholder or a mesh (100%); a swap moves the box centre < 0.05 m	Phone in hand, Station
P2-C03	Centre mark and dial hand-off	C	P2-C01, P1-B09, P1-B11	Floor disc, standing lock	Dial ignored above 0.1 m/s in 10 of 10 tries; q = 3…6 each load	Phone in headset
P2-C04	Guided 3-4-5 walk	C	P2-C03, P1-B09	Beat wiring of the `P1-B09` loop markers and read-outs, floor arrows, `walk.*` events	Closure < 0.25 m at q = 3, 4, 5; 90° ± 5° per lap at q = 5	Phone in headset
P2-C05	Mode compare	C	P2-C04, P1-B06, P1-B10	Mode chip, mini-map reveal after lap 1, home beacon	Switch keeps the tile id; beacon distance matches the kernel within 0.1 m	Phone in headset
P2-C06	Caption chips and equation cards	C	P2-C01	`CaptionChip.cs`, `EquationCard.cs`, `IEquationSource`, sources for S0, S2, S3, S4	Chip shown at 6 of 6 entrances; card refresh ≥ 10 Hz	Editor
P2-C07	Action twins	C	P2-C01	`ActionMap.cs`, dwell reticle	Full run completed three times, once per input alone	Phone in headset
P2-C08	Exhibit menu and return Home	C	P2-C01, P2-C12, P2-C15, P2-A01	`ExhibitMenu.cs`, kernel reset	After return, Home walls within 0.05 m of the RoomModel	Phone in headset
P2-C09	Tutorial prompts and FMS	C	P2-C01, P1-D06	`PromptQueue.cs`, `fms` event, `pg.fmsStop`, operator alert when an entry ≥ `pg.fmsStop`	One prompt per beat; FMS asked every 60 s ± 5 s; the entry appears in the log	Phone in headset, Station
P2-C10	Bump Guard check in every space	C	P1-C09, P2-A04	`docs/evidence/bumpguard_spaces.md`: the scripted approach test of §11.4.5 repeated in S0–S6 with the same fixed thresholds	In each space the warning starts ≥ 0.6 s before predicted contact in 10 of 10 wall approaches; no STRONG cue and no WARN on the straight sides in one clear-floor 3-4-5 walk (corner WARNs are counted and reported, as in `P1-C09`)	Phone in headset
P2-C11	Playground run (G8)	C	P2-A03, P2-C02, P2-C05, P2-C07, P2-C08, P2-C09, P2-C10, P2-C17, P1-D05, P1-D07	Run script, operator checklist	Three complete runs in two rooms with no restart; Spectator shows every beat	Phone in headset, Station

ID	Work package	Lane	Depends on	Output (files, components, interfaces)	Acceptance test (binary, with numbers)	Needs (Editor / phone in hand / phone in headset / Station)
P2-C12	Glued exhibit: wrap-walk	C	G7, P1-A09, P1-A15, P1-C06	`GluedExhibit.cs` : wrap counter, `glued.*` events, exhibit tasks (the space itself is `P1-A09` ; the cell size is read from the RoomModel)	`glued.wrap` logged on 10 of 10 crossings; the counter equals the number of crossings; cell sides equal the RoomModel `cells` value	Editor, phone in headset
P2-C13	Chiral kit	C	P2-C12	Prompts for the chiral comparison, using the landmark and the avatar of `P1-A09`	Determinant test 100%; avatar in 24 copies	Editor
P2-C14	Gluing switch and T^3	C	P2-C12	Gluing sign and input twin bound to the T^2, Klein and T^3 modes of `P1-A09`	Looking up > 60° in T^3 shows the avatar; S1 stays within its rows of §8.5	Phone in headset
P2-C15	Impossible Rooms exhibit	C	G7, P1-A11, P1-C06	`RoomsExhibit.cs` : exhibit tasks and the fallback prompt over the `P1-A11` layouts	Every footprint inside the eroded floor (WorldLint physical fit 100%) for three RoomModels	Editor
P2-C16	Loop corridor and room numbers	C	P2-C15	Room-number signs, `rooms.loop` event	Room number after four turns ≠ start in 10 of 10 runs; fallback chosen when min(Wf, Df) < 2.4 m	Phone in headset
P2-C17	Pointing probe and SEQ	C	P2-C04, P2-C09, P1-D06	`PointingProbe.cs` : after a 3-leg walk in S0 and in S2 the walker looks at where they started and presses A (gaze-dwell twin); logs `probe {ray_az, pred_geodesic, pred_euclid, err_geo, err_euc}` (Spec §13.1); Console SEQ field (1–7) logged as `seq`	In replay both predictions match the kernel values within 2° (proposed); `probe` and `seq` events appear in `events.jsonl` for S0 and S2	Phone in headset, Station

G8 = P2-C11 passed; every other package in this table is upstream of it.

10.2 Lung Mission

Phase 2, lanes B and D; the 4D medical runtime and the director of the use case (`P2-A20` to `P2-A32`) are built by lane A on its Phase-1 `Mesh4D` . Gates: G9 Lung 4D, G10 Lung airway and sky. The concept of the worldtube, slice, shadow and tilt is in §5; this section is the build spec.

10.2.1 Story and claim limits

One story: a trainee plans a biopsy of one lung lesion. The lung defeats a flat static 3D picture twice: it has too many branches to fit, and it does not hold still (R2 §E). Negative curvature answers the first, a circular fourth axis the second.

Every screen of the use case carries the footer "Education and research visualisation. Not for clinical use." Each asset carries exactly one label:

Content	Status	On-screen label
Tumour, lungs, heart, oesophagus, cord, trachea contours on 10 phases, patient 101_HM10395	Real	"Public research data: TCIA 4D-Lung, CC BY 3.0. Meshes derived by ATLAS."
"Rigid shape" tube (phase-0 shape moved along the centroid path)	Real path, simplified shape	"Simplified: rigid shape"
10 mm nodule, 15 mm motion (Lujan model)	Illustrative	"Illustrative model"
Airway tree (Weibel/Kitaoka-type generator)	Model-generated	"Model-generated airway"
Optional AeroPath proximal tree	Real, different patient	"AeroPath, CC BY 4.0, not this patient"
5 mm clearance threshold, literature figures (17.6 mm, 12 ± 2 mm)	Teaching values	"Teaching threshold" / "Literature value"

The footer, the labels and the credits panel are built by `P2-A32` ; the label strings travel in each package's `manifest.json` (§10.2.4). The motion of the real case is never exaggerated: if the exaggeration slider of §12.1 (risk 24) is used, it acts on the illustrative twin only, with its factor on screen.

Not claimed: clinical benefit, low-distortion embedding (Spec §11.3.1), that the airway belongs to patient 101, that $\mathbb{R}^3 \times S^1$ is curved, that space–time tilts are symmetries (R2 A.5). Wording: "target lesion" for the real case (locally advanced tumour), "nodule" only for the illustrative twin.

10.2.2 Beats

`LungDirector` (lane A, `P2-A32`) is a state machine over the beats below. A beat advances when the walker completes it or on an operator `SceneCmd` from the Console, which can also enter any beat directly; each change emits `Event{ev:"lung.beat"}`. Beats a and c to f are lane A's. Beats b and g are lane B's and beat h is lane D's; the director enters those three through one interface, stubbed until they exist. The lesion chamber hand-over between b and c is `P2-B09`. The full run takes the beats in the order a to h. In the long demo slot (§13.2) the order is b on the one-fork route of §10.2.3, then a, then c to f; the short slot shows d and e only.

Twins for every action: **walk/gaze** (dwell 700 ms), **gamepad**, **Console** (operator, via `ParamSet`). The marker cube is an Extended fourth twin (R2 D.8).

Beat	Space and why this geometry	Walker does	Number on screen	Twins (walk/gaze · pad · Console)	Acceptance test
a. Toybox recap	S6, tesseract. Same slicer as the tumour, so the three verbs are learnt on an object with known answers	Slices, looks down for the shadow, tilts once	Offset along w; 48 tets (the cross-section name appears too once the Playground's `P2-A02` exists)	step along lane · bumpers/triggers · sliders	Corner-first slice area 0.8660 at offset 0.5 and 6.9282 at 2.0, to 4 decimals (R2 D.10)
b. Airway wing	S2 (fallback S3). 2^g branches at constant corridor width fit only in H^2	Walks 4 forks to the lesion chamber (1 fork on the demo route of §10.2.3)	Generation g; 2^16 ≈ 66,000 terminal bronchioles (R2 C.4); insertion depth in mm	walk to door · stick + A · "choose branch"	4 forks walked by a naive walker with no rescue beacon in ≥ 9 of 10 runs (Spec §11.3.5)
c. Lesion chamber	Curvature ramps to flat while standing; S6 loads. "The map was one frozen slice"	Presses "unroll time": the phase-0 lesion becomes a tube	17.6 mm, literature value (Chen 2015, R2 A.3)	dwell on plinth · A · button	Unroll completes with the 10 phase copies at k·Λ/10 ± 1 cm
d. Breathing worldtube	$\mathbb{R}^3 \times S^1$: phase wraps, so the tube closes	Scrubs phase (slice); looks down (shadow, ρ = 0.5 shell, core)	GTV, ITV, ITV/GTV, core in cm^3; "core empty" on the twin	walk/look down past −35° · bumpers, Y · sliders	On-screen volumes equal `table.json` within 1 % (R2 D.10)
e. Breath lane	One breath = Λ = 1.5 m along the longest clear floor segment; copies wrap because time wraps	Walks through the breath; drops two flags	Gated ITV in cm^3; duty cycle in %	walk + dwell · A twice · two sliders	Tube visible in ≥ 3 copies at 60 fps; gated volume equals table entry within 1 %
f. Scanner artefact	Rotation of the slicing hyperplane in z–τ: tan θ = c/v	Turns the floor dial (5 detents)	Table speed v in mm/s; apparent length in mm	dial dwell · triggers · slider	20 mm test sphere (tube generator, not the 10 mm twin), 15 mm motion, breathing period τ = 4 s, v = 20 mm/s: apparent length spans 11.9–34.8 mm ± 0.5 mm (R2 A.3)
g. Needle sky	S^2: straight approaches to one point are the sphere of directions; head direction is a point on it	Looks at a green window; marks three directions	Minimum clearance in mm over 10 phases; robust-window share of sky in %; triangle angle sum in degrees	look + dwell · stick + A · click on sky panel	Angle sum > 180° and equal to 180° + 180°·area/π within 0.5°
h. Plan card	—	Reviews the card	All of the above	—	Card validates against `atlas.plancard/2`; every number traces to a logged event

10.2.3 Airway wing under free walking

Layout (R3 D3). The tree is snapped to the {4,5} tile graph by Rule S: each generation is one turn tile (left or right) plus one straight tile; the next fork is the tile at the end. Branch address = string over {L, R}; fork tile id = `FrameOf(address, trachea)`, exact in $\mathbb{Z}[\varphi]$. Fork spacing 2.1226 units > 1.7627, so the embedding is collision-free at every depth and greedy routing is exact (Sarkar; R3 D1). Tiles not in the tree are solid tissue.

Parameter	Value	Tag
L	loop-fit L from the RoomModel (never changed while walking)	§4.4
Tile step (corridor, centre to centre)	1.0613·L = s (2.0 m in a 3×3 m room)	derived
One generation	2 tile steps = 2s (4.0 m at s = 2.0 m)	derived
Chamber	disc, radius 0.5 m, at every tree tile centre (fork chamber or bend chamber); a floor glyph of radius 0.15 m at the centre marks where to stand	Spec §11.3.1
Corridor	drawn 0.8 m wide between 0.4 m kerbs plus a dithered membrane, so sibling branches are seen fanning out; a lit centre strip 0.2 m wide on the tile centre line marks where to walk, so a head on the strip stays at least 0.40 m from a real wall, 0.05 m clear of the WARN threshold	derived; R1 D.3
Walkable depth	16 conducting generations = 32 tile steps. Generations 17–23 appear on the map and counter only; 23 generations (46 tile steps) is the addressable depth. `long` tile ids suffice: 64-bit coefficients are exact to 42 straight tile steps, and on the airway's turning paths they need about 52 bits even at 46 steps (A.5)	R2 F9, R3 D3, derived
Lazy window	current node, 2 ancestors, descendants to depth 3: ≤ 60 nodes, ≤ 120 tiles; the drawn set is further cut by the draw rule of §8.5 (fog opaque at ρ = 3)	R3 D3
Demo route	4 forks, 8 tile steps, 8s walked (16 m at s = 2.0 m); this is the route of the G10 test. A one-fork route (2 tile steps, 2s walked) whose end tile is the lesion chamber is kept for the judged demo (§13.2) and selected from the Console (`P2-D10`); the hand-over of `P2-B09` then happens at the end of that fork	derived
Straight tiles per generation	1 (Core); 1–3 by segment-length tercile (Extended; the guarantee holds)	R3 D3

Why valve chambers. A tree has no loops, so the walker cannot follow it in a small room without being re-aimed. Teleport or stick locomotion would break "walked distance tracks insertion depth" (Spec §11.3.1). Instead every tile step is really walked, and the re-aiming happens where it cannot be seen.

Mode 1, Curved Walk (default). Chambers land on the four corners of the lap square of the loop-fit rule, corridors on its sides; the lap square lies 0.5 m from the nearest wall or scanned obstacle (§4.4), so a run of s always exists (derived).

1. The walker enters a chamber (head within 0.5 m of the tile centre). The valve behind closes and the chamber shell turns opaque.
2. At a fork chamber the two branches are offered in the endoluminal window and as two floor glyphs; the walker chooses. At a bend chamber there is one exit.
3. `ValveSolver` picks the physical direction d*: the lap-square side from the current corner whose smallest Bump Guard clearance along its whole length (static map and dynamic layer) is larger; ties alternate. If neither side is clear to 0.30 m the valve stays closed and the turn prompt asks the walker to wait.
4. With the shell opaque, U ← U·Rot(α) about the walker so that the chosen tree edge points along d*. A floor arrow (turn prompt) points along d*.
5. One iris opens at d*; the shell returns to dithered.

The tree path is never altered; only the embedding on the floor is. Bump Guard stays the only safety system.

Mode 2, Aligned Cells (fallback). Each tree tile is shown in one registered cell of side g (L_eff ≈ 0.89·g). Each tree step is a cell step. In the valve state the cell's tile frame is rotated by a multiple of 90° so the chosen tree edge faces the free neighbouring cell with the longest clear run. Needs ≥ 1×2 free cells (derived).

Criterion	Mode 1	Mode 2
Shows the claim (branches fan out at constant width)	Yes	No: each cell looks flat

Criterion	Mode 1	Mode 2
Registered to real walls	No	Yes
Comfort evidence	G2 spike only (FMS ≤ 3/20)	65 users, no sickness aborts (R1 A.6)
Walked distance per tile step	s	g

Mode 1 is the default because it has passed G2 by the time Phase 2 starts and is the only mode in which the geometry is visible. Mode 2 is selected from the Console, or automatically when the walker reports FMS > 3/20 or the free rectangle gives s < 1.5 m.

Aids and tasks.

Item	Specification
Beacon	Arrow to the target. Next step = prefix comparison of addresses (exact). Shown after 8 s of hesitation, except at the Two Compasses fork
Two Compasses	One fork chosen offline where the Euclidean compass (3D direction to the target from `pos_mm`) points down the wrong child and the beacon down the right one. Choice logged as `compass_choice`
Endoluminal window	True 3D view of the bifurcation from inside the model airway, rendered once per fork to a 512×512 texture, with clock positions ("take the 3 o'clock branch")
Route cost read-out	Insertion depth (Σ `len_mm`); sharp turns (branch angle > 45°, our definition); narrowest lumen against scope outer diameter 3.5 / 4.2 / 4.0 mm (Spec §11.3.1). In the symmetric model the lumen falls below 4.2 mm at generation 7 (derived)
Flat twins	Poincaré mini-map and a planar dendrogram, on the phone, the Spectator and the Console (`P2-D11`)

Honest limits. The H^2 layout distorts 3D branch direction, so it is always paired with the endoluminal window and the dendrogram. The share of forks where the Euclidean compass is wrong is measured offline (`greedy.json`, 1,000 random targets) and that measured number is the only one quoted.

10.2.4 Data and offline pipeline (Station)

Python CLI `atlas case <step>`, run on the Station before the demo; every step is idempotent and cached by input hash (Spec §12.1). It is not part of the Forge service and never on the pose path.

Step	Command	Does	Output
1	`fetch`	TCIA REST `getSeries` / `getImage` for 101_HM10395, study UID in R2 C.2: 10 CT + 10 RTSTRUCT series, about 0.8 GB	DICOM cache
2	`masks`	Rasterise `Tumor_c00…c90`, lungs, heart, oesophagus, cord, trachea on the CT grid (0.977 × 0.977 × 3 mm)	masks per phase
3	`tube4d`	Masks → SDFs (1 mm isotropic, σ = 1 mm) → mean-shape template of 600 triangles → Newton projection onto each phase (5 iterations) → prisms → 3 tets each with index-sorted diagonals (R2 D.2, D.3)	`tube4d.bin`, `tube4d_rigid.bin`, `phases.glb`
4	`shells`	Union, intersection and occupancy with 4 sub-steps between phases, wrapping 9 → 0	`itv.glb`, `core.glb`, `rho50.glb` (≤ 1.5k triangles each), `occ.r8` (64^3), `table.json`
5	`chest`	Lungs, heart, trachea template with 10 blend shapes	`chest.glb` (≤ 40k triangles)
6	`twin`	Lujan $z(t) = z_0 - b\cdot\cos^{\wedge}(2n)(\pi t/\tau - \varphi_0)$: 10 mm sphere, b = 15 mm, n = 2, breathing period τ = 4 s, and a hysteresis loop about 3 mm wide (anterior–posterior amplitude b_{AP} = 4 mm, phase lag δ = 24°); then steps 3–4. Expected volumes: GTV 0.52 cm^3; ITV 1.70 cm^3 with the head-to-feet motion alone and about 2.2 cm^3 with the hysteresis; core empty in both (`P2-D02`)	second case package
7	`airway`	Dichotomous generator: diameter ratio 0.794, length/diameter 3 ± 1, angle 37°, plane rotation 90° (R2 C.4); picks target and Two Compasses fork; runs the greedy study	`tree.json` (`atlas.tree/2`), `greedy.json`

Step	Command	Does	Output
8	`sky`	Per-phase distance fields of heart, oesophagus, cord, trachea; rays from the target (the argmax-ρ point) on HEALPix nside 64 (49,152 directions, astropy-healpix); risk = minimum clearance along the ray; robust = AND over 10 phases of clearance ≥ 5 mm; baked to equirectangular 512×256	`sky/risk_p0…9.r8`, `sky/robust.r8`, `sky.json`
9	`pack`	Hashes, labels, licences	`manifest.json`, `CREDITS.txt`

Package layout: `cases/<id>/{manifest.json, tube4d.bin, tube4d_rigid.bin, phases.glb, itv.glb, core.glb, rho50.glb, occ.r8, table.json, chest.glb, sky/, airway/, CREDITS.txt}`. Both packages ship inside the app build; the Hub also serves them over HTTP for updates.

File	Format
`tube4d.bin`	Header: magic `ATL4`, version u16, nVerts u32, nTets u32, layers u16, M f32, Λ f32. Then P float32 n×4 (x, y, z, τ in metres), T int32 m×4, slab id u8 per tet. Phase 10 repeats phase 0
`table.json`	GTV per phase, ITV, core (cm^3); 10×10 gating table indexed [start bin][length]; centroid path (mm); argmax-ρ point; max ρ
`manifest.json`	`atlas.case/1`: id, kind (`real` / `illustrative`), label strings, licence, credit, M = 15, Λ = 1.5 m, file hashes
`tree.json`	`atlas.tree/2`: per node `address`, `gen`, `len_mm`, `radius_mm`, `pos_mm`, `dir`, `source: "model"`; `target.address`; `compass_fork.address`. No `parent_iso`, no `R_m`

Credits (required): Hugo et al. 2016, *Data from 4D Lung Imaging of NSCLC Patients*, TCIA, doi 10.7937/K9/TCIA.2016.ELN8YGLE, CC BY 3.0, plus the TCIA citation; AeroPath (a different patient from the tumour case) under CC BY 4.0 if used (R2 F8). Shown in the app credits and the submission page. The TCIA data-use terms are acknowledged, and the citation is recorded in the licence ledger, by G9 (open item 18, §14).

10.2.5 Runtime components and budgets

Component (lane)	Budget	Tag
`TubeLoader`, `TubeSlicer` on `Mesh4D` (A)	18k tets per period; ≤ 1.0 ms at the 95th percentile, 0 ms when the hyperplane is unchanged; one slab in ten touched for a pure phase slice. Above 1.5 ms: cut the template to 300 triangles	R2 D.4, D.10
Slice mesh	≤ 36k triangles, mostly degenerate; fixed slots, `SetVertexBufferData`	R2 D.10
Shells	3 × 1.5k triangles	R2 D.10
Ghost phase meshes	10 × 600 triangles; gated shadow = selected phases drawn opaque in one colour	R2 D.6
Chest	≤ 40k triangles, one material	R2 D.10
Lane copies	5–7 × 3k triangles	R2 D.10
4D scene total	≤ 240k triangles and < 80 draw calls per frame (both eyes); a use-case scene, inside the frame budget of §8.5	R2 D.10
Memory	tube 0.4 MB, slice buffer 1.2 MB, occupancy 0.26 MB	R2 D.10
Airway wing (B)	Inside the frame budget (≤ 300 draw calls and ≤ 500k triangles per frame, both eyes); target ≤ 150 draw calls per frame; must be measured	§8.5
Needle sky (B)	One inverted sphere, 11 R8 textures (1.4 MB), 0 ms compute per frame	derived

Phase colour: one cyclic row in the shared palette, u = frac(τ/Λ), inhale cool, exhale warm, lightness varied (R2 D.7). No self-avatar copies on the lane.

10.2.6 Measures and tasks

Task	Tier	Walker does	Score	Logged (`Event`, scene `lung`)
M2 Needle dwell point	Core	Places a point	Hit fraction H = ρ(p) from `occ.r8`; best = max ρ	`m2_place {p_mm, hit, best}`
M4 Gating window	Core	Chooses a phase arc	ITV reduction against the best arc of equal duty	`m4_gate {start, len, itv_cm3, duty, best_cm3}`

Task	Tier	Walker does	Score	Logged (`Event` , scene `lung`)
M3 Spot the artefact	Core	Picks the true shape among three	Correct or not, before and after beat f	`m3_pick {choice, correct, when}`
M1 Margin fit	Extended	Sizes an ellipsoid	Coverage C = min over phases; excess O = volume/ITV; pass if C ≥ 0.95	`m1_fit {radii_mm, C, O}`

Also logged: `fork {address, choice, correct, beacon_shown, t_s}` , `compass_choice` , `sky_pick {az_deg, el_deg, clear_mm, robust}` , `gc_check {angle_sum_deg}` , walked distance, FMS each minute, and the Single Ease Question at the end of the run (`seq` , typed into the Console field of `P2-C17`). The Hub writes `events.jsonl` . A pilot compares a static 3D model with the 4D view within subjects; the only claim permitted is "in a pilot with n people, hit fraction changed from X to Y" (R2 B.3).

10.2.7 Corrections applied to the older documents

Older text	Applied here	Tag
Slice Tilt in the time–left-right plane	Scanner tilt in z–τ, tan θ = c/v	R2 F1
Seven precomputed tilted slices, blended	Live CPU slice; no blending	R2 F2, F3
GTV and organs on all phases for all patients	True for patient 101 only among those needed; patient 112 excluded	R2 F4
ITV(arc) = OR of 10 masks	4 sub-steps between phases	R2 F5
"≈ 17 mm CT-to-body divergence"	17.6 mm is nodule displacement (Chen 2015), one contributor; labelled literature value	R2 F6
1.5 m glued cell, 0.75 m ring in the 3×3 m square	Breath lane on the longest clear segment; ring as fallback	R2 F7
"≈ 23 generations"	About 16 conducting, 23 in all	R2 F9
12 mm illustrative motion expected from the real data	Real tube is subtle (6.8 mm SI); the twin is the dramatic one	R2 F10
Wedge layout at R = 2.2 m, ℓ = 1.5 m (372 crossings by depth 8); parent-relative isometries	Rule S on the tile graph; addresses	R3 point 8, 9
Sphere tracing on the phone; TotalSegmentator vessels	Precomputed sky from contoured organs; vessels Extended	derived
Roll a Lung live build, tile solver, H200 batch, kimimaro/skan skeleton	Dropped; offline generator on the Station	BP §0.5

10.2.8 Work packages

Lane length is written Λ to keep L for the curvature radius. τ is the time coordinate along the lane, in metres; where it is called the breathing period (the Lujan curve of §10.2.4 and the scanner-tilt test) it is that period in seconds.

ID	Work package	Lane	Depends on	Output (files, components, interfaces)	Acceptance test (binary, with numbers)	Needs (Editor / phone in hand / phone in headset / Station)
P2-D01	Case contract	D	G7 Phase-1 exit	`docs/case_contract.md` ; C# and Python readers for `tube4d.bin` , `table.json` , `manifest.json`	A 2-tet fixture round-trips Python → C# byte-identical	Editor, Station
P2-D02	`atlas case twin`	D	P2-D01	Illustrative package (steps 6, 3, 4)	GTV 0.52 cm³ within 3 %; with the hysteresis off (b_AP = 0) ITV 1.70 cm³ within 3 %; with b_AP = 4 mm and δ = 24° ITV about 2.2 cm³ (pin the value from the first run of the pipeline's own 0.25 mm voxel union, then hold it within 3 %); core volume 0 in both	Station
P2-D03	`atlas case fetch` , `masks`	D	P2-D01	DICOM cache; masks	20 series fetched; tumour centroid range 3.1 / 3.1 / 6.8 mm within 1 mm	Station

ID	Work package	Lane	Depends on	Output (files, components, interfaces)	Acceptance test (binary, with numbers)	Needs (Editor / phone in hand / phone in headset / Station)
P2-D04	`atlas case tube4d`	D	P2-D03	`tube4d.bin`, `tube4d_rigid.bin`, `phases.glb`	18,000 tets; no cracks (every interior face shared by exactly 2 tets); slice at each phase within 1 mm Hausdorff of that phase's contour mesh	Station
P2-D05	`atlas case shells`	D	P2-D04	shells, `occ.r8`, `table.json`	ITV 47.4 cm³ and core 17.2 cm³ within 5 % (R2 C.2)	Station
P2-D06	`atlas case chest`	D	P2-D03	`chest.glb`	≤ 40k triangles; lung volume 4.42 L at 0 % and 3.87 L at 50 % within 3 %	Station
P2-D07	`atlas case airway`	D	P2-D01	`tree.json`, `greedy.json`	16 generations addressable; diameter ratio 0.794 ± 0.08; exactly 1 Two Compasses fork on the 4-fork route	Station
P2-D08	`atlas case sky`	D	P2-D03	`sky/*`, `sky.json`	`robust.r8` equals the pixel-wise AND of the 10 phase maps (0 differing pixels); HEALPix indices match astropy-healpix on 1,000 random directions	Station
P2-D09	`atlas case pack` and delivery	D	P2-D02, P2-D05, P2-D06, P2-D07, P2-D08	Two packages in the build; Hub route `/cases/<id>/…`	Hash check passes for every file; package loads from the build in ≤ 2 s with the network off	Phone in hand, Station
P2-D10	Lung Console panel	D	P1-D06, P2-D01	Sliders for phase, tilt, gating arc; branch and sky buttons; route switch (four forks, or the one-fork demo route); label preview	Each control changes the phone state in ≤ 250 ms	Phone in hand, Station
P2-D11	Lung twins on Spectator and Console	D	P2-D07, P1-D05, P1-D07	Dendrogram and Poincaré map with the walker's address; lane strip with flags	Address shown equals the phone's within 1 frame of the `fork` event	Station, phone in hand
P2-D12	Plan card	D	P2-D10	`atlas.plancard/2` writer on the phone; card page on the Hub with print stylesheet	Card validates against the schema; contains both `not_for_clinical_use: true` and the case label; prints on one page	Station, phone in hand
P2-D13	Event logging and pilot export	D	G7, P1-D01	`events.jsonl` → `lung_summary.csv` per session	All 7 Core event types of §10.2.6 (`m2_place`, `m4_gate`, `m3_pick`, `fork`, `compass_choice`, `sky_pick`, `gc_check`) present after one full run	Station
P2-A20	`TubeLoader`	A	P1-A12, P2-D01, P2-D02	Loads `tube4d.bin` into `Mesh4D`	Illustrative tube: slice at θ = 0 has area within 2 % of $4\pi R^2$	Editor
P2-A21	Live slicer views	A	P2-A20	`TubeSlicer` with dirty flag and slab skipping	≤ 1.0 ms at the 95th percentile over 1,000 moving-slice frames; 0 allocations	Phone in hand
P2-A22	Phase colouring	A	P2-A21	Palette row, `tau` vertex channel	Colour at τ = 0 and τ = Λ identical (ΔRGB = 0)	Editor
P2-A23	Shells and look-down shadow	A	P2-A20, P2-D02	ITV, ρ = 0.5, core; fade at pitch < −35°	Displayed volumes equal `table.json` within 1 %	Phone in headset
P2-A24	Breathing chest	A	P2-D06	Blend-shape player, wrap 9 → 0	No visible jump at the wrap (vertex step ≤ 1 mm at display scale)	Editor
P2-A25	Breath lane	A	P2-A21, P1-C06	Lane on the longest clear segment; 5–7 copies; phase stripes	≥ 3 copies visible at 60 fps; lane lies inside free floor with ≥ 0.5 m clearance	Phone in headset
P2-A26	Unroll animation and "now" plane	A	P2-A25	Beat c sequence	2 of 3 naive viewers answer "the tumour over time" unprompted (R2 D.9); else enable the ring	Phone in headset
P2-A27	Gating flags	A	P2-A25, P2-A23	Two flags; gated shadow; duty	Volume equals table entry within 1 %; duty = arc length/10	Phone in headset
P2-A28	Scanner tilt	A	P2-A21	z–τ dial, 5 detents, v read-out	On a 20 mm test sphere with 15 mm motion and a breathing period τ = 4 s, from the tube generator of `P1-A12` (not the 10 mm twin): apparent length spans 11.9–34.8 mm ± 0.5 mm at v = 20 mm/s	Editor

ID	Work package	Lane	Depends on	Output (files, components, interfaces)	Acceptance test (binary, with numbers)	Needs (Editor / phone in hand / phone in headset / Station)
P2-A29	Tasks M2, M3, M4	A	P2-A23, P2-A27, P2-A28	Task UI, scoring, events	M2 hit equals `occ.r8` sample within 0.01; three event types emitted	Phone in headset
P2-A30	Ring fallback	A	P2-A22	10 phase meshes on a 0.75–1.0 m circle	Fits the largest free disc of the RoomModel; 60 fps	Phone in headset
P2-A31	Lung 4D soak	A	P2-A29, P2-A32, P2-D09	Soak report	95th-percentile frame time ≤ 17.5 ms with the slice moving, on both packages; thermal state never "Serious"	Phone in headset
P2-A32	Lung director, labels and credits	A	P2-D09, P2-A22, P2-A24, P2-A26, P2-A29, P2-A30	`LungDirector.cs` (beats a–h by condition and by `SceneCmd`, `lung.beat` event; beats b, g and h are entered through an interface that lanes B and D implement, stubbed until then), the footer on every screen, one label per asset from `manifest.json`, credits panel	Beats a and c–f advance both ways on both packages; the footer is present on every beat; every content class of §10.2.1 present in a package shows exactly its label; the credits show the TCIA citation; one `lung.beat` event per beat in `events.jsonl`	Phone in headset, Station
P2-B01	`AirwayTree`	B	P1-B04, P2-D07	Address → tile id (`FrameOf`), lazy window	32,767 nodes to depth 14 map to distinct ids; window ≤ 60 nodes	Editor
P2-B02	Corridor and chamber kit	B	G7, P1-B05	Fork, bend and straight tile meshes with kerbs, membranes, iris, the lit centre strip (0.2 m wide) and the chamber's standing glyph (radius 0.15 m)	Tile meshes come from the `P1-B05` generator and keep its levels of detail, its 16 boundary segments per edge and its triangle caps (§8.4); kerbs, membranes and iris have no edge longer than 0.25 m; wing ≤ 150 draw calls per frame (both eyes) at ρ = 3	Editor
P2-B03	`ValveSolver`, Mode 1	B	P2-B01, P1-C06, P1-C09	Steps 1–5 of §10.2.3	100 simulated 8-step routes in 3 RoomModels: head always ≥ 0.4 m from walls; rotation applied only while the shell is opaque; with one side blocked in the dynamic layer the other side is chosen, and with neither side clear to 0.30 m the valve stays closed, in 20 of 20 simulated cases	Editor
P2-B04	Valve walk, Mode 2	B	P2-B01, P1-B10	90° frame rotation per cell	Same 100 routes complete with ≥ 1×2 cells	Editor
P2-B05	Mode switch	B	P2-B02, P2-B03, P2-B04	Console switch; automatic rule	Switch at a chamber keeps the address unchanged	Phone in headset
P2-B06	Beacon and route cost	B	P2-B01	Arrow, 8 s rule, read-out	Beacon step equals the address-prefix step on 1,000 random targets	Editor
P2-B07	Endoluminal window	B	P2-D07	512×512 render per fork, clock labels	Render cost in 1 frame only; 0 ms afterwards	Phone in hand
P2-B08	Two Compasses	B	P2-B06, P2-B07	Fork UI, `compass_choice` event	Compass and beacon point to different children at the designed fork	Phone in headset
P2-B09	Lesion chamber hand-over	B	P2-B02, P2-B03, P2-A26	Curvature ramp to flat over 3–5 s while standing; S6 load; triggered at the end tile of the selected route (four forks, or the one-fork demo route)	No frame > 33 ms during the hand-over	Phone in headset
P2-B10	Needle sky	B	P2-D08	Sky sphere, phase and robust layers, dwell pick, read-outs	Clearance shown equals `sky.json` within 0.1 mm at 20 test directions	Phone in headset
P2-B11	Great-circle check	B	P2-B10	Three arcs, angle sum	Octant triangle gives 270° ± 0.5°	Editor
P2-B12	Airway run test	B	P2-B05, P2-B08	Run report	4 forks without rescue in ≥ 9 of 10 naive runs; FMS ≤ 3/20; walking the centre strip for 8 tile steps raises no WARN on straight sides	Phone in headset

G9 = P2-A31 passed with both packages (which needs P2-A32 and P2-D09). G10 = P2-B12, P2-B09, P2-B10 and P2-B11 passed, all eight beats of P2-A32 run end to end, P2-D11 and P2-D13 verified, and P2-D12 printed from a real run.

10.3 Shared House and Post Circling

10.3.1 Purpose and claim limits

Two scanned homes become the two sheets of one walkable surface, joined at a free-standing door frame. Core: **Post Circling**, solo, no network, second home pre-scanned, a recorded visitor labelled "recorded". Extended: **House Call**, a live second walker on phone B. Stretch: voice, hand-over, more than two homes.

Claims we make: the surface is an exact two-sheet branched cover; walking is 1:1 and flat. Claims we do not make: novelty over MRUnion (TVCG 2025) until it has been read (Spec §11.4); any clinical value of House Call; social-presence results.

10.3.2 Why the geometry is necessary

Flat telepresence must use the intersection of the rooms or one merged room (Spec §11.4). The cover gives each walker the whole of the other home through a door that occupies 0 m^2 of wall. The three door behaviours are not scripted: they follow from one rule, "crossing the cut swaps the sheet".

10.3.3 Maths

Item	Formula	Tag
Sheets, door	N = 2 homes, K = 1 door, σ = transposition ($\sigma = \sigma^{-1}$, so crossing direction does not matter)	Spec §11.4
Topology	$\chi = N - 2K = 0$; genus $g = K - N + 1 = 0$; two boundary circles: an annulus	Spec §11.4
Curvature	Flat except at the posts; cone angle $2\pi k = 4\pi$, curvature $2\pi - 2\pi k = -2\pi$ per post; check $2\pi N - 4\pi K = 2\pi\chi = 0$	Spec §11.4
Cut	Open segment P_1P_2 between the post centres on the floor	derived
Crossing test	With orient(a,b,c) = (b−a)×(c−a), a step p→q crosses when orient(P_1,P_2,p)·orient(P_1,P_2,q) < 0 and orient(p,q,P_1)·orient(p,q,P_2) < 0; then sheet ← 1 − sheet	derived
Winding	θ_1 += atan2((p−P_1)×(q−P_1), (p−P_1)·(q−P_1)); a closed loop round P_1 alone returns to its sheet only when θ_1 is a multiple of 4π	derived
Embedding	M = T_A · Flip^f · T_B^{-1}, where T_X is the door pose in RoomModel X and f ∈ {0,1} turns the door 180° about its centre	derived

Three consequences (derived): through the door crosses the cut once (switch); round the whole frame crosses it zero times (stay); round one post crosses it once (switch), and a second lap returns.

A straight sight line meets the cut at most once, so a depth-1 stencil portal renders the cover exactly; no recursion is needed (derived; matches §2).

10.3.4 What is registered and what is not

Thing	Registered to reality?
Sheet 0: the walker's own Home (walls, live AI furniture)	Yes
Door frame	Position yes; the frame itself is virtual (no real prop)
Sheet 1: the other home's walls and furniture	No. Virtual only
Walker position	1:1 on both sheets; no drift, no holonomy

Keeping the walker inside their own free floor F_A while on sheet 1:

1. **Door fit.** M is fixed by the door poses, so the search runs over the door pose in home B (x, z on a 0.1 m grid, yaw in 5° steps, f) and maximises area(F_A ∩ M·F_B) on the RoomModel free-floor grids. Virtual furniture standing in real free floor is harmless; virtual free floor over real obstacles is the hazard to minimise.
2. **Guest rug (not built unless the lead approves open item 13 of §14; the answer is needed by G11, before `P3-A04` starts).** Default: no rug. Sheet 1 is drawn in full, with its own floor only and no marking derived from the real

room. The door fit (item 1) keeps virtual free floor over real obstacles to a minimum, Bump Guard (item 3) is the only protection, and the spotter recommendation of §4.10 applies to every walker on sheet 1.

3. **Bump Guard** runs unchanged in tracking space on both sheets (warning cue + haptic only, D19).

Door placement in the walker's room: centre at the free-floor point of maximum signed distance; door width 0.8 m (design choice); the circling post needs ≥ 0.9 m clear radius (Spec §11.4), the other post ≥ 0.5 m (design choice). If no pose qualifies, the use case reports "room too small" and does not start.

10.3.5 Flow

Beat	Walker does	Sees	Flat twin (mini-map and Spectator)
1 Door appears	Stands in Home	Frame rises from the floor; other home visible through it	Two floor plans side by side, cut drawn on both
2 Peek	Leans head into the opening	Other home, correct per eye	Dot at the cut
3 Through	Walks through	Whole surround is the other home; own room visible back through the frame	Dot jumps to plan B
4 Round the frame	Walks round both posts	Stays on the same sheet	Loop encloses both posts; counter 0
5 Round one post	Circles the marked post once	Sheet switches without using the opening	Winding dial about P_1 reads 2π
6 Second lap	Circles again	Back home	Dial reads 4π, "cone angle 4π" caption
7 Visitor	Meets recorded or live visitor	Capsule-and-head avatar; label "recorded" or "live"	Second dot with sheet colour

10.3.6 Data, Station and link

Need	Detail
Second RoomModel	`roommodel_B.json` plus Forge GLBs of a team member's room, used with their consent, or of a synthetic room, never a pilot walker's room; pre-scanned and bundled, or scanned by phone B and fetched over HTTP from the Hub
Recorded visitor	`housecall_recorded.bytes` : pose and sheet at 30 Hz in sheet-B coordinates
Shared state	{sheet id, pose in sheet coordinates}; sheet k coordinates = RoomModel k frame; each client applies its own M (Spec §11.4)
Link additions	`State.sheet` (already in Appendix B.1) carries the sheet id; a new message `PeerState` (id 0x11, reserved in Appendix B.2; the Hub relays a peer's `State`); JSON `house.door` carries the door pose. Peer playback buffer 80 ms (Spec: 60–100 ms). The message, the relay and `house.door` are `P3-D07` (lane D); the playback buffer is `P3-C04` ; both packages are Extended
Spectator	Renders the active sheet and both plans
Privacy	Only RoomModel geometry and Forge meshes are shared; no camera images (Spec §11.4)

Post Circling must run with the Station unreachable.

10.3.7 Measures logged

Sheet switches with cause (opening or post); θ_1 and θ_2; time on each sheet; Bump Guard warnings per sheet; overlap area of the door fit (m^2); peer latency (ms); FMS on exit.

10.3.8 Work packages

Lanes A and C build the use case. Lane D adds two packages, because `Atlas.Link` , the Hub, the mini-map and the Spectator are its code (§9.2): the peer relay (`P3-D07` , Extended) and the two-plan flat twin (`P3-D08`).

ID	Work package	Lane	Depends on	Output (files, components, interfaces)	Acceptance test (binary, with numbers)	Needs (Editor / phone in hand / phone in headset / Station)
P3-A01	Sheet state machine	A	G7 Phase-1 exit	`SheetState.cs` (crossing test, winding, per-eye sheet index)	Unit tests: 1,000 random segments match brute force; scripted paths give through = 1 switch, frame loop = 0, one-post loop = 1, two laps = 0 net	Editor
P3-A02	Two-sheet door rendering	A	P3-A01, P1-A10	`SheetDoor.cs` , door volume 0.1 m deep, stencil mask + depth reset + plane clip	Head moved through the opening at 1.5 m/s in simulation: 0 frames show the wrong sheet in either eye; ≤ 300 draw calls	Editor
P3-A03	Second home loader	A	P3-A02, P1-C06, P1-D10	`SheetHome.cs` builds sheet 1 from `roommodel_B.json` under M, own pooled material	Home B wall corners land within 0.01 m of M·(RoomModel B corners)	Editor
P3-A04	(Blocked on open item 13; do not start) Guest rug and unlit floor	A	P3-A03, P3-C02, and the lead's answer to open item 13	`GuestRug.cs` mesh from the fit mask	Rug area equals the mask area within 2 %; no rug vertex nearer than 0.3 m to a real obstacle	Editor
P3-A05	Winding counter and flat-twin data	A	P3-A01	Winding counter and the flat-twin data source (two plans, the cut, the winding angle) exposed through `ISpaceModule`	The counter reads 4π ± 0.1 rad after two laps of one post on a scripted path	Editor
P3-A06	Visitor avatar, recorded track, labels	A	P3-A03	`Visitor.cs` , recorder, "recorded"/"live" label	Avatar drawn only on its own sheet in 100 % of 200 sampled frames; label always visible	Editor
P3-A07	Phone performance pass	A	P3-A02, P3-A03, P3-A05, P3-A06	Soak log	60 fps median, never below 30, ≤ 300 draw calls, ≤ 500k triangles over a 10-crossing run	Phone in headset
P3-C01	Second scan path	C	G7, P1-C06, P1-D01	Export and import of `roommodel_B.json` , bundled and phone-B routes	Phone B scan appears on phone A with identical free-floor grid (0 differing cells)	Phone in hand, Station
P3-C02	Door placement and door fit	C	P3-C01	`DoorFit.cs` returning door poses, f, overlap mask	Post clearances met (0.9 m, 0.5 m) on 3 test rooms; result equals exhaustive search optimum	Editor
P3-C03	Bump Guard on sheet 1	C	P3-A03, P1-C09	Test protocol and log	In 10 of 10 approaches to a real obstacle on sheet 1 the cue and the haptic start ≥ 0.6 s before predicted contact, as on sheet 0 (the lead-time row of §11.2)	Phone in headset
P3-C04	Peer link (Extended)	C	P3-A06, P3-D07	Peer playback buffer (80 ms), avatar driven from `PeerState` , latency log	Two phones: peer avatar error ≤ 0.1 m at rest; latency logged; the use case is unaffected by a Hub shutdown	Phone in hand, Station
P3-C05	G12 walk-through	C	P3-A07, P3-C02, P3-C03, P3-D08	Run log and capture (with `P3-C04` when the Extended House Call is built)	All seven beats completed by 3 walkers with 0 contacts	Phone in headset, Station
P3-D07	Peer relay (Extended)	D	G7, P1-D02	`PeerState` message id and codec in `Atlas.Link` ; the Hub relays a peer's `State` as `PeerState` ; `house.door` JSON; Appendix B updated	Golden round trip for `PeerState` ; the relay adds < 2 ms	Station
P3-D08	Two-plan flat twin	D	P3-A05, P1-D07	Two floor plans, the cut and the winding dial in the mini-map and on the Spectator	The dial reads 4π ± 0.1 rad after two laps of one post	Editor, Station

G12 = P3-C05 passed, with the door placed by the door fit of P3-C02. Packages marked Extended or Blocked feed no gate: the House Call (P3-C04, P3-D07) is checked at G12 only if it is built, and the guest rug (P3-A04) is not part of G12.

10.3.9 Open risks

Risk	Handling
Rooms too small for 0.9 m post clearance	Report and do not start; measure on real rooms
Guest rug read as a ghost of the real room	Not built by default (open item 13); Bump Guard alone protects on sheet 1, tested by `P3-C03`

Risk	Handling
Phone B has no headset	Phone B runs a mono handheld view; open item 14 (§14)
Two walkers in one physical room	Not supported; walkers use separate rooms

10.4 FieldFit

10.4.1 Purpose and claim limits

FieldFit shows what a restricted visual field is like and how a field-remapping lens trades sharpness for coverage. It is a research prototype and a teaching tool, not a low-vision aid (Spec §11.5). Display acuity is about 14–17 px/deg (Spec §11.5), so we claim nothing about acuity.

What changes under VR only: there is no camera image. The lens remaps the **rendered Home** (the walker's scanned room with its live AI furniture). The source field half-width E is therefore our own render parameter, not a camera limit, and camera-to-photon latency no longer applies. The old Lens-standing and Lens-6DoF camera modes are dropped.

10.4.2 Why the geometry is necessary

On the sphere of view directions, with $s = 2\cdot\tan(e/2)$, any conformal lens that fits half-width E into a residual radius ρ_0 has central magnification $\le \tan(\rho_0/2)/\tan(E/2)$ (Spec §11.5, Schwarz lemma). For $E = 54°$, $\rho_0 = 10°$ that is 0.172: the centre shrinks 5.8×. A sharp centre plus the whole field forces a quasiconformal lens, and the distortion K is the price shown to the walker.

10.4.3 Maths

Item	Formula	Tag
Constants	$S = 2\cdot\tan(E/2)$, $\sigma_0 = 2\cdot\tan(\rho_0/2)$, $s_1 = 2\cdot\tan(e_1/2)$	Spec §11.5
Uniform lens	$\sigma = s\cdot\sigma_0/S$; conformal; attains the bound	derived
Power-law lens	$\sigma = s$ for $s \le s_1$; $\sigma = s_1\cdot(s/s_1)^{(1/K^*)}$ beyond; $K^* = \ln(S/s_1)/\ln(\sigma_0/s_1)$	Spec §11.5
Inverse, per output pixel	$s = \sigma$ for $\sigma \le s_1$, else $s_1\cdot(\sigma/s_1)^{K^*}$; $e = 2\cdot\operatorname{atan}(s/2)$	Spec §11.5
Default	$E = 54°$, $\rho_0 = 10°$, $e_1 = 4°$ gives $K^* = 2.92$; rings at 10°, 20°, 30° land at 5.5°, 7.0°, 8.0°	Brief §3.7, re-derived
Slider	$\sigma_t = (1 - t)\cdot\sigma_{uniform} + t\cdot\sigma_{power}$, monotone, inverted through a 1,024-sample radial table	derived
Cautionary lens	$R = \rho_0\cdot\tanh(e/\rho_0)$; $K \approx 34$ at 30°	Spec §11.5
Source lookup	$uv = 0.5 + 0.5\cdot(\tan e / \tan E)\cdot(\cos\alpha, \sin\alpha)$, α = azimuth	derived
Centre density	Source target of N pixels gives N/157.7 px/deg at the centre for E = 54°: 9.7 at N = 1,536 (the default) and 13.0 at N = 2,048 (the high tier)	derived
Scotoma River	Lines follow $\mathrm{Im}(w) = \text{const}$, $w = z + a^2/z$; letter scale $1/\lvert 1 - a^2/z^2\rvert$ (0.5 directly above), clamped to ≤ 2	Spec §11.5, scale derived

Render path. Each eye renders Home to a square target of half-width E: $1{,}536^2$ with 4× MSAA by default (9.7 px/deg), $2{,}048^2$ only on the high tier; multisampled colour and depth are memoryless. The source size follows the thermal ladder (1,536 → 1,280 → 1,024). A lookup texture per eye (RG16F, 512^2) stores, for each screen pixel, the source uv after: inverse headset pre-distortion → view direction (e_{out}, α) → inverse lens → source uv. Pixels with $e_{out} > \rho_0$ store uv = −1 and are drawn black: this is the head-locked field loss. The lookup takes the place of the barrel-distortion step in the ordinary final blit (`P1-A05`), and that blit samples the two source targets in place of the two eye textures, so the frame still ends in one blit. The texture is rebuilt only when a parameter changes, not every frame.

10.4.4 Flow

Beat	Walker does	Sees	Twin on the Spectator
1 Blind-Spot Finder	One eye lit, fixates a cross, taps when a dot is seen	Dots on a grid 8°–22° temporal, ±8° vertical (design choice); found spot re-probed as catch trials	Field map filling in

Beat	Walker does	Sees	Twin on the Spectator
2 Struggle	Finds 5 marked objects in Home through a 10° tunnel	Black beyond 10°	Undistorted view with the tunnel outlined
3 Fit and see	Moves the slider, repeats the search	Uniform → power-law; tanh shown once as the warning	Same view with rings at 10°, 20°, 30°
4 Inspect	Looks around	"LENS ON" badge	K heatmap and the Schwarz bound
5 Scotoma River (Extended)	Reads text with one eye	Text parting round their own blind spot	Streamline plot

10.4.5 Registration and safety

Home stays registered; the lens bends only directions. Lens beats are standing or seated (Spec §11.5). The lens switches off when the head moves more than 0.5 m from its start point (design choice), on one tap, or from the Console. Bump Guard stays on. The badge "LENS ON" is always visible while a lens is active.

10.4.6 Data and Station

Need	Detail
Field presets	`fieldloss/*.png` masks, 512^2: 10° tunnel, hemianopia, one UWHVF field (BSD-3); Core uses the tunnel only
Console	Parameters `lens.E`, `lens.rho0`, `lens.e1`, `lens.t`, `lens.profile`, `lens.on` in the registry
Teichmüller lookup (Stretch)	Offline solve on the Station for one UWHVF field, exported as RG16F 512^2
Search targets	5 tokens placed on RoomModel object boxes, positions seeded per trial

10.4.7 Measures logged

Blind-spot centre and time to find (s); catch-trial false positives; search time per target (s); head-turn amplitude (degrees, sum of absolute yaw); lens profile, K*, t; blit cost (ms); lens auto-off events; FMS on exit.

10.4.8 Work packages

ID	Work package	Lane	Depends on	Output (files, components, interfaces)	Acceptance test (binary, with numbers)	Needs (Editor / phone in hand / phone in headset / Station)
P3-B01	Lens maths library	B	G7 Phase-1 exit	`LensMath.cs` : profiles, inverses, K(e)	K* = 2.918 ± 0.001; rings at 5.5°, 7.0°, 8.0° ± 0.05°; 54° maps to 10.00°; tanh K at 30° between 30 and 40	Editor
P3-B02	Lookup builder	B	P3-B01, P1-A05	`LensLut.cs` (Burst), two RG16F 512^2 textures	Rebuild < 16 ms; round trip through forward lens within 0.1° at 1,000 samples	Editor
P3-B03	Wide source render and composed final blit	B	P3-B02, P1-C07	`FieldFitFeature.cs`, shader `FieldFitBlit`	With lens off, output differs from the ordinary blit by < 1 pixel; exactly one final blit per frame	Editor
P3-B04	Field-loss mask and slider	B	P3-B03, P3-D01	Mask sampling, `lens.t` morph	Tunnel edge at 10° ± 0.2° measured on a ring test scene	Editor
P3-B05	Lens guard	B	P3-B03	Badge, tap-off, 0.5 m auto-off	Auto-off within 1 frame of 0.5 m displacement in 10 of 10 trials	Phone in headset
P3-B06	Blind-Spot Finder	B	G7, P1-A04	`BlindSpot.cs`, one-eye stimulus, catch trials	Blind spot found in ≤ 30 s by ≥ 4 of 5 testers (Spec §11.5)	Phone in headset
P3-B07	Scotoma River (Extended)	B	P3-B06	`RiverText.cs`	No glyph centre inside radius a; scale directly above = 0.5 ± 0.02	Phone in headset
P3-B08	Phone performance pass	B	P3-B03, P3-B04, P3-B05	Soak log	Blit < 1 ms (Spec §11.5); with the lens on, GPU p95 ≤ 9 ms cold and every "must" frame-time row of §11.2 holds over a full search trial	Phone in headset, Station
P3-D01	Field presets	D	G7, P1-D08	`fieldloss/` masks and generator script	Tunnel mask area equals a 10° disc within 1 %	Station

ID	Work package	Lane	Depends on	Output (files, components, interfaces)	Acceptance test (binary, with numbers)	Needs (Editor / phone in hand / phone in headset / Station)
P3-D02	Console lens panel	D	G7, P1-D03	Six `lens.*` parameters, stop-lens button	Each change reaches the phone in < 100 ms on the LAN	Station, phone in hand
P3-D03	Spectator heatmap and bound	D	P3-B01, P1-D05	K heatmap, bound readout	Heatmap reads K = 1 inside e_1 and 2.92 ± 0.05 outside; bound reads 0.172	Station
P3-D04	Search task and logging	D	P3-B04, P1-D04	`SearchTask.cs`, CSV columns above	5 targets logged per trial with 0 missing rows over 10 trials	Phone in headset, Station
P3-D05	Teichmüller lookup (Stretch)	D	P3-B02	Offline solver output for one UWHVF field	Loads and renders; maximum K reported	Station
P3-D06	G13 measurement run	D	P3-B06, P3-B08, P3-D02, P3-D03, P3-D04	Result table	Median search time with the power-law lens is lower than with the tunnel alone for ≥ 4 of 5 testers	Phone in headset, Station

G13 = P3-D06 passed and, with the lens on, FieldFit meets the "must" performance rows of §11.2 (the soak of P3-B08 supplies the numbers). Packages marked Extended or Stretch feed no gate.

10.4.9 Open risks

Risk	Handling
Per-eye radial remap changes binocular disparity; fusion may fail	Test in `P3-B03`; fallback renders both eyes from the centre eye while the lens is on
1,536² source targets exceed the frame budget	Step to 1,280² or 2× MSAA; measured in `P3-B08`
Fewer than 5 scanned objects in Home	Tokens also placed on wall positions
No collision-judgement evidence for walking under a lens	Walking stays excluded (Spec §11.5)

11. Verification: tests, metrics and gates

ATLAS is proved by automated tests (11.1), a metrics register filled with measured values (11.2) and gate checklists (11.3); 11.4 gives the measurement protocols. Rules throughout:

- Published numbers come from a Release build with Frame Timing Stats on, phone A in the closed headset (R5 §E2). "Cold" = thermal Nominal at the start; "hot" = after the 15-minute soak.
- A gate passes only when every box is ticked on one commit. A missed "must" blocks the gate; a missed "should" is recorded with its cause.
- Where a number appears both in §11 and in a task row, the §11.2 Must column and the §11.3 checklist win.
- "(proposed)" marks a value we chose with no measured or published source; it is confirmed or replaced at the gate where it is first checked.

11.1 Automated tests

EditMode, double precision, unless stated. All run before every merge to `main`. Two tests have a half that turns green at a later gate: the pose-jump half of `Safety_Independence` at G5, and the on-device half of `TileId_Seed` at G6.

Test	What it proves	Pass condition	Owner lane
`Kernel_ExpLog`	exp∘log = identity for κ < 0, = 0, > 0	Error < 1e-9 on 1,000 random points with ρ ≤ 4; < 1 mm after the float cast	B
`Kernel_EtaOrthonormal`	Kernel matrices stay on the group	$\|\|M^{T}\eta M - \eta\|\|$ < 1e-12 for generators, < 1e-9 after renormalisation (R3)	B
`Kernel_Inverse`	Inverse = $\eta M^{T}\eta$	$\|\|M\cdot\eta M^{T}\eta - I\|\|$ < 1e-12 on 1,000 random M (derived)	B
`Holonomy_Triangle`	Rotation round a geodesic triangle = angle defect	Error < 0.1° on 100 random triangles; sign flips with κ (R5)	B
`Closure_345`	The 3-4-5 walk closes	$(\text{Step}\cdot\text{Rot } 90°)^{n} = I$ to 1e-12 for n = 5 at s/L = 1.0613, n = 3 at s/R = π/2, n = 4 flat; no smaller n closes (derived)	B
`Lorentz_Drift`	Renormalisation holds over long walks	$\|\|C^{T}\eta C - \eta\|\|$ (Frobenius) < 1e-9 after 100,000 random boosts; the unrenormalised value is logged only (R5)	B
`Float32_Upload`	The float cast is safe for the farthest tile that is drawn	$\|\langle p,p\rangle_{\eta} + 1\|$ < 1e-3 for tile-relative anchors at ρ = 4.35, the farthest tile centre drawn on the high tier (3.5 + 0.8425) (R5; radius derived)	B
`TileId_RingInvariant`	Ids are exact	$\varphi(X^2+Y^2) - T^2 = -1$ for every tile to depth 12, evaluated in `Int128` or `BigInteger`, never in `long` (the invariant multiplies coefficients, and in `long` its products overflow from about 22 straight steps) (R3; derived)	B
`TileId_Growth`	The graph is {4,5}	Tiles at graph distance 0…8 = 1, 4, 12, 28, 64, 148, 340, 780, 1792 (OEIS A377322; R3 T7)	B
`TileId_VertexClosure`	Five squares meet at a vertex	A walk round a vertex returns the same tile after exactly 5 steps, not 4; still exact 35 tiles from home (R3)	B
`TileId_Unique`	No duplicates, no order dependence	Depth-6 ball by BFS, DFS and 10 random walks gives one id set; `Neighbour(Neighbour(t,k), back) == t` (R3)	B
`TileId_Seed`	Content is deterministic	Golden file of 64 (id, seed) pairs identical under Mono and IL2CPP (device PlayMode) (R3)	B
`VisibleSet_Counts`	The tile counts and the draw rule are correct	Tile centres within ρ = 2 / 3 / 3.5 / 4 of a viewer at a tile centre = 13 / 45 / 81 / 129, and 105 within 3.8425; under the draw rule of §8.5 at ρ = 3, on 500 random poses with the head pitched anywhere from 20° up to straight down, at most 71 tiles are drawn (59 when the gaze is level) and no undrawn tile has a point that is nearer than ρ and inside either eye's frustum (derived)	B

Test	What it proves	Pass condition	Owner lane
`TileMesh_SeamsEdges`	Tile meshes join and warp cleanly at every level of detail	Shared-edge vertices of neighbouring tiles coincide to ≤ 1 mm after the warp for every pair of levels (n = 16, 8 and 4 per side, each with the same 16 boundary segments per edge); worst screen-space chord error ≤ 0.15° from an eye height of 1.6 m; both at L = 1.4, 1.9 and 3.0 m; ≤ 1,200 / 500 / 250 triangles for the three levels at any L; five corner wedges round a post leave no gap and no overlap (A.6)	B
`Mode2_SightLines`	Aligned Cells instance list is complete	3×3 cell grid: 13 / 15 / 17 instances with the user in the centre / an edge / a corner cell; ≤ 4 tiles per physical cell (R1 §A.3)	B
`WorldLint_All`	A generated world is usable	100% of tiles reachable; no tile with 4 closed edges; landmark within ρ ≤ 2.5 of every tile centre; simulated draw calls ≤ 300 and triangles ≤ 500k from every tile centre × 8 headings under the draw rule of §8.5; enlarged `localBounds` on every warped renderer; every scan box has a generated mesh, a fallback mesh or a placeholder; the region of every tile to depth 6 (597 tiles) is the same on two runs, and so on the phone and on the Station; same seed → identical world (R6 §5.3)	B
`Glued_Handedness`	Klein copies mirror correctly	Determinant sign of copy (row, col) = (−1)^row; torus generators commute (R6 §5.3, R3 §C3)	A
`Portal_Graph`	Room graphs are valid	BFS from the start reaches 100% of rooms; every portal has a twin; allocated stencil references ≤ 15 per frame; every room cell inside the eroded free floor; 1,000 seeds × 5 floor shapes pass or return the fallback layout (R3)	A
`Slicer_Tesseract`	Mesh4D slices are correct	Corner-first slice area 0.8660 at x+y+z+w = 0.5 and 6.9282 at 2.0, to four decimals (R2 §D.10)	A
`Slicer_Ball`	Curved 4D bodies slice correctly	Unit 4D ball at offset c gives radius $\sqrt{(1 - c^2)}$ within 1% at c = 0, 0.5, 0.9 (R2 §D.10; tolerance proposed)	A
`Slicer_MovingSphere`	A tilted slice reproduces the scanner artefact	For a sphere moving at constant speed u, apparent length = D·v/(v − u) within 2%; at θ = 0 area within 2% of $4\pi R^2$ (R2 §D.5, §D.10)	A
`Shader_Contract`	The look shader keeps the draw path cheap	SRP Batcher compatible; no MaterialPropertyBlock in `Atlas.*`; ≤ 8 compiled variants (proposed)	A
`FloorScan_Synthetic`	Floor analysis and loop-fit	Rectangle 3×3 m (free rectangle Wf × Df = 2.4 × 2.4 m) → s = 2.0 m, L = 1.885 m; 4×5 m (3.4 × 4.4 m) → s = 3.0 m, L = 2.83 m; L-shape and room-with-table: free rectangle within 0.05 m of the analytic answer, table cells not free; floor outside the confirmed outline is not free (derived)	C
`BumpGuard_Replay`	Warning logic on replayed poses	20 approaches: cue fires ≥ 0.6 s before predicted contact, exactly one rising edge each (0.05 m hysteresis); on 5 replays of the lap square no STRONG cue and no WARN on the straight sides (more than 0.5 m from a lap corner, as in 11.4.5); a cue when tracking is not Normal for > 0.5 s; a 0.25 m pose jump injected in replay produces no STRONG from velocity (R1 §C.4; counts proposed)	C
`Safety_Independence`	Bump Guard never touches the network	Assembly half (green from G0): the `Atlas.Safety` assembly definition references neither `Atlas.Link` nor `Atlas.Telemetry`, and no source file in it uses a type from either. Pose-jump half (green from G5): an injected 0.25 m pose jump leaves the geometry frame unchanged but reaches Bump Guard (§0.5, rule 1; §6.2)	C
`Link_RoundTrip`	Every message survives encode → decode	Byte-exact golden round trip for every Appendix B message (the reserved `PeerState` id has no layout until `P3-D07` and is not part of this test); `State` = 202 B as laid out in Appendix B.1; unknown types and trailing bytes ignored (R4 §5.3)	D
`Link_ClockSync`	Offset estimate survives a lossy link	Simulated 10% loss, 15 ms one-way delay, 0–5 ms jitter (proposed): error against the injected offset ≤ 2 ms p95 (R4 §5.4)	D
`Link_Watchdog`	State machine is correct	DEGRADED after 500 ms silence, OFFLINE at 3 s, back-off 0.5 / 1 / 2 / 4 s, RESYNC replays the parameter snapshot (R4 §5.5)	D

11.2 Metrics register

Captured by `AtlasPerfProbe` (3,600-frame ring buffer, one CSV row per second, stored on the phone and sent to the Hub) unless stated. "Per space" = each of S0–S6 alone, with the room's content, on the `budget_room` fixture (S1 in both its 5 × 5 and its 3 × 3 × 3 layout). The per-space planning figures of §8.5 are measured there at G4 and replaced by the measured values; the gate itself tests the "must" column below.

Metric	Must	Should	Stretch	How captured	Gate where first checked
Performance (R5 §D1)					
Frame time p50 / p95 / p99	≤ 16.7 / 20 / 33.3 ms	≤ 16.7 / 17.5 / 25 ms	≤ 16.7 / 16.9 / 18 ms	`FrameTimingManager`	G1; per space G4
Frames over 17.2 / 33.8 ms (one and two vsync intervals + 0.5 ms)	≤ 5% / 0.5%	≤ 1% / 0.1%	≤ 0.2% / 0	same	G4
Frame rate, hot	≥ 45 mean, never < 30	60 locked	60 locked with full-size eye textures (1398 × 1290, scale 1.0)	same + Metal HUD	G7
Draw calls, both eyes	≤ 300	≤ 200	≤ 120	`ProfilerRecorder` "Draw Calls Count"	G4
Set-pass calls	≤ 60	≤ 30	≤ 15	"SetPass Calls Count"	G4
Triangles, both eyes	≤ 500k	≤ 300k	≤ 200k	"Triangles Count"	G4
CPU main / render p95	≤ 8 / 6 ms	≤ 6 / 5 ms	≤ 4 / 3 ms	`cpuMainThreadFrameTime`, `cpuRenderThreadFrameTime`	G1, G4
GPU p95, cold / hot	≤ 9 / 14 ms	≤ 7 / 11 ms	≤ 5 / 8 ms	`gpuFrameTime`, checked against the Metal HUD	G4, G7
Resident memory	≤ 1.5 GB	≤ 1.0 GB	≤ 700 MB	"System Used Memory"	G4
Managed allocation per frame	≤ 1 kB	0	0	"GC Allocated In Frame"	G4
Thermal timeline, 15-minute soak	No Critical; Serious not before 10 min	No Serious	Nominal or Fair only	Native `thermalState` plugin, 2 s poll	G4, G7
Time-to-Fair in the soak, LiDAR depth on against off	reported	—	—	11.4.1 (the two soaks)	G4
Slicer p95: tesseract / tumour tube	≤ 0.5 / 1.0 ms	≤ 0.2 ms / —	≤ 0.05 ms / —	`ProfilerMarker("Atlas.Slicer4D")` (R5, R2 §D.10)	G4, G9
Motion-to-photon, rotation	Measured; ≤ 100 ms	≤ 50 ms	≤ 30 ms	11.4.2	G7
Geometric correctness					
3-4-5 walk closure error	< 0.25 m at loop-fit L	< 0.15 m (proposed)	< 0.10 m (proposed)	Virtual pose logged at start and end (R1 §D.8)	G2, G7
Measured holonomy against prediction	Within 5°	Within 2° (proposed)	—	World-turn read-out of the 3-4-5 walk (`P1-B09`), logged; the prediction is 90° for one lap at loop-fit L	G7
In-app Lorentz drift	< 1e-9 always	< 1e-12	—	Computed every 60 frames (R5)	G3
Holonomy update / tile-crossing frame	≤ 0.1 / 2 ms	≤ 0.02 / 0.5 ms	— / ≤ 0.1 ms	`ProfilerMarker` (R5)	G3
Phone and Spectator tile-id agreement	100% of samples on a closed loop	—	—	Hub log (R4)	G6
Scan					
Scan end → walkable Home	≤ 90 s	≤ 60 s	≤ 30 s (proposed)	Event log (R5 §E3)	G5
Free rectangle Wf, Df against a tape measure	≤ 0.10 m error	≤ 0.05 m (proposed)	—	Tape, two rooms: the clear span between walls or furniture, less 0.6 m (§4.4)	G5

Metric	Must	Should	Stretch	How captured	Gate where first checked
Loop-closure drift of raw ARKit pose, 20 m	< 2% (0.40 m)	< 1%	< 0.5%	11.4.3 (R5)	G5
Bump Guard (thresholds of §4.10; pass marks proposed)					
Warning lead distance at 1.0 m/s	≥ 0.60 m	≥ 0.80 m	—	11.4.5	G5
Warning lead time to predicted contact	≥ 0.6 s	≥ 0.8 s	—	11.4.5	G5
Missed warnings in 20 scripted approaches	0	0	0	11.4.5	G5
STRONG cues, and WARN cues on straight sides, on a 5-minute lap of the lap square (0.5 m from the walls)	≤ 2	0	0	11.4.5	G5
Corner WARN cues on that lap	reported	—	—	11.4.5	G5
Static-map error after the 5-minute lap (touched real wall against the SDF edge, less the 0.05 m offset of §4.2, three points)	≤ 0.10 m	≤ 0.05 m	—	11.4.5	G5
Frame-time cost of live LiDAR depth, p95	≤ 1.0 ms	≤ 0.5 ms	—	Probe, depth on against off; the depth-off run uses a recorded pose track with nobody wearing the headset	G5
Link (R4 §5.10)					
Connect time, launch → CONNECTED	≤ 3 s p95 of 10 launches	—	—	Hub log	G6
`State` **rate and loss**	≥ 50 Hz mean, < 2%	≥ 58 Hz mean, < 0.5%	—	Sequence numbers at the Hub	G6
Phone → Hub latency, clock-corrected	p95 ≤ 25 ms, p99 ≤ 50 ms	p50 ≤ 10 ms	—	`t_us` against receive time	G6
Phone → Spectator, screen to screen	≤ 150 ms	≤ 100 ms	≤ 60 ms	11.4.6 (R5, R4)	G6
Parameter round trip, slider → phone, p95	≤ 100 ms on a clean link; ≤ 250 ms at 10% simulated loss	≤ 100 ms at 10% simulated loss	—	`ParamSet` echo (11.4.6)	G6
Operator command, Console → effect on the phone	≤ 250 ms for each of the six commands	—	—	Hub send time against the phone's log entry, clock-corrected (test L10)	G6
Clock offset jitter	≤ 2 ms p95	—	—	Hub log	G6
Main-thread p95, link on against off	Differs ≤ 0.2 ms; 0 B allocated by `Atlas.Link`	—	—	Two probe runs (R4)	G6
Asset round trip					
Forge job start → placed mesh (`AssetAck` **), per object**	≤ 60 s	≤ 30 s (proposed)	≤ 20 s	`Event` timestamps (test L15); the wait in the queue is reported separately	G6

Metric	Must	Should	Stretch	How captured	Gate where first checked
Triangles after clean-up	1,500–5,000 per object and no more than the target sent with the job; ≤ 60,000 for all generated meshes in the room; edges ≤ 0.25 m	—	—	Forge report (BP §5; target rule in §4.5)	G6
Scale error against the scanned box	≤ 10% of the longest side (proposed)	≤ 5%	—	Forge report against RoomModel	G6
Worst frame while a GLB loads	< 20 ms	≤ 17.2 ms (no frame over one vsync interval + 0.5 ms)	—	Frame log	G6
Comfort and usability (R5 §D2)					
FMS (Fast Motion Sickness scale, 0–20, spoken each minute), median peak	≤ 5	≤ 3	≤ 2	Operator types into the Console	G2, G7
Pointing-to-origin error after a 3-leg walk	Measured in S0 and S2, n ≥ 5	S2 within 1.5× of S0	Within 1.2×	`probe` event of `P2-C17`: head ray at a button press against the geodesic and the Euclidean prediction	G8 (first values); n ≥ 5 in `P4-C05`
Single Ease Question (1–7), median	≥ 5	≥ 5.5	≥ 6	Console SEQ field of `P2-C17`, one question per use case	G8 (first values); n ≥ 5 in `P4-C05`
Onboarding: launch → first completed walk	≤ 90 s	≤ 60 s	≤ 30 s	Event log	G7

11.3 Gate checklists

G0 Contracts frozen (present: all four lanes)

- [ ] RoomModel fields, the `Atlas.Link` schema (Appendix B), parameter registry ids and the look-shader property list are merged and tagged; the three sample rooms and the performance fixture `budget_room` are in the repository and validate against the RoomModel schema.
- [ ] Assembly definitions exist; the assembly-reference half of `Safety_Independence` is green (the pose-jump half is checked at G5).

G1 First light (present: A, C, D)

- [ ] Stereo rig, empty scene, phone A: 60 fps on the Metal HUD; CPU main ≤ 3 ms, GPU ≤ 2 ms (R5 §E3).
- [ ] Unity ≥ 6000.3.21f1, thermal frame-rate setting off; the thermal plugin (delivered by `P1-A02`) returns a value.
- [ ] Lowest-resolution 60 Hz ARKit configuration shown in the start-up log; the environment depth image is non-null at the chosen video format.
- [ ] The phone pairs with the Hub; `State` packets from the phone appear in the Hub log, and the dashboard shows one telemetry row per second.
- [ ] One command installs the current `main` on a phone, and every engineer has installed a build.
- [ ] Both phones report LiDAR (`P1-X04`); rear camera and LiDAR are unobstructed in the headset.

G2 Curved-walk spike (present: B, C, three walkers from the team, a spotter)

- [ ] Safety for the spike is in place, because Bump Guard does not exist yet: the walkers are team members only, the floor is cleared, the lap square is marked on the real floor for the spotter (a test aid, not part of the product), and a spotter stays within arm's reach throughout.

- [ ] Protocol 11.4.4 run; pass or fail recorded per walker, with wall contacts and spotter stops counted.
- [ ] Decision recorded: Mode 1 default, Mode 1 at a larger L, or Mode 2 default; the comfort floor for L and R is recorded with it.

G3 Spaces in the Editor (present: A, B)

- [ ] Every 11.1 test owned by lanes A and B is green (the on-device half of `TileId_Seed` is checked at G6).
- [ ] S1–S6 each load in SpaceLab under AR simulation, on the three sample rooms, and expose their parameters (S0 is checked at G5).
- [ ] Mode 1 and Mode 2 switch on one tile graph with no tile-id change.

G4 Spaces on the phone at budget (present: A, B, D)

- [ ] S0–S6 each meet every "must" performance row, cold, on `budget_room` with the room's content loaded (the Home shell and all twelve stand-in objects; S1 in both its 5 × 5 and its 3 × 3 × 3 layout); the measured values are written into the register and replace the planning figures of §8.5.
- [ ] Fog is fully opaque at ρ = 3 in both eyes, and every tile that reaches inside that radius and can be in view is drawn: no hole in the floor and no popping while walking or turning, and the edge of the drawn world cannot be told from the background.
- [ ] The 15-minute soak (11.4.1), alternating S2 Curved Walk (ρ = 3, the room as tile zero) and S1 Glued 5 × 5 on `budget_room`, passes the thermal row.
- [ ] The soak is repeated once with depth off, driven by a recorded pose track with nobody wearing the headset, and the difference in time-to-Fair is recorded.
- [ ] Each thermal-ladder tier forced from the Console takes effect within 1 s and sets the eye-texture size (render scale), MSAA, ρ and Glued copies of its row in §8.6.

G5 Scan, Home and Bump Guard (present: C, B, D)

- [ ] Self-scan produces a RoomModel, the walker confirms the outline, and all scan rows are at "must" in two different rooms.
- [ ] A RoomModel with a 1.4 × 1.4 m free floor selects standing mode.
- [ ] Home (S0) loads in SpaceLab and exposes its parameters; a touched real wall lies within 0.05 m of the virtual wall at three points (should: 0.03 m).
- [ ] Pre-scan path: a saved ARWorldMap relocalises within 10 s to ≤ 5 cm (should: 5 s, 3 cm).
- [ ] Meshing and plane detection are off after the scan; LiDAR depth stays on.
- [ ] Warning cue (`P1-C14`): for WARN, STRONG and tracking-lost the indicator, sound and haptic all start within one frame of the call in 20 of 20 trials. The cue is indicator, sound and haptic only, and the world does not change.
- [ ] Protocol 11.4.5, run by an engineer who did not write Bump Guard, passes every Bump Guard row at the depth rate that ships (at least 15 Hz), with the constants confirmed by `BumpGuard_Replay`.
- [ ] Static-map error after the 5-minute lap is ≤ 0.10 m at three points.
- [ ] The pose-jump half of `Safety_Independence` is green.
- [ ] Bump Guard behaves identically with the Station switched off.
- [ ] Input and pose (`P1-C10`, `P1-C11`): every SpaceLab action can be done with the gamepad alone and with gaze dwell alone; a 0.5 m jump injected in replay is absent from the geometry pose stream and present in the safety pose stream; loss of tracking raises the tracking-lost event within 0.5 s.
- [ ] Onboarding in hand (`P1-C12`): three first-time users reach Home in the headset with at most three prompts each.

G6 Station link, four functions (present: D plus one engineer from each other lane)

- [ ] Every test in the table of §7.6 passes.
- [ ] Spectator mirror: tile-id agreement and screen-to-screen latency at "must"; the on-device half of `TileId_Seed` is green and the world hash is the same on the phone and the Station.
- [ ] Live tuning: curvature, fog, space id and 4D slice change from the Console; values clamp; all six operator commands (switch space, return to Home, recentre, quality tier, STOP (warning cue), lock) act on the phone within 250 ms.

- [ ] Telemetry: a 15-minute run plotted with no gap > 2 s; the CSV exists on the Station and the phone.
- [ ] Scan and asset round trip, in a room with at least four object boxes: every box with a candidate frame is photographed, generated by the Forge, returned and placed (at most 20 generated meshes per room, largest boxes first); asset rows at "must"; a placeholder is shown until each mesh arrives.
- [ ] Fallback set (`P1-D12`): with the Forge stopped before the scan, every box holds a built-in fallback mesh within 10 s of entering Home (proposed); after a Forge restart each is replaced by its generated mesh.
- [ ] With the Station unreachable, a parameter from the kill-switch list of §12.4 is changed from the on-device menu and the change is in the phone's log.
- [ ] Flat-twin mini-map (`P1-D07`): in replay the map shows the same tile-id sequence as the log; it costs ≤ 0.3 ms on the phone.
- [ ] Router unplugged for 10 s: OFFLINE within 3.5 s, CONNECTED within 5 s of return, parameters restored, frame time unchanged (R4).

G7 Phase-1 exit (present: all four lanes)

- [ ] G0–G6 and every 11.1 test green on one commit.
- [ ] All four Station functions shown in one session without a restart.
- [ ] S0, S1, S2, S3, S4, S5 and S6 each run on phone A at "must" budget in the room scanned for the session; the per-space table is measured again on that scan and recorded beside the G4 values.
- [ ] Self-scan → Home with live AI furniture: at least three objects generated and placed in the session.
- [ ] The 3-4-5 walk closes within 0.25 m after 3, 4 and 5 corners; curvature never changes while head speed exceeds 0.1 m/s.
- [ ] Mode 1 and Mode 2 are both walked from the same tile zero.
- [ ] Room content in the registered spaces: in Glued (S1) the room's walls (as kerbs) and furniture are drawn once, where they really are, among the copies (`P1-A15`); in Aligned Cells (S3) each object is drawn in its own cell (`P1-B12`). In both, a virtual table edge is within 5 cm of the real one at three touch points.
- [ ] Bump Guard warns at a wall and at a chair moved after the scan.
- [ ] The 15-minute in-headset soak passes; motion-to-photon and loop drift are measured.
- [ ] Every register row first checked at G1–G7 holds a measured value.

G8 Playground run (present: A, C, two first-time users)

- [ ] A first-time user completes the Geometry Playground with walk, look and one button.
- [ ] Every space shows one predicted-against-measured number.
- [ ] Comfort and usability rows hold a measured value; n ≥ 5 is reached in `P4-C05`.

G9 Lung 4D (present: A, B, D)

- [ ] Toybox, breathing worldtube, breath lane and scanner-artefact demo all run on the phone, on both case packages (real and illustrative).
- [ ] ITV, core and gated volumes on screen equal `table.json` within 1% (R2 §D.10).
- [ ] `P2-A31`: slicer p95 ≤ 1.0 ms and frame p95 ≤ 17.5 ms through a soak with the slice moving, on both packages; thermal state never Serious.
- [ ] `P2-A32`: beats a and c–f advance both ways on both packages. The footer "Education and research visualisation. Not for clinical use." is on every screen, every asset shows exactly its label of §10.2.1 ("Illustrative model" on the twin; "Public research data: TCIA 4D-Lung, CC BY 3.0. Meshes derived by ATLAS." on the real case), and the credits show the TCIA citation.

G10 Lung airway and sky (present: B, D)

- [ ] Procedural airway snapped to the {4,5} tile graph, 16 generations addressable, no overlap, labelled "model-generated".
- [ ] `P2-B12`: a first-time user walks four forks without the rescue beacon in ≥ 9 of 10 runs, comfort score ≤ 3; walking the centre strip for 8 tile steps raises no WARN on the straight sides.

- [ ] `P2-B10`: the needle sky reads precomputed maps (no per-frame compute); clearance shown equals `sky.json` within 0.1 mm at 20 directions.
- [ ] `P2-B09`, `P2-B11`, `P2-D09`, `P2-D11` and `P2-D13` are accepted, and all eight beats of `P2-A32` (a to h) run end to end on the phone; the airway run ends in the lesion-chamber hand-over.
- [ ] `P2-D12`: the plan card validates and prints from a real run.

G11 Phase-2 exit (present: all four lanes)

- [ ] G8–G10 green; performance rows at "must", hot, in both use cases.
- [ ] FMS median peak ≤ 5 with three people who did not build the use case.
- [ ] Set-pass calls ≤ 60; a kill switch exists for everything Phase 2 added.

G12 Shared House (present: A, C, both phones)

- [ ] `P3-C05`: all seven beats of Post Circling completed by three walkers with no contact: through the door switches sheet, round the frame does not, round one post does, a second lap returns.
- [ ] (Extended) House Call: a live second walker on phone B appears on the right sheet within 0.1 m.
- [ ] The recorded visitor carries a visible "recorded" label.
- [ ] `P3-C03`: on sheet 1 Bump Guard gives the cue and the haptic ≥ 0.6 s before predicted contact in 10 of 10 approaches to a real obstacle, as on sheet 0.

G13 FieldFit (present: B, D)

- [ ] Blind spot found within 30 s by ≥ 4 of 5 testers; the composed final blit costs < 1 ms.
- [ ] `P3-D06`: median search time with the power-law lens is lower than with the tunnel alone for ≥ 4 of 5 testers.
- [ ] FieldFit meets the "must" performance rows of §11.2 with the lens on.
- [ ] No clinical claim in any string.

G14 Feature freeze (present: all four lanes)

- [ ] Full demo path twice in a row on one phone without cooling: no Serious, p99 ≤ 33.3 ms (R5 §E3).
- [ ] "Should" reached on the frame-time, CPU and GPU rows, or each miss documented.
- [ ] Recovery from a forced tracking loss within 15 s.

G15 Golden build and submission (present: all four lanes)

- [ ] The golden build launches with Wi-Fi off and runs the full demo with the Station unplugged.
- [ ] The filled register, one soak chart and its CSV are in the repository.
- [ ] A fresh clone builds from the README; architecture diagram, maths note, licences and AI-asset credits are present; the comment audit (`P4-A05`) finds a summary on 100% of public `Atlas.*` types.
- [ ] No camera frame from any visitor is stored, and `FORGE_DEBUG_KEEP` is off; the booth sign's data line discloses the hosted fallback (§7.4); delete-by-id removes every file for a session id on the Hub, the Forge and the phone (`P4-D02`).

11.4 Measurement protocols

11.4.1 15-minute soak

1. Release build, phone A in the closed headset, brightness 40%, start at thermal Nominal.
2. A walker, or a recorded pose track, alternates S2 Curved Walk (ρ = 3, the room as tile zero) and S1 Glued 5 × 5 for 15 minutes with LiDAR depth on. At G4 the room is `budget_room`; at G7 it is the room scanned for the session. The tour script of `P1-A14` fixes how often the space changes, and that interval is recorded with the result.
3. `AtlasPerfProbe` writes one CSV row per second; the Console plots frame time and thermal state.
4. Pass: every "must" performance row holds; no Critical; Serious not before 10 min. Record time-to-Fair and time-to-Serious.
5. At G4, repeat the soak once with depth off, driven by a recorded pose track with nobody wearing the headset, starting again from thermal Nominal, and record the difference in time-to-Fair.
6. Commit each run's file as `perf_<session>.csv`.

11.4.2 Motion-to-photon by slow-motion video

1. The app shows a world-locked vertical line. Phone B films at 240 fps (4.17 ms per frame) with the test phone's body and screen in shot.
2. Twist the test phone sharply against a stop.
3. Count frames from first physical movement to first on-screen movement.
4. Ten trials, gyro prediction off and on; report mean and range (resolution ±4 ms) (R5 §B4).

11.4.3 20 m loop drift

1. Mark a start point on the floor; in the headset, press the button on the mark.
2. Walk a 20 m closed path (for example two laps of a 2 m × 3 m rectangle, or five laps of a 1 m square), return to the mark, press again.
3. Drift = distance between the two logged raw ARKit positions.
4. Three runs; report the worst. Pass: < 0.40 m.

11.4.4 Curved-walk spike

1. Safety for the spike: Bump Guard does not exist yet. The walkers are team members only; the floor is cleared; the lap square is marked on the real floor for the spotter (a test aid, not part of the product); a spotter stays within arm's reach of the walker throughout. Warnings are not counted; wall contacts and spotter stops are.
2. Apply the loop-fit rule to the test room, measured wall to wall with a tape (the scan does not exist yet); record s and L.
3. Three walkers each walk the pentagon: five sides of s, a 90° left turn at each corner. Closure error = distance between virtual start and end.
4. Each walker then walks freely for three minutes at each of 0.5·L, L and 1.5·L (0.9, 1.9 and 2.8 m in a 3×3 m room; R1 §D.8), speaking an FMS score each minute. Wall contacts and spotter stops are counted.
5. Pass rule: closure < 0.25 m, every FMS ≤ 3 at loop-fit L, zero wall contacts and zero spotter stops, for all three walkers → Mode 1 stays the default.
6. Otherwise raise L and repeat once; if that fails, Mode 2 becomes the default and Mode 1 a switchable demonstration.

11.4.5 Bump Guard scripted approach

Set-up: the build under test processes depth at the rate that ships (never below 15 Hz), and that rate is recorded. The vertical field of view of the depth image, with the phone in the headset, is measured and recorded.

1. Scanned room with one wall and one scanned table, plus one chair and one stool 0.45 m high placed after the scan (visible only to live LiDAR).
2. A walker in the headset, looking level and not at the floor, with a spotter, makes 20 approaches, five to each of the four obstacles: head-on and at 45°, each at about 0.5 and 1.0 m/s, and a second head-on approach at 1.0 m/s. Every approach starts at least 3 m from the obstacle and reaches it within 30 s, because a level view from 1.6 m first sees the top of a 0.45 m stool from about 2.6 m.
3. The walker stops at the cue. The log gives the lead distance (from the head to the obstacle when the cue starts; the obstacle's position is taken with a tape measure) and the lead time to predicted contact. The distance at which the walker came to rest is measured with the tape and recorded beside them.
4. One run of five minutes lapping the lap square (0.5 m from the walls) counts STRONG cues anywhere and WARN cues on the straight sides, and separately the WARN cues at the corners. A WARN is a corner cue when the head is within 0.5 m of a lap corner as the cue starts (at 1.0 m/s a corner WARN starts 0.40 m before the corner; derived); every other WARN is on a straight side. Corner cues are reported, not judged (open item 19 in §14).
5. After the lap, repeat the three-point wall touch of the G5 checklist and record the static-map error: how far the edge of the distance field lies from each touched point on the real wall, after allowing for the one-cell (5 cm) inset of that edge.
6. Pass: all Bump Guard rows at "must".

11.4.6 Link latency

1. Own router; phone in the headset; Hub and Spectator running.
2. Hub latency: the Hub logs clock-corrected `t_us` → receive time over five minutes of walking; report p50, p95, p99.
3. Screen to screen: a 240 fps camera films the phone screen and the Spectator monitor together during a sharp head turn; count frames between the two; ten trials.
4. Parameter round trip: the Console stamps each `ParamSet`, the phone echoes the applied value in an `Event`; report p95 with and without 10% simulated loss.

11.4.7 Asset round trip

1. Scan a room holding a tape-measured box and three ordinary objects.
2. For each object the log records: photo sent, Forge start, GLB ready, mesh placed.
3. Report per object: Forge start → placed time (the register row), photo sent → placed time (which includes the wait in the queue), triangle count against the target sent with the job, longest edge, and longest-side scale against the scanned box.
4. Read the frame log during each GLB load.
5. Pass: all asset rows at "must" for four objects out of four.

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12. Risks and fallbacks

Wherever possible a fallback is a parameter change, not a code change. Parameters live in the registry (§7.3.2) and are set from the Console; the parameters of §12.2 and §12.4 are also in the on-device menu (task `P1-D13`), so every ladder and every kill switch works with the Station unreachable. Bump Guard is the deliberate exception: its thresholds and depth rate are constants in code (§4.10) and change only in a rebuild. Likelihood / impact: H high, M medium, L low.

12.1 Risk register

#	Risk	Why it matters (rubric line or safety)	Likelihood / impact	Early signal (gate or metric)	Mitigation built into the plan	Fallback if it happens
Safety and comfort						
1	Bump Guard is a warning cue and haptic only; the walker never sees the room (D19)	Safety	M / H	G5 protocol 11.4.5: any missed warning in 20 approaches; more than 2 STRONG cues or straight-side WARN cues on the 5-minute lap	Two sources (static scan map, live LiDAR depth kept on); thresholds set for a lead of at least 0.6 s and confirmed on a recorded lap by `BumpGuard_Replay` before G5; walker confirms the scan outline before the headset goes on	Human spotter, recommended whenever someone outside the team walks; stay in registered spaces (S0, S1, S3, S5, S6); if the test shows it is needed, raise the WARN and STRONG distance and look-ahead constants in code and rebuild (they are not registry parameters). WARN cues at the lap corners are open item 19 (§14)
2	Live depth covers a forward cone only (about ±35°, 0.3–5 m) (Spec §7), and a headset that looks level cannot see low obstacles close by (§4.10)	Safety: side and back steps, people behind, low objects	H / H	G5 protocol 11.4.5: a missed warning at a 45° approach or at the 0.45 m stool placed after the scan	Static map covers every direction; depth covers what is new ahead, and the dynamic layer keeps what it has seen (a cell is cleared only when seen empty, or after 30 s without a new hit); cables and loose objects are removed before a session	Spotter covers the rear; onboarding says "walk forwards"
3	Furniture moved after the scan	Safety: stale static map	M / H	Depth returns inside cells the map marks free	Depth overrides the map; rescan is one user action	Rescan; spotter
4	Virtual walls and furniture drift off the real ones in Curved Walk (D9, §4.6)	Safety; UX 30%	H / M	By design: one lap turns the world (§3.3)	Bump Guard works in tracking space only; onboarding states nothing in S2 is solid	`space.id` → S3 or S0, which stay registered
5	Smooth $H^2 \times E$ walking has no published comfort study (R1)	UX 30%	M / H	G2: FMS > 3/20 or pentagon closure error ≥ 0.25 m	G2 runs before S2 content is built; no gains, no vignette (BP §0.5)	`curved.mode` → Aligned Cells as the walked default; S2 stays a short opt-in
30	Cybersickness outside Mode 1: mirrored Klein rows, portals, the FieldFit lens	UX 30%	M / H	FMS median peak above 5, the "must" of §11.2, in any space: read at G8 and G11, on exit from Shared House (§10.3.7) and in the pilot (§13.5); a lens trial stopped for discomfort at G13	Every step is 1:1, with no gains (rule 3 of §0.5): the Glued cover is fixed to the floor (D15) and portal transforms are the identity (D16); curvature changes only while standing (§10.1.4); lens beats are standing or seated, and the lens switches off when the head moves more than 0.5 m (§10.4.5); FMS asked each minute with a stop value (open item 6, §14)	Skip the space, or switch the lens off; `space.id` → S0
Geometry correctness						

#	Risk	Why it matters (rubric line or safety)	Likelihood / impact	Early signal (gate or metric)	Mitigation built into the plan	Fallback if it happens
6	64-bit tile ids are exact to 42 straight tile steps and overflow at 43. The walkable lung route is 32 tile steps (16 generations, §10.2.3); 46 is only the addressable depth (23 generations), and on the airway's paths the coefficients need about 52 bits at 46 steps, so `long` suffices there (R3, derived)	Creativity 30%: wrong tiles	L / H	`checked` overflow exception in the id arithmetic; the invariant $\varphi(X^2+Y^2) - T^2 = -1$ fails. The invariant multiplies coefficients, so it is evaluated in Int128 or BigInteger: in `long` its products overflow from about 22 straight steps	`checked` arithmetic; ids behind one struct	128-bit add-only struct (≈ 83 steps, R3); root lung ids at the lobar bronchus
7	Plain square tiles leave gaps under the log map (R6)	Graphics 25%	M / M	G3: seams at tile edges	Tile generator maps the Klein square to the chart (§3.4, A.6)	Regenerate; none needed at runtime
8	Float matrices lose precision far from the anchor (R3)	Graphics: jitter	M / M	G3: shimmer on far tiles	Double on CPU; re-anchor by tile id; no tile whose centre is farther than $\rho + 0.8425$ is drawn (3.84 at $\rho = 3$; draw rule of §8.5)	Lower `curved.cullRho`
9	Sphere antipode breaks the log map (R6)	Graphics: S4 artefacts	M / L	G3: stretched triangles behind the planet	Fragment discard near the antipode; tight fog	Lower `planet.fogRho`
10	ARKit relocalisation jump integrated as motion	Creativity: world slides	M / M	Pose-jump counter in telemetry	Jumps > 0.2 m per frame rejected for geometry; Bump Guard always uses the raw pose and resets its velocity filter on such a jump, so a relocalisation raises no false look-ahead cue (§4.10)	"Return to Home" re-anchors on tile zero
Performance and heat						
11	Thermal "Serious": Unity caps at 30 fps unless the Player Setting is off (R5)	Graphics 25%	M / H	Thermal state in telemetry; G4 soak	Unity ≥ 6000.3.21f1; thermal ladder acts at "Fair"; dark palette; meshing off after the scan	Lower rungs (§12.2); swap to phone B
12	LiDAR depth stays on while walking; heat cost not published (R1)	Heat against safety	M / M	G5 depth on/off comparison: frame-time p95 rise > 1.0 ms; G4 soak: time-to-Fair with depth on against off	Budgets are measured with depth on; the G4 soak is run once more with depth off, on a recorded pose track with nobody in the headset, to measure the difference (`P1-A14`)	Processing fewer depth frames saves CPU only (the rate is a code constant, lowered in a rebuild and never below 15 Hz, §4.10); for heat, start the thermal ladder one rung lower. Depth is never switched off while someone walks
13	Two-camera rig is CPU-bound; the tile centres within ρ number about $5(\cosh\rho - 1)$, and more tiles than that are drawn	Graphics: > 300 draw calls	M / H	G4: draw calls, render-thread ms	Fog fully opaque at $\rho = 3$; 45 tile centres lie within $\rho = 3$ and at most 71 tiles are drawn (draw rule of §8.5); levels of detail	ρ 2.5 (25 tile centres, at most 39 tiles drawn with a level gaze and 52 at any pitch) → 2 (13 centres, at most 26 level and 38 at any pitch); `RenderMeshInstanced` for shells. If the view-cone test is not built: cull centres at $\rho = 3$ and make the fog opaque at $\rho = 2.15$
14	HyperEngine's warp shader uses MaterialPropertyBlocks and instanced properties: not SRP-Batcher compatible as written (R3)	Graphics: batching lost	H / M	Frame Debugger at G3 reports "not compatible"	Port moves the tile anchor into `UnityPerMaterial` ; one pooled material per tile	Instanced variant of the same shader

#	Risk	Why it matters (rubric line or safety)	Likelihood / impact	Early signal (gate or metric)	Mitigation built into the plan	Fallback if it happens
Scan and tracking						
15	RoomPlan through AR Foundation gives furniture boxes only (R1)	UX: scan flow	H / M	G5: no wall data from the route in use	Ranked scan routes; walls from planes and classified mesh	`scan.route` → manual corner taps
16	Tracking drift or loss inside a lens headset; iOS 26.4 LiDAR drift report, said fixed in 26.6.1 (R1; re-checked under open item 3 of §14)	Safety; closure error	M / H	G2: closure error ≥ 0.25 m. G5: static-map error after the 5-minute lap > 0.10 m, or 20 m loop drift ≥ 0.40 m (§11.2)	iOS version recorded at set-up; headset exposes cameras and LiDAR; while tracking is not Normal, Bump Guard uses live depth alone and raises "stand still" at 0.5 s; a static map that disagrees with live depth by more than 0.15 m for 2 s raises the "rescan" icon (§4.10)	Rescan; team pre-scan with saved ARWorldMap (D21); update iOS
17	Phone B has no LiDAR or is slower (open item 2, §14)	No swap phone	L / M	Set-up checklist; G4 on phone B	LiDAR on both phones verified at G1 (`P1-X04`); budgets measured on both phones	Phone B never walks; it takes object photos only
Link and Station						
18	LiteNetLib 2.1.4 unverified on Unity 6.3 + IL2CPP + iOS 26 (R4)	Blocks G6	M / H	G1 link smoke test	`Atlas.Link` hides the transport behind one interface	Raw UDP with the same `State` packing
19	LiteNetLib needs Wi-Fi even in development (`iproxy` forwards TCP only) (R4)	Process	H / L	First paired session	Own router; QR pairing to a known IP	Editor client on the PC for link work
20	Local Network prompt denied; Windows Firewall drops UDP (R4)	Link silent	M / M	Hub prints "no packets received" after 10 s	Prompt raised in onboarding with retry; firewall rule in the set-up script	Phone runs standalone
21	Venue radio congestion	Mirror stutters	M / M	Packet loss and RTT in telemetry	ATLAS app never depends on the link	Lower `link.stateHz` ; recorded-state replay
Live AI mesh pipeline						
22	Forge slow, down or unreachable (D20)	UX: built-in furniture in place of the walker's own	M / M	G6: Forge job start to placed mesh takes more than 60 s per object (§11.2, test L15)	Placeholder box at once; retry; results cached by photo hash	`forge.backend` → hosted API (only if the walker was told and agreed, §7.4) → cache → built-in fallback set (`P1-D12`)
23	TripoSG output is geometry-only, over budget or the wrong shape; the RTX 5070 needs PyTorch cu128 (BP §0.5)	Graphics; warp needs a ≤ 0.25 m grid	M / M	Forge post-check: triangle count, bounds against the box	Decimate to the triangle target sent with the job, regrid, palette colour, scale to the scanned box	Reject the mesh; the job fails and the box takes its built-in fallback mesh (`P1-D12`)
Medical data and claims						
24	Real tumours barely move (median 4.3 mm SI, R2)	UX: worldtube looks static	H / M	G9: first render of patient 101	Illustrative twin (10 mm nodule, 15 mm motion) beside it, labelled (D25)	Exaggeration slider with the factor on screen, on the illustrative twin only; never on the real case
25	Over-claiming; "4D shadow" and "tilted slice" framings are ours (R2)	Presentation 15%; ethics	M / H	A clinical claim in any string at G13 (§11.3); a hit in the phrase check (`P4-D05`)	"Not for clinical use" on screen; labels "illustrative model", "model-generated"; the never-say list of §13.4, which the phrase-check script uses	Remove the claim

#	Risk	Why it matters (rubric line or safety)	Likelihood / impact	Early signal (gate or metric)	Mitigation built into the plan	Fallback if it happens
26	Licences: TCIA CC BY 3.0, AeroPath CC BY 4.0, HyperEngine MIT notice; no GPL code (HyperRogue); ATM'22 labels not redistributable; Hunyuan3D off; Asset Store packs out of the repo	Submission validity	M / M	Ledger incomplete at G14	Ledger row added with every dependency; TCIA data-use terms acknowledged and the citation recorded by G9 (open item 18, §14)	Drop the asset; procedural substitute
31	Privacy: room photos and motion logs of walkers from outside the team	Ethics; Presentation 15%	L / H	A crop file left on the Forge after its job (test L20 of §7.6, at G6); the "crops kept" banner on the Console with an outside walker; any image byte in a log (`P4-D02`)	Rule 10 of §0.5: no camera frame is stored, and crops are held in memory and deleted when the job finishes; the hosted fallback is disclosed and can be declined (§7.4); anonymous ids, motion logs kept encrypted on the Station, delete-by-id (§13.5); the data line on the booth sign	Set `forge.backend` to cache when a walker declines the hosted service; delete that walker's data by id (§13.5); run the session with `link.enabled` off, so the Station records no log
Team process						
27	One Mac is the only iOS build seat; free-profile signing expires (Spec §15.8)	Every device gate queues behind it	H / H	Build queue length; install failure	Editor simulation in all lanes; one-command build script; lane A owns the seat	Device tests batched per gate; reinstall before each demo
28	Shared scenes: four lanes touch SpaceLab	Merge conflicts	H / M	Conflicts in `.unity` files	One additive scene per space module; only lane A edits SpaceLab	Revert to last green; re-apply as a prefab
29	Shared shader: look (lane A) and warp include (lane B) in one file set	One edit breaks every space	M / H	`Shader_Contract` fails (§11.1), or the shader contract check run at every gate fails (§9.3)	Interface frozen at G0; warp is an include with one entry point	Per-space keyword disables the new path

12.2 Fallback ladders

Item	Full	Step 1	Step 2	Last resort	Switch
S0 Home	Swift RoomPlan plugin, Forge meshes	ARKit mesh, planes, RoomPlan boxes	Mesh and planes only; with no RoomPlan boxes there is no live furniture (§4.2), and the obstacles stay placeholder boxes	Manual corner taps; built-in fallback set	`scan.route` , `forge.backend`
S1 Glued	T^2/Klein 5 × 5 or T^3 3 × 3 × 3	T^2 and Klein only (no T^3), 5 × 5	3 × 3 copies	T^2 only, 3 × 3	`glued.mode` , `glued.copies`
S2 Curved Walk	ρ = 3, scale 0.8, MSAA 4×	ρ = 2.5, scale 0.7	ρ = 2, scale 0.6, MSAA 2×	Switch to S3	`curved.cullRho` , `render.scale` , `render.msaa` , `curved.mode`
S3 Aligned Cells	Per-instance clip gates	Fewer neighbour cells visible	The 2 × 2 corner demo, five tiles round one post (§4.7)	Home with the curvature dial, standing	`cells.viewDepth`
S4 Small Planet	Six tiles, loop-fit R	Tighter fog	Larger R	Standing view, no walking	`planet.fogRho` , `planet.R`
S5 Impossible Rooms	Portal depth 1	Depth 0	Fewer rooms	One room pair	`portal.depth` , `rooms.count`
S6 Slicer	Live worldtube (≈ 18k tets) and polytopes	Stepped precomputed slices, no blending (R2)	Tesseract only (48 tets)	Static slice meshes	`slicer.source`

Item	Full	Step 1	Step 2	Last resort	Switch
Station: mirror	`State` at 60 Hz	30 Hz	10 Hz	Recorded-state replay	`link.stateHz`
Station: tuner	Console	Console under `demoLock`: the phone ignores `ParamSet`, the six operator commands still work	On-device menu (`P1-D13`) for the parameters of §12.4	Compiled defaults	`demoLock`
Station: telemetry	Live dashboard and CSV	CSV on the phone, uploaded over HTTP later	In-headset HUD	—	`telemetry.mode`
Station: round-trip	TripoSG on the 5070	Hosted API, only if the walker was told and agreed (§7.4)	Cache by photo hash	Built-in fallback set (`P1-D12`)	`forge.backend`

The last-resort column is for a failure during a live demo only. It never satisfies a gate: G4 and G7 need every space S0 to S6, and both Mode 1 and Mode 2, at Step 2 or better.

The thermal ladder (§8.6; R5) drives `render.scale`, `render.msaa`, `curved.cullRho` and `glued.copies` by itself; the Console can pin any rung.

12.3 If a gate fails

Gate	Blocked	Can proceed	Lead decides
G1 First light	G2, G4 (all device work in lanes A, B and D)	Kernel, tile graph, Hub process, Forge, all Editor work	Fix the rig first; no lane starts device tests
G2 Curved-walk spike	S2 as the walked default	Everything else; S3 shares the kernel and tile graph	Raise L one step and repeat the protocol once (§11.4.4); if it still fails, make S3 the default and walk the five-corner leg of the 3-4-5 walk in S3
G3 Spaces in the Editor	G4 for the failing space only	Other spaces; scan; link	Fix the space. It may pass at ladder Step 1 or Step 2, never at the last-resort rung; no space is held past G7
G4 Spaces on the phone at budget	G7	G5, G6	Accept Step 1 or Step 2 as that space's default. A last-resort rung does not pass G4
G5 Scan, Home, Bump Guard	All walking by anyone outside the team, and all walking without a spotter within reach; G7	Walking by team members with a spotter within reach, as at G2; S6 seated; link; Forge	Nothing is traded for this; no public walking until it passes
G6 Station link	Function 4 blocks live furniture; functions 1–3 block operator tooling	All on-phone spaces	Name the failed function; D22 requires all four for G7
G7 Phase-1 exit	P2	Fixes only	No P2 work until G1–G6 are green

12.4 Kill switches

Each is one registry parameter, settable from the Console and from the on-device menu (`P1-D13`); every change is logged with the session. Bump Guard has no off switch and no registry parameter: its thresholds and depth rate are constants in code (§4.10).

- `space.id`: force any space; S0 is the safe default.
- `curved.mode`: Mode 1 or Mode 2.
- `curved.cullRho`: 2–3.5; the radius ρ at which the fog is fully opaque (draw rule of §8.5).
- `render.scale`: 0.8, 0.7 or 0.6; selects the eye-texture set (§8.2).
- `render.msaa`: 2 or 4.
- `glued.copies`: 3 or 5 per side.
- `portal.depth`: 0 or 1.
- `slicer.source`: live, stepped, tesseract.

- `pose.gyroPredict` : on or off.
- `forge.backend` : local, hosted, cache, off (§7.4). Set it to cache for a walker who declines the hosted service.
- `scan.route` : 1–4.
- `link.enabled` : off puts the ATLAS app in standalone mode.
- `link.stateHz` : 10, 30 or 60.
- `demoLock` : the phone ignores `ParamSet` ; the six operator commands still work (§7.3.2).
- `thermal.pinTier` : hold one ladder rung.

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13. Demo, documentation and presentation

This section covers Phase 4 (gates G14 Feature freeze, G15 Golden build and submission) and everything a judge sees. Paths under `docs/` and `media/` are proposed.

13.1 Rubric-to-evidence map

Rubric line	Shown live	Number or figure behind it	Lives in
Geometric Creativity 30%	Scanned room as tile zero; curvature dial; 3-4-5 walk; Mode 1 and Mode 2 on one tile graph; S1, S4, S5, S6	Corners 3 / 4 / 5, closure error < 0.25 m (derived); holonomy error < 1°, Lorentz drift < 1e-9 (R5 D1)	`docs/maths.md`, `docs/evidence/correctness.json`
UX and Interaction 30%	Self-scan onboarding; walk + look + one button; Bump Guard cue and haptic; return to Home	FMS per minute, SEQ, time to first walk, pointing error (R5 D2)	`docs/metrics.md`, `docs/pilot/`
Graphics and Technical Execution 25%	60 fps in the headset; Spectator re-render; Console telemetry	Frame time p50 / p95 / p99, draw calls ≤ 300, triangles ≤ 500k, thermal timeline (R5 D1)	`docs/metrics.md`, `docs/evidence/perf_<session>.csv`
Documentation and Presentation 15%	Dashboard reveal, honesty table, deck, video	Deliverables of 13.4 complete	`README.md`, `docs/`, `media/`
Bonus: beyond gaming	Lung Mission (hyperbolic airway walk, worldtube, breath lane, scanner artefact); FieldFit	ITV/GTV of patient 101 beside the published median 1.35 (R2)	`docs/maths.md`
Bonus: educational value	One rule per space; equation card, predicted against measured	One card per space S0–S6	App, Spectator
Bonus: polish	Shared look; audio loop per space; distinct warning cue	15-minute soak at "must" values on both phones	`docs/metrics.md`
Bonus: novel technical approach	Loop-fit rule; exact $\mathbb{Z}[\varphi]$ tile ids; two renderers on one kernel; live Forge furniture	Tile centres within ρ = 2 / 2.5 / 3 / 3.5: 13 / 25 / 45 / 81 (derived); Forge time per object, to be measured	`docs/architecture.md`

13.2 The demo

Roles, space, kit

Role	Does
Narrator	Talks to the judge; asks FMS and SEQ; relays "home" and "stop" to the operator (the Console's return-to-Home and STOP commands, §7.3.2)
Operator	At the Station: Console, space switching, FMS entry, recording
Spotter (recommended)	Walks behind the judge; a touch on the shoulder means stop. Bump Guard is warning and haptic only (D19), depth sees only forwards, and Mode 1 is not registered to the room (§4.6). Never merged with another role (Spec §14.1)
Explainer	At the Spectator screen for onlookers

Space: free rectangle with min(Wf, Df) ≥ 1.5 m (about 2.1 m wall to wall), the smallest floor that is not standing mode (§4.4). One level floor, with no stairs or step-down in or beside the walked area; glass doors and mirrors are left outside the confirmed outline; cables and loose objects are removed (§4.10). Onlookers stay outside the scanned floor. Even, flicker-free light (Spec §14.1).

Kit: phones A and B; the lens headset (a second one only if open item 14 provides it), stored lens-down away from windows; Station; Mac with cables; own 5 GHz router; chargers; desk fan; face covers, wipes, dry microfibre for lenses; booth sign; chair and water (Spec §14.1).

Onboarding (phone in hand until step 5)

1. Narrator asks: VR before, carsick easily, glasses. Consent line and baseline FMS (Brief §8.1). The operator tells a walker from outside the team that, if the Station's own generator fails, crops of the object photos go to a hosted service whose retention we do not control; if the walker declines, the operator sets `forge.backend` to cache before the scan (§7.4).
2. App launch: camera and Local Network prompts are answered in hand, never in the headset (BP §0.5).
3. Pairing: the judge scans the Hub's QR code; the Console shows "phone connected" (R4).
4. Scan: 30–60 s self-scan, the judge confirms the floor outline (D21). Fast path: team pre-scan and saved ARWorldMap relocalise instead.
5. Headset on, focus check, one button press. Spotter: "I stay behind you. If I touch your shoulder, stop. A sound and a buzz mean something real is close."

Target: first walk ≤ 90 s after launch (R5 D2).

Run-of-show

The organisers' slot length is unknown (Brief §13); both scripts are rehearsed.

Long slot (10 min, self-scan):

Time	Segment	Judge does	Spectator shows
0:00–0:40	Frame	Consent, FMS, prompts, pairing	"Connected"
0:40–1:40	Scan	Scans, confirms outline, headset on	RoomModel outline; s and L
1:40–2:40	Home and dial	Sees the room 1:1, bends it, walks the five-corner lap	3-4-5 card: corners, closure error
2:40–3:40	Curved Walk	Walks to a landmark; furniture meshes arrive; switches to Aligned Cells	Forge queue; flat-twin mini-map after the lap (Brief §12)
3:40–4:20	Airway	Walks one fork of the hyperbolic airway (two tile steps, one valve chamber) on the one-fork demo route, ending in the lesion chamber	Poincaré map and dendrogram with the walker's address; "2^16 branches, same corridor width"
4:20–5:00	Slicer	Slices and rotates the tesseract	Slice and shadow
5:00–6:10	Lung 4D	Unroll time; worldtube slice and shadow; breath lane; tilted slice	ITV number; artefact view
6:10–6:25	Exit	Headset off; exit FMS; SEQ	"Preparing your results"
6:25–7:10	Dashboard reveal	Turns to the screen	Path, comfort, performance, mathematics
7:10–10:00	Questions	Judge's choice: S1, S4, S5, Post Circling or FieldFit	Honesty table; architecture

The demo uses a one-fork route whose end tile is the lesion chamber (a Console switch); the four-fork route of §10.2.3 stays the G10 test.

Headset time 4:45, under the 5 min limit (Brief §9.4).

Short slot (5 min, pre-scan fast path): frame 0:25 → relocalise 0:20 → Home, dial, lap 0:45 → Curved Walk 0:45 → worldtube and breath lane 0:45 → exit 0:10 → dashboard 0:30 → questions 1:20. Headset time 2:45. Slicer, Aligned Cells, the airway fork and the artefact move to video.

Comfort rules

- Real walking only; identity gains; no vignette while walking (BP §0.5).
- L changes only while the judge stands still (R1).
- FMS (0–20) is spoken once a minute; the operator types it into the Console.
- Stop threshold: **to be confirmed by the team** (open item 6, §14). The older documents offer ≥ 4 (Brief §8.1); the performance research offers ≥ 10 (R5 D2). The registry key is `pg.fmsStop` (§10.1.4). At the threshold, or on any verbal discomfort: headset off, chair, water. FMS > 3 switches Mode 1 to Mode 2 (G2 rule).
- Order: flat and registered (Home) → curved (Mode 1) → stationary (Slicer, worldtube).

Fallbacks

Failure	Detected by	Immediate action	If it persists
No link	Console RTT tile; phone link icon	Continue: the app is standalone, Bump Guard never uses the network, the CSV is written on the phone	Dashboard from the phone's file; video for onlookers
Hot phone	Thermal state on the Console	Thermal ladder drops a rung; the narrator names it	Swap phones between judges
Bad scan	Outline wrong at confirmation; or the Bump Guard "rescan" icon during the walk (§4.10)	Manual corner taps; on the "rescan" icon the judge stands still, the headset comes off and the room is rescanned	Team pre-scan and ARWorldMap
Forge down	Forge tile on the Console	Hosted API if the walker was told and agreed, otherwise built-in set	Built-in set, labelled "built-in"
Tracking lost	Tracking state; pose jump > 0.2 m	Judge stands still; spotter steps in; relocalise, target ≤ 15 s (Brief §9.6)	Home; phone B; video
App will not open	Launch check before each session	Phone B; reinstall from the Mac	Video and a saved dashboard

Phone swap rule: never start a session on a phone reporting Serious. Between judges, swap if the Console shows Fair (amber). During a session, swap only when the ladder reaches Serious. The resting phone is out of the headset, case off, at the fan, screen off (BP §5). Charge it only once it is back at Nominal, and unplug it before it goes back into the headset (charging adds heat, R5).

Honesty table (on the sign; shown in questions)

Item	Real	Generated live	Pre-computed	Illustrative
Room, walls, object boxes	Scanned in the session, or team pre-scan (said aloud)			
s, L, tiling size		Loop-fit from the scan		
Furniture meshes		Forge, per object, for at most 20 objects (largest first)	Built-in set, labelled "built-in", when the Forge is unreachable or an object has no usable photo; objects beyond the twentieth stay placeholder boxes	
Curved world		Tile graph and world generator, by seed		
Tumour, 10 phases	TCIA 4D-Lung patient 101 contours	Slices by Mesh4D	Tetrahedral worldtube	
Mobile nodule (10 mm, 15 mm motion)				"illustrative model"
Airway tree				"model-generated"
Proximal airway (if shown)	AeroPath (CC BY 4.0), a different patient from the tumour case		Meshes	
Second home and visitor (Shared House)	A team member's room, used with their consent; the visitor is labelled "recorded" or "live"		Pre-scanned and bundled; the recorded visitor track	A synthetic room, if one is used, is said aloud
Performance numbers	Measured on the phone		Soak chart from an earlier run	

13.3 Narration: one rule per space

Space	Rule in ten seconds
S0 Home	"This is your room, rebuilt from your scan."
3-4-5 walk	"Same steps, same left turns: four corners close on a flat floor, five here."

Space	Rule in ten seconds
S2 Curved Walk	"Your room is tile zero of a floor with five squares at every corner."
S3 Aligned Cells	"The floor stays put; which room you enter depends on the way you came."
S4 Small Planet	"Three right-angle turns bring you home."
S1 Glued	"Every wall is a door to the opposite wall."
S5 Impossible Rooms	"More rooms than the floor has space for."
S6 Slicer	"You see one slice of a four-dimensional object."
Airway	"Every branch gets a corridor of the same width, because this floor has room for all of them."
Worldtube	"One breath, one object: a slice is an instant, the shadow is the volume to treat."
Breath lane	"Walk the lane and you walk through the breath."
Tilted slice	"A slow scanner cuts space-time at a slant; that is the artefact."

Closing line: "Every space you walked was built from your room and from a rule you can check with your feet."

13.4 Documentation deliverables

Deliverable	Content	Source
`README.md`	What ATLAS is; how to run the app, Hub, Spectator and Forge; hardware	Appendix C
`docs/CREDITS.md`	4D-Lung CC BY 3.0 (Hugo et al. 2016 plus the TCIA citation); AeroPath CC BY 4.0 if used; LiteNetLib, TripoSG, HyperEngine and Engine4D MIT (copyright notice kept in every ported file); rembg and its model weights; CC0 kits (Kenney, Quaternius, KayKit); Meshy CC BY 4.0 if any hosted free-tier mesh ships; audio; AI-generated assets labelled; HyperRogue (GPL-2.0) read only, no code copied	Licence ledger
`docs/maths.md`	1 Coordinates and kernel (hyperboloid; tile id + local matrix). 2 Rendering: the log-map vertex warp, why not Klein or Poincaré for the walked view, chart-authored tile meshes, fog and cull. 3 {4,5} tiling and $\mathbb{Z}[\varphi]$ ids. 4 Loop-fit and the 3-4-5 proof, including the sphere. 5 Mode 2 as the polyhedral model with the same curvature per tile. 6 Flat but not simple: glued covers and portals. 7 4D slicing and the worldtube. 8 The lens mathematics (FieldFit). Each formula names its unit test and figure	Appendix A
`docs/architecture.md`	Two layers, assemblies, message table, Station processes	Sections 6, 7
`docs/metrics.md`	R5 D1 and D2 rows with must, should and measured; one soak chart	`perf_<session>.csv`
`docs/evidence/test_report.md`	One line per §11.1 test: what it proves, pass condition, result on the golden commit; linked from the top of the README	CI output
`docs/prior_art.md`	Positioning table below	Brief §12
Video	Reel ≤ 3 min in this order: scan → Home → dial → 3-4-5 walk with the corner counter → Mode 1 / Mode 2 → airway fork → worldtube, breath lane, tilted slice → dashboard. One 30–60 s clip per space S0–S6, plus one clip each for the airway, the lung 4D (with the scanner artefact), Post Circling and FieldFit (Spec §14.2)	Spectator capture
Deck	Rule per space, architecture, metrics table, honesty table	`docs/`

Code-comment standard, checked at G15 (audit: `P4-A05`): every public type and method in `Atlas.*` has an XML summary; every formula carries its source tag and units; every shader property states its space and unit; each assembly has a `README.md`; no numeric literals for tunables outside the parameter registry. The one exception is Bump Guard: its thresholds and depth rate are named constants in `Atlas.Safety` and are never in the registry (§4.10).

Claims language (Brief §2.4). Say: "to our knowledge, no prior system combines…"; "education and research visualisation; not for clinical use". Never say: "first"; "intuitive" without pilot data; "generated live" for a built-in asset; "plans your procedure", "treats", "detects" a condition, "rehabilitates", "expands your visual field"; any cut-off that labels a person as impaired. The phrase-check script (`P4-D05`) uses the full Avoid list of Brief §2.4.

Prior work	What it does	How ATLAS differs
Holonomy (TU Delft)	Walks {4,5} as flat cells in a fixed 3×3 m area	Cell size from the scanned room; the same tile graph also drives a smooth mode. Its comfort evidence covers Mode 2 only (R1)
Hart–Segerman, HyperRogue VR, Hyperbolica	Correct curved rendering; no real room	Scanned room as tile zero; room-derived curvature; the 3-4-5 walk
Tea for God	Euclidean overlapping rooms fitted to the play area	Curved and glued spaces; live AI furniture
4D Toys, Miegakure	4D slicing as play	Breathing worldtube and scanner artefact on real data

Before submission, lane B reads the two 3DUI 2024 contest entries ("Beyond Euclid … Spherical Geometry", "Space-Bending Interaction…") and records any overlap (Brief §12).

13.5 Evaluation we can run

Pilot: n ≥ 5 walkers from outside the team, reached in the rehearsals (`P4-C05`); anonymous ids; no camera frames stored; motion logs are treated as health data and kept encrypted on the Station (Brief §0, Spec §15.5); the dashboard shows only this walker plus aggregates; data deleted on request (Spec §13.1).

What is kept per session: the RoomModel (floor outline, object boxes), furniture meshes, pose and event logs, and FMS scores; no images. A self-scan by a walker from outside the team is not saved as an ARWorldMap, and its RoomModel, cached GLBs and Forge room record are deleted at session end unless the walker agrees to keep them. Delete-by-id (`P4-D02`) removes the Hub logs (`perf.csv` , `state.bin` , `events.jsonl` , `phone.log`), the Forge room, mesh and assets (`DELETE /rooms/{roomId}` , Appendix B.4), and the phone's RoomModel, world map, cached GLBs and copy of the performance CSV.

Measure	Capture	Must (R5 D2)
FMS per minute: peak and rise	Console entry	Median peak ≤ 5
SEQ (1–7), once per use case	Console entry (`P2-C17`)	Median ≥ 5
Time to first walk	Event log	≤ 90 s
Pointing back to the start after three legs, Home against Curved Walk	Head ray at button press (`P2-C17`)	Measured in both
Sessions ended early	Operator flag	≤ 1 in 5
Bump Guard warnings per minute; wall contacts	Event log; spotter count	Reported

May claim: descriptive values with n stated; that a pointing gap between flat and curved shows the geometry is not cosmetic. May not claim: statistical significance, clinical benefit, comfort of Mode 1 beyond our sample, or general usability. One trial per judge is a demonstration, not a measurement (Brief §8.1).

13.6 Work packages for Phase 4

ID	Work package	Lane	Depends on	Output (files, components, interfaces)	Acceptance test (binary, with numbers)	Needs (Editor / phone in hand / phone in headset / Station)
P4-A01	Look pass per space	A	G12, G13	Palette and fog presets	In S0–S6, a 360° capture shows, in either eye, 0 surfaces beyond the distance at which the fog is fully opaque and 0 undrawn tiles inside it; draw calls ≤ 300; triangles ≤ 500k	Phone in headset
P4-A02	Audio loops and cues	A	G14 Feature freeze	One loop per space; final mix of the Bump Guard tones (built in `P1-C14`)	7 of 7 spaces have a loop; cue starts ≤ 1 frame after the event in 20 of 20 trials; allocation ≤ 1 kB per frame	Phone in hand
P4-A03	Performance hardening	A	P4-A01	`perf_<session>.csv` per phone	15-minute soak on A and B: p95 ≤ 20 ms, p99 ≤ 33.3 ms, never < 30 fps, no Serious before 10 min	Phone in headset, Station

ID	Work package	Lane	Depends on	Output (files, components, interfaces)	Acceptance test (binary, with numbers)	Needs (Editor / phone in hand / phone in headset / Station)
P4-A04	Golden build and launch check	A	P4-A03, P4-C04	Tagged build on both phones	Wi-Fi off: 7 of 7 spaces load on both phones; signing valid past the event date; 1 reinstall from the Mac succeeds; Guided Access, orientation lock and Do Not Disturb on, on both phones (Spec §15.5)	Phone in hand
P4-A05	Comment audit	A	G14 Feature freeze	Audit script; assembly READMEs	100% of public `Atlas.*` types have a summary; 0 compiler warnings	Editor
P4-B01	Maths notes	B	G11 Phase-2 exit	`docs/maths.md`	8 chapters present; every formula names a passing test	Editor
P4-B02	Correctness report	B	G12, G13, P1-B04, P1-A12	`docs/evidence/correctness.json`	Lorentz drift < 1e-9; holonomy error < 1°; tesseract slice volume error < 1e-4; tile counts 13, 25, 45, 81	Editor
P4-B03	Equation-card audit and Spectator mirror	B	P4-B01, P2-C06, P2-A03, P1-D05	Each of the 7 cards checked against `docs/maths.md` (`docs/evidence/cards.md`); the same card drawn on the Spectator from `Event` data	7 of 7 cards show a predicted and a measured value that agree with the log to 3 significant figures, on the phone and on the Spectator	Phone in headset, Station
P4-B04	Figure data	B	P4-B02	Scripts and data in `figures/`	`python figures/make_figures.py` exits with code 0 from a fresh clone and writes fig01–fig06	Editor
P4-B05	Prior-art note	B	G7 Phase-1 exit	`docs/prior_art.md`	4 table rows and 2 contest entries each have an overlap note	Editor
P4-C01	Onboarding polish with outside walkers	C	G12, G13	Revised prompt, pairing, scan and fit screens (built in `P1-C12`)	Local Network prompt precedes the headset step in 5 of 5 fresh installs; median first walk ≤ 90 s, n ≥ 5	Phone in hand
P4-C02	Comfort protocol	C	P4-D02	`pg.fmsStop` set from pilot data; the comfort protocol page in `docs/demo/` (the prompt and the alert are built in `P2-C09`)	Prompt every 60 s ± 5 s; a value ≥ `pg.fmsStop` raises the operator alert in 10 of 10 trials	Station
P4-C03	Pre-scan fast-path drill	C	G12, G13	The `P1-C13` world-map path wired into the demo flow and the runbook	Relocalises within 10 s in 5 of 5 trials	Phone in hand
P4-C04	Failure drills	C	P4-C01	`docs/demo/runbook.md`	6 of 6 fallbacks exercised; forced tracking loss recovers in ≤ 15 s in 3 of 3 trials	Phone in headset, Station
P4-C05	Rehearsals	C	P4-C04, P4-D01	`docs/demo/rehearsal_log.md`	≥ 5 outside walkers; headset time ≤ 5:00 in every run; overruns ≤ 10 s per segment	Phone in headset, Station
P4-C06	Sign and briefings	C	G14 Feature freeze	Booth sign, spotter briefing (stairs, glass and mirrors, cables, loose objects), honesty table	Sign carries minimum age 13, exclusions, the data line (with the hosted-fallback sentence of §7.4) and "not for clinical use": 4 of 4	Editor
P4-D01	Judges' dashboard	D	P4-D02	Console page `/session/<id>`	5 panels (path, comfort, performance, spaces visited, mathematics) render from log files with the phone disconnected; the mathematics panel shows this session's predicted against measured values (corners 3 / 4 / 5, closure error in m, world turn per lap against 90°) and the test summary from `correctness.json` (tests passed / total, Lorentz drift, holonomy error, tile counts 13 / 25 / 45 / 81); the page shows no other session's id, path or scores, only aggregates	Station
P4-D02	Session log and pilot data	D	G12, G13, P1-D01	`events.jsonl` per session; delete-by-id; the Hub's `logs/` folder on an encrypted volume	0 image bytes in any log; delete-by-id removes every file for the id on the Hub, the Forge and the phone; 100% of FMS entries timestamped; `logs/` is on an encrypted volume	Station, phone in hand
P4-D03	Metrics report	D	P4-A03, P4-D02	`docs/metrics.md` ; chart script	Every R5 D1 and D2 "must" row holds a measured value or "unmeasured"; chart regenerates from the CSV	Station

ID	Work package	Lane	Depends on	Output (files, components, interfaces)	Acceptance test (binary, with numbers)	Needs (Editor / phone in hand / phone in headset / Station)
P4-D04	Spectator capture	D	G12, G13, P1-D05	`media/` clips and reel	11 clips of 30–60 s (S0–S6, airway, lung 4D, Post Circling, FieldFit); reel ≤ 3 min containing the 3-4-5 counter and the artefact view	Station
P4-D05	Deck and claims check	D	P4-D03	Deck; phrase-check script	0 hits for the phrases of the Avoid list of Brief §2.4 (§13.4) in `README.md`, `docs/` and UI strings	Editor
P4-D06	Offline pack	D	P4-D04, P4-D01	Reel and a saved dashboard on the Station	Both open with the router unplugged	Station

14. Open items

Things only the team can decide or check. None of them blocks the start; each names the gate by which it must be settled.

#	Item	Why it matters	Settle by
1	**Headset model and its lens numbers** (distortion coefficients, lens separation, screen-to-lens distance); check the fit with both phones and with a glasses wearer	The lens blit (`P1-A05`) needs them. Without them straight lines bow	G4
2	**Both phones really have LiDAR** (15 Pro Max and 14 Pro Max, not Plus models), and the headset leaves the rear camera and LiDAR unobstructed	Scanning and Bump Guard depend on it	G1
3	**iOS version on both phones.** A LiDAR tracking-drift regression was reported for one recent iOS release and reported fixed in a later one (R1; re-read the thread). Run the 20 m loop drift test on the installed version	Drift moves the static map under Bump Guard	G5
4	**Unity patch and Xcode**: Unity ≥ 6000.3.21f1, Xcode 26 on the Mac	Thermal frame-rate setting; ARKit plugin	G0
5	**How outside walkers are supervised.** Bump Guard is a warning and a haptic by decision. This plan recommends a cleared floor, a confirmed scan and a human spotter whenever someone outside the team wears the headset	Safety of people who do not know the room	G7
6	**Comfort stop value.** The comfort score (0–20, spoken each minute) at which a session ends. The older documents offer 4 (Brief §8.1); the performance research offers 10 (R5 D2)	Run-of-show, Console alert	G2
7	**Comfort floor for L and R**	Set from the G2 result; provisional values are L ≥ 1.4 m and R ≥ 1.2 m	G2
8	**Hosted image-to-3D service** for when the local generator fails: which one, what budget, what licence, what data-retention terms. And the measured time per object on the RTX 5070	Forge fallback (`P1-D08`); what a walker is told before the scan and what the booth sign's data line says (§7.4)	G6
9	**Gamepad model**, and whether its rumble is reachable from Unity on iOS	Second haptic channel for Bump Guard; twins	G5
10	**Headset lever**: a single button that produces a screen touch? Are double and triple clicks acceptable?	Twin mapping in the Playground	G8
11	**Marker cube**: in scope for the Slicer and the Lung 4D, or gamepad and gaze only?	Extended input in Phase 2	G8
12	**AeroPath licence wording** differs between its two hosting pages (CC BY 4.0 and MIT). Attribute under CC BY 4.0, and confirm before shipping the optional real proximal airway	Credits	G10
13	**Shared House "guest rug"** (§10.3): a rug on the second sheet that marks where the walker's own floor is free. Is that acceptable under "no ghost of the room"? Without it only Bump Guard protects the walker on the second sheet. Default until the lead answers: no rug. Sheet 1 is drawn in full, with no marking derived from the real room; `P3-A04` is not built; the door fit still maximises the overlap of the two free floors; and the spotter recommendation of §4.10 applies to every walker on sheet 1	Phase 3 design	G11 (before `P3-A04` starts)
14	**A second headset** for a live House Call on phone B	Extended scope of Phase 3	G12
15	**The organisers' rules**: whether code and assets may be prepared before the event and what must be declared as prior work; slot length; one judge or a panel; live demo or video; the space, light and power at the venue; whether our own Wi-Fi router is allowed	Eligibility of everything built under this plan; run-of-show; the Station link needs the router	G0 (prior-work rule); G14 (the rest)
16	**H200 cluster.** Not assumed anywhere in this plan. If it is available it can run batch jobs, never anything live	Optional	—
17	**Phase 2 lane split.** Lane A works on both the Playground (4 packages) and the Lung Mission (13 packages), so the Lung Mission has three lanes and not two. Accept, or move `P2-A01` – `P2-A04` to lane C	Balance of the lanes in Phase 2 (§9.6)	G7

#	Item	Why it matters	Settle by
18	**TCIA data-use terms** acknowledged for 4D-Lung, and the required citation recorded in the licence ledger (Spec §17 Q6)	Credits; submission validity	G9
19	**Bump Guard at the lap corners.** The lap square is 0.5 m inside the walls, so a walker faster than 0.5 m/s at a corner gets a WARN there (§4.10). Accept that, or widen the inset to 0.7 m: s = min(W, D) − 1.4 m, so a 3 × 3 m room gives s = 1.6 m and L = 1.51 m, and a corner stays quiet up to 0.75 m/s (derived)	A cue at every corner of the 3-4-5 walk, or a smaller lap and stronger curvature in every room	G5

Crosswalk to the Spec's open questions (Spec §17; Brief §13 holds the same list): Q1 is item 15; Q2 is settled by D5; Q3 is item 2 (no Android client, Appendix C.3); Q4 is item 16; Q5 is item 1; Q6 is D25 and item 18; Q7 is settled by D3; Q9 by D9; Q10 by D5 and item 14; Q11 by D25; Q12 by Appendix C.1 (no entitlements; the 1.5 GB memory budget is measured on both phones at G4); Q13 by D4. Q8 (the name) stays ATLAS.

Assumptions made in this plan

- The plan is one Markdown document with figures; the six research memos it cites are kept beside it in `documents/research/`.
- Mode 1 (Curved Walk) is the default hyperbolic mode, subject to gate G2. If G2 fails, Aligned Cells becomes the default and Curved Walk a switchable demonstration.
- Phase 2 places the Glued and Impossible Rooms exhibits with lane C, and the whole medical 4D runtime, including the Lung director, labels and credits (`P2-A32`), with lane A, to balance the lanes.
- Work that sits between two lanes goes to the lane that owns the code it touches: room content in Glued to lane A (`P1-A15`) and in Aligned Cells to lane B (`P1-B12`); the warning cue and haptics to lane C (`P1-C14`); the built-in fallback set and the on-device parameter menu to lane D (`P1-D12`, `P1-D13`); the pointing probe and SEQ to lane C (`P2-C17`); the peer relay and the two-plan flat twin of the Shared House to lane D (`P3-D07`, `P3-D08`), so that use case has two lane-D packages beside those of lanes A and C.
- Phase 3 starts after G11 although its packages depend technically only on Phase 1.
- Tile ids use 64-bit integers. Their coefficients are exact for 42 straight tile steps from tile zero and overflow at step 43. The longest route in the plan, the walkable airway, is 32 tile steps (16 generations); addressing all 23 generations (46 steps) along the airway's own paths needs about 52 bits, so 64 bits are enough there too. A wider integer type is a later option.
- Bump Guard's constants are starting values, held in code (§4.10): WARN when the clearance at the head is under 0.30 m, or when the straight path of the next 0.8 s comes within 0.05 m of a hazard; STRONG likewise at 0.20 m and 0.4 s; a cue relaxes only after 0.05 m and 0.3 s of margin; an obstacle seen by live depth is remembered until it is seen empty, or for 30 s without a new hit; live depth is processed at every frame the phone delivers and never below 15 Hz. `BumpGuard_Replay` on a recorded lap confirms or changes them before G5; open item 19 covers the lap corners.
- At most 20 scanned objects receive generated meshes, largest boxes first; further objects stay placeholder boxes. The triangle target per object is clamp(60,000 ÷ n, 3,000, 5,000) for n meshed objects, so the furniture of a room never exceeds 60,000 triangles (§4.5, §8.5).
- The Glued space (S1) is a lattice of open-frame cells (corner posts, a floor grid, one chiral landmark; at most 1,500 triangles each), 5 × 5 for the torus and the Klein bottle and 3 × 3 × 3 for T^3. The real room is not repeated: its walls sink to kerbs and every scanned object is drawn once, at its true position, among the copies (§4.9).
- In Curved Walk a tile is drawn when its centre is within ρ + 0.8425 of the walker and its bounding disc on the floor is not outside the view frustum widened by 12° to each side and 4° up and down. At ρ = 3 at most 59 tiles are drawn with a level gaze and 71 at any pitch (§8.5). If the view test is not built, tile centres are culled at ρ = 3 and the fog is made fully opaque at ρ = 2.15.

Appendix A. Formulas

Every formula here was checked numerically. The figure script (`figures/make_figures.py`) recomputes the tiling counts, the loop closures and the worldtube numbers each time it runs; `research/scripts/tile_draw_rule.py` recomputes the drawn-tile counts of A.3 and `research/scripts/lap_square_check.py` the lap-square numbers of A.4. Each formula has a unit test in §11.1 or an acceptance test in its task row (§9.5).

A.1 Conventions

- Curvature $K = -1/L^2$ (hyperbolic) or $+1/R^2$ (spherical). Inside the kernel lengths are in units of L (or R); they are multiplied by L only at the boundary to metres.
- Hyperbolic plane: the hyperboloid $H^2 = \{ p = (x, z, w) : \langle p, p\rangle = -1, w > 0 \}$ with $\langle a, b\rangle = a_x b_x + a_z b_z - a_w b_w$. $\eta = \mathrm{diag}(1, 1, -1)$. Base point $o = (0, 0, 1)$.
- Sphere: the unit sphere with the ordinary dot product; base point (0, 0, 1).
- One parameter κ covers all three: $\kappa = -1$ hyperbolic, 0 flat, $+1$ spherical.

A.2 Kernel

Operation	Hyperbolic ($\kappa = -1$)	Spherical ($\kappa = +1$)
Step by a tangent vector Δ, with u = Δ/|Δ|, d = |Δ|	`Boost(Δ)` = $[[1+(c-1)u_x^2, (c-1)u_x u_z, s\cdot u_x], [(c-1)u_x u_z, 1+(c-1)u_z^2, s\cdot u_z], [s\cdot u_x, s\cdot u_z, c]]$, c = cosh d, s = sinh d	Same with c = cos d, s = sin d and last row $[-s\cdot u_x, -s\cdot u_z, c]$
Unified form	Matrix exponential of $[[0, 0, \Delta_x], [0, 0, \Delta_z], [-\kappa\Delta_x, -\kappa\Delta_z, 0]]$	same
Inverse	$M^{-1} = \eta\cdot M^{\mathsf{T}}\cdot\eta$	$M^{-1} = M^{\mathsf{T}}$
Exponential map at o	Exp(t) = (sinh r · t/r, cosh r), r = |t|	(sin r · t/r, cos r)
Logarithm map at o	Log(p) = asinh(|p_xz|) · p_xz/|p_xz|	acos(p_w) · p_xz/|p_xz|
Distance	cosh d = −⟨p, q⟩	cos d = p·q
Renormalise	Gram–Schmidt with respect to η, every frame	Ordinary Gram–Schmidt
Apparent size of width w at distance d	w / (L·sinh(d/L))	w / (R·sin(d/R))

Holonomy. Carrying a frame round a closed geodesic polygon rotates it by the polygon's area divided by L^2 (hyperbolic: against the sense of the loop; spherical: with it). By Gauss–Bonnet the area of a geodesic n-gon is $L^2\cdot((n-2)\pi - \Sigma \text{ angles})$ in H^2 and $R^2\cdot(\Sigma \text{ angles} - (n-2)\pi)$ on S^2.

A.3 Regular tilings {p, q}

p-sided tiles, q at each corner. From the right triangle (centre, edge midpoint, corner) with angles π/p, π/2, π/q:

Quantity	Formula (hyperbolic; replace cosh by cos on the sphere)	{4,5}	{4,6}	{4,3} sphere
Half-edge a	cosh a = cos(π/p) / sin(π/q)	0.6269	0.8814	0.6155
Edge 2a		1.2537	1.7627	1.2310
In-radius r	cosh r = cos(π/q) / sin(π/p)	0.5306	0.6585	π/4
Centre spacing 2r		**1.0613** (cosh = φ)	**1.3170** (cosh = 2)	**π/2**
Circumradius r_c	cosh r_c = cot(π/p)·cot(π/q)	0.8425	1.1462	0.9553
Tile area	π(p − 2) − 2πp/q (sphere: 2πp/q − π(p − 2))	2π/5	2π/3	2π/3
Corner angle	2π/q	72°	60°	120°
Klein (gnomonic) half-width of the tile	tanh r (sphere: tan r)	0.4859	0.5774	1

For four-sided tiles the centre spacing is $2\cdot\mathrm{asinh}(\sqrt{\cos(2\pi/q)})$.

Tile counts, {4,5}. Tile centres within hyperbolic distance ρ of a viewer at a tile centre: 13 (ρ = 2), 25 (2.5), 45 (3), 81 (3.5), 129 (4), 373 (5). Tiles at graph distance 0, 1, 2, …: 1, 4, 12, 28, 64, 148, 340, 780, 1792 (OEIS A377322).

Draw rule, {4,5}. Fog starts at ρ − 1 and is fully opaque at ρ (default ρ = 3). A tile whose centre lies just outside ρ still reaches 0.8425 inside it, so culling tile centres at ρ would leave holes. A tile is therefore drawn when (a) its centre is within ρ + 0.8425 of the walker (0.8425 is the circumradius r_c, so this is every tile that can reach inside the fog radius) and (b) it can be in view: the disc of radius r_c·L round the floor position of its centre (L·d along its bearing, d being the distance to its centre) is not wholly outside any of the four side planes of the view frustum, widened by 12° to each side and 4° up and down. A disc of radius r round c is wholly outside a plane with inward normal n when n·c + r·|n_floor| < 0. The log map never stretches a length (its tangential scale is d ÷ sinh d ≤ 1 and its radial scale is 1), so the tile lies inside that disc and the test is safe at any head pitch. It runs once per frame on the CPU, for both eyes. At ρ = 3, 45 tile centres lie within the fog radius when the walker stands at a tile centre (40 to 50 elsewhere in the tile), 104 to 119 tiles reach inside it, and at most 59 are drawn with a level gaze and 71 at any pitch (enumerated; `research/scripts/tile_draw_rule.py`). If the view test is not built, the fallback is to cull tile centres at ρ = 3 and make the fog opaque at ρ = 2.15, because culling by centre covers only the disc of radius ρ − r_c = 2.1575.

A.4 The loop-fit rule and the 3-4-5 walk

One "walk s, turn left 90°" is the isometry g = Boost(s/L)·Rot(90°). It is a rotation by an angle φ about a fixed point, with

```
cos(ϕ/2) = cosh(s / 2L) / √2        (hyperbolic)
cos(ϕ/2) = cos(s / 2R) / √2         (spherical)
```

The walk closes after n sides when φ = 360°/n:

n	Condition	s/L or s/R
3 (sphere)	cos(s/2R) = √2·cos(π/3)	π/2 = 1.5708
4 (flat)	always	any
5	cosh(s/2L) = √2·cos(π/5)	1.0613
6	cosh(s/2L) = √2·cos(π/6)	1.3170
7	cosh(s/2L) = √2·cos(π/7)	1.4491
n → ∞	cosh(s/2L) = √2	1.7627 (beyond this the walk never closes)

These are the centre spacings of the {4, n} tilings, which is why "rooms per corner" and "corners to close" are the same number.

After one physical lap (four sides, four turns) at the five-corner value: the walker is displaced by exactly s and their frame is turned by 90°. The same motion is a 72° rotation about the centre of the five-sided loop, which lies 0.8425·L from each of the loop's corners; five laps are the identity. That centre is a corner post only when the loop runs through tile centres. In the room the loop is walked on the lap square of tile zero, whose corners are s/√2 = 0.7504·L from the room centre: the centre of rotation is then 0.092·L from the room centre (0.17 m at L = 1.88 m), and the lap square's corners lie 0.092·L (0.17 m) inside tile zero's corner posts. For a small loop the turn per lap tends to s^2/L^2.

Loop-fit rule. s = clamp(min(W, D) − 1.0 m, 1.0 m, 3.2 m); L = s/1.0613 (q = 5); L = s/1.3170 (q = 6); R = 2s/π (q = 3). W and D are wall to wall in an empty rectangular room. The scan gives the free rectangle Wf × Df, already 0.30 m inside walls and furniture, so W = Wf + 0.6 m and the code uses s = clamp(min(Wf, Df) − 0.4 m, 1.0 m, 3.2 m). The lap square is centred on the anchor, at least 0.20 m inside the free rectangle and so at least 0.5 m from the nearest wall or scanned obstacle.

Other closures. A rectangular lap a × b closes after six sides when cosh(b/L)·(cosh(a/L) − 1) = cosh(a/L). An equiangular four-sided tile with q per corner and midline half-lengths m_1, m_2 satisfies sinh m_1 · sinh m_2 = cos(2π/q); a non-square tile can only tile when q is even.

A.5 Exact tile ids for {4,5}

```
R = rotation by 90° about the tile centre          H = diag(−1, −1, 1)  (half-turn)
T = | ϕ   0  √ϕ |                                    ϕ = (1 + √5)/2,  ϕ² = ϕ + 1
    | 0   1   0 |                                    cosh(2r) = ϕ,  sinh(2r) = √ϕ
    | √ϕ  0   ϕ |
Step through edge k:   S_k = R^k · T · H            S_0·S_0 = I;   (S_1)^5 = I  and no smaller power is
```

In this section R, T and H are the matrices defined above: R is not the sphere radius, and the matrix T is not the coordinate T of a tile centre used below. Every product of R, T and H has entries of the form u + vφ with integers u, v, apart from a common factor √φ on the entries that mix space and time. A tile centre is therefore (√φ·X, √φ·Y, T) with X, Y, T in $\mathbb{Z}[\varphi]$, and

```
tile id = (X, Y, T) = six integers           invariant:  ϕ·(X² + Y²) − T² = −1
```

The neighbour step multiplies by 0, ±1 or ±φ only, and (u + vφ)·φ = v + (u + v)φ, so it needs additions alone. Coefficients grow by a factor of about 2.9 per straight tile step: 64-bit integers are exact to 42 straight steps from tile zero and overflow at 43. The invariant multiplies coefficients, so it is evaluated in Int128 or BigInteger (or in tests only); in `long` its products overflow from about 22 straight steps. Paths that turn stay nearer tile zero. The walkable lung route is 32 tile steps (16 generations); 46 steps (23 generations) is only the addressable depth, and on the airway's paths the coefficients need about 52 bits at 46 steps, so `long` suffices there.

A.6 Tile meshes

Lay a regular grid on the Klein square $[-h, h]^2$, h = tanh r. Move each grid point k to the log chart of the tile:

```
chart position (metres) = L · atanh(|k|) · k/|k|           (sphere: R · atan(|k|) · k/|k| on the gnomonic square)
```

Straight grid lines in the Klein model are geodesics, so neighbouring tiles share their edge exactly.

Subdivision rule. By grid count, not by edge length: n = 16 grid cells per side on the walker's current tile and the tiles that touch it (12 in {4,5}); n = 8 on every other drawn tile; n = 4 only on the high tier, beyond ρ = 3. Every level keeps the same 16 boundary segments per edge, so neighbours of different level share their edge vertices exactly. Caps: 1,200, 500 and 250 triangles for the three levels, for any L. Acceptance is by error, not by length: worst screen-space chord error ≤ 0.15° from an eye height of 1.6 m at L = 1.4, 1.9 and 3.0 m. The rule "longest edge ≤ 0.25 m" stays for the Home shell, furniture and authored props.

Corner-post wedges. The log chart keeps angles only at the tile centre. At a corner, lengths across the radius are drawn at r_c / sinh r_c = 0.891 of the scale of lengths along it, so the tile's corner, which is 2π/q = 72° in the curved metric, spans 2·atan(0.891·tan 36°) = 65.8° in mesh coordinates for {4,5}. Each wedge is bounded by the tile's own two edges: it is built in Klein coordinates and mapped with the formula above, and five wedges make one post.

A.7 Aligned Cells as a polyhedral model

Cell side g, q cells per corner in the world, four on the floor.

```
cone angle at a corner post = q · 90°                          (450° for q = 5)
curvature at a post         = 2π − q·π/2                       (−π/2 for q = 5)
curvature per tile          = 4 corners × (−π/2) ÷ 5 tiles = −2π/5        (the same as the smooth tile)
equivalent radius           = g / √(2π/5) = 0.892 · g
```

A.8 Four dimensions

- **Cut of a tetrahedron** by the hyperplane n·x = w_0: with f_i = n·x_i − w_0 at each vertex, every edge whose ends have opposite signs gives the point x_i + f_i/(f_i − f_j)·(x_j − x_i). Three such points make a triangle, four a quadrilateral.
- **Prism to tetrahedra.** A surface triangle carried from phase k to k + 1 is a prism. With its vertex indices sorted a < b < c it splits into (a_k, b_k, c_k, c_k+1), (a_k, b_k, b_k+1, c_k+1), (a_k, a_k+1, b_k+1, c_k+1); sorting makes

neighbouring prisms agree on shared faces.

- **Checks.** The unit tesseract cut by $x + y + z + w = c$ is a regular tetrahedron of surface area 0.8660 at $c = 0.5$ and a regular octahedron of area 6.9282 at $c = 2$. A unit 4D ball cut at offset c is a ball of radius $\sqrt{(1 - c^2)}$.
- **Scanner tilt.** A table moving at speed v samples z at time $t(z) = t_0 + (z - z_0)/v$. With the time axis scaled by c metres per second this is a hyperplane tilted by θ, $\tan \theta = c/v$. A sphere of diameter D moving at constant speed u along the table appears with length $D \cdot v/(v - u)$; this is the unit test. A breathing tumour does not move at constant speed: the slicer gives the range of apparent lengths directly, and it is never more than D + b, with b the distance travelled in the breathing curve below.
- **Illustrative breathing curve.** $z(t) = z_0 - b \cdot \cos^4(\pi t/\tau)$. Mean position $z_0 - 0.375 \cdot b$. Gating to 30–70 % of the cycle leaves 11.9 % of the motion at a 40 % duty cycle.

Appendix B. Message schema v1

Little-endian. Byte 0 is the message id. Poses: Unity convention (left-handed, Y up, metres), quaternions `[x, y, z, w]`. `t_us` = sender's monotonic microseconds. JSON messages are id byte + UTF-8 string. `proto` = `[1, 0]`.

B.1 `State` (0x10, channel 1, binary, 202 B)

Field	Type	Meaning
`msgType`	u8	0x10
`flags`	u8	bits 0–1 tracking (normal, limited, lost); bit 2 pose jump rejected; bit 3 warning cue active; bits 4–5 quality tier; bit 6 demo lock
`sceneId`	u16	SpaceLab module or use-case scene
`seq`	u32	per send
`t_us`	i64	phone clock
`headPos`	3 × f32	physical head position, RoomModel anchor frame
`headRot`	4 × f32	physical head rotation
`spaceId`	u8	0–6 = S0–S6
`bumpLevel`	u8	0 clear, 1 warn, 2 strong, 3 tracking lost (ARKit tracking not Normal for 0.5 s, §4.10)
`tileId`	6 × i64	{4,5}: the six $\mathbb{Z}[\varphi]$ integers; S1: cell (i, j, k, 0, 0, 0); S4: face index; S5: room id; else 0
`sheet`	i16	Mode 2 cell index or portal sheet; in Shared House, the sheet id (§10.3.6)
`xform`	9 × f64	tile-local transform, row-major (identity when unused)
`p`	8 × f32	`[0]` L, `[1]` fog start ($\rho - 1$), `[2]` ρ, the radius at which the fog is fully opaque and from which the draw rule follows (`curved.cullRho`, §8.5), `[3]` slice w, `[4..7]` 4D angles or spare
`worldVer`	u16	bumps on regeneration

B.2 Other LiteNetLib messages

Message	Id	Ch	Encoding	Direction	Fields
`Hello`	0x01	0	JSON	client → Hub	`proto`, `role` (`phone`, `spectator`, `editor`), `device`, `app`, `build`, `worldHash`, `roomId`
`Welcome`	0x02	0	JSON	Hub → client	`proto`, `sessionId`, `worldHash`, `roomId`, `demoLock`, `forgeUrl`
`TPing`	0x03	–	binary	both	`seq u32`, `t1_us i64`
`TPong`	0x04	–	binary	both	`seq u32`, `t1_us`, `t2_us`, `t3_us` (i64)
`PeerState`	0x11	1	binary	Hub → phone	Reserved, Phase 3 (Extended, `P3-D07`): a peer's `State`, relayed by the Hub. `P3-D07` fixes the layout and adds it here; Phase 1 has no codec for it and a receiver ignores the id (§7.5, rule 5)
`ParamManifest`	0x20	0	JSON	phone → Hub	`params: [{id, name, group, min, max, def, val, onRelease}]`
`ParamSet`	0x21	0	binary	Hub → phone	`count u8`, then `count × {id u16, value f32}`
`SceneCmd`	0x22	0	JSON	Hub → phone	`{space: "S0"…"S6", variant, args}`
`OperatorCmd`	0x23	0	JSON	Hub → phone	`{cmd: home | recentre | quality | cue | lock, args}`; `quality` takes `{tier: 0–3}`, `lock` takes `{on}`; `cue` is the Console's STOP button (§7.3.2)

Message	Id	Ch	Encoding	Direction	Fields
`Telemetry`	0x30	0	binary	phone → Hub	`t_us i64`, `seq u32`, `sceneId u16`, `spaceId u8`, `tier u8`; f32: `fps`, `p50`, `p95`, `p99`, `max`, `over17_2`, `over33_8` (share of frames over 17.2 ms and over 33.8 ms, §11.2), `cpuMainP95`, `cpuRenderP95`, `gpuP95`, `renderScale`; u16: `drawCalls`, `setPass`, `trisK`, `memMB`, `visibleTiles`, `copies`, `rttMs`, `lossPermille`, `cueCount`; `gcAllocB u32`; u8: `thermal` (0–3), `battery`, `tracking`, `msaa`, `stateHz`, `bumpMax`, `assetsPlaced`, `assetsPending`
`Event`	0x40	0	JSON	phone → Hub	`{t_us, scene, ev, data}`
`RoomOutline`	0x41	0	binary	phone → Hub → spectators	`count u8`, `count × {x f32, z f32}`, `height f32`, `rect 5 × f32` (the free rectangle: centre x, z, Wf, Df, yaw), `boxCount u8`, `boxCount × {id u8, cat u8, cx, cz, w, d, h, yaw f32}`
`Log`	0x42	0	JSON	phone → Hub	`{t_us, level, msg}` (warnings and errors)
`JobStatus`	0x50	0	JSON	Hub → phone, spectators	`{jobId, roomId, boxId, state, source: local | hosted | cache, error}`
`AssetReady`	0x51	0	JSON	Hub → phone, spectators	`{jobId, roomId, boxId, sha256, url, bytes, tris, maxEdge, bounds[3], palette}`
`AssetAck`	0x52	0	JSON	phone → Hub	`{jobId, boxId, sha256, ok, loadMs, worstFrameMs, reason}`
`Bye`	0x7F	0	JSON	both	`{reason}`

B.3 Hub HTTP (port 8080)

Route	Request	Response
`GET /`	–	Console page
`GET /pair`	–	page with the QR of `atlas://<ip>:9050`
`WS /ws`	Console → Hub: `{t: "param", id, value}`, `{t: "scene", …}`, `{t: "op", cmd, args}`	Hub → Console: `manifest`, `telemetry`, `state` (10 Hz digest), `event`, `log`, `job`, `link` as JSON
`GET /api/session`	–	`{sessionId, phone: {state, rttMs, loss}, spectators, forge: {up, queue, debugKeep}}`; `debugKeep` true makes the Console show the red "crops kept" banner (§7.4)
`GET /api/presets` · `POST /api/presets/{name}`	– · `{values: {id: value}}`	list · 204
`GET /logs/{session}/perf.csv`	–	CSV
`POST /internal/job` **(loopback only)**	`JobStatus` or `AssetReady` body	204

B.4 Forge HTTP (port 8000)

Route	Request	Response
`GET /health`	–	`{ok, gpu, queue, pipelineVer, debugKeep}`; `debugKeep` is true when the Forge runs with `FORGE_DEBUG_KEEP=1`
`POST /rooms`	RoomModel JSON	201 `{roomId, roomHash}`
`PUT /rooms/{roomId}/mesh`	binary mesh, ≤ 2 MB	204
`GET /rooms/{roomId}`	–	RoomModel JSON

Route	Request	Response
`DELETE /rooms/{roomId}`	–	204; removes the room record, its mesh and the GLBs made for it (§13.5)
`POST /rooms/{roomId}/objects`	multipart: `meta` = `{boxId, category, dims[3], yaw, triTarget, backend, views: [{score, azimuthDeg, intrinsics[4], camToBox[16]}]}`; files `view0` … `view2` (JPEG). `triTarget` is the triangle target of §7.4 (3,000–5,000); `backend` is the phone's `forge.backend` value (`local` , `hosted` or `cache`)	202 `{jobId}` or 200 `{jobId, cached: true, sha256}`
`GET /jobs/{jobId}` · `GET /jobs?room={roomId}`	–	`JobStatus` (with the `AssetReady` fields when ready) · list
`GET /assets/{sha256}.glb`	–	GLB; `ETag` = hash; 404 if unknown

Appendix C. Set-up checklist

Three set-up tasks have no dependency and run side by side: `P1-X01` (C.1 and C.2: build seat, signing and Unity; lane A leads), `P1-X03` (the Station's tools, the first line of C.7; lane D) and `P1-X04` (phones and router: the phone lines of C.1 and C.6 and the router lines of C.7; lane C). The remaining lines are ticked by the Phase 1 tasks that build on them (§9.5). Within a section, work top to bottom.

"verify" means: confirm on the machine, then write the installed number into `docs/versions.md`. Port numbers are defaults assumed here; Appendix B wins if it differs.

C.1 Accounts and machines

Who / what	Needs
Signing owner (lane A)	Apple account, two-factor on, signed in to Xcode on the Mac as a Personal Team (Spec §2.8)
All four engineers	Repository access, Git LFS, Unity account, Unity Editor at the pinned patch with iOS and Windows Build Support (licence terms: verify)
MacBook Pro M3 Max	Xcode 26, Unity with iOS Build Support. The only iOS build seat (§9.3)
Station PC (RTX 5070)	.NET 8 SDK, Python 3.11, NVIDIA driver with CUDA 12.8, Unity with Windows Build Support
Phone A (USB-C), phone B (Lightning)	One cable each; Developer Mode; iOS version recorded

Free Personal Team rules (Spec App. E):

- [ ] One owner account. A build signed by another team cannot install over the existing app.
- [ ] One fixed bundle id, `com.<owner>.atlas`, for the whole project (limit: 10 new app ids per 7 days).
- [ ] At most 3 registered devices (phone A, phone B, one spare) and 3 apps per device. No TestFlight.
- [ ] Builds stop launching 7 days after install; any reinstall renews them.
- [ ] The app ships with no entitlements.

C.2 Versions to pin

Component	Pin	Source / status
Unity	6000.3 LTS, patch ≥ 6000.3.21f1, exact patch in `ProjectVersion.txt`	(R5 C1) thermal frame-rate Player Setting exists from this patch
URP	17.x as bundled	(Spec §2.2) verify
AR Foundation, ARKit XR Plugin	6.5.x, same version for both	(BP §0.5) verify patch
Xcode	26	(BP §0.5) required by ARKit XR Plugin 6.4+
iOS deployment target	17.0	(Spec §2.2)
iOS on the phones	Recorded; drift test run	(R1 C.1) drift regression reported in 26.4–26.6: verify
LiteNetLib	2.1.4: OpenUPM `com.revenantx.litenetlib` in Unity, NuGet `LiteNetLib` in the Hub	(R4 §1) on Unity 6.3 + IL2CPP + iOS 26: verify
.NET SDK	8	(R4 §2)
glTFast	`com.unity.cloud.gltfast`	(Spec §2.3) verify version; furniture arrives as GLB at run time
Burst, Collections, Mathematics, Input System, Test Framework	As resolved for 6000.3, committed in `packages-lock.json`	(Spec §2.3)
Python	3.11	(Spec §2.2)
PyTorch	≥ 2.7, cu128 wheels	(Spec §2.2) Blackwell needs CUDA 12.8

Component	Pin	Source / status
TripoSG	Commit hash in the Forge lock file	(BP §0.5) MIT, 8 GB VRAM; on cu128: verify
FastAPI, uvicorn, pydantic v2, trimesh, `rembg` (background removal), `gltfpack`	Lock file committed	(Spec §2.5)
TotalSegmentator	Optional, `station/pipelines` only	(BP §0.5) airway generations 1–2 only

C.3 Unity packages: keep or drop

Package (Spec §2.3 unless noted)	Decision	Reason
URP; AR Foundation, ARKit, XR Management	Keep	ARKit is the only XR loader; no display provider
Input System; Burst, Collections, Mathematics	Keep	Input; kernel, Mesh4D, Bump Guard jobs
glTFast	Keep	Run-time GLB from the Forge
meshopt decompress, KTX	Only if the Forge emits them	Palette-coloured GLB needs neither
Test Framework, `com.unity.profiling.core`	Keep	EditMode and PlayMode tests; perf probe
apple/unityplugins Core + CoreHaptics	Keep	Bump Guard haptic pulses
ProBuilder (BP §0), LiteNetLib (R4)	Add	Greybox rooms; link transport
ARCore	Drop	No Android client; re-add only if the team wants Android later
Cardboard SDK / XR plugin	Drop	Conflicts with the ARKit loader (BP §1.1); distortion is our own blit, Cardboard coefficients as numbers only
MessagePack-CSharp	Drop	Hand-packed binary plus JSON (R4 §4)
LiveKit, UniVRM, uLipSync, Steam Audio	Drop	Voice is Stretch (§10.3.1) and would not use LiveKit; lip-synced VRM avatars are not in the plan (its avatars are plain meshes); simple panning is enough
UXF, Sentry	Drop	Own CSV telemetry and `Log` messages
HoloKit Unity SDK (BP §0.5)	Not in the manifest	Optional hand-tracking spike on a branch; needs iOS 18+

C.4 Project and Player settings

- [ ] Project from the AR Mobile template (BP §6); XR Plug-in Management → iOS → ARKit only.
- [ ] Editor: Asset Serialisation = Force Text; Visible Meta Files; UnityYAMLMerge as the Git merge tool; Git LFS for `*.glb`, `*.ktx2`, `*.png`, `*.fbx` (Spec §2.7).
- [ ] Player (iOS): IL2CPP, ARM64, Metal only; target iOS 17.0; landscape left only; Frame Timing Stats on; Managed Stripping Level Low; VSync count 1 (Spec §2.7).
- [ ] Player (iOS): "Adjust iOS FPS based on thermal state" off; the thermal ladder owns quality (R5 C1).
- [ ] Signing Team ID = the Personal Team; Automatically Sign on; bundle id fixed.
- [ ] Boot script: `Application.targetFrameRate = 60`; `Screen.sleepTimeout = SleepTimeout.NeverSleep` (BP §5, R4 §3.5).
- [ ] ARKit video: the lowest-resolution 60 Hz video format from `ARCameraManager.GetConfigurations()` (BP §5). The environment depth image must be non-null at that format (checked in C.6).
- [ ] ARKit depth (`P1-C01`): `AROcclusionManager` with environment depth mode Fastest, temporal smoothing off, human stencil and human depth off. The depth image is read only with `TryAcquireEnvironmentDepthCpuImage`, at 256 × 192 (§8.2).
- [ ] ARKit after the scan: meshing, plane detection, light estimation and environment probes off; LiDAR depth stays on (D19).

URP asset `Atlas_URP_iOS` (BP §4, R5 E1; this plan's value where BP gives a range):

Setting	Value
Rendering path	Forward, Render Graph on
HDR; Depth Texture; Opaque Texture; Depth Priming	Off
MSAA	4×
Render scale	1.0 in the URP asset (the eye-texture size carries the 0.8)
Post-processing, FXAA, STP	Off
Shadows; additional lights	Off
SRP Batcher	On
GPU Resident Drawer, static batching, baked occlusion culling	Off

C.5 Info.plist and build settings

`Assets/Atlas/Editor/AtlasPostBuild.cs` (`[PostProcessBuild]`, lane A) writes every key. Nobody edits the generated Xcode project by hand.

Key	Value	Why
`NSCameraUsageDescription`	Text	Tracking, scan, object photos
`NSLocalNetworkUsageDescription`	Text	Unicast UDP to the Hub raises the prompt (R4 §3.1)
`NSAppTransportSecurity` → `NSAllowsLocalNetworking`	`YES`	HTTP to a LAN address on iOS 17+ (R4 §3.4)
`UIFileSharingEnabled`	`YES`	Local perf CSV readable from the Mac
`CFBundleURLTypes`	scheme `atlas`	QR deep link `atlas://<ip>:9050` (R4 §3.7)
`MetalHudEnabled`	`YES`, development builds only	Metal Performance HUD (R5 E2)

- [ ] Also `UIRequiresFullScreen = YES` and build setting `ENABLE_BITCODE = NO` (Spec §2.4). No `NSBonjourServices`, no microphone key, no multicast entitlement (R4 §3.2).
- [ ] `Assets/Plugins/iOS/AtlasThermal.mm` returns `NSProcessInfo.thermalState`; an Editor stub returns 0 (BP §1.4). It is delivered with the first build (`P1-A02`) and must return a value on the phone at G1.
- [ ] On a patch below 6000.3.21f1 the script also rewrites `adjustFrameRateForThermalState:` to return the target rate (R5 C1).

C.6 First build to both phones

- [ ] Xcode › Settings › Accounts shows "Name (Personal Team)".
- [ ] Each phone: connect, Trust This Computer, Settings › Privacy & Security › Developer Mode › on, restart, Enable.
- [ ] Build the empty stereo scene (AR Session, XR Origin, two-camera rig, no camera background); install with `tools/build_ios.sh` (C.8) or Xcode Run.
- [ ] First launch: Settings › General › VPN & Device Management › owner account › Trust.
- [ ] Accept the camera prompt in hand, not in the headset.

Device check (both phones)	Pass
Empty stereo scene, Release build	Metal HUD reads 60 fps (BP §6)
Tracking	Session state reaches tracking; head pose moves both eyes
LiDAR	`ARMeshManager` mesh subsystem non-null and running; environment depth image non-null at the chosen video format (C.4)

If phone B fails the LiDAR check, raise it in §14.

C.7 Station set-up

- [ ] Tools (`P1-X03`): .NET 8 SDK (`dotnet --version` reports 8.x); Python 3.11; NVIDIA driver with CUDA 12.8 and the cu128 PyTorch wheels (`nvidia-smi` and a PyTorch CUDA test pass); the `station/` folders of C.9 exist.
- [ ] Own Wi-Fi router, no internet needed, fixed non-DFS channel; Station on Ethernet; phones on 5 GHz; AirDrop and Handoff off (R4 §3.6).
- [ ] DHCP reservation: Station at `192.168.x.10`; Windows network profile Private (R4 §3.7–3.8). Check (`P1-X04`): each phone opens a test page served from that address.
- [ ] Firewall, once, as administrator (`station/setup_firewall.ps1`):

```
netsh advfirewall firewall add rule name="ATLAS Hub UDP" dir=in action=allow protocol=UDP localport=9050
netsh advfirewall firewall add rule name="ATLAS Hub HTTP" dir=in action=allow protocol=TCP localport=8080
netsh advfirewall firewall add rule name="ATLAS Forge" dir=in action=allow protocol=TCP localport=8000
```

- [ ] Hub: `dotnet run --project station/hub`. UDP 9050; Kestrel on `0.0.0.0:8080` serves the Console (R4 §3.8).
- [ ] The Hub's `logs/` folder is on an encrypted volume; motion logs are treated as health data (§13.5).
- [ ] Forge: Python 3.11 virtual environment in `station/forge`; `torch.cuda.is_available()` is true; `uvicorn forge.app:app --host 0.0.0.0 --port 8000`. One process owns the GPU (Spec §2.5). Port 8000: (derived).
- [ ] Spectator: Windows build with the `ATLAS_SPECTATOR` define, or an Editor in play mode; joins the Hub as role `spectator` (R4 §6.1).
- [ ] Pairing: open `http://192.168.x.10:8080/pair`, scan the QR with the iOS Camera app; `Application.deepLinkActivated` stores the address in `PlayerPrefs`. Fallback: typed address (R4 §3.7).
- [ ] Raise the Local Network prompt on the onboarding screen, phone in hand; the client retries the connect every 1 s (R4 §3.3). Repeat after any delete-and-reinstall.

`iproxy` forwards TCP only, so LiteNetLib traffic does not pass through a USB tunnel (R4 §3.6): on-device development needs the router as well as the cable.

Acceptance: the Console shows the phone connected; a `GET` to the Forge health route from the phone returns 200; with the Hub stopped the app runs unchanged apart from the link indicator.

C.8 Development loop and build seat

Three engineers have no Mac; their loop is the Editor (R4 §6.5):

- [ ] XR Simulation on for Editor play mode (Windows Editor: verify). It does not replace a real scan: axis-aligned planes only.
- [ ] `recordings/` holds `room_*.json` (RoomModel from a real scan) and `walk_*.poselog` (timestamped head poses). The Editor loads a RoomModel and replays a pose log into the pose pipeline in place of ARKit.
- [ ] `ReplayPhone` feeds a recorded `state.bin` to the Hub, so the Console and Spectator need no phone; the Editor also runs `LinkClient` unchanged against a Hub on loopback.
- [ ] Kernel, tile-id, FloorScan, loop-fit and message-packing logic runs as EditMode tests; native code sits behind stubs so every assembly compiles on Windows.

Build seat:

- [ ] `tools/build_ios.sh <branch>`: fetch → Unity batch-mode iOS build, Append mode → `xcodebuild -allowProvisioningUpdates` → `xcrun devicectl device install app` to each connected phone. With a Personal Team: verify (Spec §2.8). Fallback: Xcode Run.
- [ ] Any engineer can run the script. Build queue in one shared file: who, branch, what is tested, pass number. One integration build at every gate.
- [ ] Phones are booked, test protocol written first. Hand-held before head-mounted. Thermal state recorded at the start of every measurement.
- [ ] Published numbers: Release build, debugger detached, Metal API Validation off, in the headset (R5 E2).

C.9 Repository layout and ownership

One assembly definition per folder; lane in brackets.

```
atlas/
├── unity/
│   ├── Assets/Atlas/
│   │   ├── Kernel/       Atlas.Kernel (B; Mesh4D and its slicing job: A; no scene types)
│   │   ├── World/        Atlas.World (A, B, C; see §6.2)
│   │   ├── Rendering/    Atlas.Rendering (A; warp and gates includes, tile renderer: B)
│   │   │   └── Shaders/  AtlasCore.hlsl, AtlasWarp.hlsl, AtlasPortalClip.hlsl, AtlasGates.hlsl
│   │   ├── Scan/         Atlas.Scan (C)
│   │   ├── Safety/       Atlas.Safety (C; Bump Guard; must not reference Atlas.Link or Atlas.Telemetry)
│   │   ├── Input/        Atlas.Input (C; action map and twins)
│   │   ├── Link/         Atlas.Link (D; plain C#, no engine references, shared with the Hub)
│   │   ├── Telemetry/    Atlas.Telemetry (D)
│   │   ├── Spaces/       Atlas.Spaces.Home (C); .Glued, .Portal, .Slicer (A); .CurvedWalk, .AlignedCells, .SmallPlanet (B)
│   │   ├── Blades/       Atlas.Blades.Playground, .Lung, .SharedHouse, .FieldFit
│   │   ├── Spectator/    Atlas.Spectator (D)
│   │   ├── Editor/       AtlasPostBuild.cs
│   │   └── Tests/        one test assembly per area
│   ├── Assets/Plugins/iOS/   AtlasThermal.mm
│   ├── Packages/             manifest.json, packages-lock.json
│   └── ProjectSettings/
├── station/
│   ├── hub/          .NET 8 console; compiles the Atlas.Link sources; wwwroot/ = Console
│   ├── forge/        Python FastAPI, TripoSG
│   └── pipelines/    medical data preparation
├── recordings/       RoomModel JSON, pose logs, state.bin
├── tools/            build_ios.sh, ReplayPhone
├── docs/             contracts/, evidence/, versions.md, licence ledger
└── figures/
```

Who may edit which scene, prefab and shader file is set out once, in §9.3 and §8.3; the rule for the shader property ids (`AtlasShaderIDs.cs`) is in §6.4. Scene names are those of §9.3; each space's scene takes the suffix of its assembly: `Space_Home`, `Space_Glued`, `Space_Portal`, `Space_Slicer`, `Space_CurvedWalk`, `Space_AlignedCells`, `Space_SmallPlanet`.

Rules: tunables live in the parameter registry, never in scenes; no new shader keyword without lane A; the shared shader stays SRP-Batcher compatible at every gate; Asset Store packs stay out of the repository (BP §3d).

C.10 Things that bite at first set-up

Trap	Rule	Source
Frame rate stuck at 30 fps	iOS default, or the Unity thermal cap: check both C.4 settings in the built app	BP §1.3, R5 C1
Phone never connects	In order: Local Network prompt denied or raised while backgrounded; firewall rule missing; server bound to `localhost`	R4 §3.1, §3.8
LiteNetLib precompiled DLL, or its `NetSerializer`, under IL2CPP	OpenUPM package or sources only; pack messages by hand	R4 §1
App will not open after 7 days; "Untrusted Developer"; identifier mismatch	Reinstall; trust the owner; delete the copy signed by another team	Spec App. E
RoomPlan capture removes scene depth from frames	Re-apply the session configuration after capture starts: verify	R1 C.1
Older CUDA code on the RTX 5070	cu128 wheels; rebuild custom kernels	Spec §2.7

Appendix D. What changed against the older documents

Order of authority: the team lead's decisions (§2 of this plan) > Build Pipeline (BP) > Spec > Brief. The research memos R1–R6 correct BP where they verified or derived a different value. In the "older position" column, "old D-n" is a row of the decision table shared by Spec §1.2 and Brief §1.4, and "NN-n" is a Spec §0 non-negotiable. In the "decided by" column, "D-n" is a row of §2 of this plan. "Blade" is the older documents' word for a use case.

D.1 Superseded

Topic	Older position (Spec/Brief ref)	Now	Decided by
Stereo	One camera, side-by-side pass; Cardboard XR plugin as fallback (old D5, Spec §5.2)	Own two-camera rig, ±0.032 m, one render texture per eye (1118 × 1032); ARKit the only XR loader	BP §0, §1.1; R5 for the eye textures (D.2)
Optics	Cardboard C API, distortion meshes, chromatic-aberration fix (Spec §5.2)	Barrel pre-distortion as the one final blit, which samples both eye textures and upscales them; hard-coded coefficients	BP §0.5
Latency	Core Motion prediction + late rotational warp, ≤ 30 ms (old D16)	`TrackedPoseDriver` "Update only"; optional gyro prediction; no late warp	BP §4, R5
Thermal tier	30 fps render warped to 60 Hz (NN2, Spec §5.4)	No warp tier; quality ladder; plan never to reach Serious	R5 (overrides BP §1.3; see D.2)
Projection	Klein chart for walked $H^2 \times E$ (old D9)	Log-map warp for anything walked; Klein/Poincaré for maps only	BP §0
Subdivision	None (Klein), or $\lesssim 0.2R$ (Spec §6.3)	Tile levels by grid count: n = 16 per side on the walker's current tile and the 12 tiles that touch it, n = 8 on every other drawn tile, n = 4 only on the high tier beyond ρ = 3. A longest edge of 0.25 m for the Home shell, furniture and authored props	BP §3a, R3, R5
Shader source	Own `CurvedSpace.hlsl`, 4×4 isometry, $\langle p,p\rangle = -R^2$ (Spec §5.3, §6.1)	HyperEngine $H^2 \times E$ port (MIT); unit hyperboloid scaled by L; double 3×3 on the CPU; κ-unified kernel	BP §4, §6 step 5; this plan §6.2
Tile addressing	Offline double-precision lookup tables (Spec §5.3)	Exact integer ids in $\mathbb{Z}[\varphi]$; pose = (tile id, local double 3×3)	BP §0.5, R3
Fog and culling	Draw radius 3–4R, exponential fog (Spec §5.3)	Fog starts at ρ = 2 and is opaque at ρ = 3 by default; every tile that reaches inside that radius and can be in view is drawn, by the CPU draw rule of §8.5	D13, R3, R5
Instancing	One instanced draw per module type (Spec §5.3)	SRP Batcher; one pooled material instance per tile or room; no property blocks	BP §4
Display mode	Mode A see-through recommended (Spec §7.2)	VR, no camera background	BP §1.2, D19
Safety	Safe Harbour (passthrough + outline), Stop state, operator panic (NN5, Spec §7.3)	Bump Guard: warning cue and haptic only. Its thresholds and depth rate are constants in code, with no registry parameter and no off switch. The Console's STOP button fires the same cue and nothing freezes; return to Home is a user action	D19
Depth after set-up	Depth sectors drive the Stop state machine; person detection (Spec §7.2–7.3)	LiDAR environment depth stays on and feeds Bump Guard only (the seven sectors and the dynamic layer); human stencil and human depth off; meshing and planes off	D19
Scanning	Team pre-scan of the venue; kit rebuild on the PC (Spec §8.1, §8.3)	Walker self-scan is the headline; pre-scan + ARWorldMap the fast path; RoomModel built on the phone	D2, D21
Room role	Hub is a neutral Euclidean lobby (Spec §11.1)	Home is tile zero of the generated world; L from the loop-fit rule	D18

Topic	Older position (Spec/Brief ref)	Now	Decided by
Hyperbolic walking	Holonomy mapping of nine 1 m cells; portals + curved wings (old D15, Brief §7.13)	Mode 1 Curved Walk and Mode 2 Aligned Cells; one kernel, one tile graph	D9
Glued edges	1.5 m cell, 3×3 instanced copies, cull-flipped mirror draw (Spec §11.2)	Universal cover fixed to the floor. The cell is an open frame cut from the free rectangle (n = max(1, round(Wf / 1.5 m)) cells along a side, likewise for Df, which gives cells of 1.12 to 2.25 m), repeated as 5 × 5 GameObject copies (3 × 3 × 3 for T^3) with negative scale for Klein rows. The room itself is not repeated: its walls (as kerbs) and furniture are drawn once, where they really are	BP §4, D15
Portals	Custom `StencilPortalFeature`, recursion ≤ 2 (old D10)	Render Objects features, stencil refs 1–15, depth 1 ceiling, shader plane clip	BP §4
4D slicer	`Slicer4D.compute` (Spec §5.3, Brief §3.9)	`Mesh4D` on the CPU with Burst; Engine4D vertex slicer only above ~50k tetrahedra	BP §4, R2
What 4D is	Phase-binned CT only; spatial 4D on the roadmap (Spec §11.3.2)	Four pieces on one slicer: toybox, worldtube, breath lane, scanner-artefact demo	D25
Look and materials	`AtlasUber` lit shader, baked probes, per-region LUT (Spec §9.2)	One material, one 128^2 palette, vertex-colour AO, `fwidth` lines	BP §2
Post-processing	On-tile post, grading, sprite bloom (Spec §5.3)	None; no bloom, no real-time shadows	BP §4
Budgets	< 100 draws, ≤ 150k triangles per eye; scale 0.87 (Spec §5.2–5.3)	≤ 300 draws, ≤ 500k triangles per frame (both eyes); render scale 0.8 (set by the eye-texture size), 4× MSAA	BP §5, R5
Asset formats	GLB + meshopt + KTX2 throughout (Spec §8.2)	FBX at edit time for authored content; GLB only for Forge output	BP §3a, D20
AI generation	SPAR3D or Hunyuan3D-2mini; TRELLIS.2 on the H200 (old D11)	TripoSG on the RTX 5070, hosted API fallback; H200 never on the critical path	BP §0.5, §0, D20
Networking	Python FastAPI server, WebSocket + raw UDP, MessagePack (old D12, Spec §10)	`Atlas.Link` on LiteNetLib both ends; .NET 8 Hub; binary `State` at 60 Hz, JSON control, HTTP for files; Python only in the Forge	BP §0.5, R4
Discovery	Bonjour or QR (Spec §10.1)	Unicast to a known IP by QR pairing; own router	BP §0.5
PC role	Console and spectator in a later phase (Spec §15.2)	All four Station functions gate Phase 1 (G6)	D22
Knife Map	Spectator screen with the equation card, fed by a `KnifeState` message (Spec §10.2, §13.1, Appendix B)	Name retired. Its parts are the Spectator (`P1-D05`), the flat-twin mini-map (`P1-D07`), the equation cards (`P2-C06`, `P2-A03`; audited and mirrored on the Spectator in `P4-B03`) and the judges' dashboard (`P4-D01`). `State` and `Event` carry the data (Appendix B)	this plan
Crash reporting	Sentry or Crashlytics with MetricKit (Spec §15.5)	Dropped. `Log` messages and the telemetry CSV take its place (Appendix C.3)	this plan
Medical data	One real patient throughout; segmented airway with distal completion (old D13, old D17, old D18)	4D-Lung patient 101 beside a labelled illustrative twin; procedural airway, optional AeroPath proximal tree	BP §0.5, D25
Play area	Taped 3×3 m square, centre X, stanchions (old D8, Spec §6.4)	Free walk on the scanned free floor. The product marks nothing on the real floor; the lap square marked for the spotter at gate G2 is a test aid	D2
Comfort	Vignette during world motion (Spec §6.8)	No vignette while walking; gains at identity; gate G2 on Mode 1	BP §0.5, R1
Hand input	Vision hand pose, pinch, the gesture set and the hand "hero" techniques (old D7; Spec §4.5, §7.1; Brief §3.5)	Not built. Inputs are the gamepad, gaze dwell and the headset lever (`P1-C10`); the marker cube is Extended (open item 11); hand tracking exists only as an optional HoloKit spike (Appendix C.3)	BP §0.5, this plan
Use cases	Shared House and FieldFit late and first to be cut (Spec §15.2, §15.6)	All carried: P2 Playground + Lung Mission; P3 Shared House + FieldFit	D3

Topic	Older position (Spec/Brief ref)	Now	Decided by
Tier B and C blades	"Station Zero", Small Planet, Wormhole Quest, the 4D Toybox and the other Tier B and C blades; "Station Zero" as the swap for Shared House (Brief §2.3, §4.6; Spec §17 Q7)	Not built as use cases. The geometry of Small Planet is space S4 and the 4D Toybox is the toybox of the Slicer (S6); the rest stay as documented ideas in the Brief	D1, D3
House Call partner	Galaxy S24 Android client, LiveKit voice, avatar (old D3; Spec §11.4)	No Android client and no LiveKit (Appendix C.3). Core is Post Circling, solo, with a recorded visitor. A live House Call is Extended, on phone B (`P3-C04`, `P3-D07`), in a second headset if there is one (open item 14); voice is Stretch (§10.3.1)	D3, this plan
Process and timescale	Calendar-sized phases, spike phase, cut-first list, unowned areas (Spec §15)	Gate-driven (G0–G15); four lanes A–D; tiers Core / Extended / Stretch	D3, D5, D4

D.2 Build Pipeline items overridden by newer decisions or by verified research

BP item	Replaced by	Reason
Taped, roped square with a spotter (§1.2, §4)	Free walk in the scanned room	D2
"Turn off … occlusion" after set-up (§5)	Environment depth stays on; meshing, planes, probes, light estimation off	Bump Guard needs live depth (D19)
"Don't let the user cross a glued edge" (§4)	The walker crosses freely	The cover is fixed to the floor: nothing is re-centred, so no global inverted culling (D15)
Glued Room: one combined mesh per copy, under 5k triangles each (§5)	Each copy is an open-frame cell of at most 1,500 triangles, with a self-avatar in every copy but the walker's own. The room's walls (as kerbs) and furniture are drawn once, where they really are	A copy of the room with its furniture cannot be held under 5k: the furniture alone may reach 60,000 triangles (§4.5), and 25 copies of that for two eyes would be 3 million triangles a frame against a budget of 500k (derived). Drawn once, everything that looks solid is where the real thing is
Fog at 5–8 units; cull at d > ~4 (§0.5, §4)	Fog opaque at ρ = 3; every tile that reaches inside that radius and can be in view is drawn (§8.5)	5–8 is the float-precision horizon; 45 tile centres lie within ρ = 3 (at most 71 tiles drawn) but 129 within ρ = 4, over the draw budget (R3, R6)
Thermal Player Setting "on 6.5+" (§1.3)	Pin Unity ≥ 6000.3.21f1	The setting exists from that patch (R5)
Accept Unity's 30 fps cap at Serious (§1.3, §5)	Cap switched off; the ladder keeps 60 fps at Serious and locks to 30 fps only on its last rung	A silent drop to 30 fps with no reprojection is the worst outcome in a headset; a controlled quality drop is better (R5)
CC0 kits first for props (§0, §3b)	Scanned objects are generated live by the Forge, at most 20 per room and the largest first; further objects stay placeholder boxes. A built-in fallback set (one CC0 mesh per RoomPlan category, `P1-D12`) replaces the boxes when the Forge is unreachable	D20
Tile pivot at a floor corner (§3a)	Curved tiles pivot at the tile centre; flat rooms keep the corner	The centre is the chart origin (§3.4, R6)
0.1–0.25 m subdivision, undifferentiated (§3a)	Tile levels by grid count: n = 16 per side on the walker's current tile and the 12 tiles that touch it, n = 8 on every other drawn tile. The 0.25 m longest-edge rule stays for the Home shell, furniture and authored props	A fine grid on every tile is too dense when up to 57 tiles are drawn. A count per side holds the triangle caps for any L and lets tiles of different level share their edge vertices exactly (R3, R5)
MSAA 2× as the start (§4)	4× at scale 0.8; 2× on the thermal ladder	Fill rate is not the constraint (R5)
Half-screen viewports for the two eye cameras; render scale 0.75–0.8 set in the URP asset (§0, §4)	One render texture per eye, 1118 × 1032 (0.8 × the half-screen), each eye camera with a full viewport. URP Render Scale stays 1.0 and the texture size carries the 0.8. One final blit samples both textures	Two eye cameras are two render passes. Drawn into one multisampled target, the second pass would have to load what the first stored, so the multisampled colour could not be memoryless or resolved on-tile (R5)

BP item	Replaced by	Reason
HoloKit SDK spike first (§0.5)	Own rig is the baseline; HoloKit an optional hand-tracking spike	HoloKit is built for optical see-through, needs iOS 18+ (the target is iOS 17.0) and still needs our own barrel pre-distortion blit. For a lens headset it adds only hand tracking, which is optional (rule 5). BP §0.5 itself names the own rig as the fallback, and the rig is on the path to G1, so it is built first (D7)
LiteNetLib with no PC peer named (§0.5)	The .NET Hub terminates it	Own wire protocol: both ends must be .NET (R4)
Warped tiles and rooms rebuilt in Blender, with the Blender MCP (§0, §3a)	Tile shells and the Home shell generated in C# at run time (`P1-B05` , §3.4); Blender only for landmarks and props	L and the tile size come from the scanned room and q from the dial, so a tile cannot be authored for one fixed L with a 0.25 m grid; the Home shell comes from the walker's own scan. The MCP was available only to the session that wrote BP
Timed order-of-work list (§6)	Gates	D4

D.3 Still valid: read the older documents for detail

- **Interaction design space, technique catalogue, unifier results:** Brief §3.1–3.6.
- **Claims language:** Brief §2.4 (the phrase check of `P4-D05` uses its full Avoid list); Real / Pre-baked / Mocked table Spec §14.3.
- **Why each use case needs its geometry:** Brief §4.0.
- **Kernel identities and formula sheet:** Spec §6.1, Appendix A ([D] items still need checking).
- **Mapping scanned meshes into curved space:** Spec §6.3, for the distortion table and the rule "keep objects under about 0.3 L" only. Its pipeline (steps 1–6), Klein placement and H^3 honeycomb table are superseded by §3.4 and §4.6 of this plan (D.1, rows Projection and Subdivision).
- **Glued-room maths notes:** Spec §6.2, §11.2; Brief §3.7.
- **Branched-cover maths (Shared House):** Spec §11.4, Appendix A.
- **FieldFit lens maths:** Spec §11.5; Brief §4.5.
- **4D maths:** Brief §3.9 (tetrahedron ∩ hyperplane, quaternion double cover).
- **Robustness rules:** Spec §6.8 and Brief §3.4, rules R1–R5 and R7 only. They apply to the marker cube (Spec §7.1) and to any later hand input. R6 (vignette during world motion) and R8 (Safe Harbour) are superseded (D.1, rows Comfort and Safety).
- **Architecture rules:** Spec §3.4, for the principle that the phone decides everything within a frame and never waits for the Station; and the rule that safety never references the link (Spec §3.2). The Spec's assembly map (§3.2), frame loop (§3.3) and interfaces (§3.5) are superseded by §6.2, §6.3 and §6.5 of this plan.
- **Depth-sector job:** Spec §7.3 (seven sectors across ±35° of the view direction, on the 256 × 192 depth image), as a Bump Guard input only. The Spec's 10 cm occupancy grid, its distances and hysteresis, person detection and the state machine are superseded by §4.2 and §4.10 of this plan (D.1, rows Safety and Depth after set-up).
- **Router and Wi-Fi set-up:** Spec §10.1. **Signing:** Spec §2.8 and Spec App. E.
- **Evaluation instruments:** Brief §8, Spec §13. **Prior art:** Brief §12.
- **Consent, FMS and SEQ wording; booth-sign text and exclusion list:** Brief §8.1, §9.9. On the sign, "inside a taped 3×3 m square" becomes "in the room you scanned", and the data line also carries the hosted-fallback disclosure of §7.4. The spotter briefing is the one in §13.2 of this plan.

D.4 Errors in the older documents that the plan corrects

Error	Correction	Source
Slice Tilt rotates in the time–left-right plane (Spec §11.2.1); both axes lie in the slice, so nothing changes	Tilt in the z–τ plane	R2
"Precompute 7 tilted slices and blend neighbours" (Spec §11.2.1)	Tilted slices change topology: slice live, never cross-fade	R2

Error	Correction	Source
Airway layout (Spec §6.7): ℓ = 1.5 m, R = 2.2 m, 120° forks	Overlaps (372 edge crossings by depth 8); 120° forks need ℓ ≥ L·ln 3; tree snapped to the {4,5} tile graph	R3
"About 23 generations" (Brief §4.3.1)	About 16 conducting, 23 in all	R2
≈ 17 mm quoted as CT-to-body divergence (Spec §11.3.2)	It is nodule displacement between breath states; not shown as divergence	R2
"GTV + OARs on all 10 phases" (Spec §11.3.4)	Tumour contours on all phases for 20/20 patients; lungs for 5/20	R2
ITV as the OR of phase masks (Spec §11.2.1)	Under-covers small fast nodules; interpolate sub-steps	R2
Illustrative 12 mm motion implied for the real case (Brief)	Real median 4.3 mm; subtle real tube beside the labelled twin	R2
Real obstacles become props at the same place in every space (Brief §3.8)	Holds only in registered spaces (S0, S1, S3, S5, S6)	R1
Holonomy comfort evidence applied to smooth curved walking	Applies to Mode 2 only; Mode 1 has gate G2	R1
Post-processing both off and on (Spec §5.2 vs §5.3)	Off	this plan
Kit grid 0.5 m vs 1–2 m (Spec §8.2 vs §9.2)	0.5 m	this plan
Draw calls < 100 per eye vs ≈ 250 (Spec §5.3)	≤ 300 per frame, both eyes	this plan
Hyperboloid $-R^2$ (Spec §6.1) vs unit form (Spec §6.3)	Unit, scaled by L	this plan
Three hyperbolic constructions with no priority; no R for a scanned room	Modes 1 and 2, portals as S5; L from the loop-fit rule	this plan
`OperatorCmd` fields differ between Spec §10.1 and its Appendix B	Appendix B of this plan is the only schema	this plan
Swap phones at "fair" although "fair" is a full-quality tier (Spec §5.4)	One thermal ladder, defined in §8	this plan
Phone B both scans and does not scan objects; "≤ 2–3" vs "≤ 2" tracked images; 63 vs 64 techniques; "Shared House" vs "House Call"	Each stated once in this plan; the name is Shared House	this plan

D.5 Where the risks of Spec §16 went

Spec §16 lists 21 engineering risks, and Brief §11 carries the same list. Each one is now a row of the risk register (§12.1), an open item (§14), or settled by a decision or a section of this plan.

Spec risk	Now
1 Thermal throttling	Risk 11; risk 12 for the heat of live depth
2 Hands not seen	Retired: hand input is not built (D.1, row Hand input)
3 Drift or tracking loss	Risk 16
4 Cybersickness	Risk 5 for Mode 1; risk 30 for the other spaces
5 Read as "just portals"	§13.1 and §13.3: the equation cards and the 3-4-5 walk
6 Medical over-claiming	Risk 25
7 Second phone or network failure	Post Circling is Core, solo and needs no network (§10.3)
8 AI generation slow or unavailable	Risk 22
9 RTX 5070 incompatibilities	Risk 23
10 Licences	Risk 26
11 Headset fit	Open item 1

Spec risk	Now
12 Scope creep	D3 fixes the four use cases; inside them the tiers are Core / Extended / Stretch; the rest of the design space stays in the Brief (D.3)
13 Challenge rules unknown	Open item 15
14 Hyperbolic navigation	§3.5
15 Privacy	Risk 31
16 No single-pass stereo	Risk 13
17 Motion-to-photon	D8 and §8.7; measured by the motion-to-photon row of §11.2
18 Airway segmentation too shallow	Retired by D25: the airway is procedural
19 Over-claiming the hyperbolic airway	§10.2.1 (what is not claimed) and §10.2.3 (honest limits)
20 Free-account limits	Risk 27; the app ships with no entitlements (Appendix C.1)
21 Feels like a tour	§13.3: one rule per space

Appendix E. Glossary

One sentence per term. Canonical ATLAS names are in **bold** and are filed under the group they belong to.

E.1 Geometry and mathematics

Term	Meaning
3-4-5 walk	Walk s, turn left 90°, repeat: the loop closes after 3 corners on the sphere, 4 flat, 5 hyperbolic.
Aligned Cells (Mode 2, S3)	Hyperbolic mode in which square floor cells stay registered to the room and each shows one flat tile chosen by the path walked.
Angle defect	360° minus the sum of the angles meeting at a vertex; five squares give −90°, which is curvature concentrated at a point.
Branched cover	A space wrapped several times round a point, so one circuit does not bring you back.
Cone point	A point where the total angle around it is not 360°.
Curved Walk (Mode 1, S2)	Smooth $H^2 \times E$ drawn with the log-map warp; not registered to the real room.
Exp map / log map	Exp turns a direction and distance into the point reached; log is its inverse and is what the vertex warp uses.
Free rectangle	The largest rectangle of walkable floor aligned to the dominant wall direction, Wf × Df, already 0.30 m inside walls and furniture; the RoomModel stores it as `freeRect`.
Gauss–Bonnet	The angle excess of a geodesic polygon equals curvature × its area.
Geodesic	The straightest possible path in a space.
Holonomy	The rotation a direction picks up when carried round a closed loop (curvature × enclosed area; 72° per {4,5} tile).
Hyperbolic plane (H^2)	The surface of constant negative curvature, where area grows exponentially with radius.
Hyperboloid model	H^2 as the sheet $x^2 + y^2 - t^2 = -1$; the kernel stores poses as 3×3 matrices acting on it.
Hyperplane slice	The 3D cross-section of a 4D object cut by a flat 3D space.
Klein bottle	A square with one pair of edges glued straight and the other pair glued with a flip.
Klein model	Disc picture of H^2 in which geodesics are straight chords; used for maps and tile authoring only.
L, ρ, R	L is the hyperbolic curvature radius, in metres per hyperbolic unit (the Spec calls it R); ρ is hyperbolic distance in units of L; R in this plan is the sphere radius of the Small Planet.
Lap square	The square of side s walked in the 3-4-5 walk; it is centred on the anchor, at least 0.20 m inside the free rectangle and so at least 0.5 m from the nearest wall or scanned obstacle.
Loop-fit rule	s = clamp(min(W, D) − 1.0 m, 1.0 m, 3.2 m), with W and D measured wall to wall in an empty rectangular room (in code, from the RoomModel's `freeRect`: s = clamp(min(Wf, Df) − 0.4 m, 1.0 m, 3.2 m)); then L = s/1.0613 (hyperbolic) and R = 2s/π (spherical).
Lorentz boost	The hyperboloid-model matrix that translates along a geodesic; one is applied per walked step.
Poincaré disc	Disc picture of H^2 that keeps angles true; geodesics are arcs meeting the rim at 90°.
Projection (shadow)	Flattening a 4D object onto 3D by discarding one axis.
{p,q} tiling	Regular p-sided tiles, q at each corner; {4,5} is hyperbolic, {4,4} flat, {4,3} the cube on a sphere.
Registered	Said of a space in which the virtual walls and furniture are where the real ones are (§4.9); Curved Walk and the Small Planet are registered only until the first step.
Tesseract	The 4D cube: 8 cubic cells, 16 vertices.
Tile zero	The home tile of the {4,5} tiling; it carries the rebuilt room, and it is not the same thing as "the walker's current tile", the tile the walker stands on at that moment.
Torus (T^2)	A square with opposite edges glued without flips.
Universal cover	The unrolled, endlessly repeating space of which a glued space is the folded form.

E.2 Medical

Term	Meaning
4D-CT	A CT sorted into (here ten) 3D volumes, one per part of the breathing cycle.
Breath lane	Breath time laid along the longest clear floor segment, one breath per 1.5 m, repeating.
CT-to-body divergence	The mismatch between airway and lesion positions on the planning CT and in the patient during bronchoscopy.
Gating, duty cycle	Treating only inside a chosen phase window; duty cycle is the fraction of the breath the beam is on.
GTV	Gross tumour volume: the visible tumour on one image.
ITV / IGTV	The volume the tumour sweeps over a breath; IGTV is the union of the GTV over all phases.
Mid-ventilation	The phase at which the tumour is nearest its time-averaged position.
MIP	Maximum intensity projection: each voxel's brightest value over all phases.
Respiratory phase	Position within one breath, 0–100 %; a point on the circle S^1.
Worldtube	The tumour over one breath as a single 4D object in $\mathbb{R}^3 \times S^1$ (our term).

E.3 Engineering

Term	Meaning
ARWorldMap	ARKit's saved tracking map, reloaded to relocalise in a pre-scanned room.
ATLAS app	The Unity iOS app on the phone; renders everything and runs all safety.
`budget_room`	The performance fixture frozen at G0: a 4 × 5 m room with twelve stand-in objects of exactly 5,000 triangles each and no edge over 0.25 m; budgets are measured on it, and the three sample rooms stay for functional tests.
Bump Guard	On-phone safety: the static scan map and live LiDAR depth (with its dynamic layer) predict contact and give a warning cue and haptic; its thresholds and depth rate are constants in code, with no registry parameter, no off switch and no use of the network.
`BumpCue`	The component in `Atlas.Safety` that produces the warning cue (indicator, sound and haptic); its one local entry point `Fire(level, direction)` is called by Bump Guard and by the Console's STOP button (`P1-C14`).
Burst	Unity's compiler that turns C# jobs into optimised native code.
Console	The web page served by the Hub: tuner, operator buttons, telemetry.
Dynamic layer	Bump Guard's memory of obstacles that were not in the scan: 5 cm cells marked where live depth lands more than 0.3 m nearer than the static map predicts, each cleared only when it is later seen empty or after 30 s without a new hit.
Flat twin	The flat picture that rule 6 requires beside every curved or glued view: the space's flat map with the walker's path (Poincaré disc, lattice plan, cube net, room graph), drawn as the mini-map in the headset and beside the mirror on the Station; not the same as an input twin (a second way to do an action, rule 5) or the illustrative twin (the model nodule of the Lung Mission).
Forge	The Python service that turns scans and object photos into meshes.
GLB	Binary single-file glTF; the format of returned furniture meshes.
Home (S0)	The scanned room rebuilt 1:1, flat, registered to reality.
Hub	`atlas-hub` , the .NET 8 process that terminates the link and serves the Console.
LiDAR depth	Per-frame depth image from the phone's LiDAR, range about 5 m (R1).
LiteNetLib	MIT reliable-UDP library for .NET, used at both ends of the link.
Mesh4D	Tetrahedral mesh in $\mathbb{R}^4$ sliced by a hyperplane on the CPU.
MSAA	Multisample anti-aliasing; 4× to start.
Render scale	Fraction of the half-screen resolution (1398 × 1290) at which each eye is rendered before the final blit upscales it; 0.8 to start, which is an eye texture of 1118 × 1032, set by the texture size while URP Render Scale stays 1.0.
RoomModel	Scan output: free-floor grid, walkable polygon, free rectangle, walls, object boxes, anchor, s and L.
RoomPlan	Apple's room-scanning framework; through AR Foundation it yields furniture boxes and categories only.
SpaceLab	Host scene that loads one space module and exposes its parameters.
Spectator	Unity process on the Station that re-renders the walker's view from streamed state.

Term	Meaning
SRP Batcher	URP draw path that makes draws sharing a shader variant cheap.
Station	The Windows PC side: Hub, Spectator and Forge; the word always means this PC, and a stop of the Geometry Playground is an exhibit (E.4).
Stencil portal	A doorway drawn by masking pixels and drawing the far room inside the mask.

E.4 Scope, use cases and evaluation

Term	Meaning
Blade	The older documents' word for a use case; in this plan it survives only in the names `Atlas.Blades.*` and `Assets/Atlas/Blades/`.
Core / Extended / Stretch	Scope tiers of a task or work package: Core is required by a gate; Extended is built once the Core it extends has passed, and feeds no gate; Stretch is optional and feeds no gate.
Exhibit	One stop of the Geometry Playground: a Phase-1 space wrapped with its caption, tasks, equation card and events; never called a station, because the Station is the PC.
FMS	Fast Motion Sickness scale: one spoken score from 0 (no sickness) to 20, asked once a minute; the plan's comfort score.
SEQ	Single Ease Question: one rating from 1 (very difficult) to 7 (very easy), asked once per use case.

Appendix F. Sources

Every URL below appears in the research memos (R1–R4), the Build Pipeline or the Spec/Brief reference lists, except the TripoSG repository, which the Build Pipeline names without a link.

F.1 Non-Euclidean rendering and prior art

- Holonomy (TU Delft, Web3D 2024): nearest prior art; basis of Aligned Cells
- Non-Euclidean VR II: $H^2 \times E$: log-map drawing and relative motion for Curved Walk
- Weeks, body coherence: holonomy per area; track everything from one anchor
- HyperEngine: H2xE shader and tile mapper to port
- HyperRogue `vr.cpp`: headset holonomy modes; read for ideas only
- Tree structures for hyperbolic tessellations: canonical tile words and generation rules
- OEIS A377322: checks {4,5} tile counts per depth
- Slotboom 2023 thesis: stencil versus render-texture portal cost in VR
- ar-portal-arfoundation-urp: closest stencil-portal repo for AR Foundation and URP
- Engine4D: vertex-shader 4D slicer above 50k tetrahedra
- Zucconi, rendering 4D objects: tetrahedron slicing explained step by step

F.2 Walking and comfort

- Steinicke 2010: detection thresholds for rotation, distance, curvature
- Widdowson and Wang 2022: people home as if Euclidean; add beacon
- Auda, "If The Map Fits!": minimaps distract; show the flat twin late
- Suma, impossible spaces: overlap limits for Impossible Rooms
- Tea For God devlog: worlds generated to fit any play area
- VRoamer: real obstacles become virtual objects

F.3 Scanning and ARKit

- ARKit bounding boxes: RoomPlan object boxes through `ARBoundingBoxManager`
- ARKit meshing: classified LiDAR mesh for walls and floor
- RoomPlan: Swift plugin route for walls, doors, windows
- WWDC22 RoomPlan session: room size, light and scan-length limits
- Scene depth: the LiDAR depth frames Bump Guard reads
- Saving and loading world data: ARWorldMap for the pre-scan fast path
- Forum 763400: RoomPlan drops `sceneDepth` until configuration is re-run
- Forum 827240: iOS 26.4 tracking drift; check phone version

F.4 Networking

- LiteNetLib: transport library, wire format, channel types
- TN3179 local network privacy: iOS permission prompt and unicast rules
- NSAllowsLocalNetworking: plain HTTP to the Station allowed
- libusbmuxd: `iproxy` is TCP only; cannot carry LiteNetLib

F.5 Performance

- SRP Batcher: batching rules behind the one-shader draw path
- `Graphics.RenderMeshInstanced`: instanced tile shells if render thread binds
- Meta Unity performance guide: mobile-VR draw-call and triangle guidance

- Best darn grid shader: `fwidth` tile lines that do not alias

F.6 Medical 4D and data

- TCIA 4D-Lung: the real case, patient 101_HM10395; cited as Hugo et al. 2016, doi 10.7937/K9/TCIA.2016.ELN8YGLE, plus the TCIA citation
- AAPM TG-76: respiratory motion management; gating definitions
- Seppenwoolde 2002: how far lung tumours move, and the 1 to 5 mm loop between breathing in and out
- Liu 2007: share of tumours that move more than 5 and 10 mm
- Chen 2004: helical-CT motion artefacts; the tilted-slice demo
- Underberg 2004: a uniform 10 mm margin against an individual ITV
- Ezhil 2009: IGTV from all phases versus from MIP
- Wolthaus 2006: mid-ventilation and time-averaged position
- Chen 2015: nodule displacement between the planning scan and tidal breathing (17.6 mm)
- Ravikumar 2023: CT-to-body divergence in navigation bronchoscopy
- AeroPath: optional real proximal airway tree
- Kitaoka 1999: procedural airway rules; verify equations before quoting

F.7 Assets and licences

- TripoSG: the Forge's local image-to-3D model on the RTX 5070; MIT (BP §0.5)
- Meshy: a candidate for the Forge's hosted image-to-3D fallback (open item 8)
- Hunyuan3D-2 licence: territory and use restrictions to read
- Kenney: CC0 props and audio for the fallback set

F.8 Licence notes

Item	Licence	What we may do
HyperRogue	GPL-2.0	Read for ideas only; no code enters the repository.
HyperEngine, Engine4D, LiteNetLib, TripoSG	MIT	Port or use; keep the copyright notice.
TCIA 4D-Lung	CC BY 3.0	Use with credit on screen and in the README, saying that the meshes are derived; the data-use terms are acknowledged and the citation recorded in the licence ledger (open item 18).
AeroPath	CC BY 4.0 (Zenodo; the GitHub README says MIT)	Attribute under CC BY 4.0 (R2).
ATM'22	Signed agreement; redistribution prohibited	Do not ship or commit.
Hunyuan3D	Excludes the EU, UK and South Korea; bans medical decision use; outputs labelled AI-generated	Off the default path (BP §0.5).
Meshy free tier	CC BY 4.0	Credit Meshy for any free-tier mesh.
Unity Asset Store free packs	Not redistributable	Keep out of the public repository (BP §3d).